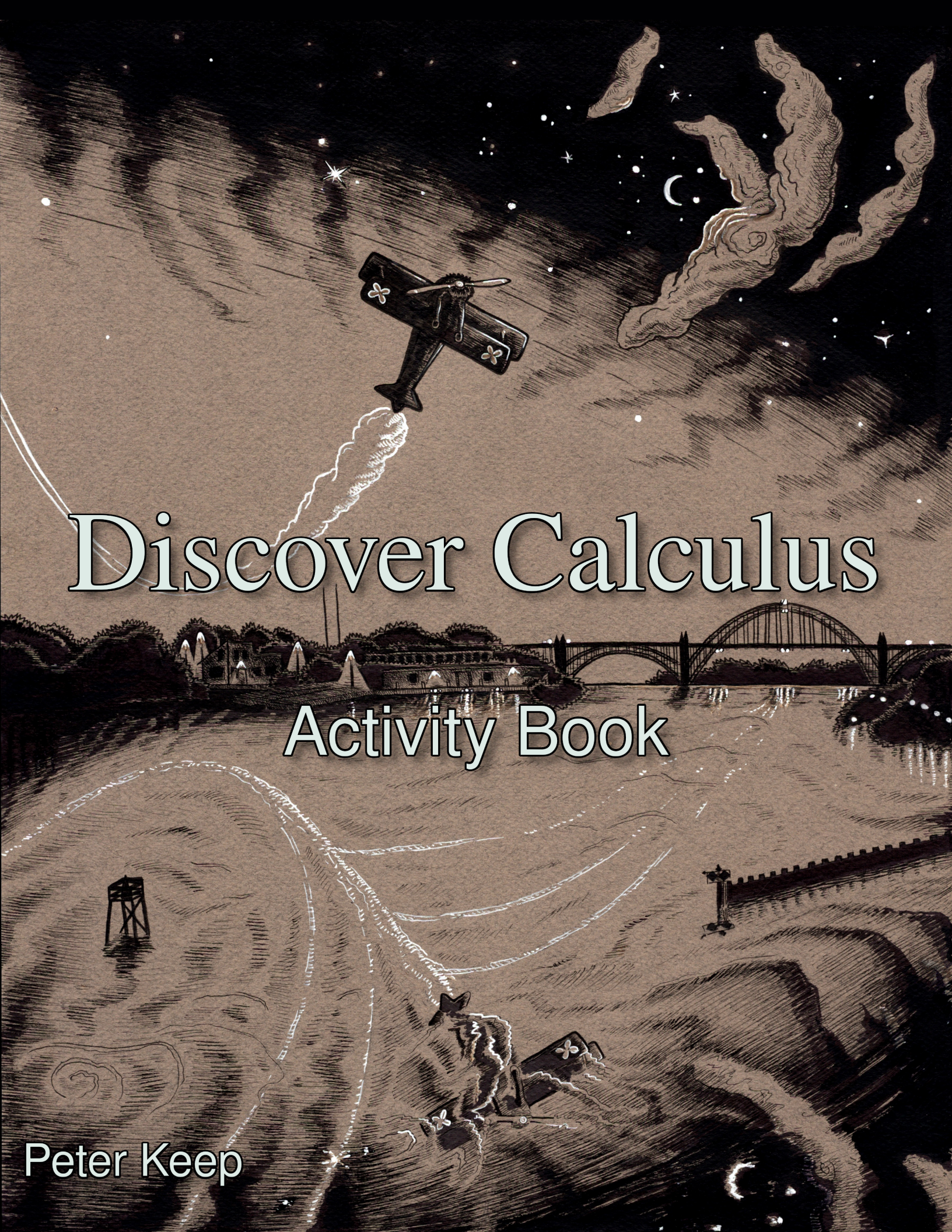

Discover Calculus

Activity Book

Peter Keep

Discover Calculus - Activity Book

All of the Activities from Each Chapter

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All of the Activities from Each Chapter

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Chapter 1

Limits

1.1 The Definition of the Limit

Activity 1.1.1 Close or Not?

We're going to try to think how we might define "close"-ness as a property, but, more importantly, we're going to try to realize the struggle of creating definitions in a mathematical context. We want our definition to be meaningful, precise, and useful, and those are hard goals to reach! Coming to some agreement on this is a particularly tricky task.

(a) For each of the following pairs of things, decide on which pairs you would classify as "close" to each other.

- You, right now, and the nearest city with a population of 1 million or higher
- Your two nostrils
- You and the door of the room you are in
- You and the person nearest you
- The floor of the room you are in and the ceiling of the room you are in

(b) For your classification of "close," what does "close" mean? Finish the sentence: A pair of objects are *close* to each other if...

(c) Let's think about how close two things would have to be in order to satisfy everyone's definition of "close." Pick two objects that you think everyone would agree are "close," if by "everyone" we meant:

- All of the people in the building you are in right now.
- All of the people in the city that you are in right now.
- All of the people in the country that you are in right now.
- Everyone, everywhere, all at once.

(d) Let's put ourselves into the context of functions and numbers. Consider the linear function $y = 4x - 1$. Our goal is to find some x-values that, when we put them into our function, give us y-value outputs that are "close" to the number 2. You get to define what close means.

First, evaluate $f(0)$ and $f(1)$. Are these y-values "close" to 2, in your definition of "close?"

(e) Pick five more, different, numbers that are "close" to 2 in your definition of "close." For each one, find the x-values that give you those y-values.

(f) How far away from $x = \frac{3}{4}$ can you go and still have y-value outputs that are "close" to 2?

To wrap this up, think about your points that you have: you have a list of x-coordinates that are clustered around $x = \frac{3}{4}$ where, when you evaluate $y = 4x - 1$ at those x-values, you get y-values that are "close" to 2. Great!
Do you think others will agree? Or do you think that other people might look at your list of y-values and decide that some of them *aren't* close to 2?
Do you think you would agree with other peoples' lists? Or you do think that you might look at other peoples' lists of y-values and decide that some of them *aren't* close to 2?

Activity 1.1.2 Approximating Limits.

For each of the following graphs of functions, approximate the limit in question. When you do so, approximate the values of the relevant one-sided limits as well.

(a) Approximate $\lim_{x\to 1} f(x)$ using the graph of the function $f(x)$ below.

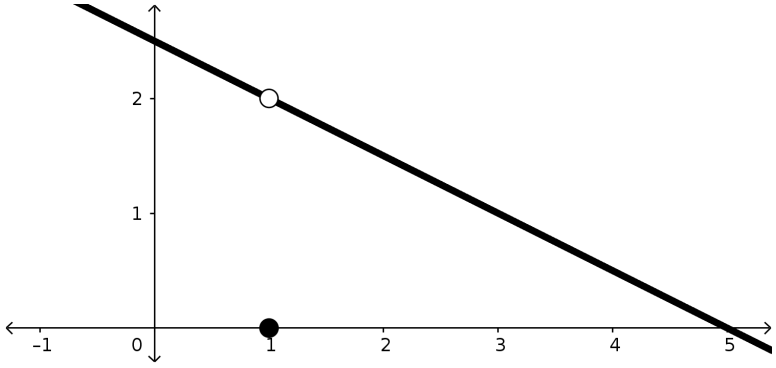


Figure 1.1.3

(b) Approximate $\lim_{x\to 2} g(x)$ using the graph of the function $g(x)$ below.

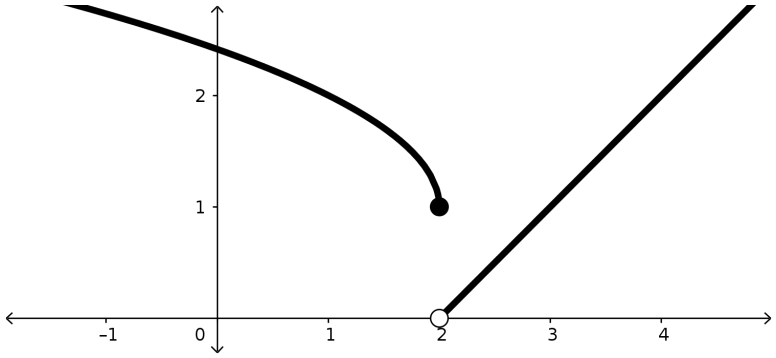


Figure 1.1.4

(c) Approximate the following three limits using the graph of the function $h(x)$ below.

- $\lim_{x\to -1} h(x)$
- $\lim_{x\to 0} h(x)$
- $\lim_{x\to 2} h(x)$

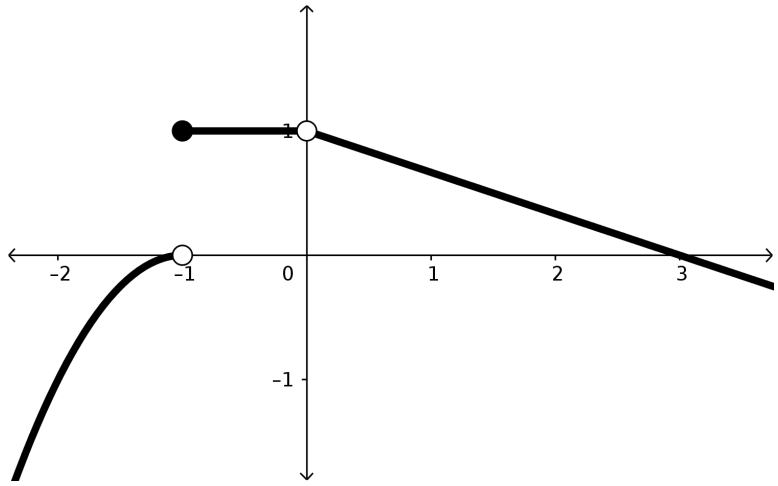


Figure 1.1.5

(d) Why do we say these are "approximations" or "estimations" of the limits we're interested in?

(e) Are there any limit statements that you made that you are 100% confident in? Which ones?

(f) Which limit statements are you least confident in? What about them makes them ones you aren't confident in?

(g) What extra details would you like to see to increase the confidence in your estimations? Are there changes we could make to the way these functions are represented that would make these approximations better or easier to make?

Activity 1.1.6 Approximating Limits Numerically.

For each of the following tables of function values, approximate the limit in question. When you do so, approximate the values of the relevant one-sided limits as well.

(a) Approximate $\lim_{x\to 1} f(x)$ using the table of values of $f(x)$ below.

Table 1.1.7

x	0.5	0.9	0.99	1	1.01	1.1	1.5
$f(x)$	8.672	9.2	9.0001	-7	8.9998	9.5	7.59

(b) Approximate $\lim_{x\to -3} g(x)$ using the table of values of $g(x)$ below.

Table 1.1.8

x	-3.5	-3.1	-3.01	-3	-2.99	-2.9	-2.5
$g(x)$	-4.41	-3.89	-4.003	-4	7.035	2.06	-4.65

(c) Approximate $\lim_{x\to \pi} h(x)$ using the table of values of $h(x)$ below.

Table 1.1.9

x	3.1	3.14	3.141	π	3.142	3.15	3.2
$h(x)$	6	6	6	undefined	5.915	6.75	8.12

(d) Are you 100% confident about the existence (or lack of existence) of any of these limits?

(e) What extra details would you like to see to increase the confidence in your estimations? Are there changes we could make to the way these functions are represented that would make these approximations better or easier to make?

Practice Problems

1. Explain in your own words the meaning of:

$$\lim_{x\to a^-} f(x) = L.$$

2. Explain in your own words the meaning of:

$$\lim_{x\to a^+} f(x) = L.$$

3. Explain in your own words the meaning of:

$$\lim_{x\to a} f(x) = L.$$

4. Say we know that $\lim_{x\to 3^-} f(x) = 2$ and $\lim_{x\to 3^+} f(x) = 2$. What do we know (specifically or in general, if anything) about each of the following?

(a) $f(3)$

(b) $f(2.999)$

(c) $f(3.001)$

(d) $\lim_{x\to 3} f(x)$

5. Which of the following is possible? Explain why or why not, and any other conclusions that we can draw.

(a) For some function $f(x)$, $\lim_{x\to 5} f(x) = 6$ and $f(5) = -3$

(b) For some function $g(x)$, $\lim_{x\to 4^-} g(x) = -\frac{3}{2}$ and $\lim_{x\to 4^+} g(x) = \frac{4}{7}$.

(c) For some function $\ell(t)$, $\lim_{t\to \alpha} \ell(t) = 2$ and $\lim_{t\to \alpha^+} \ell(t) = 1$.

(d) For some function $r(\theta)$, $\lim_{\theta\to 0} r(\theta)$ does not exist, $\lim_{\theta\to 0^-} r(\theta) = \pi$, and $\lim_{\theta\to 0^+} r(\theta) = -\frac{\pi}{2}$.

(e) For some function $j(w)$, $j(4) = \pi$ while $\lim_{w\to 4} j(w)$ does not exist.

6. Fill in the following tables in order to satisfy the requirements listed. Afterwards, include a sentence or two justifying your choices.

(a) *Requirements:* $\lim_{x\to 1} f(x) = 3$

x	_____	0.93	_____	1	_____	_____	1.04
$f(x)$	_____	_____	_____	_____	_____	_____	_____

(b) *Requirements:* $\lim_{x\to -5^-} f(x) = 2$, $f(-5) = 6$, and $\lim_{x\to -5} f(x)$ doesn't exist.

x	-5.2	_____	_____	-5	_____	4.98	_____
$f(x)$	_____	_____	_____	_____	_____	_____	_____

(c) *Requirements:* $f(7)$ does not exist and $\lim_{x\to 7} f(x) = 3$

x	_____	_____	6.985	7	_____	_____	7.002
$f(x)$	_____	_____	_____	_____	_____	_____	_____

(d) *Requirements:* $\lim_{x\to 0^-} f(x) = \pi$ and $\lim_{x\to 0^+} f(x) = e$.

x	_____	-0.14	_____	0	_____	_____	0.5
$f(x)$	_____	_____	_____	_____	_____	_____	_____

7. From the following tables, estimate/report each of the requested values. Explain your choices.

(a) *Requested:* $\lim_{x\to 1^-} f(x)$, $\lim_{x\to 1^+} f(x)$, $\lim_{x\to 1} f(x)$, and $f(1)$

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	2.4	2.48	2.4998	9	2.5004	2.52	2.8

(b) *Requested:* $\lim_{x\to 8^-} f(x)$, $\lim_{x\to 8^+} f(x)$, $\lim_{x\to 8} f(x)$, and $f(8)$

x	7.9	7.99	7.999	8	7.001	7.01	7.1
$f(x)$	-1.5	-1.9	-1.999	-2	7.0001	7.2	7.5

(c) *Requested:* $\lim_{x\to \pi^-} f(x)$, $\lim_{x\to \pi^+} f(x)$, $\lim_{x\to \pi} f(x)$, and $f(\pi)$

x	3.1	3.14	3.141	π	3.142	3.15	3.2
$f(x)$	-3	-3	-3	does not exist	-3	-3	-3

8. For each of the listed requirements, sketch a graph of a function that satisfies each. Afterwards, include a sentence or two justifying you sketch.

(a) *Requirements:* $f(6) = 0$, $\lim_{x\to 6} f(x) = -2$, $\lim_{x\to -2^-} f(x) = 1$, and $\lim_{x\to -2} f(x)$ does not exist.

(b) *Requirements:* $\lim_{\omega\to 0} \rho(\omega) = 8$, $\lim_{\omega\to 2} \rho(\omega) = -2$, and $\rho(2)$ does not exist.

(c) *Requirements:* $\lim_{t\to -3^-} q(t) = 0$, $\lim_{t\to -3^+} q(t) = 4$, $\lim_{t\to -1} q(t) = 9$, and $q(-1) = 9$.

9. From the graph of $f(x)$ below, estimate of each of the requested values. Explain each of your choices.

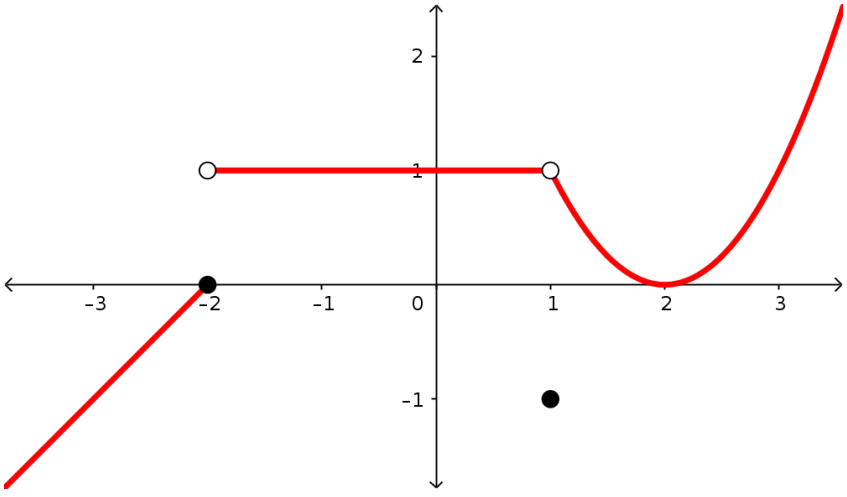


Figure 1.1.10 The function $f(x)$.

(a) $\lim_{x\to -2^-} f(x)$

(b) $\lim_{x\to -2^+} f(x)$

(c) $\lim_{x \to -2} f(x)$

(d) $\lim_{x \to 0^-} f(x)$

(e) $\lim_{x \to 0^+} f(x)$

(f) $\lim_{x \to 0} f(x)$

(g) $\lim_{x \to 1^-} f(x)$

(h) $\lim_{x \to 1^+} f(x)$

(i) $\lim_{x \to 1} f(x)$

(j) $\lim_{x \to 2^-} f(x)$

(k) $\lim_{x \to 2^+} f(x)$

(l) $\lim_{x \to 2} f(x)$

1.2 Evaluating Limits

Activity 1.2.1 From Estimating to Evaluating Limits (Part 1).

Let's consider the following graphs of functions $f(x)$ and $g(x)$.

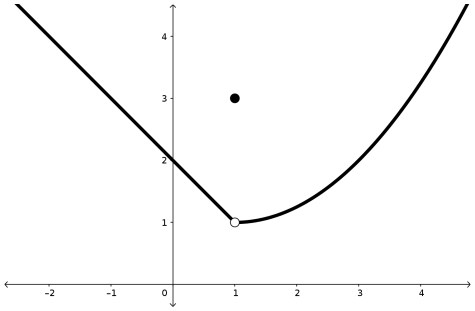


Figure 1.2.2 Graph of the function $f(x)$.

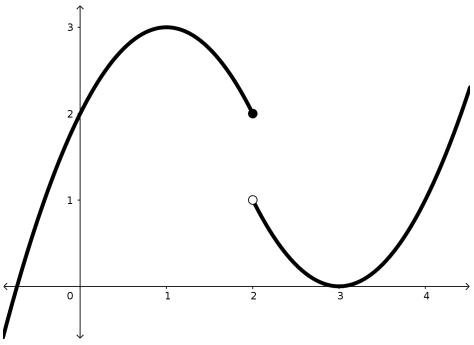


Figure 1.2.3 Graph of the function $g(x)$.

(a) Estimate the values of the following limits. If you believe that the limit does not exist, say so and explain why.

- $\lim_{x\to 1^-} f(x)$
- $\lim_{x\to 1^+} f(x)$
- $\lim_{x\to 1} f(x)$

(b) Estimate the values of the following limits. If you believe that the limit does not exist, say so and explain why.

- $\lim_{x\to 2^-} g(x)$
- $\lim_{x\to 2^+} g(x)$
- $\lim_{x\to 2} g(x)$

(c) Find the values of $f(1)$ and $g(2)$.

(d) For the limits and function values above, which of these are you most confident in? What about the limit, function value, or graph of the function makes you confident about your answer?

Similarly, which of these are you the least confident in? What about the limit, function value, or graph of the function makes you not have confidence in your answer?

Activity 1.2.4 From Estimating to Evaluating Limits (Part 2).

Let's consider the following graphs of functions $f(x)$ and $g(x)$, now with the added labels of the equations defining each part of these functions.

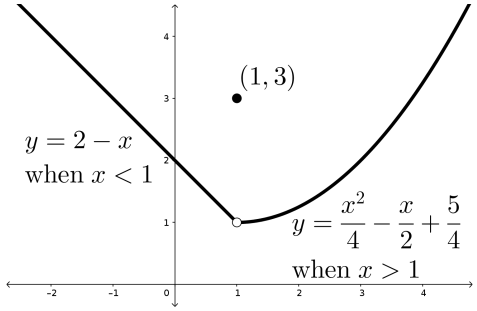


Figure 1.2.5 Graph of the function $f(x)$.

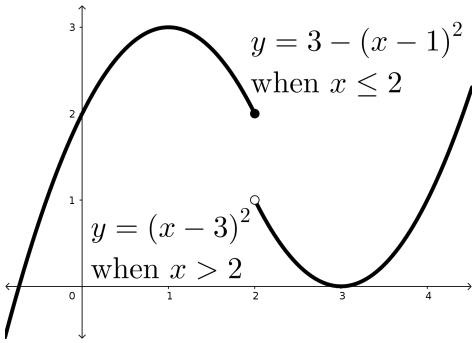


Figure 1.2.6 Graph of the function $g(x)$.

(a) Estimate the values of the following limits. If you believe that the limit does not exist, say so and explain why.

- $\lim_{x\to 1^-} f(x)$
- $\lim_{x\to 1^+} f(x)$
- $\lim_{x\to 1} f(x)$

(b) Estimate the values of the following limits. If you believe that the limit does not exist, say so and explain why.

- $\lim_{x\to 2^-} g(x)$
- $\lim_{x\to 2^+} g(x)$
- $\lim_{x\to 2} g(x)$

(c) Does the addition of the function rules change the level of confidence you have in these answers? What limits are you more confident in with this added information?

(d) Consider these functions without their graphs:

$$f(x) = \begin{cases} 2-x & \text{when } x < 1 \\ 3 & \text{when } x = 1 \\ \frac{x^2}{4} - \frac{x}{2} + \frac{5}{4} & \text{when } x > 1 \end{cases}$$

$$g(x) = \begin{cases} 3-(x-1)^2 & \text{when } x \le 2 \\ (x-3)^2 & \text{when } x > 2 \end{cases}$$

Find the limits $\lim_{x\to 1} f(x)$ and $\lim_{x\to 2} g(x)$. Compare these values of $f(1)$ and $g(2)$: are they related at all?

Activity 1.2.7 Combinations of Functions.

We want to remind ourselves how we can combine functions using different operations, and how we might find outputs based on the different combinations. Our goal is to then think about how this might work with limits: how can we summarize the behavior of combinations of functions around some point?

Let's consider some functions $f(x) = x^2 + 3$ and $g(x) = x - \frac{1}{x}$. We'll say that the domain of both functions is $(0, \infty)$ for our own convenience.

(a) Let's consider the function $h(x) = f(x) + g(x)$. Describe at least two different ways of finding the value of $h(2)$.

(b) If we instead define the function $h(x) = f(x) - g(x)$, how would you describe at least two different ways of finding the value of $h(2)$?

(c) What about a scaled version of one of these functions? If we let $h(x) = 4f(x)$ and $j(x) = \frac{g(x)}{3}$, can you describe more than one way to find the value of $h(3)$ and $j(3)$?

(d) You can probably guess where we're going: we're going to define a function that is the product of f and g: $h(x) = f(x) \cdot g(x)$. Describe more than one way of evaluating $h(4)$.

(e) And finally, let's define $h(x) = \frac{f(x)}{g(x)}$. Now describe more than one way of finding $h(4)$.

(f) If $h(x) = \frac{f(x)}{g(x)}$, then are there any x-values that are in the domain of f and g (the domain is $x > 0$) that $h(x)$ cannot be defined for? Why?

Ok, we can confront this big idea: when we combine functions, we can either evaluate the combination of the functions at some x-value or evaluate each function separately and just combine the answers! Of course, there are some limitations (like when the combination isn't nicely defined because of division by 0 or something else), but this is a good framework to move forward with!

Activity 1.2.8 Limits of Polynomial Functions.

We're going to use a combination of properties from Theorem 1.2.8 and Theorem 1.2.9 to think a bit more deeply about polynomial functions. Let's consider a polynomial function:

$$f(x) = 2x^4 - 4x^3 + \frac{x}{2} - 5$$

(a) We're going to evaluate the limit $\lim_{x\to 1} f(x)$. First, use the properties from Theorem 1.2.8 to rewrite this limit as 4 different limits added or subtracted together.

(b) Now, for each of these limits, rewrite them as products of things until you have only limits of constants and identity functions, as in Theorem 1.2.9. Evaluate your limits.

(c) Based on the definition of a limit (Definition 1.1.2), we normally say that $\lim_{x\to 1} f(x)$ is not dependent on the value of $f(1)$. Why do we say this?

(d) Compare the values of $\lim_{x\to 1} f(x)$ and $f(1)$. Why do these values feel connected?

(e) Come up with a new polynomial function: some combination of coefficients with x's raised to natural number exponents. Call your new polynomial function $g(x)$. Evaluate $\lim_{x\to -1} g(x)$ and compare the value to $g(-1)$. Explain why these values are the same.

(f) Explain why, for any polynomial function $p(x)$, the limit $\lim_{x\to a} p(x)$ is the same value as $p(a)$.

Practice Problems

1. Given $\lim_{x\to 3} f(x) = 5$ and $\lim_{x\to 3} g(x) = -2$, evaluate the following limits. If the limit doesn't exist, explain why. Write out a few steps and explanations to justify your work.

(a) $\lim_{x\to 3}\left(6f(x) - \frac{g(x)}{3}\right)$

(b) $\lim_{x\to 3}(f(x))^2 g(x)$

(c) $\lim_{x\to 3^-}\left(\frac{4g(x)}{f(x)} + 3f(x)\right)$

2. Evaluate each limit. Justify your answers.

(a) $\lim_{x\to 0}(4x^3 - 6x^2 + 7x - 10)$

(b) $\lim_{x\to -2}(9 - 3x + x^2 + 3x^3 + x^4)$

(c) $\lim_{t\to a}\left(9t^2 + 3at - 1\right)$ where a is some real number

(d) $\lim_{s\to 1}\left(\frac{5s^2 - 6s + 1}{s^2 - 4}\right)$

(e) $\lim_{t\to 2}\left(\frac{4t - 5}{6 + t^2}\right)$

(f) $\lim_{z\to 2}\left(\frac{z^2 + z - 4}{z + 2}\right)$

3. Evaluate each limit. If the limit does not exist, explain why not.

(a) Let $f(x) = \begin{cases} 3x - 2 & \text{if } x < -1 \\ x^2 + x - 4 & \text{if } x \geq -1 \end{cases}$.

Evaluate $\lim_{x\to -1^-} f(x)$, $\lim_{x\to -1^+} f(x)$, and $\lim_{x\to -1} f(x)$.

(b) Let $g(x) = \begin{cases} 6 + x & \text{if } x < -3 \\ 6 & \text{if } x = -3 \\ \frac{2x+15}{3} & \text{if } x > -3 \end{cases}$.

Evaluate $\lim_{x\to -3^-} g(x)$, $\lim_{x\to -3^+} g(x)$, and $\lim_{x\to -3} g(x)$.

(c) Let $s(t) = \begin{cases} t^2 + 1 & \text{if } t < 1 \\ 1 - t^2 & \text{if } t \geq 1 \end{cases}$.

Evaluate $\lim_{t\to 1^-} s(t)$, $\lim_{t\to 1^+} s(t)$, and $\lim_{t\to 1} s(t)$

(d) Let $r(\theta) = \begin{cases} \theta - 1 + \theta^3 & \text{if } \theta < 2 \\ 3\theta + 2 & \text{if } \theta \geq 2 \end{cases}$.

Evaluate $\lim_{\theta\to 2^-} r(\theta)$, $\lim_{\theta\to 2^+} r(\theta)$, and $\lim_{\theta\to 2} r(\theta)$.

1.3 First Indeterminate Forms

Activity 1.3.1 Limits of (Slightly) Different Functions.

(a) Using the graph of $f(x)$ below, approximate $\lim_{x\to 1} f(x)$.

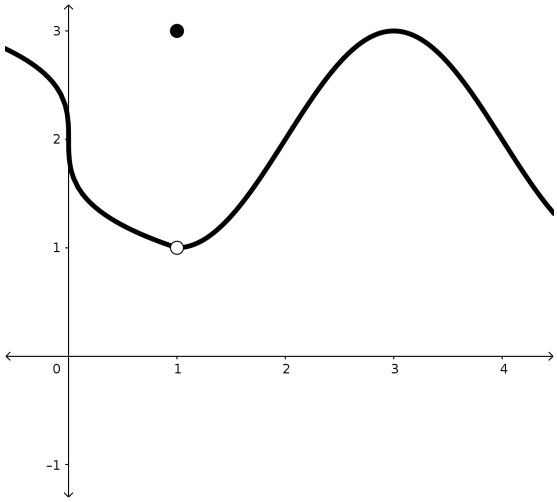


Figure 1.3.2

(b) Using the graph of the slightly different function $g(x)$ below, approximate $\lim_{x\to 1} g(x)$.

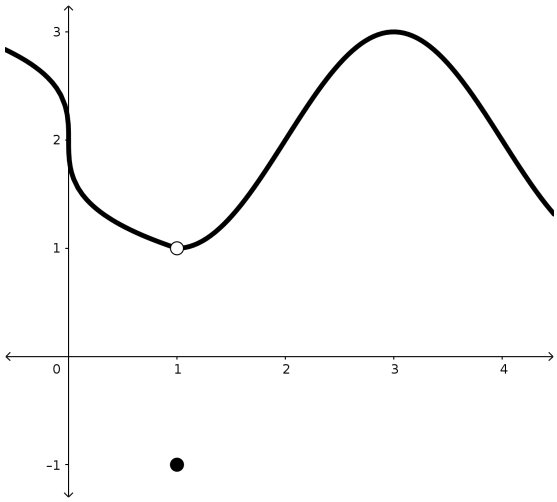


Figure 1.3.3

(c) Compare the values of $f(1)$ and $g(1)$ and discuss the impact that this difference had on the values of the limits.

(d) For the function $r(t)$ defined below, evaluate the limit $\lim_{t\to 4} r(t)$.

$$r(t) = \begin{cases} 2t - \frac{4}{t} & \text{when } t < 4 \\ 8 & \text{when } t = 4 \\ t^2 - t - 5 & \text{when } t > 4 \end{cases}$$

(e) For the slightly different function $s(t)$ defined below, evaluate the limit $\lim_{t\to 4} s(t)$.

$$s(t) = \begin{cases} 2t - \frac{4}{t} & \text{when } t \leq 4 \\ t^2 - t - 5 & \text{when } t > 4 \end{cases}$$

(f) Do the changes in the way that the function was defined impact the evaluation of the limit at all? Why not?

Activity 1.3.4

(a) Were going to evaluate $\lim_{x\to 3}\left(\dfrac{x^2-7x+12}{x-3}\right)$.

- First, check that we get the indeterminate form $\frac{0}{0}$ when $x \to 3$.
- Now we want to find a new function that is equivalent to $f(x) = \dfrac{x^2-7x+12}{x-3}$ for all x-values other than $x = 3$. Try factoring the numerator, $x^2 - 7x + 12$. What do you notice?
- "Cancel" out any factors that show up in the numerator and denominator. Make a special note about what that factor is.
- This function is equivalent to $f(x) = \dfrac{x^2-7x+12}{x-3}$ except at $x = 3$. The difference is that this function has an actual function output at $x = 3$, while $f(x)$ doesn't. Evaluate the limit as $x \to 3$ for your new function.

(b) Now we'll evaluate a new limit: $\lim_{x\to 1}\left(\dfrac{\sqrt{x^2+3}-2}{x^2-5x+4}\right)$.

- First, check that we get the indeterminate form $\frac{0}{0}$ when $x \to 1$.
- Now we want a new function that is equivalent to $g(x) = \dfrac{\sqrt{x^2+3}-2}{x^2-5x+4}$ for all x-values other than $x = 1$. Try multiplying the numerator and the denominator by $(\sqrt{x^2+3}+2)$. We'll call this the "conjugate" of the numerator.
- In your multiplication, confirm that $(\sqrt{x^2+3}-2)(\sqrt{x^2+3}+2) = (x^2+3)-4$.
- Try to factor the new numerator and denominator. Do you notice anything? Can you "cancel" anything? Make another note of what factor(s) you cancel.
- This function is equivalent to $g(x) = \dfrac{\sqrt{x^2+3}-2}{x^2-5x+4}$ except at $x = 1$. The difference is that this function has an actual function output at $x = 1$, while $g(x)$ doesn't. Evaluate the limit as $x \to 1$ for your new function.

(c) Our last limit in this activity is going to be $\lim_{x\to -2}\left(\dfrac{3-\frac{3}{x+3}}{x^2+2x}\right)$.

- Again, check to see that we get the indeterminate form $\frac{0}{0}$ when $x \to -2$.
- Again, we want a new function that is equivalent to $h(x) = \dfrac{3-\frac{3}{x+3}}{x^2+2x}$ for all x-values other than $x = -2$. Try completing the subtraction in the numerator, $3 - \frac{3}{x+3}$, using "common denominators."
- Try to factor the new numerator and denominator(s). Do you notice anything? Can you "cancel" anything? Make another note of what factor(s) you cancel.
- For the final time, we've found a function that is equivalent to $h(x) = \dfrac{3-\frac{3}{x+3}}{x^2+2x}$ except at $x = -2$. The difference is that this function has an actual function output at $x = -2$, while $h(x)$ doesn't. Evaluate the limit as $x \to -2$ for your new function.

(d) In each of the previous limits, we ended up finding a factor that was shared in the numerator and denominator to cancel. Think back to each example and the factor you found. Why is it clear that these *must* have been the factors we found to cancel?

(e) Let's say we have some new function $f(x)$ where $\lim_{x\to 5} f(x) \overset{?}{\to} \frac{0}{0}$. You know, based on these examples, that you're going to apply *some* algebra trick to rewrite your function, factor, and cancel. Can you predict what you will end up looking for to cancel in the numerator and denominator? Why?

Activity 1.3.5

Let's consider a new limit:

$$\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta}.$$

This one is strange!

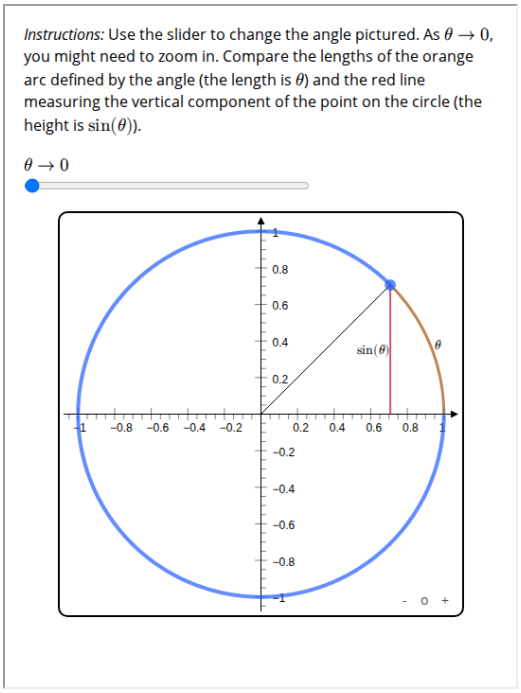
(a) Notice that this function, $f(\theta) = \dfrac{\sin(\theta)}{\theta}$, is resistent to our algebra tricks:

- There's nothing to "factor" here, since our trigonometric function is not a polynomial.
- We can't use a trick like the "conjugate" to multiply and rewrite, since there's no square roots and also only one term in the numerator.
- There aren't any fractions that we can combine by addition or subtraction.

(b) Be frustrated at this new limit for resisting our algebra tricks.

(c) Now let's think about the meaning of $\sin(\theta)$ and even θ in general. In this text, we will often use Greek letters, like θ, to represent angles. In general, these angles will be measured in radians (unless otherwise specified). So what does the sine function *do* or *tell us*? What is a radian?

(d) Let's visualize our limit, then, by comparing the length of the arc and the height of the point as $\theta \to 0$.



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(e) Explain to yourself, until you are absolutely certain, why the two lengths *must* be the same in the limit as $\theta \to 0$. What does this mean about $\lim_{\theta \to 0} \dfrac{\sin(\theta)}{\theta}$?

Practice Problems

1. Explain, in your own words, why Theorem 1.3.4 is true.

2. Consider the following limit:

$$\lim_{x\to 3}\left(\frac{2x^2-5x-3}{x^2+5x-24}\right).$$

(a) Confirm that this limit has an indeterminate form.

(b) Evaluate the limit.

(c) When you were evaluating the limit, you likely "cancelled" a factor of $(x-3)$ from the numerator and denominator. Why might you have known, before factoring anything, that $(x-3)$ would be a factor shared in the numerator and denominator?

For instance, how did you know it wasn't going to be $(x-4)$ or $(x+1)$ or something else?

3. Consider the following limit:

$$\lim_{x\to 4}\left(\frac{\frac{2}{x+1}-\frac{x-2}{5}}{x-4}\right).$$

(a) Confirm that this limit has an indeterminate form.

(b) Evaluate the limit.

(c) When you were evaluating the limit, you likely "cancelled" a factor of $(x-4)$ from the numerator and denominator. Why might you have known, before factoring anything, that $(x-4)$ would be a factor shared in the numerator and denominator?

For instance, how did you know it wasn't going to be $(x-3)$ or $(x+1)$ or something else?

4. Use the algebra tricks from Algebra Tricks for Indeterminate Forms to evaluate each limit.

(a) $\lim_{x\to 2}\left(\frac{x^2-4}{x^2+3x-10}\right)$

(b) $\lim_{x\to -1}\left(\frac{2x^2+x-1}{x+1}\right)$

(c) $\lim_{x\to 4}\left(\frac{4-x}{x-4}\right)$

(d) $\lim_{x\to 1}\left(\frac{\sqrt{x+3}-2}{x-1}\right)$

(e) $\lim_{x\to 2}\left(\frac{\sqrt{x^2-1}-\sqrt{x+1}}{x-2}\right)$

(f) $\lim_{h\to 0}\left(\frac{\sqrt{a+h}-\sqrt{a}}{h}\right)$ where a is some non-negative real number

(g) $\lim_{t\to 3}\left(\frac{\frac{1}{t}-\frac{1}{3}}{t-3}\right)$

(h) $\lim_{t\to 6}\left(\frac{\frac{t+1}{t-1}-\frac{7}{11-t}}{t-6}\right)$

(i) $\lim_{h\to 0}\left(\frac{\frac{1}{a+h}-\frac{1}{a}}{h}\right)$ where a is some real number

1.4 Limits Involving Infinity

Activity 1.4.1 What Happens When We Divide by 0?

First, let's make sure we're clear on one thing: there is no real number that is represented as some other number divided by 0.
When we talk about "dividing by 0" here (and in Section 1.3), we're talking about the behavior of some function in a limit. We want to consider what it might look like to have a function that involves division where the denominator *gets arbitrarily close to 0* (or, the limit of the denominator is 0).

(a) Remember when, once upon a time, you learned that dividing a number by a fraction is the same as multiplying the first number by the reciprocal of the fraction? Why is this true?

(b) What is the relationship between a number and its reciprocal? How does the size of a number impact the size of the reciprocal? Why?

(c) Consider $12 \div N$. What is the value of this division problem when:

- $N = 6$?
- $N = 4$?
- $N = 3$?
- $N = 2$?
- $N = 1$?

(d) Let's again consider $12 \div N$. What is the value of this division problem when:

- $N = \frac{1}{2}$?
- $N = \frac{1}{3}$?
- $N = \frac{1}{4}$?
- $N = \frac{1}{6}$?
- $N = \frac{1}{1000}$?

(e) Consider a function $f(x) = \frac{12}{x}$. What happens to the value of this function when $x \to 0^+$? Note that this means that the x-values we're considering most are very small and positive.

(f) Consider a function $f(x) = \frac{12}{x}$. What happens to the value of this function when $x \to 0^-$? Note that this means that the x-values we're considering most are very small and negative.

Activity 1.4.2 What Happens When We Divide by Infinity?

Again, we need to start by making something clear: if we were really going to try divide some real number by infinity, then we would need to rebuild our definition of what it means to divide. In the context we're in right now, we only have division defined as an operation for real (and maybe complex) numbers. Since infinity is neither, then we will not literally divide by infinity.
When we talk about "dividing by infinity" here, we're again talking about the behavior of some function in a limit. We want to consider what it might look like to have a function that involves division where the denominator *gets arbitrarily large (positive or negative)* (or, the limit of the denominator is infinite).

(a) Let's again consider $12 \div N$. What is the value of this division problem when:

- $N = 1$?
- $N = 6$?
- $N = 12$?
- $N = 24$?
- $N = 1000$?

(b) Let's again consider $12 \div N$. What is the value of this division problem when:

- $N = -1$?
- $N = -6$?
- $N = -12$?
- $N = -24$?
- $N = -1000$?

(c) Consider a function $f(x) = \frac{12}{x}$. What happens to the value of this function when $x \to \infty$? Note that this means that the x-values we're considering most are very large and positive.

(d) Consider a function $f(x) = \frac{12}{x}$. What happens to the value of this function when $x \to -\infty$? Note that this means that the x-values we're considering most are very large and negative.

(e) Why is there no difference in the behavior of $f(x)$ as $x \to \infty$ compared to $x \to -\infty$ when the sign of the function outputs are opposite ($f(x) > 0$ when $x \to \infty$ and $f(x) < 0$ when $x \to -\infty$)?

Practice Problems

1. Use the graph of the function $f(x)$ below to determine each limit.

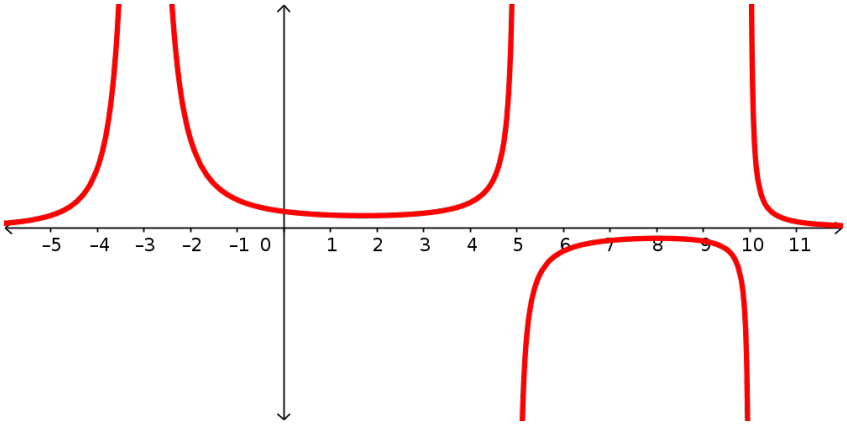


Figure 1.4.3 The function $f(x)$.

(a) $\lim_{x \to -3^-} f(x)$

(b) $\lim_{x \to -3^+} f(x)$

(c) $\lim_{x \to -3} f(x)$

(d) $\lim_{x \to 5^-} f(x)$

(e) $\lim_{x \to 5^+} f(x)$

(f) $\lim_{x \to 5} f(x)$

(g) $\lim_{x \to 10^-} f(x)$

(h) $\lim_{x \to 10^+} f(x)$

(i) $\lim_{x \to 10} f(x)$

2. Use the graph of the function $g(x)$ below to determine each limit.

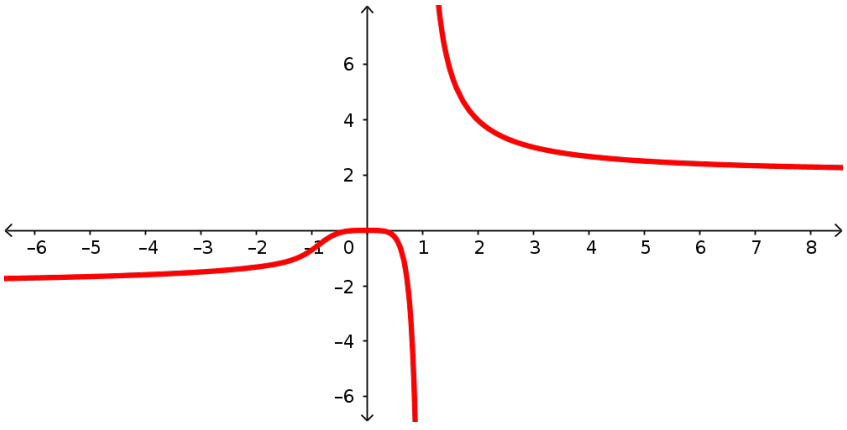


Figure 1.4.4 The function $g(x)$.

(a) $\lim_{x\to-\infty} g(x)$

(b) $\lim_{x\to\infty} g(x)$

3. Use the graph of the function $h(x)$ below to determine each limit.

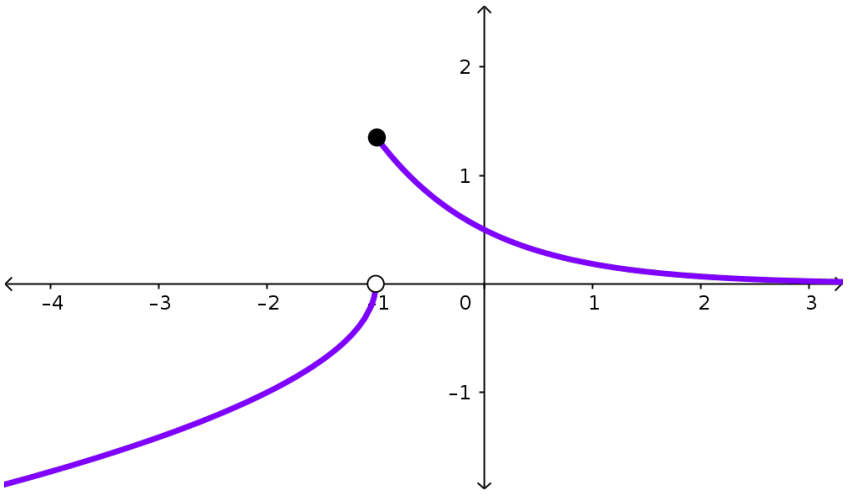


Figure 1.4.5 The function $h(x)$.

(a) $\lim_{x\to-\infty} h(x)$

(b) $\lim_{x\to\infty} h(x)$

4. For each limit, fill in the table in a way that determines the limit, and report your answer.

(a) $\lim_{x\to4^-}\left(\dfrac{3x}{x-4}\right)$

x	_____	_____	_____	4
$\frac{3x}{x-4}$	_____	_____	_____	does not exist

(b) $\lim_{x\to4^+}\left(\dfrac{3x}{x-4}\right)$

x	4	_____	_____	_____
$\frac{3x}{x-4}$	does not exist	_____	_____	_____

(c) $\lim_{x\to1}\left(\dfrac{x-3}{(2x-4)(x-1)^2}\right)$

x	_____	_____	_____	1	_____	_____	_____
$\frac{x-3}{(2x-4)(x-1)^2}$	_____	_____	_____	does not exist	_____	_____	_____

5. Evaluate the following limits analytically. Also evaluate the relevant one-sided limits when needed.

(a) $\lim_{x\to-\infty}\left(2+4x-5x^2+\dfrac{x^3}{7}-\dfrac{8x^4}{9}\right)$

(b) $\lim_{x\to\infty}\left(\dfrac{5x^3-4x+3}{2x+6}\right)$

(c) $\lim_{x \to -3^-} \left(\dfrac{5x^3 - 4x + 3}{2x + 6} \right)$

(d) $\lim_{x \to -\infty} \left(\dfrac{x - 10}{3x + 4} \right)$

(e) $\lim_{x \to 2} \left(\dfrac{4x - 1}{x^4 - 16} \right)$

(f) $\lim_{x \to \infty} \left(\dfrac{6 - 5x^5}{2 + 3x^2} \right)$

(g) $\lim_{x \to 1} \left(\dfrac{x^2 + 1}{x^3 - 2x^2 + x} \right)$

(h) $\lim_{\theta \to 3\pi/2} \sin(\theta) \tan(\theta)$

6. For each of the following functions, find the locations of any vertical asymptotes and report the behavior of the functions around those asymptotes using limit statements.

(a) $f(x) = \dfrac{|x - 4|}{x + 2}$

(b) $g(x) = \dfrac{\sin(x)}{x^2 - 1}$

(c) $h(t) = \dfrac{4}{t} + \dfrac{9}{t - 1}$

(d) $s(w) = \dfrac{w^3 - 3w^2 + w - 1}{w^2 - 5w + 6}$

(e) $f(x) = \dfrac{e^x}{(x - 3)^4}$

7. For each of the following functions, find the locations of any horizontal asymptotes using limit statements (as $x \to \infty$ and $x \to -\infty$).

(a) $f(x) = 2 + \dfrac{10}{x^2}$

(b) $g(x) = \dfrac{4x^5 - 3x^3 + 5x - 1}{10x^2 + 5x^3 - 5x^4}$

(c) $h(x) = \dfrac{3x^2 + 4x + \sqrt{x}}{1 - \sqrt{x} - 2x^2}$

(d) $j(x) = \dfrac{3}{5 + 10^{1/x}}$

1.5 The Squeeze Theorem

Activity 1.5.1 A Weird End Behavior Limit.

In this activity, we're going to find the following limit:

$$\lim_{x\to\infty}\left(\frac{\sin^2(x)}{x^2+1}\right).$$

This limit is a bit weird, in that we really haven't looked at trigonometric functions that much. We're going to start by looking at a different limit in the hopes that we can eventually build towards this one.

(a) Consider, instead, the following limit:

$$\lim_{x\to\infty}\left(\frac{1}{x^2+1}\right).$$

Find the limit and connect the process or intuition behind it to at least one of the results from this text.

(b) Let's put this limit aside and briefly talk about the sine function. What are some things to remember about this function? What should we know? How does it behave?

(c) What kinds of values doe we expect $\sin(x)$ to take on for different values of x?

$$\qquad\qquad \leq \sin(x) \leq \qquad\qquad$$

(d) What happens when we square the sine function? What kinds of values can that take on?

$$\qquad\qquad \leq \sin^2(x) \leq \qquad\qquad$$

(e) Think back to our original goal: we wanted to know the end behavior of $\dfrac{\sin^2(x)}{x^2+1}$. Right now we have two bits of information:

- We know $\lim_{x\to\infty}\left(\dfrac{1}{x^2+1}\right)$.
- We know some information about the behavior of $\sin^2(x)$. Specifically, we have some bounds on its values.

Can we combine this information?

In your inequality above, multiply $\left(\frac{1}{x^2+1}\right)$ onto all three pieces of the inequality. Make sure you're convinced about the direction or order of the inequality and whether or not it changes with this multiplication.

$$\underbrace{\frac{\qquad\qquad}{x^2+1}}_{\text{call this } f(x)} \leq \frac{\sin^2(x)}{x^2+1} \leq \underbrace{\frac{\qquad\qquad}{x^2+1}}_{\text{call this } h(x)}$$

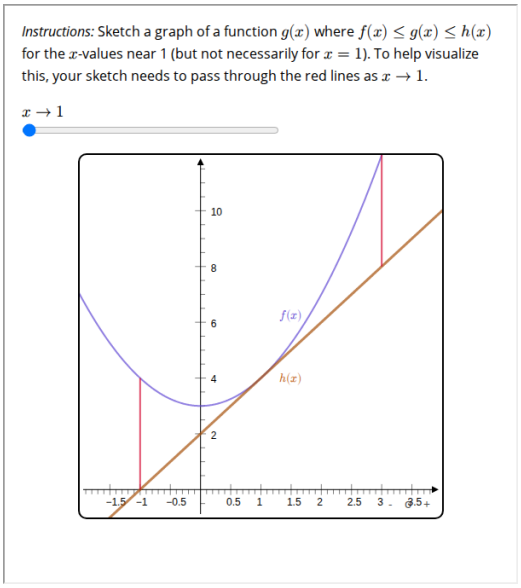
(f) For your functions $f(x)$ and $h(x)$, evaluate $\lim_{x\to\infty} f(x)$ and $\lim_{x\to\infty} h(x)$.

(g) What do you think this means about the limit we're interested in, $\lim_{x\to\infty}\left(\dfrac{\sin^2(x)}{x^2+1}\right)$?

Activity 1.5.2 Sketch This Function Around This Point.

(a) Sketch or visualize the functions $f(x) = x^2 + 3$ and $h(x) = 2x + 2$, especially around $x = 1$.

(b) Now we want to add in a sketch of some function $g(x)$, all the while satisfying the requirements of the Squeeze Theorem.



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(c) Use the Squeeze Theorem to evaluate and explain $\lim_{x \to 1} g(x)$ for your function $g(x)$.

(d) Is this limit dependent on the specific version of $g(x)$ that you sketched? Would this limit be different for someone else's choice of $g(x)$ given the same parameters?

(e) What information must be true (if anything) about $\lim_{x \to 3} g(x)$ and $\lim_{x \to 0} g(x)$?

Do we know that these limits exist? If they do, do we have information about their values?

Practice Problems

1. For each of the following statements, discuss the possible existence of the limit in question. Explain in detail how we might know whether the limit exists or not, or why it is impossible to tell.

(a) We want to know about $\lim_{x\to 4} f(x)$, and we know that $a(x) \leq f(x) \leq b(x)$ for all x-values, with $a(4) = 3$ and $b(4) = 3$.

(b) We want to know about $\lim_{x\to 0} f(x)$, and we know that $\lim_{x\to 0} g(x) = 5$ for the function $g(x) \geq f(x)$ for x-values around 0.

(c) We want to know about $\lim_{x\to -1} f(x)$, and we know that $a(x) \leq f(x) \leq b(x)$ for x-values around -1, with $\lim_{x\to -1} a(x) = 9$ and $\lim_{x\to -1} b(x) = 9$.

2. We want to use the Squeeze Theorem to evaluate $\lim_{x\to 0} \big(x^2 \sin(\pi/x)\big)$.

(a) Explain why we know that $-1 \leq \sin(\pi/x) \leq 1$ for any non-zero x-values. Why does this inequality not hold for $x = 0$?

(b) Use the inequality from (a) to build an inequality of functions $f(x) \leq x^2 \sin(\pi/x) \leq g(x)$ for non-zero x-values.

(c) With these functions $f(x)$ and $g(x)$ from (b), find $\lim_{x\to 0} f(x)$ and $\lim_{x\to 0} g(x)$.

(d) Explain how the Squeeze Theorem tells us what the value of $\lim_{x\to 0} x^2 \sin(\pi/x)$ is.

1.6 Continuity and the Intermediate Value Theorem

Activity 1.6.1 Classification and Continuity.

Let's consider the following functions, graphed below.

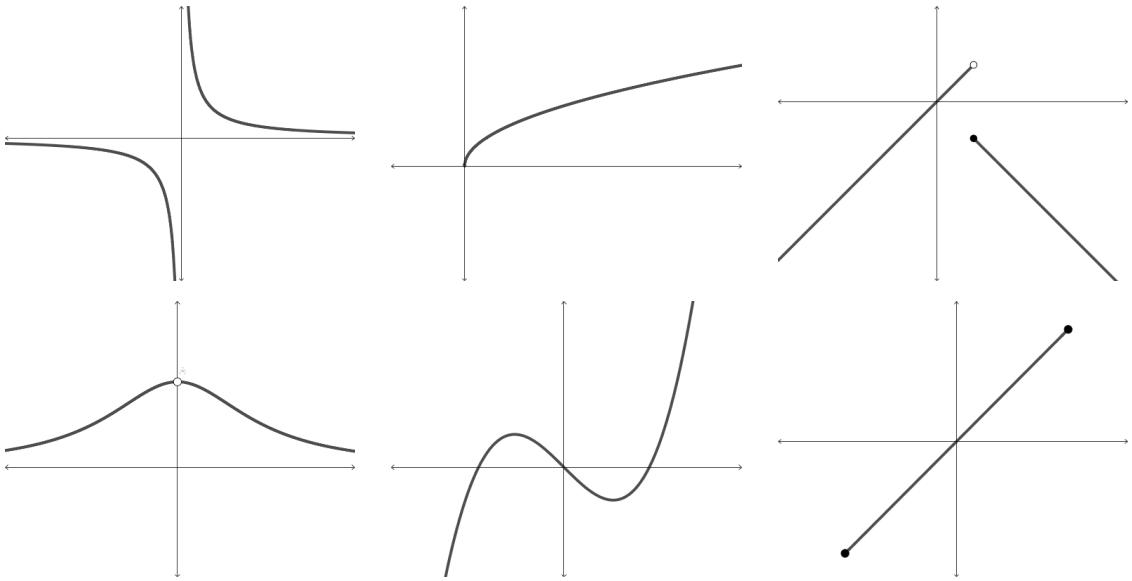


Figure 1.6.2 A variety of graphs for us to use to think about continuity.

(a) Can you point out any points on the functions above that seem like the functions might not be continuous? Note that we're not classifying each function as "continuous" or "not continuous"!

(b) We want to build towards a definition of continuity using "connectedness" as the key: a function is continuous at a point if it is connected to itself. How does that work when we think about function values and limits? Use the graphs above and the points you looked at to help!

Practice Problems

1. Explain how the definition of continuity (Definition 1.6.3) describes a function being "connected to itself" at the point.

2. What does it mean for a function to be continuous on the left at a point? What about on the right?

3. Use the definition of continuity at a point to determine whether each function is continuous at the point listed. If it is not, is the function continuous on one side at the point? Explain.

(a) $f(x) = \begin{cases} 4x+1 & \text{if } x < 1 \\ x^2+5 & \text{if } x \geq 1 \end{cases}$

Is $f(x)$ continuous at $x = 1$?

(b) $g(x) = \begin{cases} 6x^2-2 & \text{if } x < 0 \\ 4 & \text{if } x = 0 \\ 1-3\cos(x) & \text{if } x > 0 \end{cases}$

Is $g(x)$ continuous at $x = 0$?

(c) $h(x) = \begin{cases} \left(\dfrac{x^2-1}{x+1}\right) & \text{if } x \neq -1 \\ -2 & \text{if } x = -1 \end{cases}$

Is $h(x)$ continuous at $x = -1$?

4. Find the values of a that make the function continuous at the point that is listed. If no such value of a exists, say so and explain why.

(a) $f(x) = \begin{cases} x-6 & \text{if } x < 3 \\ ax^2+1 & \text{if } x \geq 3 \end{cases}$ at $x = 3$

(b) $f(x) = \begin{cases} \dfrac{x^2-4}{x+2} & \text{if } x \neq -2 \\ a & \text{if } x = -2 \end{cases}$ at $x = -2$

5. Use the properties of composition and continuity to evaluate the following limits.

(a) $\lim\limits_{x\to 0} \left(\sin\left(e^{\sqrt{x^2}}\right)\right)$

(b) $\lim\limits_{x\to 0} \ln\left(\dfrac{\cos(x^2)}{x^2+1}\right)$

(c) $\lim\limits_{x\to 0} \sqrt[3]{\dfrac{x}{\sqrt{16x+1}-1}}$

6. "Even a broken clock is right twice a day," is a common phrase used to say that even an unreliable person can have a good point on accident. Explain how we can use the Intermediate Value Theorem to prove that a broken (analog) clock is guaranteed right twice per day. (Here, we're assuming that "broken" means stuck.)

7. Explain why this argument doesn't work to show that an unreliable person will be guaranteed to have a good point on accident.

8. Prove that at some point in your life you were exactly π feet tall.

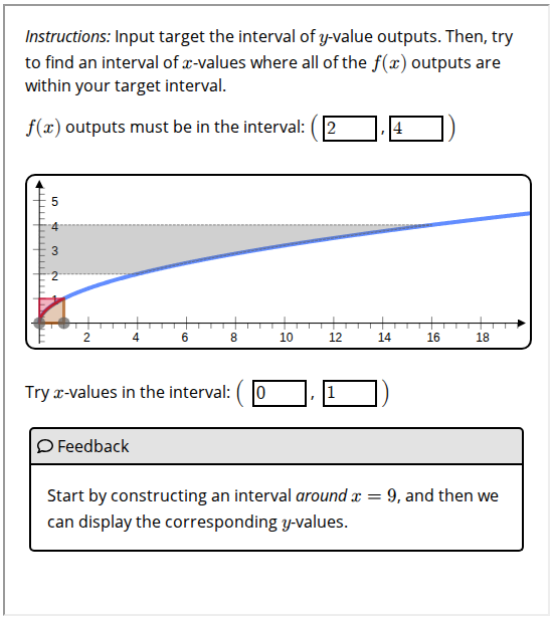
1.7 Exploration: Limits

Exploration 1.7.1 A First Function to Think About.

Let's consider the function $f(x) = \sqrt{x}$. We're going to mostly be interested in the x-values around 9, since we'll be investigating $\lim_{x \to 9} \sqrt{x}$. We know that the y-value outputs of $f(x)$ are going to be around 3, giving us:

$$\lim_{x \to 9} \sqrt{x} = 3.$$

Now we're going to perform some tasks that will hopefully help us add some precision to our definition of a limit. You can use the graph and inputs below while you work.



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(a) We want to come up with points on the graph of $f(x)$ whose y-values are within the interval $(2, 4)$. Can you find an interval of x-values around $x = 9$ where the function outputs $f(x)$ are all in the interval $(2, 4)$?

(b) Can you describe this in terms of distance? How far can x-values be from $x = 9$ where the corresponding $f(x)$ values are all in $(2, 4)$?

Your answer should fill in the blank in the following sentence:

"If I construct an interval of x-values that are within ____ units of $x = 9$, then the corresponding function outputs will all be within $(2, 4)$."

Note that another way of describing the interval, $(2, 4)$ is to say that the function outputs are all within 1 unit of $y = 3$.

(c) Repeat this process, but for the interval $(2.9, 3.1)$. Can you find an interval of x-values around $x = 9$ where the function outputs $f(x)$ are all in the interval $(2.9, 3.1)$?

(d) Can you describe this in terms of distance? How far can x-values be from $x = 9$ where the corresponding $f(x)$ values are all in $(2.9, 3.1)$?

Your answer should fill in the blank in the following sentence:

"If I construct an interval of x-values that are within ____ units of $x = 9$, then the corresponding function outputs will all be within 0.1 units of $y = 3$."

(e) Come up with an even smaller interval. Change the distance between the function outputs and $y = 3$ to something smaller than 0.1. Construct a new target interval of y-values, and see if you can find an interval of x-values such that all of the function outputs are in your target interval.

(f) Have you convinced yourself that you can always do this? Do you think it really matters what the target interval of y-values is? Can you *always* find an interval of x-values around $x = 9$ (or x-values that are some non-zero distance from $x = 9$) where the outputs are in within that target distance from $y = 3$?

(g) Let's formalize this: If ε is some real number with $\varepsilon > 0$, then can you find an interval of x-values around $x = 9$ where the function outputs $f(x)$ are all in the interval $(3 - \varepsilon, 3 + \varepsilon)$?

(h) Can you describe this in terms of distance? How far can x-values be from $x = 9$ where the corresponding $f(x)$ values are all in $(3 - \varepsilon, 3 + \varepsilon)$?

Your answer should fill in the blank in the following sentence:

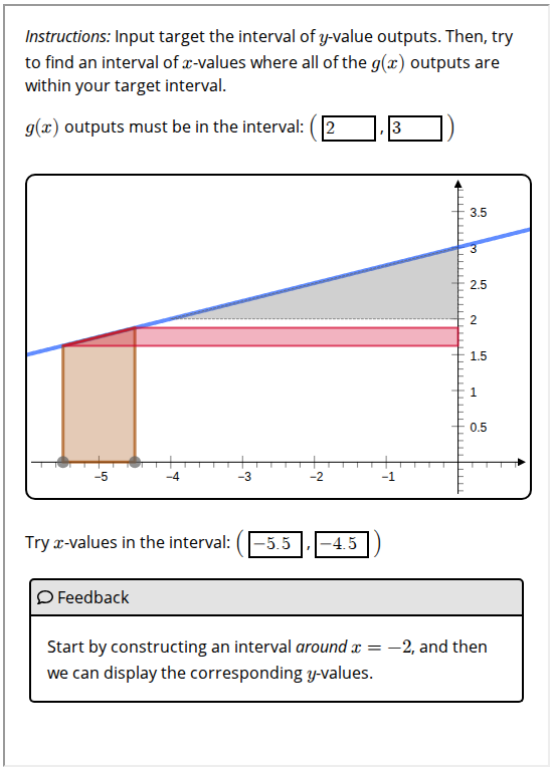
"If I construct an interval of x-values that are within _____ units of $x = 9$, then the corresponding function outputs will all be within ε units of $y = 3$."

Exploration 1.7.2 A Linear Function.

We'll try this again, but with a new function. This will change a bit of how we think about constructing our intervals. Consider the function $g(x) = \frac{x}{4} + 3$. We're going to mostly be interested in the x-values around -2, since we'll be investigating:

$$\lim_{x \to -2} \left(\frac{x}{4} + 3 \right) = \frac{5}{2}.$$

You can, again, use the graph and inputs below while you work.



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(a) We want to come up with points on the graph of $g(x)$ whose y-values are within the interval $(2, 3)$. Can you find an interval of x-values around $x = -2$ where the function outputs $g(x)$ are all in the interval $(2, 3)$?

(b) Can you describe this in terms of distance? How far can x-values be from $x = -2$ where the corresponding $g(x)$ values are all in $(2, 3)$?

Your answer should fill in the blank in the following sentence:

"If I construct an interval of x-values that are within _____ units of $x = -2$, then the corresponding function outputs will all be within $(2, 3)$."

Note that another way of describing the interval, $(2, 3)$ is to say that the function outputs are all within $\frac{1}{2}$ unit of $y = \frac{5}{2}$.

(c) Repeat this process, but for the interval $\left(\frac{19}{8}, \frac{21}{8} \right)$. Can you find an interval of x-values around $x = -2$ where the function outputs $g(x)$ are all in the interval $\left(\frac{19}{8}, \frac{21}{8} \right)$?

(d) Can you describe this in terms of distance? How far can x-values be from $x = -2$ where the corresponding $g(x)$ values are all in $\left(\frac{19}{8}, \frac{21}{8}\right)$?

Your answer should fill in the blank in the following sentence:

"If I construct an interval of x-values that are within _____ units of $x = -2$, then the corresponding function outputs will all be within $\frac{1}{8}$ unit of $y = -2$."

(e) If ε is any real number with $\varepsilon > 0$, then can you find an interval of x-values around $x = -2$ where the function outputs $g(x)$ are all in the interval $\left(\frac{5}{2} - \varepsilon, \frac{5}{2} + \varepsilon\right)$?

(f) Can you describe this in terms of distance? How far can x-values be from $x = -2$ where the corresponding $f(x)$ values are all in $\left(\frac{5}{2} - \varepsilon, \frac{5}{2} + \varepsilon\right)$?

Practice Problems

1. Consider the $\varepsilon - \delta$ definition of a limit. What does ε represent? Connect it with the standard definition of a limit that we've been using. Similarly, what does δ represent? Connect it with the standard definition of a limit we've been using.

2. Draw a picture of the following situations:

(a) A function $f(x)$ where if $0 < |x - 3| < 2$ then $|f(x) + 4| < 1$.

(b) A function $g(x)$ where if $0 < |x + 1| < \frac{1}{2}$ then $|g(x) - 6| < 2$.

3. For the following, find δ or ε. Include a sketch that illustrates the values you found with the function.

(a) For $f(x) = 3x + 1$, if $0 < |x - 2| < \delta$, then $|f(x) - 7| < 1$.

(b) For $g(x) = \sqrt{x + 3}$, if $0 < |x - 6| < 5$, then $|g(x) - 3| < \varepsilon$.

(c) For the function $f(x) = -3x + 1$, explain how to find a value for δ that corresponds with any ε so that when $0 < |x - 2| < \delta$ we have $|f(x) + 5| < \varepsilon$.

(d) Consider the open interval definition of a limit. What do (c, d) and (p, q) represent. Connect them to the standard definition of a limit that we've been using.

4. Draw a picture of the following situations:

(a) A function $f(x)$ with $a = 1$ and $L = 8.5$ where if x is in $(0, 2)$ then $f(x)$ is in $(8, 9)$.

(b) A function $g(x)$ with $a = -4$ and $L = 10$ where if x is in $(-4.5, -3.5)$ then $g(x)$ is in $(9.8, 10.2)$.

5. For the following, find values for (c, d) or (p, q). Include a sketch that illustrates the intervals you found with the function.

(a) For $f(x) = 2x - 1$, if x is in $(1, 2)$, then $f(x)$ is in (p, q).

(b) For $g(x) = x^3 + 1$, if x is in (c, d), then $g(x)$ is in $(8, 9)$.

6. For the function $f(x) = x^2 + 4$, explain how to find an interval (c, d) around $x = 2$ if we are given some interval (p, q) around $L = 8$ such that if x is in (c, d) then $f(x)$ is in (p, q).

Chapter 2

Derivatives

2.1 Introduction to Derivatives

Activity 2.1.1 Thinking about Slopes.

We're going to calculate and make some conjectures about slopes of lines between points, where the points are on the graph of a function. Let's define the following function:

$$f(x) = \frac{1}{x+2}.$$

(a) We're going to calculate a lot of slopes! Calculate the slope of the line connecting each pair of points on the curve of $f(x)$:

- $(-1, f(-1))$ and $(0, f(0))$
- $(-0.5, f(-0.5))$ and $(0, f(0))$
- $(-0.1, f(-0.1))$ and $(0, f(0))$
- $(-0.001, f(-.001))$ and $(0, f(0))$

(b) Let's calculate another group of slopes. Find the slope of the lines connecting these pairs of points:

- $(0, f(0))$ and $(1, f(1))$
- $(0, f(0))$ and $(0.5, f(0.5))$
- $(0, f(0))$ and $(0.1, f(0.1))$
- $(0, f(0))$ and $(0.001, f(0.001))$

(c) Just to make it clear what we've done, lay out your slopes in this table:

Between $(0, f(0))$ **and...**	**Slope**
$(1, f(1))$	
$(0.5, f(0.5))$	
$(0.1, f(0.1))$	
$(0.01, f(0.01))$	
$(-0.01, f(-0.01))$	
$(-0.1, f(-0.1))$	
$(-0.5, f(-0.5))$	
$(-1, f(-1))$	

(d) Now imagine a line that is tangent to the graph of $f(x)$ at $x = 0$. We are thinking of a line that touches the graph at $x = 0$, but runs alongside of the curve at that point instead of through it.

Make a conjecture about the slope of this line, using what we've seen above.

(e) Can you represent the slope you're thinking of in the table above with a limit? What limit are we approximating in the slope calculations above? Set up the limit and evaluate it, confirming your conjecture.

Activity 2.1.2 Finding a Tangent Line.

Let's think about a new function, $g(x) = \sqrt{2-x}$. We're going to think about this function around the point at $x = 1$.

(a) Ok, we are going to think about this function at this point, so let's find the coordinates of the point first. What's the y-value on our curve at $x = 1$?

(b) Use a limit similar to the one you constructed in Activity 2.1.1 to find the slope of the line tangent to the graph of $g(x)$ at $x = 1$.

(c) Now that you have a slope of this line, and the coordinates of a point that the line passes through, can you find the equation of the line?

Activity 2.1.3 Calculating a Bunch of Slopes.

Let's do this all again, but this time we'll calculate the slope at a bunch of different points on the same function.
Let's use $j(x) = x^2 - 4$.

(a) Start calculating the following derivatives, using the definition of the Derivative at a Point:

- $j'(-2)$
- $j'(0)$
- $j'(1/3)$
- $j'(-1)$

(b) Stop calculating the above derivatives when you get tired/bored of it. How many did you get through?

(c) Notice how repetitive this is: on one hand, we have to set up a completely different limit each time (since we're looking at a different point on the function each time). On the other hand, you might have noticed that the work is all the same: you factor and cancel over and over. These limits are all ones that we covered in Section 1.3 First Indeterminate Forms, and so it's no surprise that we keep using the same algebra manipulations over and over again to evaluate these limits.

Do you notice any patterns, any connections between the x-value you used for each point and the slope you calculated at that point? You might need to go back and do some more.

(d) Try to evaluate this limit in general:

$$\begin{aligned} j'(a) &= \lim_{x \to a} \left(\frac{j(x) - j(a)}{x - a} \right) \\ &= \lim_{x \to a} \left(\frac{(x^2 - 4) - (a^2 - 4)}{x - a} \right). \end{aligned}$$

Remember, you know how this goes! You're going to do the same type of algebra that you did earlier!

What is the formula, the pattern, the way of finding the slope on the $j(x)$ function at any x-value, $x = a$?

(e) Confirm this by using your new formula to recalculate the following derivatives:

- $j'(-2)$
- $j'(0)$
- $j'(1/3)$
- $j'(-1)$

We're going to try to think about the derivative as something that can be calculated in general, as well as something that can be calculated at a point. We'll define a new way of calculating it, still a limit of slopes, that will be a bit more general.

Practice Problems

1. We define a derivative of $f(x)$ at $x = a$ as $f'(x) = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)$. What is this limit? What does it represent in terms of $f(x)$? Feel free to sketch some pictures in order to discuss it visually.

2. For some function $f(x)$, we have that $f'(3) = -4$. What does this mean? Interpret this value in the context of $f(x)$.

3. For each of the following functions, use the definition of the derivative at a point to find the slope of the line tangent to the function's curve at the specified point.

(a) $f(x) = 6x + 1$ at $(1, 7)$

(b) $g(x) = x^2 - 5$ at $(-1, -4)$

(c) $j(x) = \sqrt{x}$ at $(4, 2)$

(d) $f(x) = \dfrac{1}{4 - x}$ at $(3, 1)$

(e) $g(x) = 4 + 3x - 2x^2$ at $(2, 2)$

(f) $j(x) = \dfrac{1}{x^2}$ at $(1, 1)$

4. We have talked about two limit-based definitions of derivatives:

$$f'(a) = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)$$
$$f'(x) = \lim_{\Delta x \to 0} \left(\frac{f(x + \Delta x) - f(x)}{\Delta x} \right)$$

(a) What are some similarities between these definitions? It might help to compare visual depictions of each.

(b) What are some differences between these definitions? Focus on differences in how they are built as well as how we use them.

(c) Pick a situation where it might make sense to use each definition. Explain your choice.

5. For each of the functions below, find the derivative function $f'(x)$, and evaluate that function at $x = 1$, $x = 2$, $x = 3$, and $x = 4$. Interpret these values.

(a) $f(x) = 5x + 2$

(b) $f(x) = \sqrt{x + 4}$

(c) $f(x) = x^3 - x$

(d) $f(x) = \dfrac{x + 1}{x}$

6. Find the derivative function for each of the following functions.

(a) $f(x) = 4 - \sqrt{x}$

(b) $f(x) = \dfrac{5}{x - 2}$

(c) $f(x) = 3\sqrt{x - 1}$

(d) $f(x) = x^2 + 3x$

(e) $f(x) = \dfrac{1}{x^2+1}$

(f) $f(x) = \sqrt{x^2+1}$

2.2 Interpreting Derivatives

Activity 2.2.1 Interpreting the Derivative as a Slope.

In Activity 2.1.1 Thinking about Slopes and Activity 2.1.2 Finding a Tangent Line, we built the idea of a derivative by calculating slopes and using them. Let's continue this by considering the function $f(x) = \frac{1}{x^2}$.

(a) Use Definition 2.1.3 Derivative at a Point to find $f'(2)$. What does this value represent?

(b) We want to plot the line that would be tangent to the graph of $f(x)$ at $x = 2$.

Remember that we can write the equation of a line in two ways:

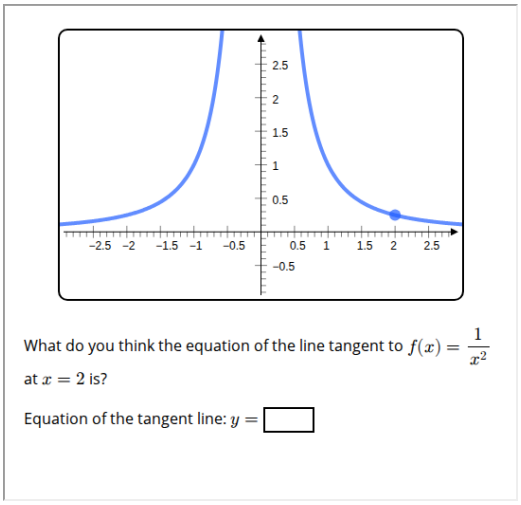
- The equation of a line with slope m that passes through the point $(a, f(a))$ is:

$$y = m(x - a) + f(a).$$

- The equation of a line with slope m that passes the point $(0, b)$ (this is another way of saying that the y-intercept of the line is b) is:

$$y = mx + b.$$

Find the equation of the line tangent to $f(x)$ at $x = 2$. Add it to the graph of $f(x) = \frac{1}{x^2}$ below to check your equation.



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(c) This tangent line is very similar to the actual curve of the function $f(x)$ near $x = 2$. Another way of saying this is that while the slope of $f(x)$ is not always the value you found for $f'(2)$, it is close to that for x-values near 2.

Use this idea of slope to predict where the y-value of our function will be at 2.01.

(d) Compare this value with $f(2.01) = \frac{1}{2.01^2}$. How close was it?

Activity 2.2.2 Interpreting the Derivative as a Rate of Change.

This is going to feel somewhat redundant, since we hopefully know that a slope is really just a rate of change. But we'll be able to explore this a bit more and see how we can use a derivative to tell us information about some specific context.

Let's say that we want to model the speed of a car as it races along a strip of the road. By the time we start measuring it (we'll call this time 0), the position of the car (along the straight strip of road) is:

$$s(t) = 73t + t^2,$$

where t is time measured in seconds and $s(t)$ is the position measured in feet. Let's say that this function is only relevant on the domain $0 \leq t \leq 15$. That is, we only model the position of the car for a 15-second window as it speeds past us.

(a) How far does the car travel in the 15 seconds that we model it? What was the car's average velocity on those 15 seconds?

(b) Calculate $s'(t)$, the derivative of $s(t)$, using Definition 2.1.5 The Derivative Function. What information does this tell us about our vehicle?

(c) Calculate $s'(0)$. Why is this smaller than the average velocity you found? What does that mean about the velocity of the car?

(d) If we call $v(t) = s'(t)$, then calculate $v'(t)$. Note that this is a derivative of a derivative.

(e) Find $v'(0)$. Why does this make sense when we think about the difference between the average velocity on the time interval and the value of $v(0)$ that we calculated?

(f) What does it mean when we notice that $v'(t)$ is constant? Explain this by interpreting it in terms of both the velocity of the vehicle as well as the position.

Activity 2.2.3 Interpreting the Derivative as a Function.

In Activity 2.1.4 Calculating a Bunch of Slopes, we calculated the derivative function for $j(x) = x^2 - 4$. Using the definition of The Derivative Function, we can see that $j'(x) = 2x$. Let's explore that a bit more.

(a) Sketch the graphs of $j(x) = x^2 - 4$ and $j'(x) = 2x$. Describe the shapes of these graphs.

(b) Find the coordinates of the point at $x = \frac{1}{2}$ on both the graph of $j(x)$ and $j'(x)$. Plot the point on each graph.

(c) Think back to our previous interpretations of the derivative: how do we interpret the y-value output you found for the j' function?

(d) Find the coordinates of another point at some other x-value on both the graph of $j(x)$ and $j'(x)$. Plot the point on each graph, and explain what the output of j' tells us at this point.

(e) Use your graph of $j'(x)$ to find the x-intercept of $j'(x)$. Locate the point on $j(x)$ with this same x-value. How do we know, visually, that this point is the x-intercept of $j'(x)$?

(f) Use your graph of $j'(x)$ to find where $j'(x)$ is positive. Pick two x-values where $j'(x) > 0$. What do you expect this to look like on the graph of $j(x)$? Find the matching points (at the same x-values) on the graph of $j(x)$ to confirm.

(g) Use your graph of $j'(x)$ to find where $j'(x)$ is negative. Pick two x-values where $j'(x) < 0$. What do you expect this to look like on the graph of $j(x)$? Find the matching points (at the same x-values) on the graph of $j(x)$ to confirm.

(h) Let's wrap this up with one final pair of points. Let's think about the point $(-3, 5)$ on the graph of $j(x)$ and the point $(-3, -6)$ on the graph of $j'(x)$. First, explain what the value of -6 (the output of j' at $x = -3$) means about the point $(-3, 5)$ on $j(x)$. Finally, why can we not use the value 5 (the output of j at $x = -3$) means about the point $(-3, -6)$ on $j'(x)$?

Practice Problems

1. For each of the following functions, use the definition of the derivative at a point to find $f'(a)$ and then find the equation of the line tangent to the graph of $f(x)$ at $x = a$.

Remember: The equation of a line given the slope m and (x_1, y_1), a point on the line is $y = m(x - x_1) + y_1$.

(a) $f(x) = x^2 + x - 1$ with $a = 0$

(b) $f(x) = x^3 + 1$ with $a = 2$

(c) $f(x) = \dfrac{2}{\sqrt{x}}$ with $a = 6$

(d) $f(x) = \sqrt{3x - 1}$ with $a = 2$

(e) $f(x) = x - x^2$ with $a = 2$

(f) $f(x) = \dfrac{x}{x + 1}$ with $a = 1$

2. We start watching a car driving down a straight road towards a stop sign. We can tell that their position function is $s(t) = -7.5t^2 + 81t$ with position measured in feet from their location when we began observing them and time measured in seconds.

(a) How fast is the car driving at $t = 0$?

(b) How fast is the car driving at $t = 2$?

(c) At what value of t does the car stop (their velocity is 0)?

(d) How far did they travel before they stopped?

(e) At $t = 0$ what is the acceleration of the car?

(f) At $t = 2$ what is the acceleration of the car?

3. The monetary value of an ancient mathematical artifact (the pencil that Isaac Barrow used to prove the Fundamental Theorem of Calculus in 1667) follows the function $m(t) = \sqrt{1600t} + 10$ where t is the number of years since 1667 and $m(t)$ is the monetary value of the pencil in US Dollars.

(a) What was the overall change in value from 1667 to 1967 (from $t = 0$ to $t = 300$)?

(b) What was the average rate of change of the value from 1667 to 1967?

(c) What was the instantaneous rate of change of the value in 1967?

(d) Find $m(354)$ and $m'(354)$, and interpret both of these values.

(e) Use $m(354)$ and $m'(354)$ to approximate the value of the pencil in 2022.

(f) Use $m(t)$ to find the value in 2022. Was the approximation close or not? Explain.

2.3 Some Early Derivative Rules

Activity 2.3.1 Derivatives of Power Functions.

We're going to do a bit of pattern recognition here, which means that we will need to differentiate several different power functions. For our reference, a power function (in general) is a function in the form $f(x) = a(x^n)$ where n and a are real numbers, and $a \neq 0$.
Let's begin our focus on the power functions x^2, x^3, and x^4. We're going to use Definition 2.1.5 The Derivative Function a lot, so feel free to review it before we begin.

(a) Find $\frac{d}{dx}\left(x^2\right)$. As a brief follow up, compare this to the derivative $j'(x)$ that you found in Activity 2.1.4 Calculating a Bunch of Slopes. Why are they the same? What does the difference, the -4, in the $j(x)$ function do to the graph of it (compared to the graph of x^2) and why does this not impact the derivative?

(b) Find $\frac{d}{dx}\left(x^3\right)$.

(c) Find $\frac{d}{dx}\left(x^4\right)$.

(d) Notice that in these derivative calculations, the main work is in multiplying $(x+\Delta x)^n$. Look back at the work done in all three of these derivative calculations and find some unifying steps to describe how you evaluate the limit/calculate the derivative *after* this tedious multiplication was finished. What steps did you do? Is it always the same thing?

Another way of stating this is: if I told you that I knew what $(x+\Delta x)^5$ was, could you give me some details on how the derivative limit would be finished?

(e) Finish the following derivative calculation:

$$\begin{aligned}\frac{d}{dx}\left(x^5\right) &= \lim_{\Delta x\to 0}\left(\frac{(x+\Delta x)^5 - x^5}{\Delta x}\right)\\ &= \lim_{\Delta x\to 0}\left(\frac{(x^5+5x^4\Delta x+10x^3\Delta x^2+10x^2\Delta x^3+5x\Delta x^4+\Delta x^5)-x^5}{\Delta x}\right)\\ &= \rightsquigarrow \ldots\end{aligned}$$

(f) Make a conjecture about the derivative of a power function in general, $\frac{d}{dx}(x^n)$.

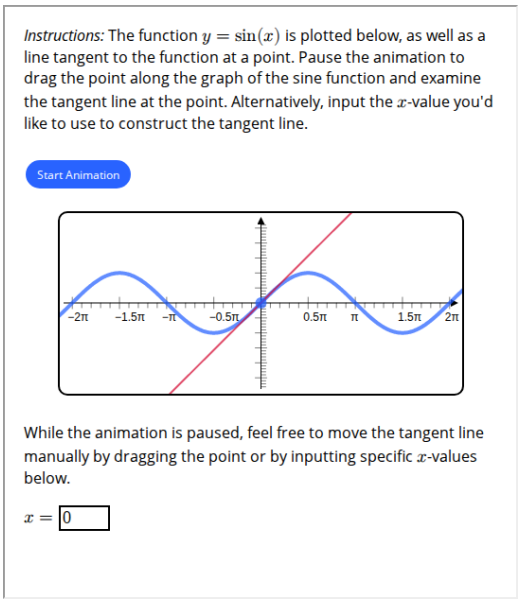
Something to notice here is that the calculation in this limit is really dependent on knowing what $(x+\Delta x)^n$ is. When n is an integer with $n \geq 2$, this really just translates to multiplication. If we can figure out how to multiply $(x+\Delta x)^n$ in general, then this limit calculation will be pretty easy to do. We noticed that:

1. The first term of that multiplication will combine with the subtraction of x^n in the numerator and subtract to 0.
2. The rest of the terms in the multiplication have at least one copy of Δx, and so we can factor out Δx and "cancel" it with the Δx in the denominator.
3. Once this is done, we've escaped the portion of the limit that was giving us the $\frac{0}{0}$ indeterminate form, and so we can evaluate the limit as $\Delta x \to 0$. The result is just whatever terms still have at least one remaining copy of Δx in them "go to" 0, and we're left with just the terms that do not have any copies of Δx in them.

Activity 2.3.2 Derivatives of Trigonometric Functions.

Let's try to think through the derivatives of $y = \sin(\theta)$ and $y = \cos(\theta)$. In this activity, we'll look at graphs and try to collect some information about the derivative functions. We'll be practicing our interpretations, so if you need to brush up on Section 2.2 before we start, that's fine!

(a) The following plot includes both the graph of $y = \sin(x)$, and the line tangent to $y = \sin(x)$. Watch as the point where we build the tangent line moves along the graph, between $x = -2\pi$ and $x = 2\pi$.



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Collect as much information about the derivative, y', as you can. What do you know about it? What are some facts about the slopes of the tangent lines in this animation?

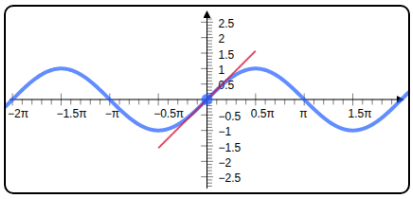
(b) We're going to get more specific here: let's find the coordinates of points that are on both the graph of $y = \sin(x)$ and it's derivative y'. Remember, to get the values for y', we're really looking at the slope of the tangent line at that point.

Instructions: Fill in values in the following table. As you plot points on the graph of both $y = \sin(x)$ and y', try to think about what function y' could be.

Input values of y and y' at the different x-values.

x	$y = \sin(x)$	y'
-2π		
$-\frac{3\pi}{2}$		
$-\pi$		
$-\frac{\pi}{2}$		
0		
$\frac{\pi}{2}$		
π		
$\frac{3\pi}{2}$		
2π		

Move the point to remind yourself of the value of y' at some point.



Do you recognize any curves that might connect these dots? Try inputting some possibilities for y' below to check!

$y' =$

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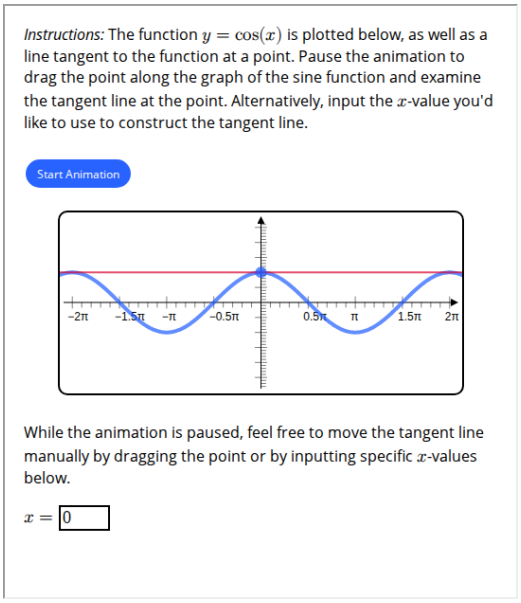
(c) Let's repeat this process using the $y = \cos(x)$ function instead.

The following plot includes both the graph of $y = \cos(x)$, and the line tangent to $y = \cos(x)$. Watch as the point where we build the tangent line moves along the graph, between $x = -2\pi$ and $x = 2\pi$.



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Collect as much information about the derivative, y', as you can. What do you know about it? What are some facts about the slopes of the tangent lines in this animation?

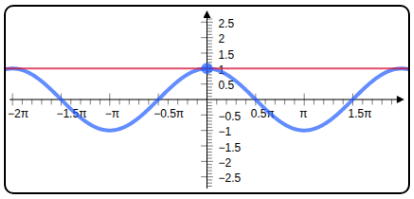
(d) We're going to get more specific here: let's find the coordinates of points that are on both the graph of $y = \cos(x)$ and it's derivative y'. Remember, to get the values for y', we're really looking at the slope of the tangent line at that point.

Instructions: Fill in values in the following table. As you plot points on the graph of both $y = \cos(x)$ and y', try to think about what function y' could be.

Input values of y and y' at the different x-values.

x	$y = \cos(x)$	y'
-2π		
$-\frac{3\pi}{2}$		
$-\pi$		
$-\frac{\pi}{2}$		
0		
$\frac{\pi}{2}$		
π		
$\frac{3\pi}{2}$		
2π		

Move the point to remind yourself of the value of y' at some point.



Do you recognize any curves that might connect these dots? Try inputting some possibilities for y' below to check!

$y' =$

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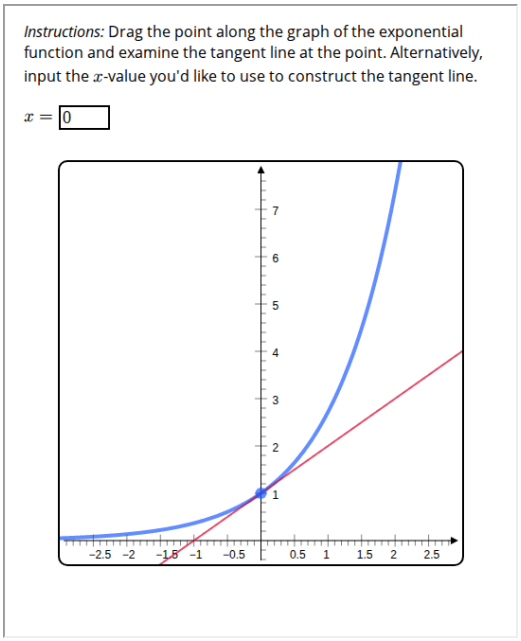
Activity 2.3.3 Derivative of the Exponential Function.

We're going to consider a maybe-unfamiliar function, $f(x) = e^x$. We'll explore this function similarly to how we thought of the derivatives of sine and cosine in Activity 2.3.7: we'll look at a tangent line at different points, and think about the slope.

(a) The plot below includes both the graph of $y = e^x$ and the line tangent to $y = e^x$. Watch as the point moves along the curve.



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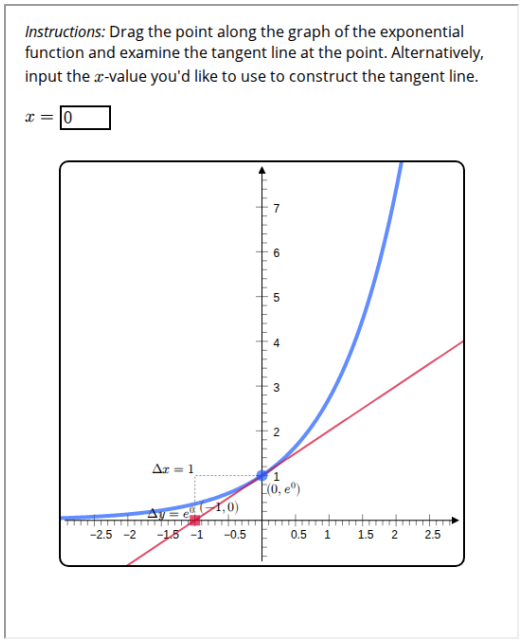
Collect as much information about the derivative, y', as you can. What do you know about it? What are some facts about the slopes of the tangent lines in this animation?

(b) There is a slightly hidden fact about slopes and tangent lines in this animation. In the following animation, we'll add (and label) one more point. Let's look at this again, this time noting the point at which this tangent line crosses the x-axis. This will make it easier to think about slopes!



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What information does this reveal about the slopes?

(c) Make a conjecture about the slope of the line tangent to the exponential function $y = e^x$ at any x-value. What do you believe the formula/equation for y' is then?

Practice Problems

1. Consider the following table of values of the derivative for f and g:

Table 2.3.4

x	-1	0	1	2
$f'(x)$	4	$\frac{1}{3}$	$-\frac{5}{2}$	0
$g'(x)$	-3	-7	$\frac{1}{2}$	-2

For each version of $h(x)$, find the derivative indicated.

(a) $h(x) = 3f(x)$; Find $h'(0)$

(b) $h(x) = -2g(x)$; Find $h'(-1)$

(c) $h(x) = \frac{1}{3}f(x) - \frac{2}{5}g(x)$; Find $h'(2)$

(d) $h(x) = g(x) - \frac{f(x)}{6}$; Find $h'(1)$

2. For each of the following functions, find $\frac{dy}{dx}$.

(a) $y = 3x^2$

(b) $y = 5x^3 + 1$

(c) $y = 4x^3 - 5x^{-2}$

(d) $y = \sqrt{x} + 4x^{15} + 400$

(e) $y = e^3 + e^\pi - e^x$

(f) $y = \dfrac{\sqrt{x^3}}{5} - \sqrt[3]{x^2}$

(g) $y = \dfrac{5}{x^3} + \dfrac{x^3}{5}$

(h) $y = \sqrt[3]{x^2} - 23$

(i) $y = e^x - x^e$

(j) $y = 4e^x + \dfrac{2}{\sqrt{x^5}}$

3. We have not yet learned how to differentiate products and quotients of functions.

For each of the following functions including products or quotients, rewrite the function and find the derivative.

(a) $f(x) = \dfrac{x^2 + 1}{x}$

(b) $g(x) = (3x + 1)^2$

(c) $p(x) = \dfrac{x^2 - 9}{x + 3}$

(d) $r(h) = \dfrac{h - 2}{\sqrt{h} - \sqrt{2}}$

(e) $\Gamma(x) = \sqrt{x}(x^{5/2} - 5)$

(f) $w(t) = \dfrac{te^t - 1}{t}$

(g) $g(w) = w(w+3)^2(2w-5)^2$

(h) $s(t) = \dfrac{6t^{3/2} + 4t^{5/3}}{5t^{1/3}}$

4. For each of the following functions, find $f'(x)$, and then $\frac{d}{dx}\left(f'(x)\right) = f''(x)$.

(a) $f(x) = 7x^5 - 2x^4 + x^3 - 7x^2 + 14x - 10$

(b) $f(x) = 4e^x - \dfrac{3}{x^5} + x$

5. For each of the following functions, use $f'(a)$ to find the equation of the line tangent to $f(x)$ at a (for the given value of a).

(a) $f(x) = 4x^2 - x^3$ for $a = 3$

(b) $f(x) = e^x - 4x$ for $a = 0$

(c) $f(x) = x^2 - x + 1$ for $a = -2$

(d) $f(x) = 4\sqrt{x} - x$ for $a = 9$

6. Consider the function $f(x) = 4e^x - 5x$.

(a) Find the derivative $f'(x)$.

(b) Interpret $f'(x)$. What does it tell us about $f(x)$?

(c) Find the value of x where the slope of the line tangent to $f(x)$ is 6.

(d) Find the value for x where $f(x)$ has a horizontal tangent line.

7. Consider the function $f(x) = \sqrt[3]{x} + x$.

(a) Find the derivative $f'(x)$.

(b) Find the value for x where the slope of the line tangent to $f(x)$ is 2.

(c) Find the value for x where $f(x)$ has a vertical tangent line.

8. Give an example of a function that meets the following requirements.

(a) The equation of the line tangent to the curve at $x = 2$ is $y = -3x + 4$.

(b) The tangent line is horizontal at $x = 5$, and the function also passes through the point $(1, -2)$.

(c) The derivative is positive when $x < -1$ and negative when $x > -1$.

2.4 The Product and Quotient Rules

Activity 2.4.1 Thinking About Derivatives of Products.

Let's start with two quick facts:

$$\frac{d}{dx}\left(x^3\right) = 3x^2 \text{ and } \frac{d}{dx}\left(\sin(x)\right) = \cos(x).$$

(a) What is $\frac{d}{dx}\left(x^3 + \sin(x)\right)$? What about $\frac{d}{dx}\left(x^3 - \sin(x)\right)$?

(b) Based on what you just explained, what is a reasonable assumption about what $\frac{d}{dx}\left(x^3 \sin(x)\right)$ might be?

(c) Let's just think about $\frac{d}{dx}\left(x^3\right)$ for a moment. What *is* x^3? Can you write this as a product? Call one of your functions $f(x)$ and the other $g(x)$. You should have that $x^3 = f(x)g(x)$ for whatever you used.

(d) Use your example to explain why, in general, $\frac{d}{dx}\left(f(x)g(x)\right) \neq \frac{d}{dx}\left(f(x)\right) \cdot \frac{d}{dx}\left(g(x)\right)$.

(e) Let's show another way that we know this. Think about $\sin(x)$. We know two things:

$$\sin(x) = (1)(\sin(x)) \text{ and } \frac{d}{dx}\left(\sin(x)\right) = \cos(x).$$

What, though, is $\frac{d}{dx}\left(1\right) \cdot \frac{d}{dx}\left(\sin(x)\right)$?

(f) Use all of this to reassure yourself that even though the derivative of a sum of functions is the sum of the derivatives of the functions, we will need to develop a better understanding of how the derivatives of products of functions work.

Activity 2.4.2 Building a Product Rule.

Let's investigate how we might differentiate the product of two functions:

$$\frac{d}{dx}\left(f(x)g(x)\right).$$

We'll use an area model for multiplication here: we can consider a rectangle where the side lengths are functions $f(x)$ and $g(x)$ that change for different values of x. Maybe x is representative of some type of time component, and the side lengths change size based on time, but it could be anything. If we want to think about $\frac{d}{dx}\left(f(x)g(x)\right)$, then we're really considering the change in area of the rectangle.

(a) Find the area of the two rectangles. The second rectangle is just one where the input variable for the side length has changed by some amount, leading to a different side length.

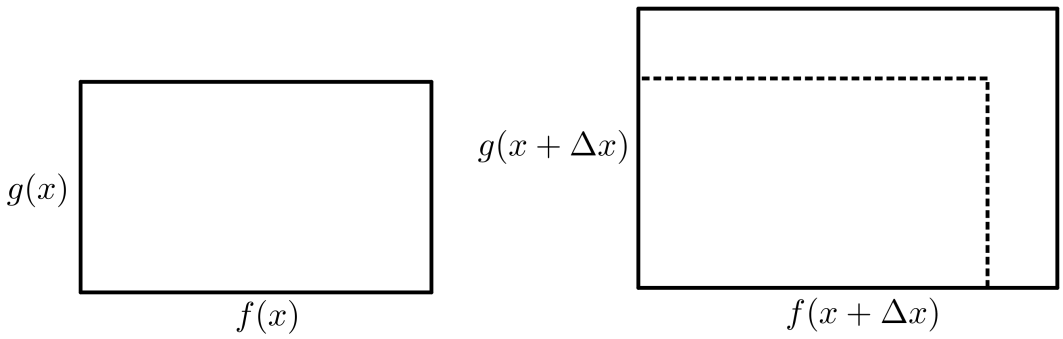


Figure 2.4.3

(b) Write out a way of calculating the difference in these areas.

(c) Let's try to calculate this difference in a second way: by finding the actual area of the region that is new in the second rectangle.

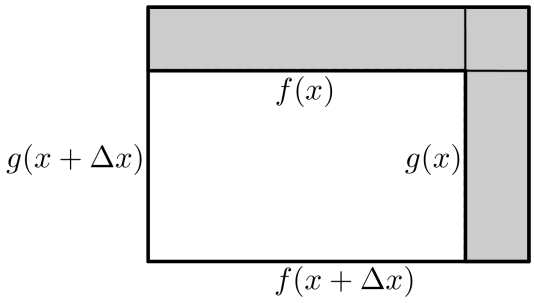


Figure 2.4.4

In order to do this, we've broken the region up into three pieces. Calculate the areas of the three pieces. Use this to fill in the following equation:

$$f(x+\Delta x)g(x+\Delta x) - f(x)g(x) = \qquad .$$

(d) We do not want to calculate the total change in area: a derivative is a *rate of change*, so in order to think about the derivative we need to divide by the change in input, Δx.

We'll also want to think about this derivative as an *instantaneous* rate of change, meaning we will consider a limit as $\Delta x \to 0$. Fill in the following:

$$\frac{d}{dx}\left(f(x)g(x)\right) \lim_{\Delta x \to 0}\left(\frac{f(x+\Delta x)g(x+\Delta x) - f(x)g(x)}{\Delta x}\right)$$

$$= \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad\qquad\qquad}{\Delta x} \right)$$

We can split this fraction up into three fractions:

$$\begin{aligned} \frac{d}{dx}\left(f(x)g(x)\right) = & \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \\ & + \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \\ & + \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \end{aligned}$$

(e) In any limit with $f(x)$ or $g(x)$ in it, notice that we can factor part out of the limit, since these functions do not rely on Δx, the part that changes in the limit. Factor these out.

In the third limit, factor out either $\lim_{\Delta x \to 0} \left(f(x + \Delta x) - f(x)\right)$ or $\lim_{\Delta x \to 0} \left(g(x + \Delta x) - g(x)\right)$.

$$\begin{aligned} \frac{d}{dx}\left(f(x)g(x)\right) = & f(x) \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \\ & + g(x) \lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \\ & + \lim_{\Delta x \to 0} \left(\qquad\qquad\qquad\qquad \right) \left(\lim_{\Delta x \to 0} \left(\frac{\qquad\qquad\qquad\qquad}{\Delta x} \right) \right) \end{aligned}$$

(f) Use Definition 2.1.5 The Derivative Function to rewrite these limits. Show that when $\Delta x \to 0$, we get:

$$f(x)g'(x) + g(x)f'(x) + 0.$$

Activity 2.4.5 Constructing a Quotient Rule.

We're going to start with a function that is a quotient of two other functions:

$$f(x) = \frac{u(x)}{v(x)}.$$

Our goal is that we want to find $f'(x)$, but we're going to shuffle this function around first. We won't calculate this derivative directly!

(a) Start with $f(x) = \frac{u(x)}{v(x)}$. Multiply $v(x)$ on both sides to write a definition for $u(x)$.

$$u(x) =$$

(b) Find $u'(x)$.

(c) Wait: we don't care about $u'(x)$, right? We care about finding $f'(x)$!
Use what you found for $u'(x)$ and solve for $f'(x)$.

$$f'(x) =$$

(d) This is a strange formula: we have a formula for $f'(x)$ written in terms of $f(x)$! But we said earlier that $f(x) = \frac{u(x)}{v(x)}$.

In your formula for $f'(x)$, replace $f(x)$ with $\frac{u(x)}{v(x)}$.

$$f'(x) =$$

Practice Problems

1. A student encounters the problem $\frac{d}{dx}\left(e^x(x^2+1)\right)$ and notes that since $\frac{d}{dx}\left(e^x\right) = e^x$ and $\frac{d}{dx}\left(x^2+1\right) = 2x$, then $\frac{d}{dx}\left(e^x(x^2+1)\right) = 2xe^x$. Explain not only why the student is incorrect, but also what kinds of pitfalls can occur if we use this "the derivative of a product is the product of the derivatives" method. Differentiate correctly, explaining your process for the student's benefit.

2. Consider the following table of values of f, g, f', and g'.

Table 2.4.6

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	-3	$\frac{1}{2}$	2	-4
2	$\frac{1}{7}$	-1	$-\frac{5}{2}$	$\frac{7}{3}$

Find the following derivatives, using the table.

(a) $\left.\dfrac{d}{dx}\left(\dfrac{g(x)}{f(x)}\right)\right|_{x=2}$

(b) $\left.\dfrac{d}{dx}\left(f(x)g(x)\right)\right|_{x=0}$

(c) $\left.\dfrac{d}{dx}\left(\dfrac{f(x)+1}{g(x)-2}\right)\right|_{x=0}$

(d) $\left.\dfrac{d}{dx}\left(f(x)g(x)\right)\right|_{x=2}$

3. Use the product and quotient rules to differentiate each of the following.

(a) $f(x) = (4x+4)(7x^2-6)$

(b) $g(x) = \dfrac{x^2+1}{x}$

(c) $j(x) = \dfrac{e^x}{x-1}$

(d) $s(t) = e^t(t^2+t-1)$

(e) $\Gamma(\theta) = 4e^\theta\sqrt{\theta}$

(f) $r(t) = e^{2t}$

(g) $w(x) = (5x^5-4x^2+5x-1)^2$

(h) $p(\alpha) = \dfrac{\alpha^3+1}{\alpha^2-1}$

(i) $f(x) = \dfrac{x+e}{x^3-4x}$

(j) $g(x) = \dfrac{x+4}{e^x}$

4. Differentiate the following.

(a) $y = e^x(x+4)^2(4x-3)$

(b) $y = \dfrac{xe^x+1}{x^2-1}$

(c) $y = \dfrac{x^2 + 4x - 1}{xe^x}$

(d) $y = \left(\dfrac{x^2 - 4}{x + 5}\right)(e^x(x + 3))$

5. Find the equation of the line tangent to the function $f(x)$ at the given point.

(a) $f(x) = \dfrac{x^2 - 5}{x^2 + 1}$ at $(1, -2)$

(b) $f(x) = \dfrac{e^x}{x + 1}$ at $(0, 1)$

(c) $f(x) = e^x(x^2 + 6)$ at $(1, 7e)$

(d) $f(x) = \dfrac{xe^x}{x + 2}$ at $(0, 0)$

2.5 The Chain Rule

Activity 2.5.1 Composition (and Decomposition) Pictionary.

This activity will involve a second group, or at least a partner. We'll go through the first part of this activity, and then connect with a second group/person to finish the second part.

(a) Build two functions, calling them $f(x)$ and $g(x)$. Pick whatever kinds of functions you'd like, but this activity will work best if these functions are in a kind of sweet spot between "simple" and "complicated," but don't overthink this.

(b) Compose $g(x)$ inside of $f(x)$ to create $(f \circ g)(x)$, which we can also write as $f\left(g(x)\right)$.

(c) Write your composed $f\left(g(x)\right)$ function on a separate sheet of paper. Do not leave any indication of what your chosen $f(x)$ and $g(x)$ are. Just write your composed function by itself.

Now, pass this composed $f\left(g(x)\right)$ to your partner/a second group.

(d) You should have received a new function from some other person/group. It is different than yours, but also labeled $f\left(g(x)\right)$ (with different choices of $f(x)$ and $g(x)$).

Identify a possibility for $f(x)$, the outside function in this composition, as well as a possibility for $g(x)$, the inside function in this composition. You can check your answer by composing these!

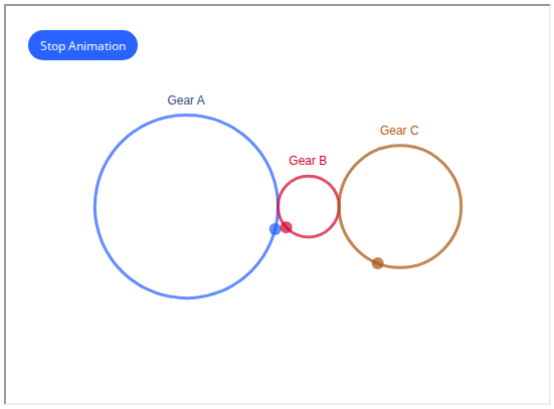
(e) Write a different pair of possibilities for $f(x)$ and $g(x)$ that still will give you the same composed function, $f\left(g(x)\right)$.

(f) Check with your partner/the second group: did you identify the pair of functions that they originally used?

Did whoever you passed your composed function to correctly identify your functions?

Activity 2.5.2 Gears and Chains.

Let's think about some gears. We've got three gears, all different sizes. But the gears are linked together, and a small motor works to spin one of the gears. Since the gears are linked, when one gear spins, they all spin. But since they are different sizes, they complete a different number of revolutions: the smaller ones spin more times than the larger ones, since they have a smaller circumference.



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For our purpose, let's say that Gear A is being driven by the motor.

(a) Let's try to quantify how much "faster" Gear B is spinning compared to Gear A. How many revolutions does Gear B complete in the time it takes Gear A to complete one revolution?

(b) Now quantify the speed of Gear C compared to its neighbor, Gear B. How many revolutions does Gear C complete in the time it takes Gear B to complete one revolution?

(c) Use the above relative "speeds" to compare Gear C and Gear A: how many revolutions does Gear C complete in the time it takes Gear A to complete one revolution?

More importantly, how do we find this?

(d) Now, let's translate this into some derivative notation: we've really been finding rates at which one thing changes (the speed of the gear spinning) relative to another's.

Call the speed of Gear B compared to Gear A: $\frac{dB}{dA}$. Now, call the speed of Gear C compared to Gear B: $\frac{dC}{dB}$. Come up with a formula to find $\frac{dC}{dA}$.

Practice Problems

1. For functions $f(x)$ and $g(x)$, explain how to use the Chain Rule to find $\frac{d}{dx}\left(f(g(x))\right)$.

2. For functions $f(x)$ and $g(x)$, explain how to use the Chain Rule to find $\frac{d}{dx}\left(g(f(x))\right)$.

3. Explain the differences in the two derivatives above.

4. For functions $f(x)$ and $g(x)$, explain how to use the Chain Rule to find $\frac{d}{dx}\left(f(g(f(g(f(g(x)))))))\right)$.

5. Consider the following table of values of f, g, f' and g'.

Table 2.5.3 Function and derivative values at selected points.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	3	0	-2
0	1	6	-1	-4
1	0	-4	1	$\frac{7}{3}$

Find the following derivatives, using the table.

(a) $\dfrac{d}{dx}\left(f(g(x))\right)\Big|_{x=-1}$

(b) $\dfrac{d}{dx}\left(f(f(x))\right)\Big|_{x=0}$

(c) $\dfrac{d}{dx}\left(g(f(x))\right)\Big|_{x=1}$

(d) $\dfrac{d}{dx}\left(g(g(x))\right)\Big|_{x=-1}$

6. Find the following derivatives.

(a) $\dfrac{d}{dx}\left(\tan(x^2)\right)$

(b) $\dfrac{d}{dx}\left(e^{x^3+1}\right)$

(c) $\dfrac{d}{dx}\left(\sin^4(x)\right)$

(d) $\dfrac{d}{dx}\left(\sqrt{\sec(x)}\right)$

(e) $\dfrac{d}{dx}\left(\sqrt{e^{\cos(x)}}\right)$

(f) $\dfrac{d}{dx}\left(\dfrac{4}{\sqrt[3]{x^2+x+1}}\right)$

(g) $\dfrac{d}{dx}\left(\sec(e^x)\right)$

7. Let's introduce a new kind of function: hyperbolic trigonometric functions. For now, let's just defined $y = \sinh(x)$ and $y = \cosh(x)$ (the hyperbolic sine and hyperbolic cosine functions) this way:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

(a) Show that $\frac{d}{dx}\left(\sinh(x)\right) = \cosh(x)$.

(b) Show that $\frac{d}{dx}\left(\cosh(x)\right) = \sinh(x)$.

(c) Let $\tanh(x) = \dfrac{\sinh(x)}{\cosh(x)}$. Find $\frac{d}{dx}\left(\tanh(x)\right)$.

(d) Find $\frac{d}{dx}\left(\sinh(x^2)\right)$

(e) Find $\frac{d}{dx}\left(\sqrt{\cosh(x)}\right)$.

8. Find the following derivatives.

(a) $\dfrac{d}{dx}\left(\dfrac{\sin(e^x)}{\sqrt{x}}\right)$

(b) $\dfrac{d}{dx}\left(e^{x\cos(x)}\right)$

(c) $\dfrac{d}{dx}\left(\tan\left(\dfrac{x+1}{x^2+1}\right)\right)$

(d) $\dfrac{d}{dx}\left((x+5)^2 e^{\tan(x)}\right)$

Chapter 3

Implicit Differentiation

3.1 Implicit Differentiation

Activity 3.1.1 Thinking about the Chain Rule.

(a) Explain to someone how (and why) we use The Chain Rule to find the following derivative:

$$\frac{d}{dx}\left(\sqrt{\sin(x)}\right).$$

(b) Let's say that $f(x) = \sin(x)$. Explain how we find the following derivative:

$$\frac{d}{dx}\left(\sqrt{f(x)}\right).$$

How is this different, or not different, than the previous derivative?

(c) Let's say that we have some other function, $g(x)$. Explain how we find the following derivative:

$$\frac{d}{dx}\left(\sqrt{g(x)}\right).$$

How is this different, or not different, than the previous derivatives?

(d) What is the difference between the following derivatives:

$$\frac{d}{dx}\left(\sqrt{x}\right) \qquad \frac{d}{dx}\left(\sqrt{y}\right) \qquad \frac{d}{dy}\left(\sqrt{y}\right)$$

Activity 3.1.2 Slopes on a Circle.

Visualize the unit circle. Feel free to draw one, or find the picture above. We're going to think about slopes on this circle.

(a) Point out the locations on the unit circle where you would expect to see tangent lines that are perfectly horizontal. What do you think the value of the derivative, $\frac{dy}{dx}$, would be at these points?

(b) Point out the locations on the unit circle where you would expect to see tangent lines that are perfectly vertical. What do you think the value of the derivative, $\frac{dy}{dx}$, would be at these points?

(c) Find the point(s) where $x = \frac{1}{2}$. What do you think the value of the derivative, $\frac{dy}{dx}$, would be at these points?

(d) For the unit circle defined by the equation $x^2 + y^2 = 1$, apply the derivative to both sides of this equation to get the following:

$$\frac{d}{dx}\left(x^2 + y^2\right) = \frac{d}{dx}(1)$$
$$\frac{d}{dx}\left(x^2\right) + \frac{d}{dx}\left(y^2\right) = \frac{d}{dx}(1)$$

Carefully consider each of these derivatives (each of the terms). Which of these will you need to apply the Chain Rule for?

(e) Differentiate. Solve for $\frac{dy}{dx}$ or y', whichever notation you decide to use.

(f) Go back to the first few questions, and try to answer them again:

(a) Find the locations of any horizontal tangent lines (where $\frac{dy}{dx} = 0$).

(b) Find the locations of any vertical tangent lines (where $\frac{dy}{dx}$ doesn't exist, or where you would divide by 0).

(c) Find the values of $\frac{dy}{dx}$ for the points on the circle where $x = \frac{1}{2}$.

Activity 3.1.3 A New Curve.

Let's consider a new curve:

$$\sin(x) + \sin(y) = x^2y^2.$$

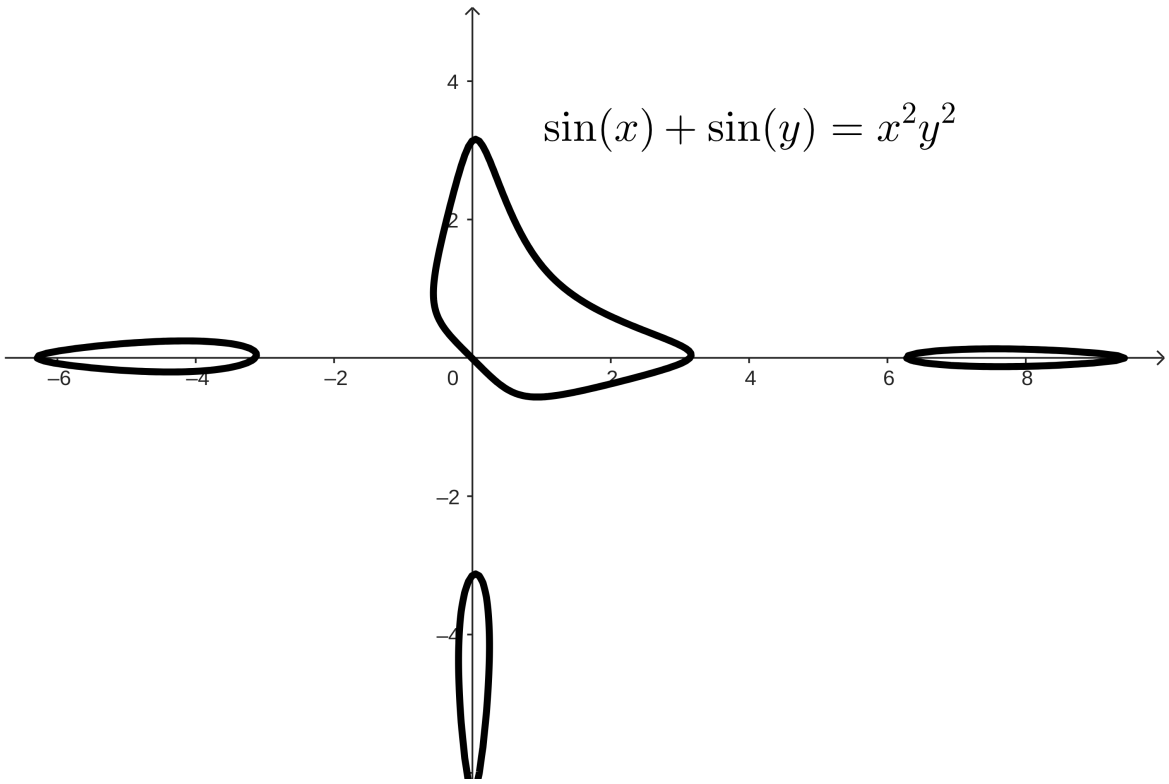


Figure 3.1.4

(a) We are going to find $\frac{dy}{dx}$ or y'. Let's dive into differentiation:

$$\frac{d}{dx}\left(\sin(x) + \sin(y)\right) = \frac{d}{dx}\left(x^2y^2\right)$$
$$\frac{d}{dx}\left(\sin(x)\right) + \frac{d}{dx}\left(\sin(y)\right) = \frac{d}{dx}\left(x^2y^2\right)$$

Think carefully about these derivatives. For each of the three, how will you approach it? What kinds of nuances or rules or strategies will you need to think about? Why?

(b) Implement your ideas or strategies from above to differentiate each term.

(c) Now we need to solve for $\frac{dy}{dx}$ or y', whichever you are using. While this equation can look complicated, we can notice something about the "location" of $\frac{dy}{dx}$ or y' in our equation.

Why do we always know that $\frac{dy}{dx}$ or y' will be *multiplied* on a term whenever it shows up?

(d) Now that we are confident that we will *always* know that we are multiplying this derivative, we can employ a consistent strategy:

(a) Rearrange our equation so that every term with a $\frac{dy}{dx}$ or y' is on one side, and everything without is on the other.

(b) Now we are guaranteed that $\frac{dy}{dx}$ or y' is a common factor: factor it out.

(c) Now we can solve for $\frac{dy}{dx}$ or y' by dividing!

Solve for $\frac{dy}{dx}$ or y' in your equation!

(e) Build the equation of a line that lays tangent to the curve at the origin. Does the value of $\frac{dy}{dx}$ at $(0, 0)$ look reasonable to you?

Activity 3.1.5 Dripping Water.

Two people are sitting at a table. Person A reaches across the table, grabs a water bottle, and begins dripping water onto the table right in front of Person B.
A puddle starts forming, in the shape of a circle.

(a) If the puddle has a thickness of 0.5cm, what is the formula for the volume of the puddle?

(b) Find a formula for how quickly the volume of the puddle is increasing per second.

(c) Convince yourself that of *course* $\frac{dV}{dt}$ is connected to the change in the radius of the puddle, $\frac{dr}{dt}$.

(d) If Person A is dripping the water at a rate of $0.2\text{cm}^3/\text{s}$ and the puddle is currently 13cm wide (in diameter), then how quickly is the edge of the puddle moving along the table?

(e) How does this change when the puddle is larger? For instance, how does this change when the puddle is 20cm wide (in diameter)?

Practice Problems

1. Explain the difference between $\frac{d}{dx}\left(x^2\right)$ and $\frac{d}{dx}\left(y^2\right)$. How is the differentiation process different in each? Why?

2. Explain why we need to use the product rule to differentiate $\frac{d}{dx}\left(ye^x\right)$.

3. Use implicit differentiation to find $\frac{dy}{dx}$ or y'. Make a note of when you are using the product or chain rule, and make sure you know why you are using them.

(a) $x^2 - y^2 = 0$

(b) $4x + \sin(y) = 0$

(c) $ye^x + 1 = 0$

(d) $x^3 \tan(y) + e^y - 10x = 1$

(e) $y^3 - xy = x^2 - 2x$

(f) $4xy - y\sec(x) + x^5 = 0$

(g) $\cos(xy) - x^3y^2 = 0$

(h) $4x^3 = y^2(4 - x)$

4. Pick 2 problems from Practice Problem 3, above, that you can solve for y explicitly. Find the derivative $\frac{dy}{dx}$ or y' this way, and confirm that your answers are the same.

5. Pick 2 problems from Practice Problem 3, above, that you cannot solve for y explicitly. Explain why implicit differentiation is a useful method here.

6. For the following curves, find the equation of the line tangent at the point given.

(a) $3x^3 + 7y^3 = 10y$ at the point $(1, 1)$

(b) $x^3 + y^3 = 9xy$ at the point $(4, 2)$

(c) $y(x^2 + 4) = 8$ at the point $(0, 2)$

(d) $(x^2 + y^2)^2 = \frac{25}{4}xy^2$ at $(1, 2)$

7. Find the locations of the horizontal and vertical tangent lines for the curve:

$$x^2 + y^2 + 2xy = 12.$$

8. Consider the following curve:

$$(xt + 1)^3 = t - x^2 + 7.$$

(a) Find $\frac{dx}{dt}$. Explain where you need to use the chain rule and why.

(b) Find $\frac{dt}{dx}$. Explain where you need to use the chain rule and why.

Related Rates. The rest of the exercises are related rates problems.

9. Consider the equation:

$$3r + 5y^2 - 4 = 3\sin(x).$$

Differentiate both sides with regard to the variable t, and solve for $\frac{dr}{dt}$.

10. Consider the equation:

$$3r + 5y^2 - 4 = 3\sin(x).$$

Differentiate both sides with regard to the variable t, and solve for $\frac{dy}{dt}$.

11. A crab is walking along a path defined by the parabola $y = x^2 + 1$. It moves horizontally forward at a rate of 2 units/second, $\frac{dx}{dt} = 2$. If it is currently at the point $(2, 5)$, what is the crab's vertical speed, $\frac{dy}{dt}$?

12. Person A is holding a balloon 30m away from another Person B. If the balloon is released and floats into the air, rising straight up at a speed of 0.5m/s. Person B watches the balloon, their eye-line forming an angle θ with the ground. When the balloon is 10m up in the air, what is the speed at which θ changes, $\frac{d\theta}{dt}$?

13. A paper cup shaped like a cone is leaking water out of the bottom. The cup is 4 inches tall, and the radius is 1/3 of the height. If the water is leaking at a rate of 1.5 in^3/s while the the cup is full, how quickly is the height of the water in the cup decreasing?

14. A 12 foot ladder is leaning against a vertical wall. Someone is pushing the base of the ladder closer to the wall at a speed of 0.5 ft/s. If the ladder is currently 6 feet away from the wall, how quickly does the top of the ladder move up the wall?

15. A spherical water balloon is leaking water. The radius of the balloon is currently 2 inches, and we notice that it is decreasing at a rate of 1 in/min. How fast is the water leaking, in in^3/min?

3.2 Derivatives of Inverse Functions

Activity 3.2.1 Building the Derivative of the Logarithm.

We're going to accomplish two things here:

1. By the end of this activity, we'll have a nice way of thinking about $\frac{d}{dx}(\ln(x))$, and we will now be able to differentiate functions involving logarithms!
2. Throughout this activity, we're going to develop a way of approaching derivatives of inverse functions more generally. Then we can apply this framework to other functions!

Let's think about this logarithmic function!

(a) We have stated (a couple of times now) how we define the log function:

$$y = e^x \longleftrightarrow x = \ln(y).$$

This arrow goes both directions: the log function is the inverse of the exponential, but the exponential is the inverse of the log function!

Can you rewrite the relationship $y = \ln(x)$ using its inverse (the exponential)?

(b) For your inverted $y = \ln(x)$ from above (it should be $x =$), apply a derivative operator to both sides, and use implicit differentiation to find $\frac{dy}{dx}$ or y'.

(c) A weird thing that we can notice is that when we use implicit differentiation, it is common to end up with our derivative written in terms of both x and y variables. This makes sense for our earlier examples: we needed specific coordinates of the point on the circle, for instance, to find the slope there.

But if $y = \ln(x)$, we want $\frac{dy}{dx}$ or y' to be a function of x:

$$f(x) = \ln(x) \longrightarrow f'(x) = \quad .$$

Your derivative is written with only y values.

Since $y = \ln(x)$, replace any instance of y with the log function. What do you have left?

(d) Remember that $y = \ln(x)$. Substitute this into your equation for $\frac{dy}{dx}$. Can you write this in a pretty simplistic way?

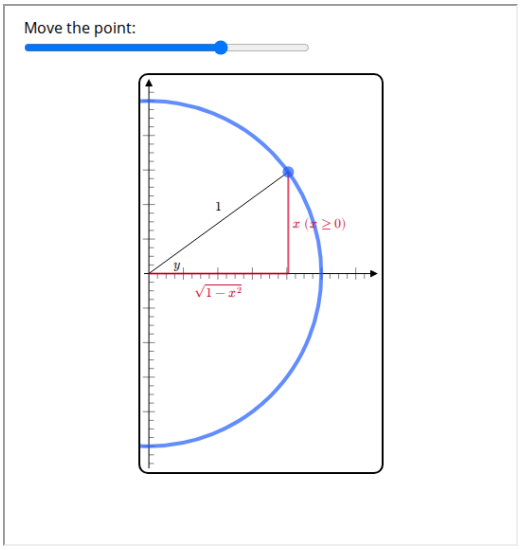
(e) Before we go much further, we should be a bit careful. What is the domain of this derivative? What are the values of x where $\frac{d}{dx}(\ln(x))$ makes sense to think about?

Activity 3.2.2 Finding the Derivative of the Inverse Sine Function.

We're going to do the same trick, except that there will be a couple of small differences due to thinking specifically about trigonometric functions.
Let's think about the function $y = \sin^{-1}(x)$. We know that this is equivalent to $x = \sin(y)$ (for y-values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

(a) Move the point around the portion of the unit circle in the graph below. Convince yourself that:

- $\sin(y) = x$
- $\sin(y) \geq 0$ when $0 \leq y \leq \frac{\pi}{2}$
- $\sin(y) < 0$ when $-\frac{\pi}{2} \leq y < 0$



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In Context
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What is $\cos(y)$ in this figure? Does the sign change depending on the value of y?

(b) Use implicit differentiation and the equation $x = \sin(y)$ to find $\frac{dy}{dx}$ or y'

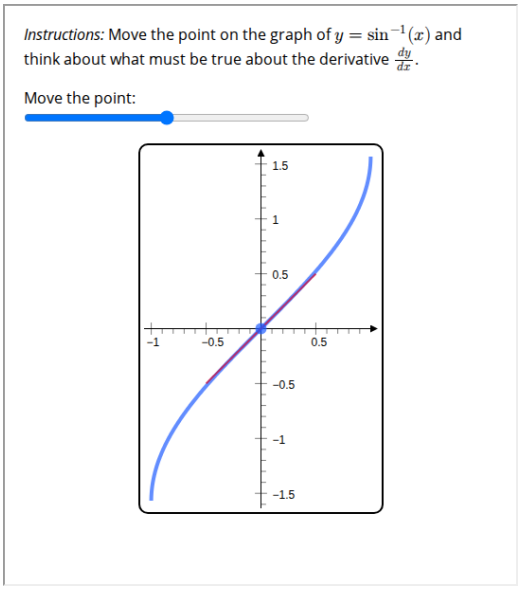
(c) If you still have your derivative written in terms of y, make sure to write $\cos(y)$ in terms of x!

(d) Let's think about the domain of this derivative: what x-values make sense to think about?

Think about this both in terms of what x-values reasonably fit into your formula of $\frac{d}{dx}\left(\sin^{-1}(x)\right)$ as well as the domain of the inverse sine function in general.

(e) Notice that in the denominator of $\frac{d}{dx}\left(\sin^{-1}(x)\right)$, you have a square root. Based on that information (and the visual above), what do you expect to be true about the sign of the derivative of the inverse sine function?

Confirm this by playing with the graph of $y = \sin^{-1}(x)$ below.



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(f) Investigate the behavior of $\frac{dy}{dx}$ at the end-points of the function: at $x = -1$ and $x = 1$. What do the slopes look like they're doing, graphically?

How does this work when you look at the function you built above? What happens when you try to find $\left.\frac{dy}{dx}\right|_{x=-1}$ or $\left.\frac{dy}{dx}\right|_{x=1}$?

Activity 3.2.3 Building the Derivatives for Inverse Tangent and Secant.

(a) Consider the triangle representing the case when $y = \tan^{-1}(x)$.

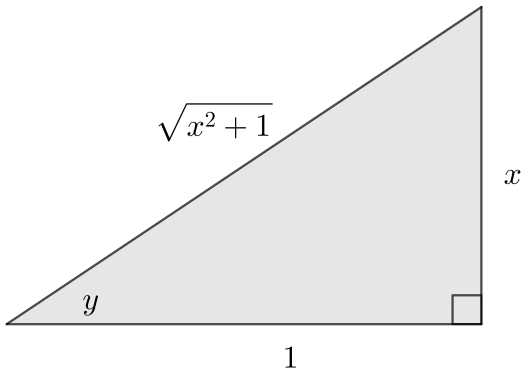


Figure 3.2.4

For $x = \tan(y)$, find $\frac{dy}{dx}$ using implicit differentiation. Find an appropriate expression for $\sec(y)$ based on the triangle above, but we will refer back to the version with the $\sec(y)$ in it later.

(b) Consider the triangle representing the case when $y = \sec^{-1}(x)$.

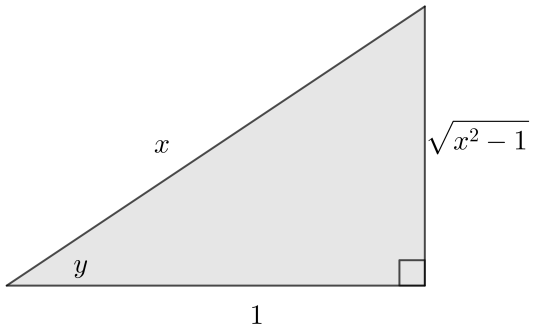
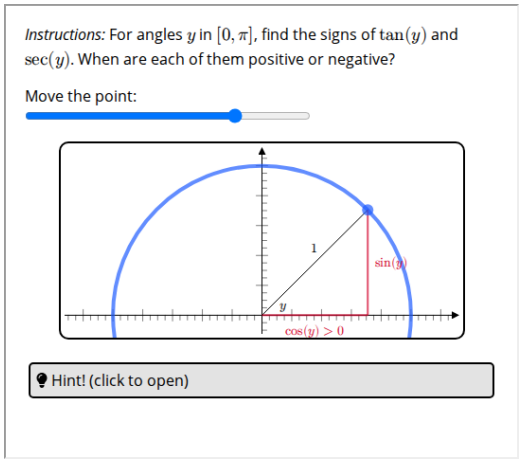


Figure 3.2.5

For $x = \sec(y)$, find $\frac{dy}{dx}$ using implicit differentiation. Find an appropriate expression for $\sec(y)$ and $\tan(y)$ based on the triangle above, but we will refer back to the version with the functions of y in it later.

(c) Here's a graph of just the unit circle for angles $[0, \pi]$. We are choosing to focus on this region, since these are the angles that the inverse tangent and inverse secant functions will return to us. We want to investigate the signs of $\tan(y)$ and $\sec(y)$.



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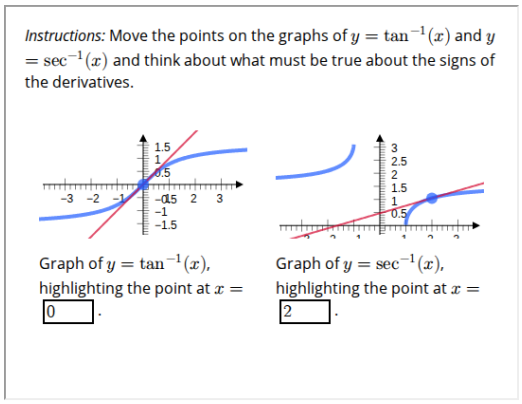
(d) Go back to our derivative expressions for both the inverse tangent and inverse secant functions. What do we know about the signs of these derivatives?

(e) Confirm your idea about the sign of the derivatives by investigating the graphs of each function.



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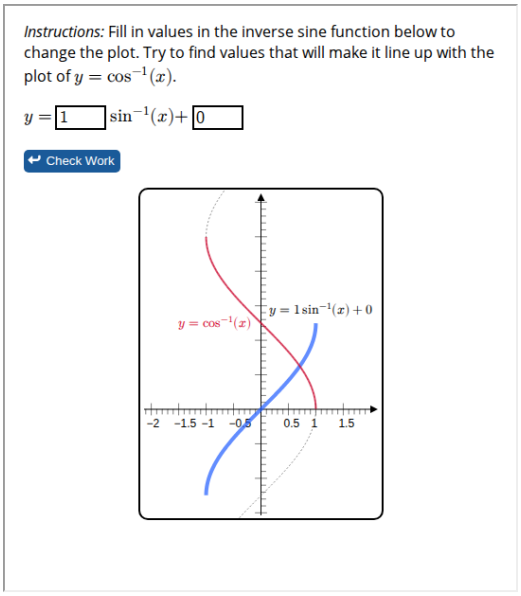
(f) How do we need to write these derivatives, when we write them in terms of x to account for the sign of the derivative?

Activity 3.2.6 Connecting These Inverse Functions.

We're going to look at a graph of $y = \cos^{-1}(x)$, but we're specifically going to try to compare it to the graph of $y = \sin^{-1}(x)$. We'll use some graphical transformations to make these functions match up, and then later we'll think about derivatives.

(a) Ok, consider the graph of $y = \cos^{-1}(x)$ and a transformed version of the inverse sine function. Apply some graphical transformations to make these match!



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(b) It might be fun to think about another reason that this connection between $\sin^{-1}(x)$ and $\cos^{-1}(x)$ exists.

Consider this triangle:

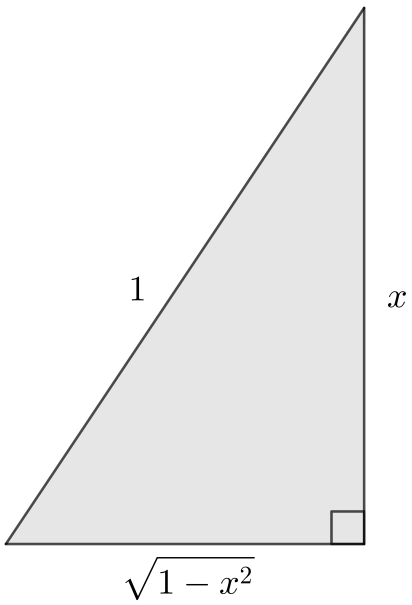


Figure 3.2.7

We're going to think about these inverse trigonometric functions as angles: let $\alpha = \cos^{-1}(x)$

and $\beta = \sin^{-1}(x)$. We can rewrite these as:

$$\cos(\alpha) = x$$
$$\sin(\beta) = x.$$

Can you fill in your triangle using this information?

Why does $\alpha + \beta = \frac{\pi}{2}$? Convince yourself that this is what we did with the graphical transformations above, as well.

(c) Use this equation above, rewriting $\cos^{-1}(x)$ as some expression involving the inverse sine function, and then find

$$\frac{d}{dx}\left(\cos^{-1}(x)\right).$$

We could repeat this task to try to connect the graph of $y = \tan^{-1}(x)$ with $y = \cot^{-1}(x)$ as well as the graph of $y = \sec^{-1}(x)$ with $y = \csc^{-1}(x)$, but we can hopefully see what will happen. In each case, we have the same kind of connection that we saw in the triangle, since these are cofunctions of each other!
We can summarize by believing that:

$$\frac{d}{dx}\left(\cos^{-1}(x)\right) = -\frac{d}{dx}\left(\sin^{-1}(x)\right) = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\left(\cot^{-1}(x)\right) = -\frac{d}{dx}\left(\tan^{-1}(x)\right) = -\frac{1}{x^2+1}$$
$$\frac{d}{dx}\left(\csc^{-1}(x)\right) = -\frac{d}{dx}\left(\sec^{-1}(x)\right) = -\frac{1}{|x|\sqrt{x^2-1}}$$

Practice Problems

1. Find the derivatives of the following functions.

(a) $y = 7^x$

(b) $y = \log_3(x)$

(c) $A(t) = 400(1.001)^{12t}$

(d) $f(x) = \dfrac{3^x}{3^x + 1}$

(e) $g(x) = \ln(x^2 + 1)$

(f) $s(x) = (e^x)(x^e)$

(g) $y = \ln(4x^3)$

(h) $y = \ln(\sqrt{x^4 + x^2})$

(i) $y = x\ln(x) - x$

(j) $y = \sin(4^x)$

(k) $y = e^x \ln(x)$

(l) $r(\theta) = (\theta^2)(5^\theta)$

(m) $y = \ln(\ln(x))$

(n) $y = -\ln(\cos(x))$

2. Find the derivatives of the following functions.

(a) $f(x) = \sin^{-1}(5x)$

(b) $f(x) = x^2 \tan^{-1}(x)$

(c) $f(x) = (\sec^{-1}(x))^3$

(d) $f(x) = \cos^{-1}\left(\frac{1}{x}\right)$

(e) $f(x) = \ln(\sin^{-1}(x))$

(f) $f(x) = \cot^{-1}(e^{2x})$

3. Find the equation of the line tangent to the graph of the following functions at the point listed.

(a) $f(x) = \tan^{-1}(x)$ at $\left(1, \frac{\pi}{4}\right)$

(b) $f(x) = \sin^{-1}\left(\frac{x}{4}\right)$ at $\left(2, \frac{\pi}{6}\right)$

(c) $f(x) = \cos^{-1}(x^2)$ at $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{3}\right)$

(d) $f(x) = \sec^{-1}(e^x)$ at $\left(\ln(2), \frac{\pi}{3}\right)$

4. Graph the function $f(x) = \tan^{-1}(x) - x^3 - x$ using a graphing utility like Desmos.

(a) Find $f'(x)$.

(b) Graph $f'(x)$ using a graphing utility like Desmos.

(c) Where is $f'(x) = 0$?

(d) Where does $f(x)$ have a horizontal tangent line?

3.3 Logarithmic Differentiation

Activity 3.3.1 Returning to the Power Rule.

Back in Section 2.3 we built an explanation for why $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$ that relied on thinking about exponents as repeated multiplication: it relied on n being some positive integer. We said, at the time, that the Power Rule generalizes and works for *any* integer, but did so without explanation. Let's consider $y = x^n$ where n is just some real number without any other restrictions.

(a) Apply a logarithm to both sides of this equation:

$$\ln(y) = \ln(x^n)$$

Now use one of the Properties of Logarithms to rewrite this equation.

(b) Use implicit differentiation to find $\frac{dy}{dx}$ or y'.

(c) Explain to yourself why this is equivalent to the Power Rule that we built so long ago:

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}.$$

(d) Let's get weird. What if we have a not-quite-power function? Where the thing in the exponent isn't simply a number, but another variable?

Let's use the same technique to think about $y = x^x$ and its derivative. First, though, confirm that this is not a power function (and so we cannot use the Power Rule to find the derivative) and is also not an exponential function (and so the derivative isn't itself or itself scaled by a log).

(e) Now apply a log to both sides:

$$\ln(y) = \ln(x^x).$$

Rewrite this using the same log property as before, and then use implicit differentiation to find $\frac{dy}{dx}$ or y'.

(f) Explain to yourself why logs are friends, especially when trying to differentiate functions in the form of $y = (f(x))^{g(x)}$.

Activity 3.3.2 Logarithmic Differentiation with Products and Quotients.

Let's say we've got some function that has products and quotients in it. Like, a lot. Consider the function:

$$y = \frac{(x-4)^2\sqrt{3x+1}}{(x+1)^7(x+5)^3}.$$

(a) Work out a general strategy for how you would find y' directly. Where would you have to use Quotient Rule? What are the pieces? Where would you have to use Product Rule? What are the pieces? Where would you have to use the Chain Rule? What are the pieces?

To be clear: do not actually differentiate this. Just look at it in horror and try to outline a plan that some other fool would use.

(b) Let's instead use logarithmic differentiation. First, apply the log to both sides to get:

$$\ln(y) = \ln\left(\frac{(x-4)^2\sqrt{3x+1}}{(x+1)^7(x+5)^3}\right).$$

Since this function is just a bunch of products of things with exponents all put into some big quotient, we can use our log properties to rewrite this!

(c) We should have:

$$\ln(y) = 2\ln(x-4) + \frac{1}{2}\ln(3x+1) - 7\ln(x+1) - 3\ln(x+5).$$

Confirm this.

(d) Now differentiate both sides! You'll have to use some Chain Rule (but not a lot)! Refer back to Fact 3.3.1 for help here.

(e) Solve for $\frac{dy}{dx}$ or y'.

(f) While this is not a *nice* looking expression for the derivative, spend some time confirming that this was a nicer *process* than differentiating directly. This is because logs are friends.

Practice Problems

1. For each of the following functions, find the derivatives in two ways: directly, and after using properties of logs or exponentials to rewrite the function and differentiate.

(a) $f(x) = \ln\left(\sqrt{5x^2e^x}\right)$

(b) $g(x) = \left(e^{5x^3+\ln(x)}\right)^7$

2. Use logarithmic differentiation to find the derivative of each of the following.

(a) $y = (4x)^{10x}$

(b) $y = (\sin(x))^{\cos(x)}$

(c) $y = (x+5)^6(x-5)^9(3x+1)^3$

(d) $y = \dfrac{(x+2)^{2/3}(x+5)^{7/3}}{(x-4)^{5/3}}$

3. For the function $f(x) = x^2e^{-x^2}$, find the points where the graph of f has horizontal tangent lines.

Chapter 4

Applications of Derivatives

4.1 Mean Value Theorem

Practice Problems

1. For each of the following scenarios, explain whether the Mean Value Theorem applies. If it does, explain what it tells us.

(a) A student in a class is distracted by someone else tapping their fingers on their desk. The student decides to count the taps. In the 3 minutes that they counted, the other student tapped on their desk 245 times.

(b) On a sunny, Chicago day in May, the temperature was 58° at 7:15am. Later on, at 3:45pm, the temperature was 79°.

(c) In a small oceanside town, residents notice that the tide goes out and in every day. High tide is at 10am, and then again at 10pm, and, wow this is pretty incredible, the water level is exactly the same at both times.

2. For each function:

- Sketch (or look up) a graph of the curve on the interval.
- Calculate the slope of the line connecting the end points.
- Find the location of any points that exist from the Mean Value Theorem.

(a) $f(x) = x^2 - 4x + 1$ on the interval $[0, 3]$.

(b) $g(x) = \dfrac{1}{\sqrt{x-3}}$ on the interval $[4, 7]$.

(c) $r(x) = \sin(x)\cos(x)$ on the interval $\left[0, \frac{\pi}{2}\right]$.

4.2 Increasing and Decreasing Functions

Activity 4.2.1 How Should We Think About Direction?

Our goal in this activity is to motivate some new terminology and results that will help us talk about the "direction" of a function and some interesting points on a function (related to the direction of a function). For us to do this, we'll look at some different examples of functions and try to think about some unifying ideas.

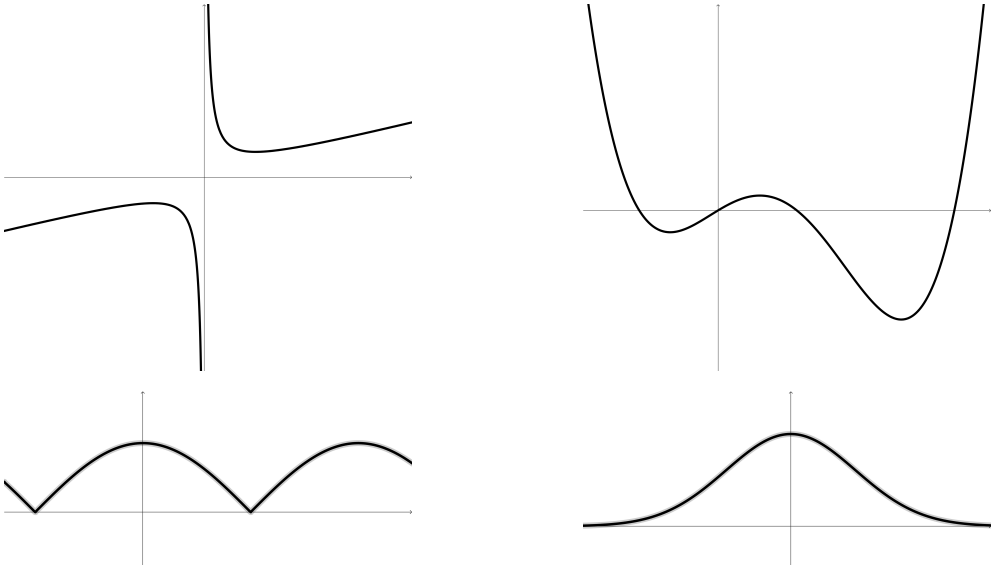


Figure 4.2.2

These examples do not cover all of the possibilities of how a function can act, but will hopefully provide us enough fertile ground to think about some different situations.

(a) In each graph, find and identify:

- The intervals where the function is increasing.
- The intervals where the function is decreasing.
- The points (or locations) around and between these intervals, the points where the function changes direction or the direction terminates.

(b) Make a conjecture about the behavior of a function at any point where the function changes direction.

(c) Look at the highest and lowest points on each function. You can even include the points that are highest and lowest just compared to the points around it. Make a conjecture about the behavior of the function at these points.

Activity 4.2.3 Comparing Critical Points.

Let's think about four different functions:

- $f(x) = 4 + 3x - x^2$
- $g(x) = \sqrt[3]{x+1} + 1 + x$
- $h(x) = (x-4)^{2/3}$
- $j(x) = 1 - x^3 - x^5$

Our goal is to find the critical points on the interval $(-\infty, \infty)$ and then to try to figure out if these critical points are local maximums or local minimums or just points that the function increases or decreases through.

(a) To start, we're going to be finding critical points. Without looking at a picture of the graph of the function, find the derivative.

Are there any x-values (in the domain of the function) where the derivative doesn't exist? We are normally looking for things like division by 0 here, but we could be finding more than that. Check out When Does a Derivative Not Exist? to remind yourself if needed.

Are there any x-values (in the domain of the function) where the derivative is 0?

(b) Now that we have the critical points for the function, let's think about where the derivative might be positive and negative. These will correspond to the direction of a function, based on Theorem 4.1.5 Sign of the Derivative and Direction of a Function.

Write out the intervals of x-values around and between your list of critical points. For each interval, what is the sign of the derivative? What do these signs mean about the direction of your function?

(c) Without looking at the graph of your function:

- What changes about how your function increases up to or decreases down to a critical point based on whether the derivative was 0 or the derivative didn't exist?
- Does your function change direction at a critical point? What will that look like, whether it does or does not change direction?

(d) Give a basic sketch of your graph. It might be helpful to find the y-values for any critical points you've got. Then you can sketch your function increasing/decreasing in the intervals between these points.

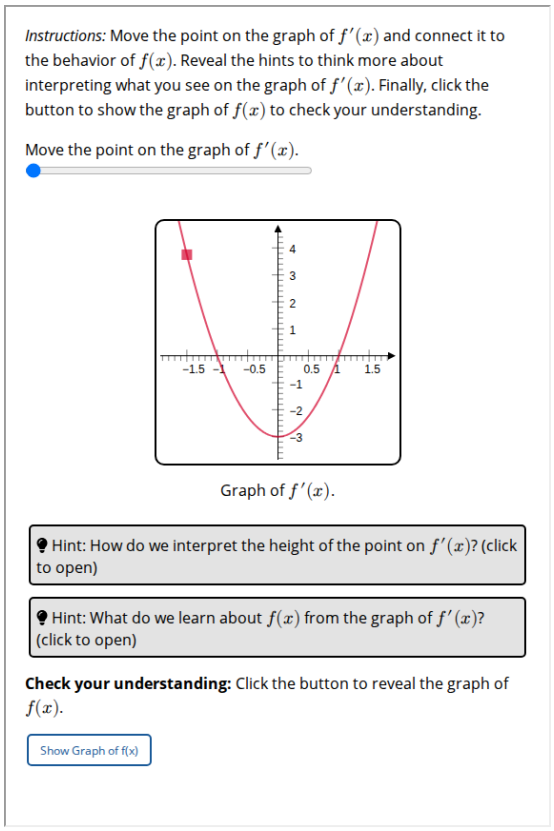
In your sketch, include enough detail to tell whether the derivative is 0 or doesn't exist at each critical point.

(e) Compare your sketch to the actual graph of the function (you can find all of the graphs in the hint).

Activity 4.2.4 First Derivative Test Graphically.

Let's focus on looking at a picture of a derivative, $f'(x)$, and trying to collect information about the function $f(x)$. This is what we've done already, except that we've done it by thinking about the representation of $f'(x)$ as a function rule written out with algebraic symbols. Here we'll focus on connecting all of that to the picture of the graphs.

For all of the following questions, refer to the plot below. You can add information with the hints whenever you need to. Don't reveal the picture of $f(x)$ until you're really ready to check what you know.



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(a) Based on the graph of $f'(x)$, estimate the interval(s) of x-values where $f(x)$ is increasing.

(b) Based on the graph of $f'(x)$, estimate the interval(s) of x-values where $f(x)$ is decreasing.

(c) Find the x-values of the critical points of $f(x)$. Once you've estimated these, classify them as local maximums, local minimums, or neither. Explain your reasoning.

(d) What do you think the graph of $f(x)$ looks like? Do your best to sketch it or explain it before revealing it!

(e) Why could we estimate the x-values of the critical numbers of $f(x)$, but not find the actual coordinates? How come we can't find the y-value based on looking at the graph of $f'(x)$?

Practice Problems

1. Explain what a critical number of $f(x)$ is. What are the things we're looking for?

2. For the following functions, find the critical numbers and explain each one. Is it one where the derivative is 0, or one where the derivative doesn't exist? Feel free to confirm these by graphing the function!

(a) $f(x) = x^2 + 5x - 4$ on $[-4, 2]$

(b) $f(x) = \sqrt[3]{x+4}$ on $[-6, -2]$

(c) $f(x) = \dfrac{x^2+1}{x-3}$

(d) $f(x) = x^2 e^{-x}$

(e) $f(x) = x^2 e^{-x}$ on $[1, 10]$

(f) $f(x) = \ln\left(x^4 + x + 1\right)$

3. Find the critical numbers of the function $f(x)$, graphed below. Explain each one.

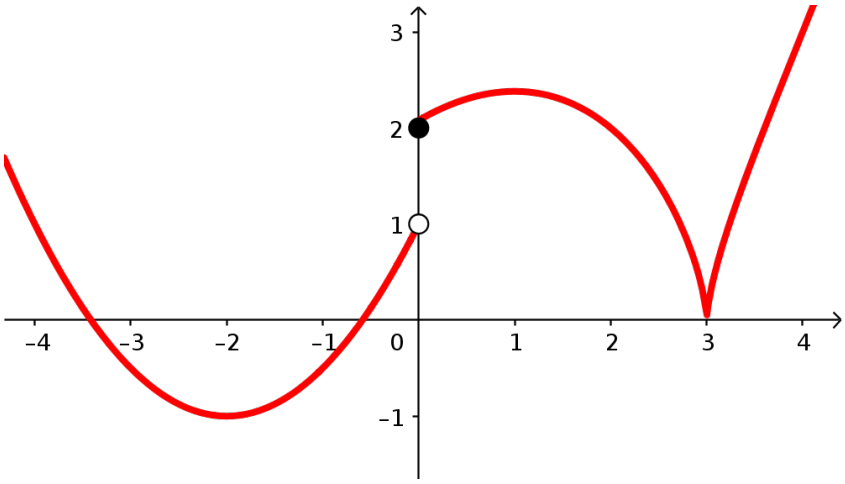


Figure 4.2.5 Graph of $f(x)$.

4. Give an example of a function that has $f'(x) > 0$ on $(-\infty, 4)$ and $f'(x) < 0$ on $(4, \infty)$. Show graphically where the function is increasing and decreasing. Explain how to find out the location of a local maximum or minimum for your example.

5. Sketch a graph of a function that increases everywhere, but has $f'(4) = 0$. Explain what sign the derivative is when $x \neq 4$.

6. For each of the following functions, use the first derivative to find the intervals where the function is increasing or decreasing.

(a) $f(x) = x^2 e^x$

(b) $f(x) = x\ln(x) - x$

(c) $f(x) = \dfrac{1}{x^2}$

(d) $f(x) = \ln(x^4 + 1)$

(e) $f(x) = \dfrac{x-4}{x^2+1}$

(f) $f(x) = e^{-x^2-2x}$

(g) $f(x) = \dfrac{\sqrt{x}}{e^x}$

(h) $f(x) = \sin(x)$ on $\left[0, \frac{\pi}{2}\right]$

(i) $f(x) = \cos^3(x)$ on $(-2\pi, 2\pi)$

7. Explain how the First Derivative Test helps us classify critical numbers as local maximums or minimums.

8. Find any local maximums or local minimums for each of the following functions. Note that these are the same functions that you've already worked with in Practice Problem 6.

(a) $f(x) = x^2 e^x$

(b) $f(x) = x\ln(x) - x$

(c) $f(x) = \dfrac{1}{x^2}$

(d) $f(x) = \ln(x^4 + 1)$

(e) $f(x) = \dfrac{x-4}{x^2+1}$

(f) $f(x) = e^{-x^2-2x}$

(g) $f(x) = \dfrac{\sqrt{x}}{e^x}$

(h) $f(x) = \sin(x)$ on $\left[0, \frac{\sqrt{\pi}}{2}\right]$

(i) $f(x) = \cos^3(x)$ on $(-2\pi, 2\pi)$

9. Sketch a graph of a function $f(x)$ that is continuous on $(-\infty, \infty)$ and has the following characteristics. Label the intervals where f is increasing/decreasing, and find the locations of any local maximums/minimums.

- $f'(x) > 0$ on $(-\infty, 3)$
- $f'(3)$ does not exist
- $f'(x) < 0$ on $(3, 6)$
- $f'(6) = 0$
- $f'(x) < 0$ on $(6, \infty)$

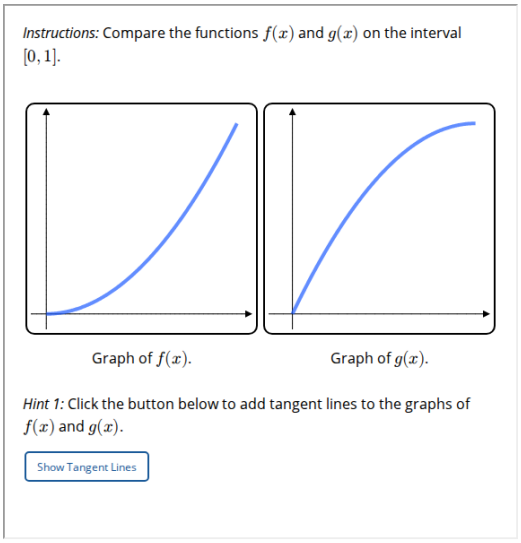
10. Sketch a graph of a function $f(x)$ and has the following characteristics. Label the intervals where f is increasing/decreasing, and find the locations of any local maximums/minimums.

- $f'(x) < 0$ on $(-\infty, 3)$
- $f'(3) = 0$
- $f'(x) > 0$ on $(3, 6)$
- $f(x)$ has a vertical asymptote at $x = 6$
- $f'(x) < 0$ on $(6, \infty)$

4.3 Concavity

Activity 4.3.1 Compare These Curves.

(a) Consider the two curves pictured below. Compare and contrast them. What characteristics of these functions are the same? What characteristics of these functions are different?



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(b) Explain the similarities you found by only talking about the slopes of each function (the values of $f'(x)$ and $g'(x)$).

(c) Explain the differences you found by only talking about the slopes of each function (the values of $f'(x)$ and $g'(x)$).

(d) Make a conjecture about the *rate of change* of both f' and g'. We'll call these things **second derivative** functions, $f''(x)$ and $g''(x)$.

Activity 4.3.2 Finding a Function's Concavity.

We're going to consider a pretty complicated function to work with, and employ a strategy similar to what we did with the first derivative:

- Our goal is to find the sign of $f''(x)$ on different intervals and where that sign *changes*.
- To do' this, we'll find the places that $f''(x) = 0$ or where $f''(x)$ doesn't exist. These are the critical points of $f'(x)$.
- From there, we can build a little sign chart, where we split up the x-values based on the domain of f and the critical numbers of f'. Then we can attach each interval of x-values with a sign of the second derivative, f'', on that interval.
- We'll interpret these signs. For any intervals where $f''(x) > 0$, we know that $f'(x)$ must be increasing and so $f(x)$ is concave up. Similarly, for any intervals where $f''(x) < 0$, we know that $f'(x)$ must be decreasing and so $f(x)$ is concave down.

Let's consider the function

$$f(x) = \ln(x^2 + 1) - \frac{x^2}{2}.$$

(a) Find the first derivative, $f'(x)$, and use some algebra to confirm that it is really:

$$f'(x) = -\frac{x(x+1)(x-1)}{x^2+1}.$$

While we have this first derivative, we *could* find the critical points of $f(x)$ and then classify those critical points using the First Derivative Test.

For our goal of finding the intervals where $f(x)$ is concave up and concave down, and then finding the inflection points of f, let's move on.

(b) Find the second derivative, $f''(x)$, and confirm that this is really:

$$f''(x) = -\frac{x^4 + 4x^2 - 1}{(x^2+1)^2}.$$

(c) Our goal is to find the x-values where $f''(x) = 0$ or where $f''(x)$ doesn't exist.

Note that in order to find where $f''(x) = 0$, we are really looking at the x-values that make the numerator 0:

$$x^4 + 4x^2 - 1 = 0.$$

Then, to find where $f''(x)$ doesn't exist, we are finding the x-values that make the denominator 0:

$$(x^2+1)^2 = 0.$$

Solve these equations.

(d) You have two critical points of $f'(x)$ (let's just call them x_1 and x_2). These are possible inflection points of $f(x)$, but we need to check to see if the concavity changes at these points.

Fill in the following sign chart and interpret.

x	$(-\infty, x_1)$	x_1	(x_1, x_2)	x_2	(x_2, ∞)
f''					
f					

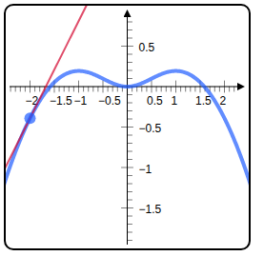
Let's confirm what we've just calculated graphically. Below, we have three graphs:

1. $f(x) = \ln(x^2+1) - \dfrac{x^2}{2}$
2. $f'(x) = -\dfrac{x(x+1)(x-1)}{x^2+1}$
3. $f''(x) = -\dfrac{x^4+4x^2-1}{(x^2+1)^2}$

Move the point on any graph and make sure the statements about signs, directions, and concavity match what you found! You should notice that signs of the first and second derivative change at the local maximums/minimums and the inflection points that we found.

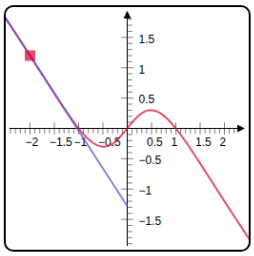
Instructions: Move the point to see the behavior of $f(x)$, $f'(x)$, and $f''(x)'$ graphically. Compare what you've found, and turn on or off information using the checkboxes underneath the graphs.

Change the location of the point on each graph.



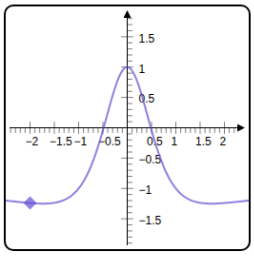
- The outputs of $f(x)$ are increasing and concave down.
- This is because the the slopes of $f(x)$ are positive and decreasing.
- This is because the concavity of $f(x)$ is negative.

Graph of $f(x)$.



- The outputs of $f'(x)$ are positive and decreasing.
- This is because the the slopes of $f'(x)$ are negative.

Graph of $f'(x)$.



- The outputs of $f''(x)$ are negative.

Graph of $f''(x)$.

☐ Show the local maximums and local minimums of $f(x)$.

☐ Show the inflection points of $f(x)$.

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Practice Problems

1. Consider a function $f(x)$ that is continuous on $(-\infty, \infty)$ with $f''(x) > 0$ on $(-\infty, 5)$ and $f''(x) < 0$ on $(5, \infty)$. What do we know about $f'(x)$ on these intervals? What do we know about $f(x)$ on these intervals?

2. Consider a function $g(x)$ that is continuous on $(-\infty, \infty)$ that is concave down on $(-\infty, -2)$, concave up on $(-2, 4)$, and concave down on $(4, \infty)$. Where are the inflection points of $g(x)$? What is happening to $g'(x)$ at these points?

3. Explain (with words and/or sketches) why the Second Derivative Test requires $f'(a) = 0$ for the second derivative to classify the critical number a as a local maximum or minimum. Why can't the second derivative classify a critical point where $f'(a)$ does not exist?

4. For each of the following functions, use the second derivative to find the intervals where the function is concave up or concave down, and find the location of any inflection points.

(a) $f(x) = e^x - x^2$

(b) $f(x) = \ln(x^2 + 1)$

(c) $f(x) = e^{-x^2}$

(d) $f(x) = 1 - 2x + \dfrac{x^5}{20} - \dfrac{x^4}{12}$

(e) $f(x) = \dfrac{4x^{5/2}}{15} - x^2$

(f) $f(x) = 5\cos(x) - 2x + 1$ on $(0, 2\pi)$

(g) $f(x) = x^2 e^x$

(h) $f(x) = \tan^{-1}(x)$

(i) $f(x) = \dfrac{1}{(x-1)(x-2)}$

(j) $f(x) = 2x^4 + x$

5. Find critical points of the following functions, and apply the Second Derivative Test to classify those critical points as local maximums/minimums. If the test fails, explain why.

(a) $f(x) = 2x^4 + x$

(b) $f(x) = \dfrac{4}{x} + 3x$ on $(0, \infty)$

(c) $f(x) = (x-1)e^x$

(d) $f(x) = \sqrt{x^3} - x$

6. Sketch a graph of a function $f(x)$ and has the following characteristics. Label the intervals where f is concave up/down, and find the locations of any local inflection points.

- $f''(x) < 0$ on $(-\infty, -1)$
- $f''(x) > 0$ on $(-1, 4)$
- $f'(x) < 0$ on $(-\infty, 2)$
- $f'(x) > 0$ on $(2, 6)$
- $f'(6) = 0$

- $f''(x) < 0$ on $(4, 6)$
- $f'(x) > 0$ on $(6, \infty)$
- $f''(x) > 0$ on $(6, \infty)$

7. Consider a function $f(x)$ that is differentiable on $(-\infty, \infty)$. We graph the *first derivative*, $f'(x)$, below.

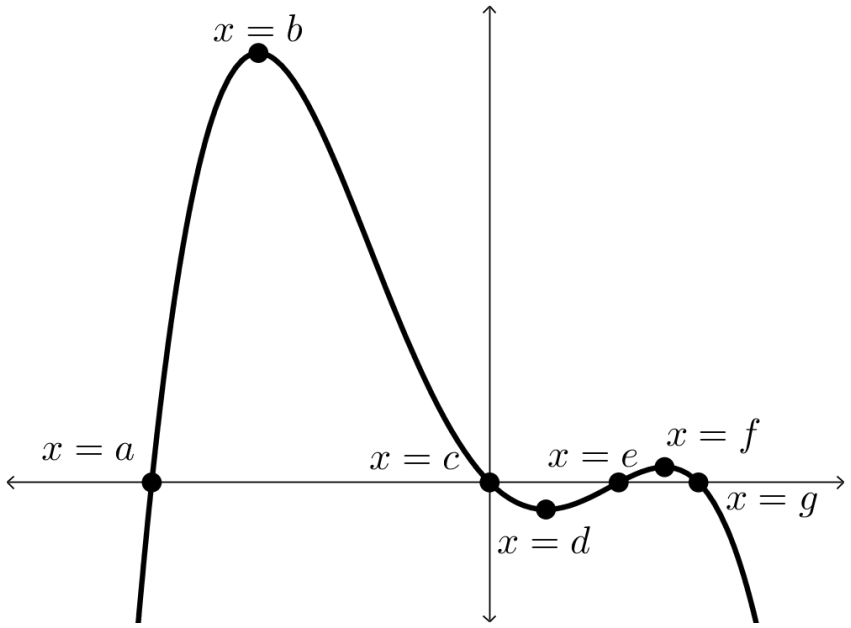


Figure 4.3.3 Graph of $f'(x)$.

(a) Where is the function $f(x)$ concave up?

(b) Where is the function $f(x)$ concave down?

(c) Where are the inflection points of $f(x)$? What do they correspond with on the graph of $f'(x)$? Explain.

(d) Explain whether $x = a$ is the location of a local max of $f(x)$, a local min of $f(x)$, or neither. Reference both the First and Second Derivative Test.

(e) Where are the other local maximums/minimums of $f(x)$?

8. Consider a function $g(x)$ that is differentiable on $(-\infty, \infty)$. We graph the *second derivative*, $g''(x)$, below.

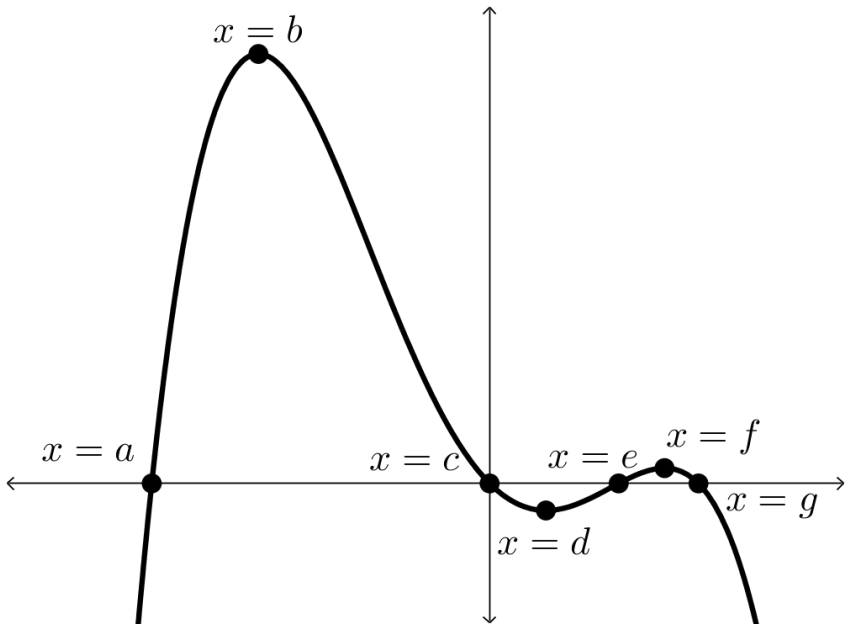


Figure 4.3.4 Graph of $g''(x)$.

(a) Where is the function $g(x)$ concave up?

(b) Where is the function $g(x)$ concave down?

(c) Where are the inflection points of $g(x)$? What do they correspond with on the graph of $g''(x)$? Explain.

(d) If there is some x-value, (call it x_0) with $a < x_0 < b$ where we know that $g'(x_0) = 0$, then what else can we say about this point on the graph of $g(x)$?

4.4 Global Maximums and Minimums

Activity 4.4.1 When Would We Not Have Maximums or Minimums?

In this section, we're going to define these global maximums and then, most importantly, try to predict when these global maximums or global minimums might actually exist for a function.
To start, maybe we should come up with some examples of functions that do not have them!

(a) Come up with some situations where a function does not have some combination of global maximum/minimums. Sketch some graphs!

(b) Come up with some examples of graphs of functions that are bounded (do not ever have y-values that tend towards infinity in a limit) that do not have some combination of global maximum/minimums.

(c) For the examples of graphs that you have built or collected, what are the features of the functions that allow for the examples you picked? If you could impose some requirements that would "fix" the examples you found (so that they had both a global maximum and a global minimum), what requirements could you use?

Practice Problems

1. What does the Extreme Value Theorem say? Why is it helpful?

2. Where, in general, do we find global/absolute extrema?

3. For each function, describe whether or not the Extreme Value Theorem applies (and explain why or why not). If it does apply, find the absolute maximum and absolute minimum.

(a) $f(x) = x^2 + 5x - 4$ on $[-4, 2]$

(b) $f(x) = \sqrt[3]{x+4}$ on $[-6, -2]$

(c) $f(x) = |x - 1|$ on $[-4, 0]$

(d) $f(x) = \dfrac{1}{x^2}$ on $[-3, 1]$

(e) $f(x) = x^{2/3}(2 - x)^2$ on $[-1, 1]$

(f) $f(x) = \begin{cases} 4x & \text{if } 0 \le x < 1 \\ \sqrt{x} & \text{if } 1 \le x \le 4 \end{cases}$

4.5 Optimization

Activity 4.5.1 Constructing a Can.

A typical can of pop is 355 ml, and has around 15 ml of headspace (air). This results in a can that can hold approximately 23 cubic inches of volume.
Let's say we want to construct this can in the most efficient way, where we use the least amount of material. How could we do that? What are the dimensions of the can?

(a) First, let's think of our can and try to translate this to some geometric shape with variable names. Collect as much information as we can about this setup! What is the shape? What are the names of the dimensions?

(b) What is the actual measurement that we are trying to optimize? Are we finding a maximum or a minimum?

(c) What other information about the can do we know? How do we translate this into a constraint, or a connection between our variables?

(d) What kinds of values can our variables take on? Is there a smallest value for either? A largest? Are there other limitations to these?

(e) Now we need to set up a calculus statement. This part mostly relies on us finding a way to build a single-variable function defining the surface area. Build that function, and then write down the calculus statement.

(f) Do some calculus to find the global maximum or minimum, and solve the optimization problem.

(g) What is the relationship between r and h, here? How do they compare? What about the height and diameter of our can?

Is this relationship noticeable in a typical can of pop?

Practice Problems

1. When we set up an optimization problem, we normally look for some formula connecting information that we have with the thing we want to optimize, some sort of constraint or link between variables, and an clue to whether we want to find a maximum or minimum. From this, we work to create a "Calculus Statement" which includes a function (of one variable) that we want to find the absolute maximum or minimum of on a domain.

(a) Why is it important to create a function of one variable? How do we normally do this?

(b) Why is it important to find a domain for the function? How do we normally do this?

(c) Why is it important to note whether we're looking for an absolute maximum or an absolute minimum?

2. Let's say we're looking at two positive numbers that add up to 7. What is the maximum product that we can get when we multiply these two numbers together?

3. Out of all of the rectangles that have an area of 24 square units, which one has the smallest perimeter? Report the dimensions and the perimeter.

4. Consider a box created from an 18in square piece of material. We form the box by cutting an h-inch square out of each corner, and folding the sides up along the dotted lines.

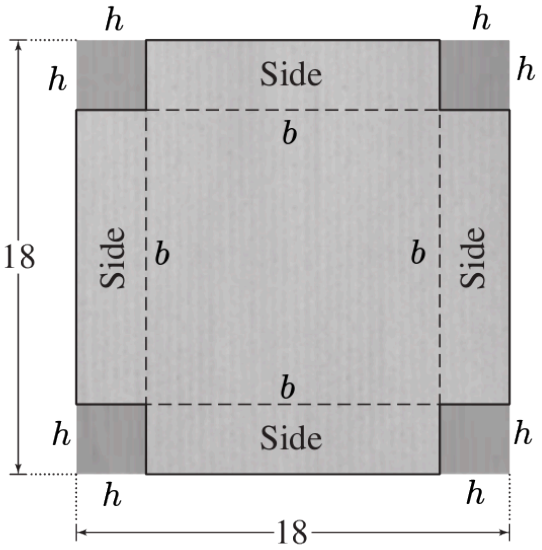


Figure 4.5.2 Material for the box.

(a) What are the dimensions of the resulting open-topped box?

(b) How are the values b and h related to each other?

(c) What is the largest possible value for h? What is the corresponding value of b?

(d) What is the smallest possible value for h? What is the corresponding value of b?

(e) Find the value for h that maximizes the volume of the box. What is the maximum volume? What are the dimensions of the box?

5. A trucking service is looking to add a route to their plan, but the road that they are considering passes through a tunnel. The top of the opening above the driving lane is formed by the curve $y = 4 - \frac{4}{27}x^3$. The width from the center line to the edge of the tunnel is 3 meters. What are the dimensions for the largest rectangular trailer the trucking service could pull through the tunnel?

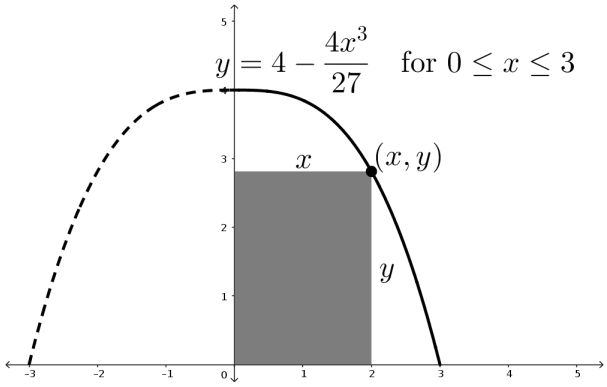


Figure 4.5.3 The truck in the tunnel.

6. A circular area of radius 20ft. is surrounded by a paved pathway. A lamp post is placed in the middle of the circular area. What height should the light be to illuminate the walk most strongly? The intensity of illumination is $I = \dfrac{\sin\theta}{s}$ where s is the distance form the light source and θ is the angle at which the light hits the surface.

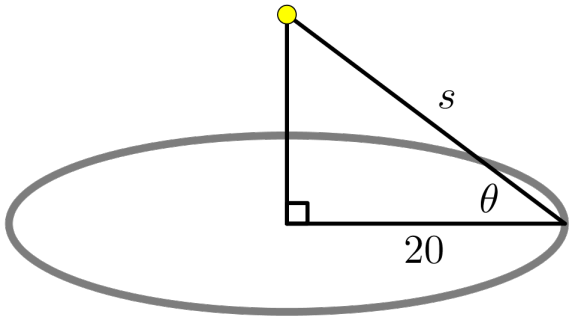


Figure 4.5.4 A solitary lampost.

4.6 Linear Approximations

Activity 4.6.1 Approximating an Exponential Function.

Let's consider the function $f(x) = e^x$. We're going to build the linear approximation, $L(x)$, but we also want to focus on understanding what the "center" is, and how we think about accuracy of our estimations.

(a) We first need to find a "good" center for our linear approximation. We have two real requirements here:

(a) We need the center to be some x-value that will be "close" to the inputs we're most interested in. We know that $L(x) \approx f(x)$ for x-values "near" the center, so we should keep this in mind. We don't have a specific input that we're interested in (we are not specifically focused on estimating $f(7.35)$ for instance), so we don't need to worry about this for now.

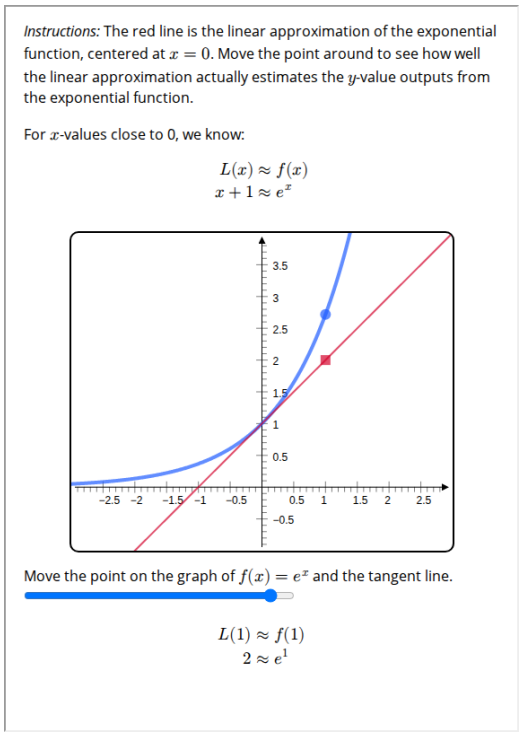
(b) We are going to need to evaluate the function and its derivative at the center: we use $f'(a)$ to find the slope and $f(a)$ to find a y-value for a point on the line. We'd like to choose a center that will make evaluating these functions reasonable, if we can!

We are going to choose a center of 0: why?

(b) Build a linear approximation of $f(x) = e^x$ centered at $x = 0$.

(c) Use your linear approximation to estimate the value of $\sqrt[10]{e} = e^{\frac{1}{10}}$.

(d) Let's visualize this approximation a bit:



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Are you confident in your approximation of $\sqrt[10]{e}$? Would you be more or less confident in an approximation of $\frac{1}{e}$? Why?

(e) Is your estimate of $\sqrt[10]{e}$ too big or too small? How can you tell, without even calculating the actual value of $\sqrt[10]{e}$?

How can you tell that *every* estimate that you get out of *any* linear approximation of e^x (no matter what the center is) is going to be too small?

Activity 4.6.2 Approximating some Values.

Pick one of the following values to approximate:

- $\sin(-0.023)$
- $\ln(2)$
- $\sqrt{8}$
- $\sqrt[3]{10}$

Throughout the rest of this activity, use your value to build a linear approximation of some relevant function and estimate the value you chose.

(a) To build a linear approximation of some function at some center, we need two things:

(a) A function.

(b) A center.

What function will you be using for $f(x)$? Why that one?

(b) What center are you choosing? Why that one?

(c) Build your linear approximation at your center! You should end up with an actual linear function. It might be helpful to plot this linear function and your actual function to confirm that you have actually built a tangent line.

(d) Use your linear approximation function to estimate your value! Report the estimate, and comment on the accuracy of your estimate. Without calculating the actual value, can you tell if this is close or not? Do you have an overestimate or underestimate?

Activity 4.6.3 Walking in the Footsteps of Ancient Mathematicians.

Let's travel all the way back to the first (or maybe second) century AD and re-create Heron's method to approximate the value of $\sqrt{2}$. We'll develop this using modern calculus, and simple linear approximation.

We're going to reframe the problem, and instead we're going to try to use a linear approximation of $f(x) = x^2 - 2$ to approximate the x-value where $f(x) = 0$. We know enough about quadratic functions to know that there are two values: $x = -\sqrt{2}$ and $x = \sqrt{2}$.

(a) We're going to build a linear approximation of $f(x) = x^2 - 2$, and we need a reasonable center. Honestly, any integer will work, since we can evaluate f and f' really easily, but we want to find one that is close to $\sqrt{2}$. Let's center our approximation at $x = 2$.

Find $f'(x)$, and then construct the linear approximation:

$$L(x) = f'(2)(x - 2) + f(2).$$

(b) Now we know that $L(x) \approx f(x)$ for x-values near our center, $x = 2$. What if we estimate the x-value where $f(x) = 0$ by solving $L(x) = 0$ instead? Since $L(x) \approx f(x)$, the x-value where $L(x) = 0$ should make $f(x)$ pretty close to 0 at least.

Solve $L(x) = 0$.

(c) Ok, this might be kind of close to the value of $\sqrt{2}$, right? Let's visualize this.

Hm...so this isn't that good of an approximation yet. We can check this by looking at the actual value of our function at $x = \frac{3}{2}$ and seeing if it's close to 0.

$$\begin{aligned} f\left(\frac{3}{2}\right) &= \left(\frac{3}{2}\right)^2 - 2 \\ &= \frac{9}{4} - 2 \\ &\\ &\frac{1}{4} \end{aligned}$$

This...isn't that close to 0.

So let's try this again. This time, though, let's center our *new* linear approximation at $x = \frac{3}{2}$.

(d) Now set this *new* linear approximation equal to 0 and solve $L(x) = 0$ to estimate the solution to $f(x) = 0$.

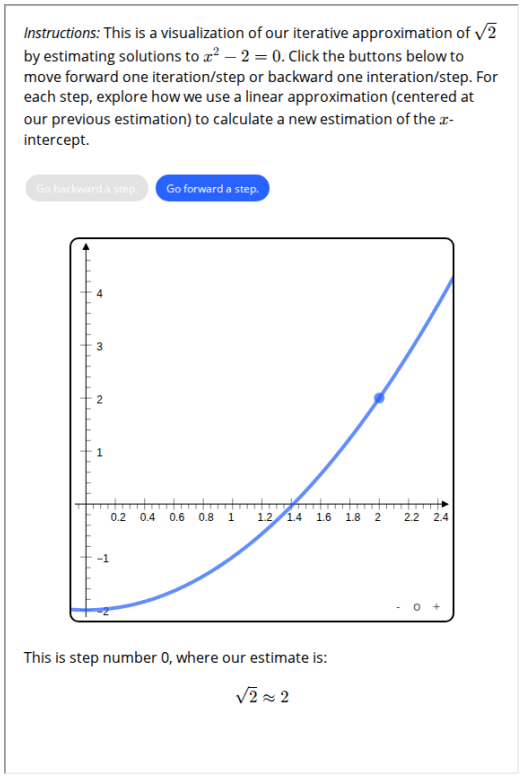
(e) We can keep repeating this process, and that's exactly what the mathematicians we talked about discovered.

Say we've built a linear approximation at some x-value (we'll call it x_{old}).

$$L(x) = f'(x_{\text{old}})(x - x_{\text{old}}) + f(x_{\text{old}}).$$

Set this equal to 0 and solve.

(f) Let's visualize these calculations.



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Something kind of strange happens in the last two steps. Why does the value of our estimation not change? What happens to our estimate?

Practice Problems

1. What is the formula for the Linear Approximation of a function $f(x)$ centered at some x-value $x = a$? Explain what this is.

2. or some function $f(x)$ that is differentiable everywhere, how could we use the concepts of linear approximation to find an approximate value $f(0.0001)$?

3. Let's say that we have a function $f(x)$ that is differentiable at $x = 5$ and a Linear Approximation $L(x) = f'(5)(x - 5) + f(5)$. Give examples of two different x-values where $L(x)$ does a good job of approximating $f(x)$ accurately, and two examples of x-values where $L(x)$ does not do a good job of approximating $f(x)$ accurately.

4. For each of the following functions and centers, build a Linear Approximation function.

(a) $f(x) = \sqrt{x}$ centered at $x = 4$

(b) $f(x) = \sin x$ centered at $x = 0$

(c) $f(x) = \cos x$ centered at $x = 0$

(d) $f(x) = \ln x$ centered at $x = 1$

(e) $f(x) = \sqrt[3]{x}$ centered at $x = 8$

(f) $f(x) = \dfrac{1}{x}$ centered at $x = 4$

5. Use the linear approximating functions you created above to estimate the following values. Comment on the accuracy of these approximations.

(a) $\sqrt{3.9}$

(b) $\sin(-0.0002)$

(c) $\cos(0.001)$

(d) $\ln(2)$

(e) $\sqrt[3]{9}$

(f) $\dfrac{1}{4.0001}$

6. Approximate each of the following values using a Linear Approximation. For each one:

- Report the function $f(x)$ and center.
- Write out the Linear Approximation function $L(x)$.
- Approximate the value.
- Justify whether your approximation is an over-approximation or under-approximation without referring to the actual value.

(a) $(5.985)^2$

(b) $(2.013)^3$

(c) $\cos(3)$

(d) $e^{0.01}$

7. Explain how Newton's method uses Linear Approximation to approximate the x-intercept of a function.

8. Explain how we cam use Newton's method to approximate:

(a) the intersection point of two functions.

(b) local maximums or minimums of a function.

(c) inflection points of a function.

9. Use Newton's method to find a zero for each function using the initial value given. Stop approximating when you get successive approximations that are within 0.001 of each other.

(a) $f(x) = x^3 - 5$ with $x_0 = 2$

(b) $f(x) = x\cos(x)$ with $x_0 = 2$

10. Use Newton's method to find an intersection point between the two curves. Stop approximating when you get successive approximations that are within 0.001 of each other. Feel free to use a graphing utility like desmos.com to graph the curves and pick a reasonable stating value.

(a) $y = x^2$ and $y = \sin(x)$ (Note: these curves intersect at $x = 0$. Find the second intersection point.)

(b) $y = e^x$ and $y = \sqrt{x} + 2$

11. Find/approximate the absolute maximum and absolute minimum of $y = x\sin(x)$ on $[0, 6]$ using Newton's method (and any other techniques or concepts we've used).

12. Find/approximate the inflection point of $y = \dfrac{x^3}{10} + e^x$ using Newton's method (and any other techniques or concepts we've used).

4.7 L'Hôpital's Rule

Activity 4.7.1 Building L'Hôpital's Rule.

We're going to take a closer look at the indeterminate form, $\frac{0}{0}$, and use our new ideas of linear approximation to think about how these types of things work.
We're going to be working with the following limit:

$$\lim_{x\to a} \frac{f(x)}{g(x)}$$

where $f(x)$ and $g(x)$ are differentiable at $x = a$ (since we're going to want to build linear approximations of them).

(a) Write out the linear approximations for both $f(x)$ and $g(x)$, both centered at $x = a$. We'll call them $L_f(x)$ and $L_g(x)$.

(b) Describe how well or how poorly these linear approximations estimate the values from our functions $f(x)$ and $g(x)$? What happens to these approximations as we get close to the center $x = a$? What happens in the limit as $x \to a$?

(c) Let's rewrite our limit. We can replace $f(x)$ with our formula for its linear approximation, $L_f(x)$ and replace $g(x)$ with its linear approximation, $L_g(x)$:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \left(\qquad \right)$$

(d) Up until now, we have not thought about indeterminate forms at all. Let's start now.

If this limit is a $\frac{0}{0}$ indeterminate form, then that means that $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$.

Since our functions are, by definition, differentiable at $x = a$, then they also have to be continuous at $x = a$. What does this mean about the values of $f(a)$ and $g(a)$?

(e) Use this new information about the values of $f(a)$ and $g(a)$ to revisit the limit. We rewrote $\lim_{x\to a} \frac{f(x)}{g(x)}$ by replacing each function with its linear approximation. What happens with the algebra when we know this information about $f(a)$ and $g(a)$?

So we have a really nice result here! In the $\frac{0}{0}$ indeterminate form, we can replace the ratio of the y-values from our functions with the ratio of slopes (coming from the first derivatives) of our functions.
In general, we'll put a step in between, where we find $f'(x)$ and $g'(x)$ first before trying to evaluate these derivatives at $x = a$.

Practice Problems

1. Explain how to use L'Hôpital's Rule to evaluate a limit with the following indeterminate forms:

(a) $0/0$

(b) ∞/∞

(c) $0 \cdot \infty$

(d) 0^0, 1^∞, or ∞^0

2. Consider the following limit: $\lim\limits_{x\to -2}\left(\dfrac{x+2}{x^2-4}\right)$

(a) Evaluate this limit using L'Hôpital's Rule.

(b) Evaluate this limit without using L'Hôpital's Rule. Instead, use some algebraic manipulation of the function.

3. Consider the following limit: $\lim\limits_{x\to 3}\left(\dfrac{2+\frac{3}{x}-x}{x-3}\right)$

(a) Evaluate this limit using L'Hôpital's Rule.

(b) Evaluate this limit without using L'Hôpital's Rule. Instead, use some algebraic manipulation of the function.

4. Consider the following limit: $\lim\limits_{x\to \infty}\left(\dfrac{2x^3-x^2+9x-1}{10-5x+3x^2-8x^4}\right)$

(a) Evaluate this limit using L'Hôpital's Rule.

(b) Evaluate this limit without using L'Hôpital's Rule. Instead, use some algebraic manipulation of the function.

5. Give an example of:

(a) A limit with a $0/0$ indeterminate form that we wouldn't be able to evaluate using the basic algebraic manipulations we have. Use L'Hôpital's Rule to evaluate that limit.

(b) A limit with an ∞/∞ indeterminate form, and evaluate it using L'Hôpital's Rule.

(c) A limit with a $0/0$ indeterminate form that equals 6.

(d) A limit with an ∞/∞ indeterminate form that equals -2

6. Evaluate the following limits using L'Hôpital's Rule.

(a) $\lim\limits_{x\to\infty}\left(\dfrac{e^{2x}+4x-2}{x^2+1}\right)$

(b) $\lim\limits_{x\to 3}\left(\dfrac{x^3-x^2-4x-6}{x^2-5x+6}\right)$

(c) $\lim\limits_{x\to\infty}\left(\dfrac{\ln x}{x^2}\right)$

(d) $\lim\limits_{x\to 0}\left(\dfrac{\sin(x)+x}{\cos(x)-2x}\right)$

(e) $\lim\limits_{x\to 0}\left(\dfrac{e^x-\frac{x^3}{6}-\frac{x^2}{2}-x-1}{\cos x-\sin x-\frac{x^3}{6}+\frac{x^2}{2}+x-1}\right)$

(f) $\lim_{x\to\infty}\left(\dfrac{x^2+1}{e^{4x+1}}\right)$

7. Evaluate the following limits using L'Hôpital's Rule.

(a) $\lim_{x\to 0^+}(x\ln x)$

(b) $\lim_{x\to 0^+}(\sin(x)\ln(x))$

(c) $\lim_{x\to\infty}(x^2e^{-x})$

(d) $\lim_{x\to\infty}(e^{-x}\sqrt{x})$

(e) $\lim_{x\to\infty}x^{1/x}$

(f) $\lim_{x\to 0^+}x^x$

(g) $\lim_{x\to 0}(\cos(x))^{(1/x)}$

(h) $\lim_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x$

(i) $\lim_{x\to\infty}\left(1+\dfrac{a}{x}\right)^x$ with a a real number

(j) $\lim_{x\to\infty}(\ln(x))^{1/x}$

4.8 Exploration: Applications of Derivatives

We have a lot of practice with calculating derivatives, and we hopefully have a good idea of how we might interpret what this derivative is: a function that tells us slopes of a line tangent to a curve at a specific x-value.

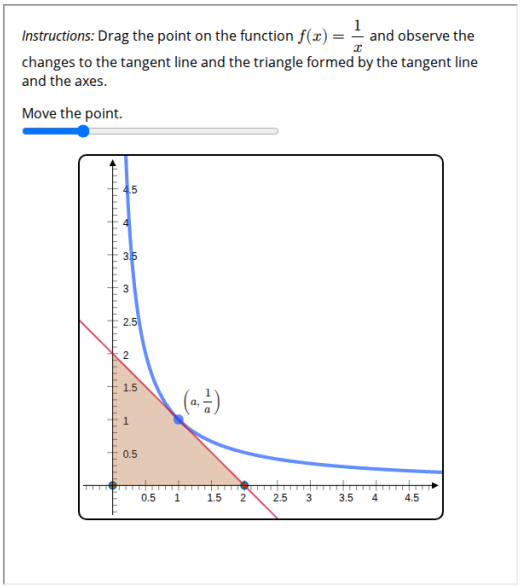
Let's think about this idea.

Exploration 4.8.1 A Tangent Triangle.

Let's consider the function $f(x) = \dfrac{1}{x}$.

(a) Consider a point on this curve at some x-value $x = a$ where $a > 0$. Find the equation of the line tangent to $f(x)$ at this point.

(b) We can graph this line, as well as the function $f(x)$.



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Find the area of the triangle formed by the tangent line and the axes.

(c) Hopefully you found something surprising about this area. Show that this surprising fact also occurs for the function $g(x) = \dfrac{k}{x}$, where $k > 0$.

(d) Are there other functions that have this interesting property about the area of a triangle formed by the tangent line and the axes? What are some requirements that these functions would need to have?

Chapter 5

Antiderivatives and Integrals

5.1 Antiderivatives and Indefinite Integrals

Activity 5.1.1 Find a Function Where....

For each of the following derivatives, find a function $f(x)$ whose first (or second) derivative matches the listed derivative.

(a) $f'(x) = 4x^3 + 6$

(b) $f'(x) = 8x^7 - x^4 + x$

(c) $f'(x) = \sqrt{x} - \dfrac{1}{x^2}$

(d) $f''(x) = x^3 + \cos(x)$

(e) $f''(x) = e^x - \dfrac{1}{\sqrt[3]{x}}$

(f) Go back through each of the above derivatives, and find a different option for $f(x)$ that still works. Make sure that it is something completely unique, and not just an equivalent function that is written differently.

Why are you able to find multiple answers in these questions, but not when we are given a function and need to find a derivative?

Activity 5.1.2 A File Sorting Speed Test.

A computer program is trying to sort a group of computer files based on their size. The program isn't very efficient, and the time that it takes to sort the files increases when it tries to sort more files.

The time that it takes, measured in seconds, based on the total, cumulative size of the files g, measured in gigabytes, is modeled by a function $T(g)$. We don't know the function, but we do know that the time increases at an instantaneous rate of $0.0001g$ seconds when the total size, g increases slightly. That is, the rate of change is a function of the file size, g, itself.

(a) We can build a function for $T'(g)$. What is it?

(b) Find all of the possibilities for the function modeling the time, T, that it takes the computer program to sort files with a total size of g.

(c) What does your constant C represent, here? You can interpret it graphically, interpret it by thinking about derivatives, but you should also interpret it in terms of the time that it takes this program to sort these files by size.

(d) Let's say that we feed some number of files totaling up to 1.4GB in size into this program. It takes 0.24 seconds to sort the files by size.

Find the function, $T(g)$, that models how quickly this program sorts these files.

Practice Problems

1. Our definition of an antiderivative is that $F(x)$ is *an* antiderivative of $f(x)$ if $F'(x) = f(x)$. Why does this not mean that $F(x)$ is *the* antiderivative of $f(x)$?

2. Let's consider the connection between derivatives, functions, antiderivatives, and indefinite integrals.

(a) For a first derivative $f'(x) = 3x^2 + \cos(x) + 2$, find a suitable function $f(x)$.

(b) For a second derivative $f''(x) = 3x^2 + \cos(x) + 2$, find a suitable first derivative $f'(x)$.

(c) For a function $f(x) = 3x^2 + \cos(x) + 2$, find a suitable antiderivative function $F(x)$.

(d) For a function $f(x) = 3x^2 + \cos(x) + 2$, find the family of antiderivatives.

(e) Find $\displaystyle\int \left(3x^2 + \cos(x) + 2\right)\, dx$.

3. Explain the difference between a single antiderivative of a function $g(x)$ and the family of antiderivatives of $g(x)$.

4. What does a family of antiderivatives look like, graphically?

5. For each of the following indefinite integrals, find the family of antiderivatives.

(a) $\displaystyle\int \left(4x^3 - \frac{1}{x}\right)\, dx$

(b) $\displaystyle\int \left(8x - \sec^2(x) - \frac{1}{x^2}\right)\, dx$

(c) $\displaystyle\int \left(x^5 - 3\sin(x) + 2e^x\right)\, dx$

(d) $\displaystyle\int \left(\sqrt[3]{t} + \frac{4}{t^3} + 2\right)\, dt$

(e) $\displaystyle\int \left(\frac{4y}{3} - \frac{3}{4y}\right)\, dy$

(f) $\displaystyle\int \left(5\sec(x)\tan(x) + 2\sqrt[5]{x^3}\right)\, dx$

(g) $\displaystyle\int \left(\frac{1}{\sqrt{1-x^2}} + \frac{4x^2}{3}\right)\, dx$

(h) $\displaystyle\int \left(\frac{1}{1+x^2} - \cos(x)\right)\, dx$

6. For each of the following indefinite integrals, rewrite the integrand (the function inside the integral) before finding a family of antiderivatives.

(a) $\displaystyle\int \left(\frac{x^2-1}{x-1}\right)\, dx$

(b) $\displaystyle\int \left(x(x^2+1)\right)\, dx$

(c) $\displaystyle\int \left(\frac{\sec(x)+\tan(x)}{\cos(x)}\right)\, dx$

(d) $\displaystyle\int \left(\frac{xe^x+1}{x}\right)\, dx$

(e) $\displaystyle\int \left(\ln(e^{2x})\right)\, dx$

(f) $\displaystyle\int \left(\frac{x^2+2}{x^2+1}\right)\, dx$

7. For each of the functions below, find the antiderivative that satisfies the condition given.

(a) $f(x) = x^2 - 4\sin(x)$ with $F(0) = 3$

(b) $g(x) = x^3 - 4x^2 + 5x$ with $G(1) = 0$

(c) $m(x) = \dfrac{9}{x} - 3e^x$ with $M(1) = 4$

5.2 Riemann Sums and Area Approximations

Activity 5.2.1 Approximating Areas.

We're going to consider two different functions, and some areas based on them. Let's think about two functions: $f(x) = 2x + 1$ and $g(x) = x^2 + 1$. For both of these functions, we'll focus on the interval $[0, 2]$. Instead of thinking about the function only, we'll be considering the two-dimensional region bounded between the graph of our function and the x-axis between $x = 0$ and $x = 2$.

(a) Find the area of the region bounded between the graph of $f(x) = 2x+1$ and the x-axis between $x = 0$ and $x = 2$.

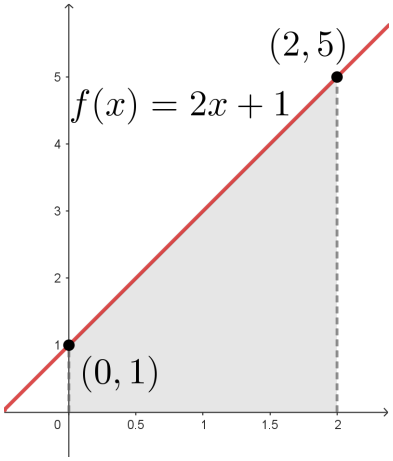


Figure 5.2.2

How did you evaluate this area? What kind(s) of shape(s) did you think about?

(b) Estimate the area of the region bounded between the graph of $g(x) = x^2 + 1$ and the x-axis between $x = 0$ and $x = 2$.

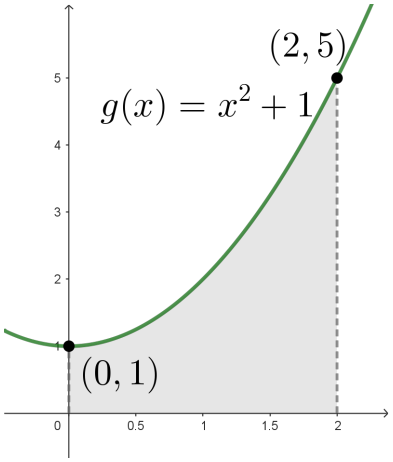


Figure 5.2.3

How did you estimate this area? What kind(s) of shape(s) did you think about?

(c) Come up with an upper and lower bound for this area. In other words, give an underestimate and overestimate for the actual area we would like to know.

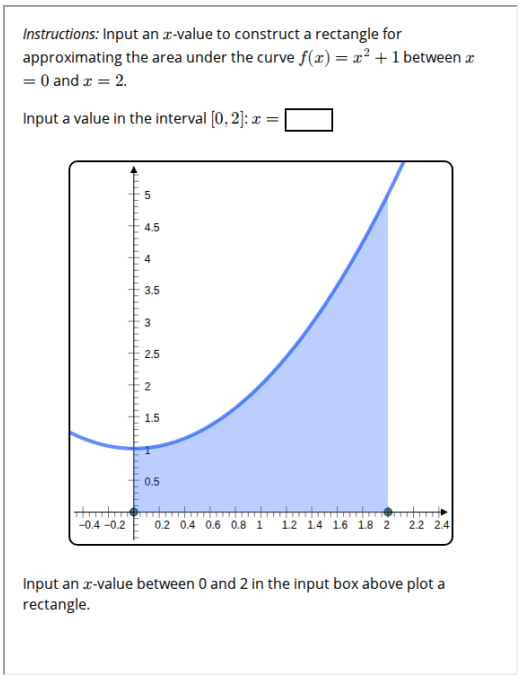
How did you come up with these estimates? How "good" do you think your estimates are? Can you come up with "better" (or closer) ones?

Activity 5.2.4 Approximating the Area using Rectangles.

We're going to stick with the function $g(x) = x^2 + 1$ on the interval $[0, 2]$, and keep thinking about the area bounded by the curve and the x-axis on this interval. We're going to approximate the area in a couple of different tries, each one more accurate than the one before. By the end of this activity, we'll have a pretty good process built!

(a) Let's start with approximating this region with a single rectangle. We're going to define the rectangle by picking some x-value in the interval $[0, 2]$. Then, we'll use the point at that x-value to define the height of our rectangle.

Essentially, we are picking a single point on our function on the interval, and our approximation is pretending that the single point we picked is representative of the whole function on the interval.



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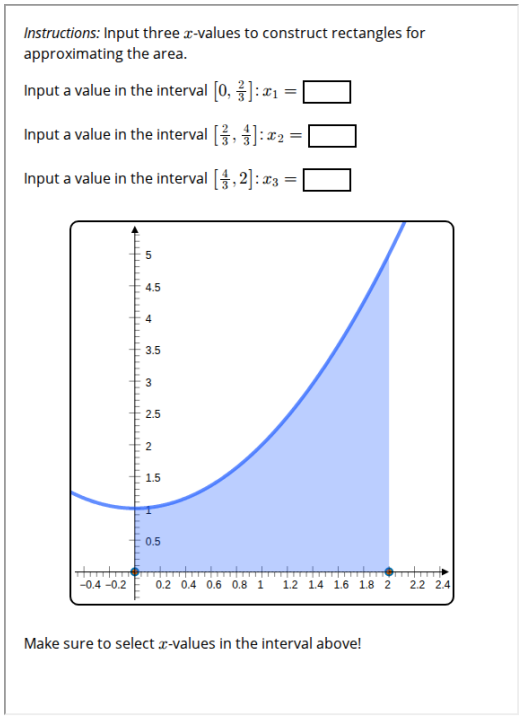
(b) Can you try repicking an x-value, and try to find one that gets you an area approximation that is pretty good?

(c) We're going to use more rectangles. Let's jump up to 3 rectangles. If we split up the interval between $x = 0$ and $x = 2$ into 3 rectangles, we can make them all the same width, and pick an x-value that we can use to get a representative point for each of the 3 rectangles.

We'll need to pick 3 x-values this time.



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(d) Can you try repicking your x-values, and trying to find one that gets you an area approximation that is pretty good?

(e) Let's scale this up a bit. Pick a good number for your number of rectangles. We'll call this value n.

(If you're working in a classroom, maybe it would be good to pick the number of groups or the number of students, or some other number between 10 and 20 or something like that.)

For your value n, we're going to divide up the interval between $x = 0$ and $x = 2$ into n pieces. These will be the intervals that we pick from to get our rectangles. What are the subintervals? What are the widths of each subinterval (and then the widths of the rectangles)? Call this with Δx.

(f) For each subinterval, pick an x-value in the subinterval to represent it.

(g) Evaluate the function $f(x) = x^2 + 1$ at each of the x-values you picked. These are the heights of your rectangles!

(h) Find the areas of each rectangle by multiplying the height of each rectangle by Δx, the width of each rectangle.

(i) Add these areas up to get a total approximation of the actual area!

What do you think: is it worth fiddling with what x-value to pick from each subinterval to try to get a better approximation? If n is large, do you think it matters how we pick the x-values from each subinterval?

Practice Problems

1. Explain in words how we will approximate the area bounded between a function and the x-axis between two x-values, $x = a$ and $x = b$.

2. Consider the function $f(x) = x^2 + 1$ on the interval $[0, 2]$.

(a) Divide the interval $[0, 2]$ into 5 equally sized subintervals.

(b) What is the width of each subinterval, Δx?

(c) Pick an x-value from each of the subintervals. Label them $x_1^*, x_2^*, ..., x_5^*$.

(d) Find the function outputs at each of the x_k^* values for $k = 1, 2, ..., 5$, $f(x_1^*), f(x_2^*), ..., f(x_5^*)$.

(e) Multiply each of the function values above by Δx.

(f) Add each of the values above together.

(g) Interpret this sum. What does this represent? What does it approximate?

(h) Draw the Riemann sum you built above. Your graph should include the graph of $f(x) = x^2 + 1$ as well as 5 rectangles, with heights defined by the points you found above.

3. Let's consider a function $f(x)$ that is continuous on some interval $[a, b]$.

(a) In the construction of a Riemann sum, we use x-values notated x_k^*. What are these? Explain them.

(b) What does it mean to construct a Left Riemann sum?

(c) What does it mean to construct a Right Riemann sum?

(d) What does it mean to construct a Midpoint Riemann sum?

4. Consider the function $g(x)$ with the following function values.

Table 5.2.5 Function values for $g(x)$

x	0	0.5	1	1.5	2	2.5	3	3.5	4
$g(x)$	0	2	-4	7	0.5	-1	3.5	-3	1

(a) Approximate the area bounded between the curve $y = g(x)$ and the x-axis between $x = 0$ and $x = 4$ using a Right Riemann sum with $n = 4$ rectangles (R_4).

(b) Approximate the area bounded between the curve $y = g(x)$ and the x-axis between $x = 0$ and $x = 4$ using a Left Riemann sum with $n = 4$ rectangles (L_4).

(c) Approximate the area bounded between the curve $y = g(x)$ and the x-axis between $x = 0$ and $x = 4$ using a Midpoint Riemann sum with $n = 4$ rectangles (M_4).

(d) Approximate the area bounded between the curve $y = g(x)$ and the x-axis between $x = 0$ and $x = 4$ using a Right Riemann sum with $n = 8$ rectangles (R_8).

(e) Approximate the area bounded between the curve $y = g(x)$ and the x-axis between $x = 0$ and $x = 4$ using a Left Riemann sum with $n = 8$ rectangles (L_8).

5. Evaluate the following Riemann sums to approximate areas described. Sketch a picture of the region and Riemann sum for each.

(a) The area bounded by the graph of $f(x) = \dfrac{x^3}{2} - 3x^2 + 4x + 2$ and the x-axis between $x = 0$ and $x = 2$ using an L_5 Riemann sum.

(b) The area bounded by the graph of $g(x) = \frac{x^3}{2} - 3x^2 + 4x + 2$ and the x-axis between $x = 2$ and $x = 4$ using an R_3 Riemann sum.

(c) The area bounded by the graph of $j(x) = \sqrt{x} - x$ and the x-axis between $x = 0$ and $x = 1$ using an L_4 Riemann sum.

(d) The area bounded by the graph of $j(x) = \sqrt{x} - x$ and the x-axis between $x = 0$ and $x = 1$ using an M_4 Riemann sum.

(e) The area bounded by the graph of $j(x) = \sqrt{x} - x$ and the x-axis between $x = 0$ and $x = 2$ using an R_5 Riemann sum.

6. Consider a function that is increasing on the interval $[a, b]$. Will a Left Riemann sum overapproximate or underapproximate the true area bounded between the curve of $f(x)$ and the x-axis on the interval $[a, b]$? What about a Right Riemann sum?

7. Consider a function that is decreasing on the interval $[a, b]$. Will a Left Riemann sum overapproximate or underapproximate the true area bounded between the curve of $f(x)$ and the x-axis on the interval $[a, b]$? What about a Right Riemann sum?

8. For each pair of Riemann sums, explain which one you think most accurately estimates the area under the curve $f(x)$ from $[a, b]$, or explain why it is impossible or difficult to tell.

(a) L_5 or R_5

(b) L_8 or M_8

(c) M_6 or R_{10000}

5.3 The Definite Integral

Activity 5.3.1 Weird Areas.

Let's think about a simple linear function, $f(x) = 4 - 3x$. We'll both approximate and evaluate the area bounded between $f(x)$ and the x-axis from $x = 0$ to $x = 3$:

$$\int_{x=0}^{x=3} (4 - 3x)\, dx$$

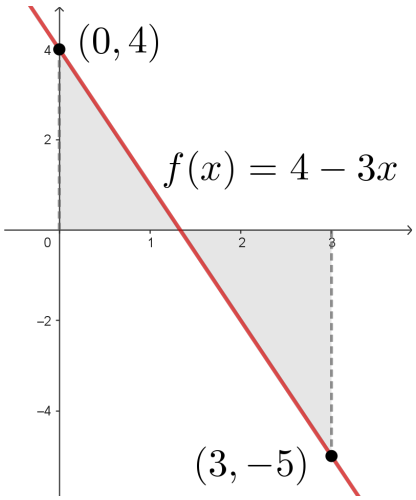


Figure 5.3.2

(a) Explain why we do not need to think about Riemann sums in order for us to calculate the shaded area. How would you calculate this without using calculus?

Calculate the area!

(b) Let's approximate this area using a Riemann sum. Calculate L_3, the Left Riemann sum with $n = 3$ rectangles.

(c) Let's approximate this area a second time, but with a different selection strategy for our x-values. Calculate R_3, the Right Riemann sum with $n = 3$ rectangles.

(d) Compare your answers for L_3 and R_3. They should be *very* different. Why? What is happening that makes R_3 specifically such a weird value?

(e) Do you need to go back and recalculate the area geometrically (from the first part of this activity)? Explain why your answer for $\int_{x=0}^{x=3} (4 - 3x)\, dx$ *should* be negative, based on the Riemann sums we calculated.

(f) Find a value $x = b$ such that:

$$\int_{x=0}^{x=b} (4 - 3x)\, dx = 0.$$

(g) Find a *different* value $x = b$ such that:

$$\int_{x=0}^{x=b} (4 - 3x)\, dx = 0.$$

Is there a second way of making this area 0?

Activity 5.3.3 Weird Areas - Part 2.

We're going to think about the same region, kind of.

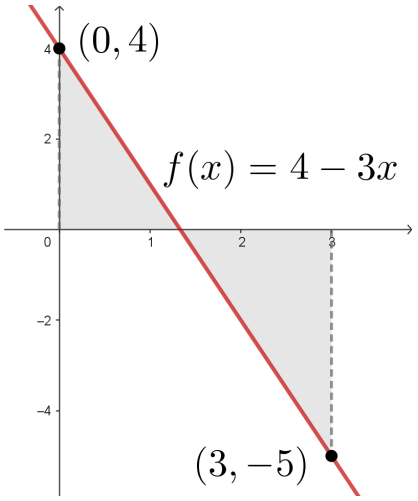


Figure 5.3.4

Let's think about the same linear function, $f(x) = 4 - 3x$, but this time we'll approximate and evaluate the area bounded between $f(x)$ and the x-axis from $x = 3$ to $x = 0$:

$$\int_{x=3}^{x=0} (4 - 3x)\, dx$$

(a) Use geometry to calculate the area. Compare this to the result from Activity 5.3.3.

(b) Let's approximate this using a Riemann sum. Calculate M_3, the Midpoint Riemann sum with $n = 3$ rectangles.

(c) Do you need to go back and recalculate the area geometrically (from the first part of this activity)? Explain why your answer for $\int_{x=3}^{x=0} (4 - 3x)\, dx$ *should* be positive, based on the Riemann sums we calculated.

Practice Problems

1. Explain what $\int_{x=a}^{x=b} f(x)\ dx$ means geometrically/graphically.

2. What does "signed" area or "net" area mean?

3. Sketch the integral $\int_{x=0}^{x=2\pi} (\sin(x))\ dx$. Estimate this value from the sketch you drew. Explain your answer.

4. Using the idea of Riemann sums, explain why area "under" the x-axis is considered negative.

5. Using the idea of Riemann sums, explain why the $\int_{x=a}^{x=b} f(x)\ dx = -\int_{x=b}^{x=a} f(x)\ dx$.

6. For each of the following definite integrals, sketch the integral and use geometry to evaluate the definite integral. Feel free to use desmos to visualize the graphs of functions.

(a) $\int_{x=0}^{x=4} (5)\ dx$

(b) $\int_{x=0}^{x=3} (2x)\ dx$

(c) $\int_{x=1}^{x=2} (x+6)\ dx$

(d) $\int_{x=-3}^{x=3} \left(\sqrt{9-x^2}\right)\ dx$

(e) $\int_{x=0}^{x=2} \left(-\sqrt{4-x^2}\right)\ dx$

7. Consider the function $f(x)$, graphed below, with areas of different regions noted (without noting sign of the area).

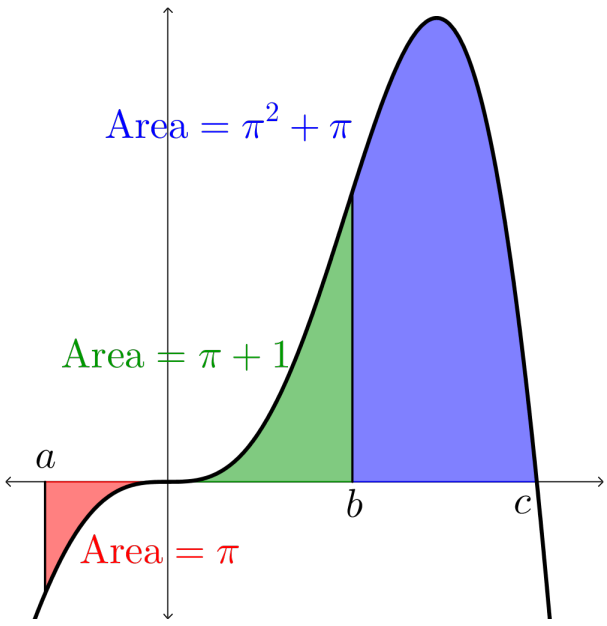


Figure 5.3.5 Graph of $f(x)$ with labeled areas.

Evaluate the following definite integrals.

(a) $\int_{x=a}^{x=0} f(x)\ dx$

(b) $\int_{x=a}^{x=b} f(x)\ dx$

(c) $\int_{x=0}^{x=c} f(x)\ dx$

(d) $\int_{x=a}^{x=c} f(x)\ dx$

(e) $\int_{x=a}^{x=c} |f(x)|\ dx$

(f) $\int_{x=a}^{x=c} (f(x)+1)\ dx$

8. Say we are given that $\int_{x=0}^{x=5} f(x)\ dx = 10$, $\int_{x=5}^{x=9} f(x)\ dx = -2$, and $\int_{x=0}^{x=9} g(x)\ dx = -6$. Evaluate each of the following.

(a) $\int_{x=0}^{x=9} (f(x)-g(x))\ dx$

(b) $\int_{x=0}^{x=5} \left(\frac{f(x)}{3}\right)\ dx$

(c) $\int_{x=0}^{x=9} (3g(x)+4)\ dx$

(d) $\int_{x=0}^{x=9} (2g(x)+f(x))\ dx$

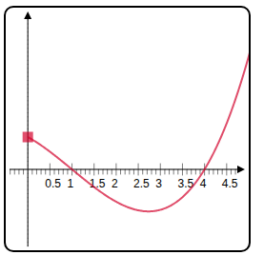
5.4 The Fundamental Theorem of Calculus

Activity 5.4.1 Interpreting the Graph of a Derivative.

Let's look at a picture of a graph of the first derivative, $f'(x)$, and try to get some information about $f(x)$ from it. Use the following graph of $f'(x)$, the first derivative, to answer the questions about $f(x)$.

Instructions: Move the point on the graph of $f'(x)$ and connect it to the behavior of $f(x)$.

Move the point.



Graph of $f'(x)$.

Check your understanding: Click the button to reveal the graph of $f(x)$.

Show Graph of f(x)

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Since we don't have a huge amount of detail, you'll likely have to estimate the x-values for intervals and points in the following questions, but that's ok! Estimate away! Just make sure you know what you're looking for in the graph of $f'(x)$ to answer these questions.

(a) List the intervals on which $f(x)$ is increasing. What about decreasing?

(b) Find the x-values of any local maximums and/or local minimums of $f(x)$.

(c) List the intervals on which $f(x)$ is concave up. What about concave down?

(d) Find the x-values of any inflection points of $f(x)$.

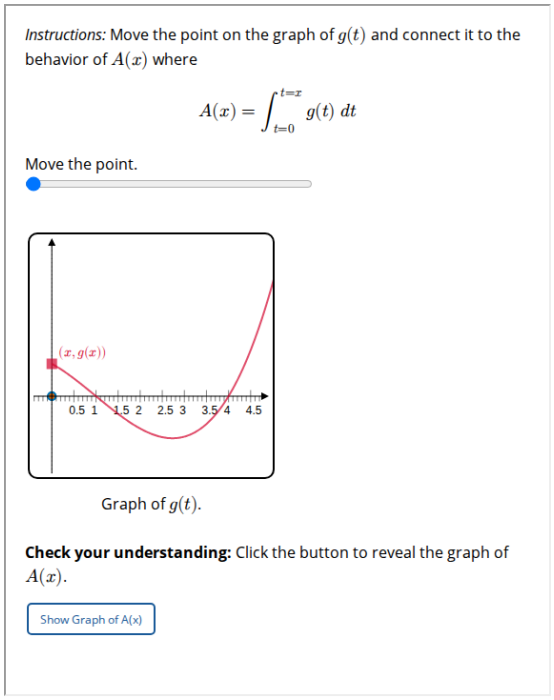
Activity 5.4.2 Interpreting Area.

First, we're going to define a bit of a weird function. Sometimes it's called the **Area function**:

$$A(x) = \int_{t=0}^{t=x} g(t)\, dt.$$

This is a strange function, because we're defining the function as an integral of another function. Specifically, note that the *input* for our area function $A(x)$ is the ending limit of integration: we're calculating the signed area "under" the curve of $g(t)$ from $t = 0$ up to some variable ending point $t = x$.

We can visualize this function by looking at the areas we create as we change x. For now, get used to just seeing the area "under" g when we move the point around. The areas themselves are the outputs of the function $A(x)$.



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Now we can think about this area function, and try to connect it to the graph of $g(t)$.

(a) List the intervals on which $A(x)$ is increasing. What about decreasing?

(b) Find the x-values of any local maximums and/or local minimums of $A(x)$.

(c) List the intervals on which $A(x)$ is concave up. What about concave down?

(d) Find the x-values of any inflection points of $A(x)$.

(e) Compare your answers here to your answers about the behavior of $f(x)$ based on the (same) graph of $f'(x)$ in Activity 5.4.1.

What does this mean about the connection between areas and derivatives, or areas and anti-derivatives?

Activity 5.4.3 Evaluating Areas and Antiderivatives.

In this short activity, we'll just collect information about antiderivatives and this new area function,

$$A(x) = \int_{t=a}^{t=x} f(t)\, dt$$

for a function $f(t)$ that is continuous on the interval $a \leq t \leq x$.
For our purposes in this activity, let's say that $f(x) = x + \cos(x)$.

(a) From the Fundamental Theorem of Calculus (Part 1), we know that $A(x)$ is an antiderivative of $f(x)$, since $A'(x) = f(x)$.

Write out the function $A(x)$, and then name/write out one *other* antiderivative of $f(x)$, some $F(x)$.

(b) We know that all of the antiderivatives of a function are connected to each other.

Describe the connection between $A(x)$ and your $F(x)$.

(c) What is the value of $A(a)$? What is the value of $F(a)$? How are they different from each other?

(d) What is the value of $A(b)$? What is the value of $F(b)$? How are they different from each other?

(e) What about the differences: $A(b) - A(a)$ compared to $F(b) - F(a)$?

Practice Problems

1. The Fundamental Theorem of Calculus comes in two parts. Explain why the Fundamental Theorem of Calculus (Part 1) guarantees that any function that is continuous on an interval has an antiderivative on that interval.

2. The Fundamental Theorem of Calculus comes in two parts. Explain what the Fundamental Theorem of Calculus (Part 2) says about evaluating definite integrals.

3. Consider the function $f(x) = x$.

(a) Sketch the graph of $f(x)$.

(b) Use geometry to find the area under the curve $f(x) = x$ from $x = 0$ to $x = 1$.

(c) Use geometry to find the area under the curve $f(x) = x$ from $x = 0$ to $x = 4$.

(d) Use geometry to find the area under the curve $f(x) = x$ from $x = 0$ to some unknown x-value with $x > 0$.

(e) Write your area formula from above as a function $A(x)$.

(f) Find $A'(x)$. How does this connect with the function $f(x) = x$?

4. Consider the function $f(x) = 4x + 1$.

(a) Sketch the graph of $f(x)$.

(b) Use geometry to find the area under the curve $f(x) = x$ from $x = 2$ to $x = 3$.

(c) Use geometry to find the area under the curve $f(x) = x$ from $x = 2$ to $x = 5$.

(d) Use geometry to find the area under the curve $f(x) = x$ from $x = 2$ to some unknown x-value with $x > 2$.

(e) Write your area formula from above as a function $A(x)$.

(f) Find $A'(x)$. How does this connect with the function $f(x) = 4x + 1$?

5. Consider the function $f(x) = 3x^2 + 6$ on the interval $x = -1$ to $x = 3$.

(a) Find an antiderivative of $f(x)$. Call it $F(x)$.

(b) Evaluate $F(3)$ and $F(-1)$.

(c) Find a different antiderivative of $f(x)$. Call it $\tilde{F}(x)$.

(d) Evaluate $\tilde{F}(3)$ and $\tilde{F}(-1)$.

(e) Compare your answers in (b) and (d). What of these is similar? What of these are different?

(f) Evaluate the definite integral $\displaystyle\int_{x=-1}^{x=3} \left(3x^2 + 6\right)\, dx$ using the Fundamental Theorem of Calculus.

(g) When evaluating a definite integral using the Fundamental Theorem of Calculus, why does it not matter which antiderivative of $f(x)$ you use?

6. Consider the graph of the first derivative $f'(x)$ below.

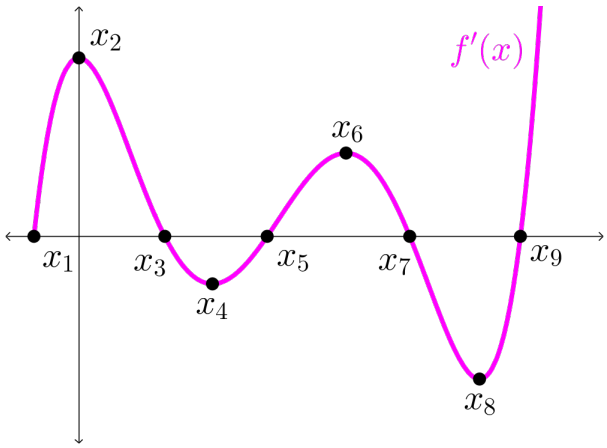


Figure 5.4.4 Graph of a first derivative, $f'(x)$.

(a) Where is the function $f(x)$ increasing? Where is it decreasing?

(b) Where are the local maximums and/or local minimums of $f(x)$?

(c) Where is the function $f(x)$ concave up? Where is it concave down?

(d) Where are the inflection points of $f(x)$?

(e) Sketch a graph of $f(x)$.

7. Consider the graph of the function $g(t)$ below with the Area function $A(x) = \int_{t=x_1}^{t=x} g(t)\,dt$ shaded in.

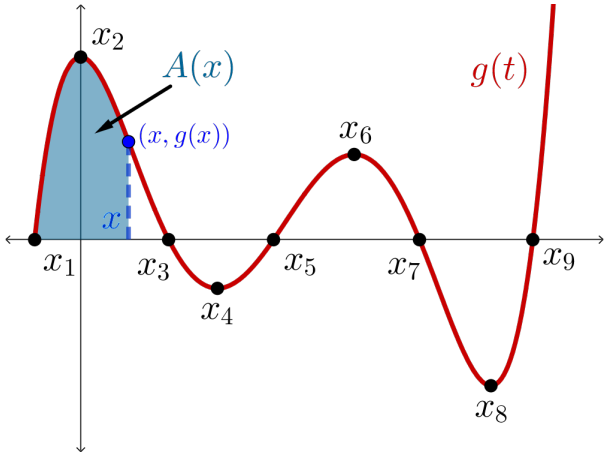


Figure 5.4.5 Graph of $g(t)$ and the area function, $A(x)$.

(a) Where is the function $A(x)$ increasing? Where is it decreasing?

(b) Where are the local maximums and/or local minimums of $A(x)$?

(c) Where is the function $A(x)$ concave up? Where is it concave down?

(d) Where are the inflection points of $A(x)$?

(e) Sketch a graph of $A(x)$.

(f) Compare your answers to the answers above. What can you conclude about the relationship between $A(x)$ and $g(x)$?

8. For each of the definite integrals below, evaluate using the Fundamental Theorem of Calculus.

(a) $\displaystyle\int_{x=1}^{x=5} \left(3x^2 + 6\right)\, dx$

(b) $\displaystyle\int_{x=1}^{x=4} \sqrt{x}\, dx$

(c) $\displaystyle\int_{x=0}^{x=\pi/4} \left(\cos(x) + 4\sin(x)\right)\, dx$

(d) $\displaystyle\int_{x=-1}^{x=1} \left(\frac{1}{1+x^2}\right)\, dx$

(e) $\displaystyle\int_{x=0}^{x=\pi} \left(\sin(x)\right)\, dx$

(f) $\displaystyle\int_{t=-2}^{t=1} (t^5 + 2t^4 - 2t^3 + t^2 - 4t - 2)\, dt$

(g) $\displaystyle\int_{x=1}^{x=3} \left(\frac{1}{x^2} + \frac{1}{x}\right)\, dx$

(h) $\displaystyle\int_{x=0}^{x=5} \left(e^x\right)\, dx$

(i) $\displaystyle\int_{x=\pi/4}^{x=\pi/3} \left(-5\sec^2(x)\right)\, dx$

(j) $\displaystyle\int_{x=1}^{x=2} \left(\frac{xe^x - 1}{x}\right)\, dx$

(k) $\displaystyle\int_{x=1}^{x=6} \left(\frac{3}{\sqrt{x}} + \frac{1}{\sqrt[3]{x^2}}\right)\, dx$

(l) $\displaystyle\int_{x=-1}^{x=2} \left(x(x-1)(x-4)\right)\, dx$

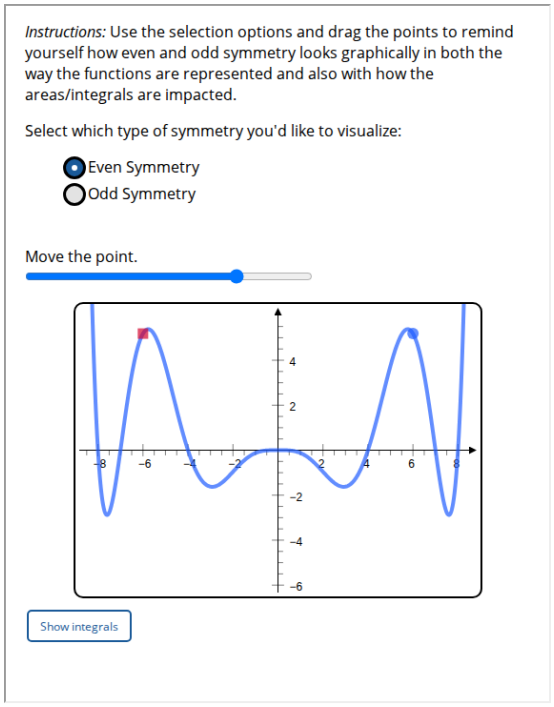
(m) $\displaystyle\int_{x=2}^{x=4} \left(\frac{4}{3x} - \frac{3x}{4}\right)\, dx$

(n) $\displaystyle\int_{\theta=\pi/3}^{\theta=\pi/2} \left(\frac{\cos\theta}{\sin^2\theta}\right)\, d\theta$

5.5 More Results about Definite Integrals

Activity 5.5.1 Symmetry in Functions and Integrals.

First, let's take a moment to remind ourselves (or see for the first time) what two types of "symmetry" we'll be considering. We call them "even" and "odd" symmetry, but sometimes we think of them as a "reflective" symmetry and a "rotational" symmetry in the graphs of our functions.



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(a) Convince yourself that you know what we mean when we say that a function is **even symmetric** on an interval if $f(-x) = f(x)$ on the interval.

Similarly, convince yourself that you know what we mean when we say that a function is **odd symmetric** on an interval if $f(-x) = -f(x)$ on the interval.

(b) Now let's think about areas. Before we visualize too much, let's start with a small question: How does the height of a function impact the area defined by a definite integral? It should be helpful to think about Riemann sums and areas of rectangles here.

The important question then, is how does a function being even or odd symmetric tell us information about areas defined by definite integrals of that function?

Activity 5.5.2 Connecting Symmetric Integrals.

We're going to do some sketching here, and I want you to be clear about something: your sketches can be absolutely terrible. It's ok! They just need to embody the kind of symmetry we're talking about. You will probably sketch something and notice that your areas aren't to scale (or maybe even the wrong sign!), and that's fine.
It might be helpful to practice sketching graphs accurately, but don't worry if that part is a struggle.

(a) Sketch a function $f(x)$ with the following properties:

- $f(x)$ is *even symmetric* on the interval $[-6, 6]$
- $\int_{x=0}^{x=6} f(x)\,dx = 4$
- $\int_{x=-6}^{x=-2} f(x)\,dx = -1$

(b) Find the values of the following integrals:

- $\int_{x=0}^{x=2} f(x)\,dx$
- $\int_{x=-6}^{x=6} f(x)\,dx$

(c) Sketch a function $g(x)$ with the following properties:

- $g(x)$ is *odd symmetric* on the interval $[-9, 9]$
- $\int_{x=0}^{x=4} g(x)\,dx = 5$
- $\int_{x=-9}^{x=0} g(x)\,dx = 2$

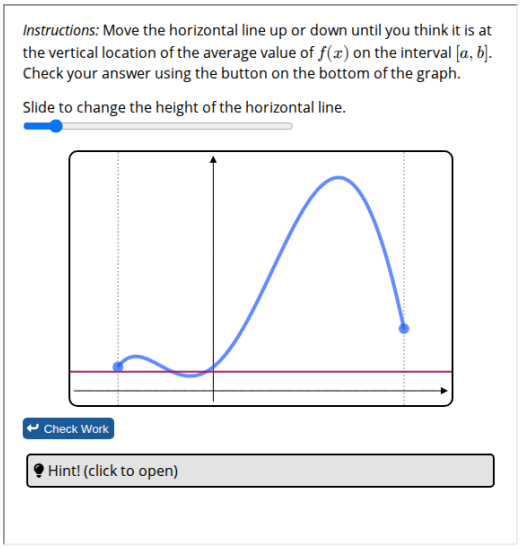
(d) Find the values of the following integrals:

- $\int_{x=-9}^{x=-4} g(x)\,dx$
- $\int_{x=-4}^{x=9} g(x)\,dx$

Activity 5.5.3 Visualizing the Average Height of a Function.

We are going to build a formula to find the"average height" or "average value" of a function $f(x)$ on the interval $[a, b]$. We're going to look at a function and try to find the average height. Along the way, we'll think a bit about areas!

(a) Consider the following function. Find the average height of the function on the interval pictured!



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(b) How does the area "under" the curve $f(x)$ on the interval compare to the area of the rectangle formed by the average height line?

(c) How do you define the two areas?

(d) Set up an equation connecting the two areas, and solve for the average height of $f(x)$.

Practice Problems

1. Explain how we can use the different types of symmetry can help us as we evaluate the definite integral $\int_{x=-a}^{x=a} f(x)\,dx$.

2. Use the definition of *even symmetric* (where $f(-x) = f(x)$ for all x-values) to explain why $\int_{x=-4}^{x=4} f(x)\,dx = 2\int_{x=0}^{x=4} f(x)\,dx$ for an even symmetric function $f(x)$.

3. Use the definition of *odd symmetric* (where $f(-x) = -f(x)$ for all x-values) to explain why $\int_{x=-4}^{x=4} f(x)\,dx = 0$ for an odd symmetric function $f(x)$.

4. Assume that $f(x)$ is some even symmetric function with $\int_{x=0}^{x=6} f(x)\,dx = 4$ and $\int_{x=-3}^{x=0} f(x)\,dx = 9$.

(a) Sketch a possible graph of the function $f(x)$, and shade in the areas from the integrals listed.

(b) Evaluate $\int_{x=0}^{x=3} f(x)\,dx$.

(c) Evaluate $\int_{x=3}^{x=6} f(x)\,dx$.

(d) Evaluate $\int_{x=-6}^{x=-3} f(x)\,dx$.

(e) Evaluate $\int_{x=-6}^{x=6} f(x)\,dx$.

5. Assume that $g(x)$ is some odd symmetric function with $\int_{x=0}^{x=6} g(x)\,dx = 4$ and $\int_{x=-3}^{x=0} g(x)\,dx = 9$.

(a) Sketch a possible graph of the function $g(x)$, and shade in the areas from the integrals listed.

(b) Evaluate $\int_{x=0}^{x=3} g(x)\,dx$.

(c) Evaluate $\int_{x=3}^{x=6} g(x)\,dx$.

(d) Evaluate $\int_{x=-6}^{x=-3} g(x)\,dx$.

(e) Evaluate $\int_{x=-6}^{x=6} g(x)\,dx$.

6. Explain how to find the average height of a function on an interval. Why does this work?

7. Note that $y = \sin(x)$ is odd symmetric and $y = x^2$ is even symmetric.

(a) Show that $f(x) = x^2\sin(x)$ is odd symmetric by showing that $f(-x) = -f(x)$.

(b) Evaluate $\int_{x=-\pi}^{x=\pi} x^2\sin(x)\,dx$.

(c) Find the average height of $f(x)$ on the interval $[-\pi, \pi]$.

8. Note that $y = \cos(x)$ is even symmetric and $y = x^2$ is even symmetric.

(a) Show that $f(x) = x^2 \cos(x)$ is even symmetric by showing that $f(-x) = f(x)$.

(b) Given that $\displaystyle\int_{x=0}^{x=\pi} x^2 \cos(x)\, dx = -2\pi$, evaluate $\displaystyle\int_{x=-\pi}^{x=\pi} x^2 \cos(x)\, dx$.

(c) Find the average height of $f(x)$ on the interval $[-\pi, \pi]$.

9. Find the average height of the following functions on the intervals listed.

(a) $f(x) = e^x - x$ on $[0, 3]$

(b) $g(x) = \cos(x) + x$ on $\left[0, \frac{\pi}{2}\right]$

(c) $j(x) = \dfrac{1}{1+x^2}$ on $[-1, 1]$

(d) $\ell(x) = \dfrac{1}{x^2}$ on $[1, 4]$

5.6 Introduction to u-Substitution

Activity 5.6.1 A Hard Integral.

We're going to be thinking about two integrals here, but before we do, we should remind ourselves about how we can "rephrase" an integration question.

If we are asked to find $\int f(x)\,dx$, then we are really being asked to find some function $F(x)$ whose derivative is $f(x)$. Of course, we're actually being asked to find *all* of the possible functions that fit this requirement, but we know that the constant of integration covers all of the differences.

This means that we can (and maybe should) check our answers pretty consistently: just find a derivative and check that it matches the integrand!

(a) Find $\int 3x^2 e^{x^3}\,dx$.

(b) Explain the role of $3x^2$ in the integral. Explain why the positioning of x^3 matters.

(c) Let's try another one.

$$\int 7x^6 \cos(x^7)\,dx$$

(d) How can you tell, in general, that the derivative you're looking at is one that was produced through the Chain Rule and not the Product Rule?

Activity 5.6.2 Picking the Pieces of a Substitution.

We're going to look at three integrals. Instead of working through them one-at-a-time, we'll look at all three simultaneously, where we can practice identifying, substituting, and antidifferentiating all at the same time.

(a) Let's consider these three integrals:

- $\displaystyle\int \frac{3x^2+1}{(x^3+x-2)^2}\,dx$
- $\displaystyle\int \cos(x)\sqrt{\sin(x)}\,dx$
- $\displaystyle\int \frac{(\ln(x))^3}{x}\,dx$

For each of these integrals, identify the substitution: define u as some function of x.

(b) For each substitution, define $du = u'\,dx$.

(c) For each integral, use your substitution (for both u and the differential du) to rewrite the integral.

(d) Antidifferentiate each integral, and then use your substitution to write each integral back in terms of x.

Activity 5.6.3 Compare Two Integrals.

Let's compare two integrals, and use this to build a more general strategy for performing u-substitution.

(a) Consider the following integral:

$$\int -4x^3 \sec^2(1-x^4)\, dx$$

Select and justify a choice for u.

(b) Perform the u-substitution and antidifferentiate, and then substitute back to write your antiderivative in terms of x.

(c) Compare that integral to this one:

$$\int x^3 \sec^2(1-x^4)\, dx$$

What is different about this new integral? What has remained the same? How does that impact your choice for u, or *does* it impact your choice for u?

(d) Has that changed what du should be?

(e) Ok, so we've noticed an issue here. There are *plenty* of good ways of solving this problem, where du doesn't "show up" perfectly in our integral. In this case, we have that we're missing a necessary coefficient. We have the x^3 part, but we are missing the -4.

Try to rewrite our integral with a -4 coefficient in there. We'll do that by multiplying the integrand function by 1, disguised as $\frac{-4}{-4}$ or $\left(-\frac{1}{4}\right)(-4)$.

(f) Now we can use the same u-substitution as before, and integrate in a similar way! Notice, though, that we will retain the coefficient of $-\frac{1}{4}$.

(This should be reasonable: our integral is $-\frac{1}{4}$ of the original one, since our coefficient was 1 to the original's -4.)

Go ahead and integrate!

Practice Problems

1. The u-substitution integration method was designed to undo a specific derivative rule. Explain.

2. The u-substitution integration method deals with composition of functions. If you're looking at a composite function $f(g(x))$, how do you identify the "inside" function, $g(x)$?

3. For each composite function $f(g(x))$ below, identify two different possibilities for both the "inside" $g(x)$ function and the "outside" $f(x)$ function.

(a) $f(g(x)) = \sqrt{x^2+1}$

(b) $f(g(x)) = \sin(e^x+1)$

(c) $f(g(x)) = e^{\cos(x)}$

(d) $f(g(x)) = (\tan(x)+x)^6$

4. Use the chain rule to differentiate each of the following. Carefully note for each problem which part of the composite function acts as the "inside"piece.

(a) $\dfrac{d}{dx}\left(e^{x^2}\right)$

(b) $\dfrac{d}{dx}\left(\ln(\sin(x))\right)$

(c) $\dfrac{d}{dx}\left(-\cos(\sqrt{x})\right)$

(d) $\dfrac{d}{dx}\left((5x+1)^{10}\right)$

(e) $\dfrac{d}{dx}\left(\tan(e^x+\pi)\right)$

(f) $\dfrac{d}{dx}\left(\sqrt{x^3+1}\right)$

5. Integrate the following indefinite integrals using the suggested u-substitution.

(a) $\displaystyle\int 2xe^{x^2}\,dx;\ u=x^2$

(b) $\displaystyle\int\left(\frac{\cos(x)}{\sin(x)}\right)dx;\ u=\sin(x)$

(c) $\displaystyle\int\left(\frac{\sin(\sqrt{x})}{2\sqrt{x}}\right)dx;\ u=\sqrt{x}$

(d) $\displaystyle\int 50(5x+1)^9\,dx;\ u=5x+1$

(e) $\displaystyle\int e^x\sec^2(e^x+\pi)\,dx;\ u=e^x+\pi$

(f) $\displaystyle\int\left(\frac{3x^2}{2\sqrt{x^3+1}}\right)dx;\ u=x^3+1$

6. Use u-substitution to integrate the following indefinite integrals.

(a) $\displaystyle\int\left(\frac{2x+1}{(x^2+x+1)^3}\right)dx$

(b) $\displaystyle\int \left(\frac{x}{x^2+1}\right)\,dx$

(c) $\displaystyle\int xe^{-x^2}\,dx$

(d) $\displaystyle\int \tan(x)\,dx$

(e) $\displaystyle\int \frac{1}{x\ln(x)}\,dx$

(f) $\displaystyle\int \sin^6(x)\cos(x)\,dx$

(g) $\displaystyle\int x\sec(x^2)\tan(x^2)\,dx$

(h) $\displaystyle\int \left(\frac{x}{1+x^4}\right)\,dx$

7. For each of the following definite integrals, identify a u-substitution and write the definite integral with regard to u, including new (translated) limits of integration.

(a) $\displaystyle\int_{x=0}^{x=3} \left(\frac{1}{x+1}\right)\,dx$

(b) $\displaystyle\int_{\theta=\sqrt{\pi/2}}^{\theta=\sqrt{\pi}} \left(\theta\cos(\theta^2)\right)\,d\theta$

(c) $\displaystyle\int_{x=0}^{x=1} \left(x\sqrt{1-x^2}\right)\,dx$

(d) $\displaystyle\int_{t=-1}^{t=1} \left(\frac{2t+1}{\sqrt[3]{t^2+t+2}}\right)\,dt$

8. Evaluate all of the definite integrals above.

Chapter 6

Applications of Integrals

6.1 Integrals as Net Change

Activity 6.1.1 Estimating Movement.

We're observing an object traveling back and forth in a straight line. Throughout a 5 minute interval, we get the following information about the velocity (in feet/second) of the object.

Table 6.1.2 Velocity of an Object

t	$v(t)$
0	0
30	2
60	4.25
90	5.75
120	3.5
150	0.75
180	-1.25
210	-3.5
240	-2.75
270	-0.5
300	-0.25

(a) Describe the motion of the object in general.

(b) When was the acceleration of the object the greatest? When was it the least?

(c) Estimate the total displacement of the object over the 5 minute interval. What is the overall change in position from the start to the end?

(d) Is this different than the total distance that the object traveled over the 5 minute interval? Why or why not?

(e) If we know the initial position of the object, how could we find the position of the object at some time, t, where t is a multiple of 30 between 0 and 300?

Activity 6.1.3 A Friendly Jogger.

Consider a jogger running along a straight-line path, where their velocity at t hours is $v(t) = 2t^2 - 8t + 6$, and velocity is measured in miles per hour. We begin observing this jogger at $t = 0$ and observe them over a course of 3 hours.

(a) When is the jogger's acceleration equal to 0 mi/hr^2?

(b) Does this time represent a maximum or minimum velocity for the jogger?

(c) When is the jogger's velocity equal to 0 mi/hr?

(d) Describe the motion of the jogger, including information about the direction that they travel and their top speeds.

Activity 6.1.4 Tracking our Jogger.

Let's revisit our jogger from Activity 6.1.4.

(a) Calculate the total displacement of the jogger from $t = 0$ to $t = 3$.

(b) Think back to our description of the jogger's movement: when is this jogger moving backwards? Split up the time interval from $t = 0$ (the start of their run) to $t = c$ (where c is the time that the jogger changed direction) to $t = 3$. Calculate the displacements on each of these two intervals.

(c) Calculate the total distance that the jogger traveled in their 3 hour run.

Practice Problems

1. Explain the following terms in reference to an object moving along a straight path from time $t = a$ to time $t = b$.

(a) **Position** of the object at time t.

(b) **Displacement** of the object.

(c) **Distance** traveled by the object.

(d) **Velocity** of the object at time t.

(e) **Speed** of the object at time t.

2. Consider the graph of a velocity function, $v(t)$, of some object moving along a line on the time interval $[0, 7]$.

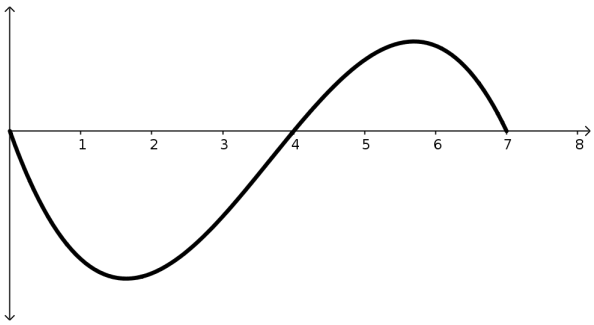


Figure 6.1.5

(a) Do you expect the displacement of the object from $t = 0$ to $t = 7$ to be positive, negative, or 0?

(b) Write two different expressions that represent the total displacement of the object from $t = 0$ to $t = 7$.

(c) Do you expect the distance traveled by the object from $t = 0$ to $t = 7$ to be positive, negative, or 0?

(d) Write two different expressions that represent the total distance traveled by the object from $t = 0$ to $t = 7$.

3. Let's consider an animal running along a straight path with the velocity function:

$$\begin{aligned} v(t) &= \frac{t^4}{10} - t^3 + \frac{27t^2}{10} - \frac{9t}{5} \\ &= \frac{t}{10}(t-1)(t-3)(t-6) \end{aligned}$$

on the time interval $[0, 6]$.

(a) What is the total displacement of the animal on the time interval $[0, 1]$?

(b) What is the total displacement of the animal on the time interval $[1, 3]$?

(c) What is the total displacement of the animal on the time interval $[3, 6]$?

(d) What is the total displacement of the animal on the time interval $[0, 6]$?

(e) What is the total distance traveled by the animal on the time interval $[0, 6]$?

(f) Write a short summary of the animal's movement, including notes about direction, speed, and where the animal travels.

4. Consider an object with velocity function $v(t) = t^2 - 4t + 2$ on the interval $[0, 100]$ with the initial position $s(0) = 3$.

(a) Determine the position function, $s(t)$, for $0 \leq t \leq 100$ using the Future Position of an Object.

(b) Determine the position function, $s(t)$, for $0 \leq t \leq 100$ using the Solving Initial Value Problems strategy.

(c) Compare the results from both methods. Explain why these are equivalent.

5. Consider an object with an acceleration function $a(t) = t + \sin(2\pi t)$ for $t \geq 0$ with $v(0) = 5$.

(a) Determine the velocity function, $v(t)$, for $t \geq 0$ using the Future Position of an Object.

(b) Determine the velocity function, $v(t)$, for $t \geq 0$ using the Solving Initial Value Problems strategy.

(c) Can you obtain the position function, $s(t)$? Explain why or why not, based on the information given.

6. During a brake test for a heavy truck, the truck decelerates from an initial velocity of 88 ft/s with the acceleration function $a(t) = -17$ ft/s^2. Assume that the initial position of the truck is $s(0) = 0$.

(a) Find the velocity function for the truck.

(b) When does the truck stop? In this situation, the truck won't have a negative velocity (since it's just braking and not eventually going in reverse). What time interval is the velocity function relevant on?

(c) What is the total displacement of the truck on this time interval?

(d) Safety standards say that for a truck like this, it needs to be able to stop (from a speed of 88ft/s) in, at most, 200 feet.

Do we need to make changes to the braking mechanism, in order to have the acceleration function change? If so, what does the acceleration need to be (assuming it is constant and we are just replacing it with a new negative number)?

6.2 Area Between Curves

Activity 6.2.1 Remembering Riemann Sums.

Let's start with the function $f(x)$ on the interval $[a, b]$ with $f(x) > 0$ on the interval. We will construct a Riemman sum to approximate the area under the curve on this interval, and then build that into the integral formula.

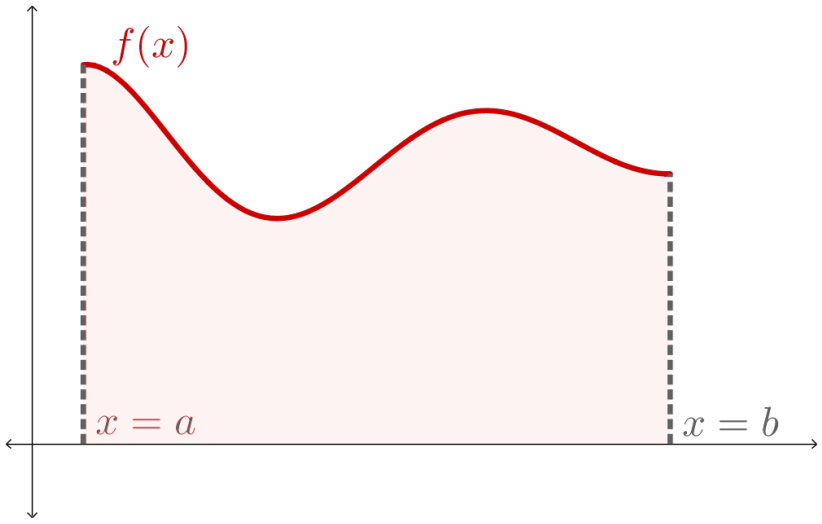


Figure 6.2.2

(a) Divide the interval $[a, b]$ into 4 equally-sized subintervals.

(b) Pick an x_k^* for $k = 1, 2, 3, 4$, one for each subinterval. Then, plot the points $(x_1^*, f(x_1^*))$, $(x_2^*, f(x_2^*))$, $(x_3^*, f(x_3^*))$, and $(x_4^*, f(x_4^*))$.

(c) Use these 4 points to draw 4 rectangles. What are the dimensions of these rectangles (the height and width)?

(d) Find the area of each rectangle by multiplying the heights and widths for each rectangle.

(e) Add up the areas to construct a Riemman sum. Is this sum very accurate? Why or why not?

(f) Now we will generalize a little more. Let's say we divide this up into n equally-sized pieces (instead of 4). Instead of trying to pick an x_k^* for the unknown number of subintervals (since we don't have a value for n yet), let's just focus on one of these: the arbitrary kth subinterval.

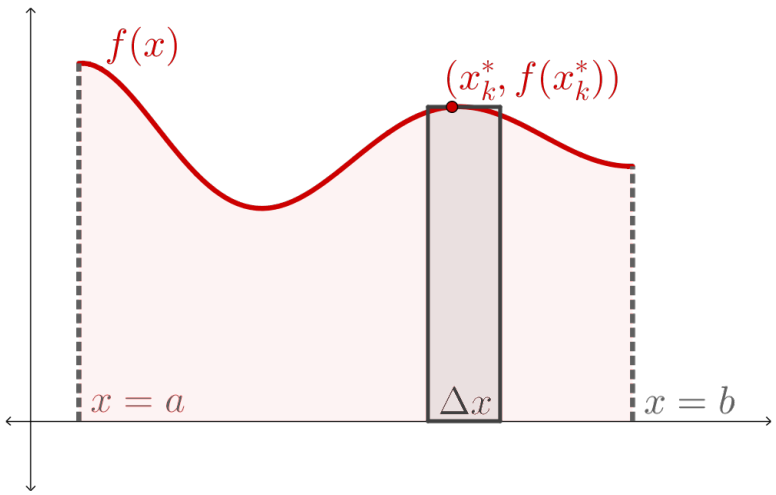


Figure 6.2.3

What are the dimensions of this kth rectangle?

(g) Find A_k, the area of this kth rectangle.

(h) Add up the areas of A_k for $k = 1, 2, 3, ..., n$ to approximate the total area, A

(i) Apply a limit as $n \to \infty$ to this Riemann sum in order to construct the integral formula for the area under the curve $f(x)$ from $x = a$ to $x = b$.

Activity 6.2.4 Area Between Curves.

Let's start with our same function $f(x)$ on the same interval $[a, b]$m but also add the function $g(x)$ on the same interval, with $f(x) > g(x) > 0$ on the interval. We will construct a Riemann sum to approximate the area between these two curves on this interval, and then build that into the integral formula.

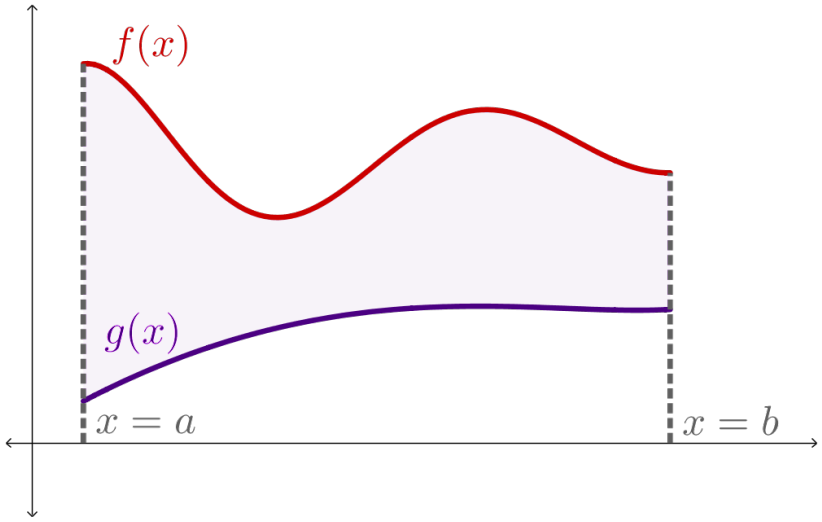


Figure 6.2.5

(a) Divide the interval $[a, b]$ into 4 equally-sized subintervals.

(b) Pick an x_k^* for $k = 1, 2, 3, 4$, one for each subinterval. Plot the points $(x_1^*, f(x_1^*))$, $(x_2^*, f(x_2^*))$, $(x_3^*, f(x_3^*))$, and $(x_4^*, f(x_4^*))$. Then plot the corresponding points on the g function: $(x_1^*, g(x_1^*))$, $(x_2^*, g(x_2^*))$, $(x_3^*, g(x_3^*))$, and $(x_4^*, g(x_4^*))$.

(c) Use these 8 points to draw 4 rectangles, with the points on the f function defining the tops of the rectangles and the points on the g function defining the bottoms of the rectangles. What are the dimensions of these rectangles (the height and width)?

(d) Find the area of each rectangle by multiplying the heights and widths for each rectangle.

(e) Add up the areas to construct a Riemman sum.

(f) Now we will generalize a little more. Let's say we divide this up into n equally-sized pieces (instead of 4). Instead of trying to pick an x_k^* for the unknown number of subintervals (since we don't have a value for n yet), let's just focus on one of these: the arbitrary kth subinterval.

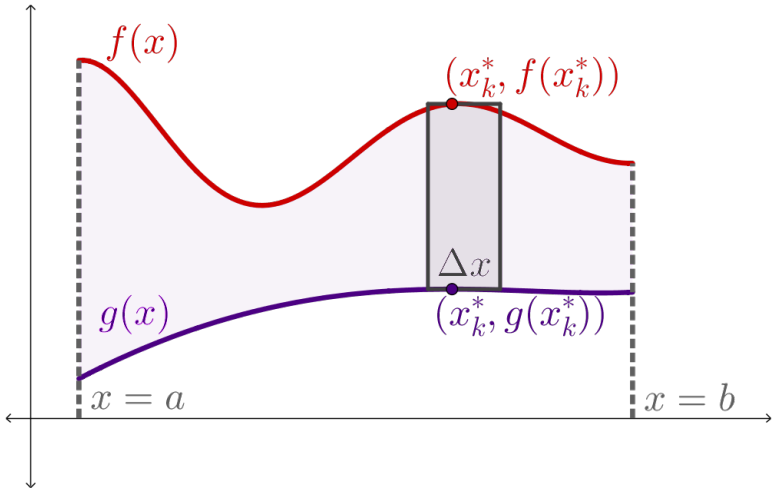


Figure 6.2.6

What are the dimensions of this kth rectangle?

(g) Find A_k, the area of this kth rectangle.

(h) Add up the areas of A_k for $k = 1, 2, 3, ..., n$ to approximate the total area, A

(i) Apply a limit as $n \to \infty$ to this Riemann sum in order to construct the integral formula for the area between the curves $f(x)$ and $g(x)$ from $x = a$ to $x = b$.

Activity 6.2.7 Trying for a Single Integral.

Let's consider the same setup as earlier: the region bounded between two curves, $y = x + 6$ and $y = x^3$, as well as the x-axis (the line $y = 0$). We'll need to name these functions, so let's call them $f(x) = x^3$ and $g(x) = x + 6$. But this time, we'll approach the region a bit differently: we're going to try to find the area of the region using only a single integral.

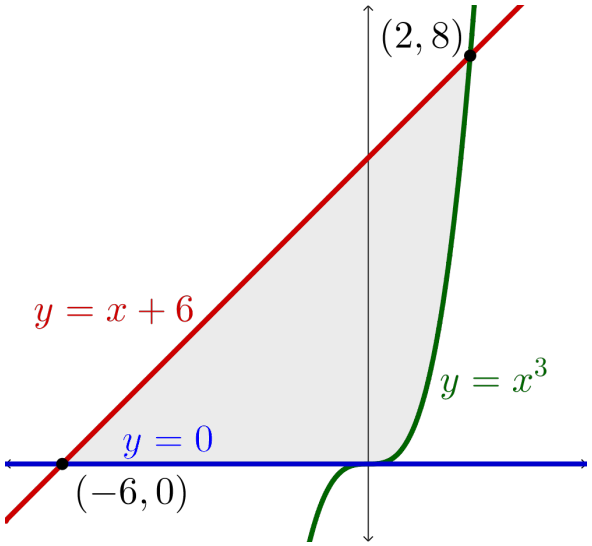


Figure 6.2.8

(a) The range of y-values in this region span from $y = 0$ to $y = 8$. Divide this interval evenly into 4 equally sized-subintervals. What is the height of each subinterval? We'll call this Δy.

(b) Pick a y-value from each sub-interval. You can call these y_1^*, y_2^*, y_3^*, and y_4^*.

(c) Find the corresponding x-values on the $f(x)$ function for each of the y-values you selected. These will be $f^{-1}(y_1^*)$, $f^{-1}(y_2^*)$, $f^{-1}(y_3^*)$, and $f^{-1}(y_4^*)$.

(d) Do the same thing for the g function. Now you have 8 points that you can plot: $\left(f^{-1}(y_1^*), y_1^*\right)$, $\left(f^{-1}(y_2^*), y_2^*\right)$, $\left(f^{-1}(y_3^*), y_3^*\right)$, and $\left(f^{-1}(y_4^*), y_4^*\right)$ as well as $\left(g^{-1}(y_1^*), y_1^*\right)$, $\left(g^{-1}(y_2^*), y_2^*\right)$, $\left(g^{-1}(y_3^*), y_3^*\right)$, and $\left(g^{-1}(y_4^*), y_4^*\right)$. Plot them.

(e) Use these points to draw 4 rectangles with points on f and g, determining the left and right ends of the rectangle. What are the dimensions of these rectangles (height and width)?

(f) Find the area of each rectangle by multiplying the height and widths of each rectangle.

(g) Add up the areas to construct a Riemann sum.

(h) Again, we'll generalize this and think about the kth rectangle, pictured below.

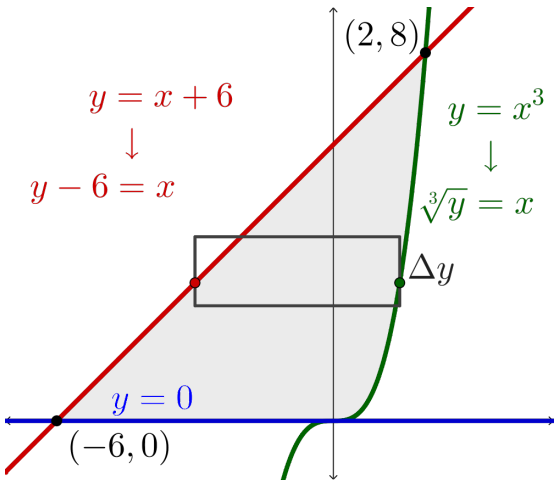


Figure 6.2.9

Which variable defines the location of the kth rectangle, here? That is, if you were to describe *where* in this graph the kth rectangle is laying, would you describe it with an x or y variable? This will act as our general input variable for the integral we're ending with.

(i) What are the dimensions of the kth rectangle?

(j) Find A_k, the area of this kth rectangle.

(k) Add up the areas of A_k for $k = 1, 2, 3, ..., n$ to approximate the total area, A.

(l) Apply a limit as $n \to \infty$ to this Riemann sum in order to construct the integral formula for the area between the curves $f(x)$ and $g(x)$ from $x = a$ to $x = b$.

(m) Now that you have an integral, evaluate it! Find the area of this region to compare with the work we did previously, where we used multiple integrals to measure the size of this same region.

Practice Problems

1. Explain how we use the "slice and sum" method to build an integral formula for the area bounded between curves. Give some details, enough to make sure you understand how the Riemann sums are constructed and how they turn into our integral formula.

2. What are some changes/considerations that we need to make when we decide to set up our integral in terms of y instead of x?

3. Set up (and practice evaluating) an integral expression representing the area of each of the regions described below.

The hint for each problem will open a graph of the region.

(a) The region bounded by the curves $y = x^2 + 1$ and $y = 4x + 1$ between $x = 0$ and $x = 2$.

(b) The region bounded by the curves $y = x$ and $y = 4 - x$ between $x = 0$ and $x = 2$

(c) The region bounded by the curves $y = \sqrt{x} + 2$ and $y = x$ and the line $x = 0$.

(d) The region bounded by the curves $y = \dfrac{2}{x^2 + 1}$ and $y = x^2$.

4. Set up and evaluate an integral representing the area of each of the regions described below. Explain whether you chose to integrate with respect to x or y, and why you made that choice.

(a) The region bounded by the curves $y = \sin(x)$ and $y = \cos(x)$ and the line $y = 0$ between $x = 0$ and $x = \frac{\pi}{2}$.

(b) The region bounded by the curves $y = x$ and $y = x^2 - 2$ and the line $y = 0$ in the first quadrant.

(c) The region bounded by the curves $y = x$ and $y = x^2 - 2$ and the line $y = 0$ in the third quadrant.

(d) The region bounded by the curves $y = 3x$, $y = 4 - x^2$, and $y = x^2$ in the first quadrant with $0 \leq x \leq \sqrt{2}$.

(e) The region bounded by the curves $y = \sqrt{32x}$, $y = 2x^2$, and $y = -4x + 6$ in the first quadrant.

(f) The *other* region bounded by the curves $y = \sqrt{32x}$, $y = 2x^2$, and $y = -4x+6$ in the first quadrant.

(g) The region bounded by the curves $x = 2y$ and $x = y^2 - 3$.

(h) The region(s) bounded by the curves $y = x^3$ and $y = x$.

6.3 Volumes of Solids of Revolution

Activity 6.3.1 Carving out a Hole in the Center.

We're going to look at the same solid as in Figure 6.3.2. But this time, when we define the 2-dimensional region that we're going to revolve around the x-axis, we're going to add a lower boundary function, $g(x)$.

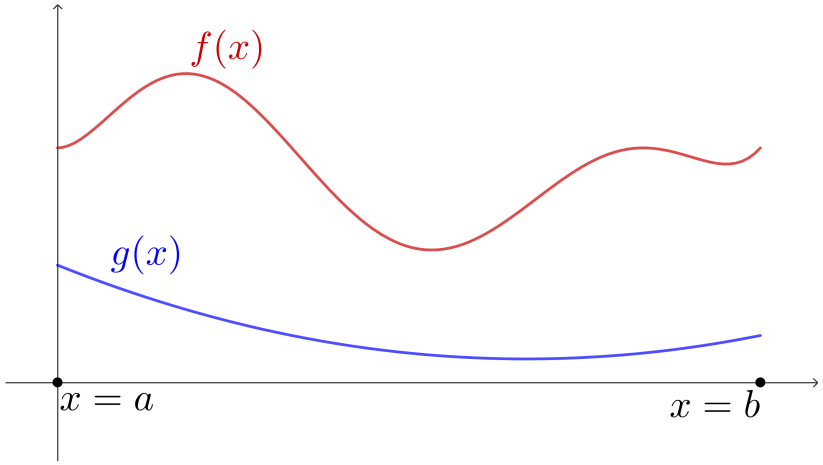


Figure 6.3.2

When we revolve this region around the x-axis, we get the following 3-dimensional solid.

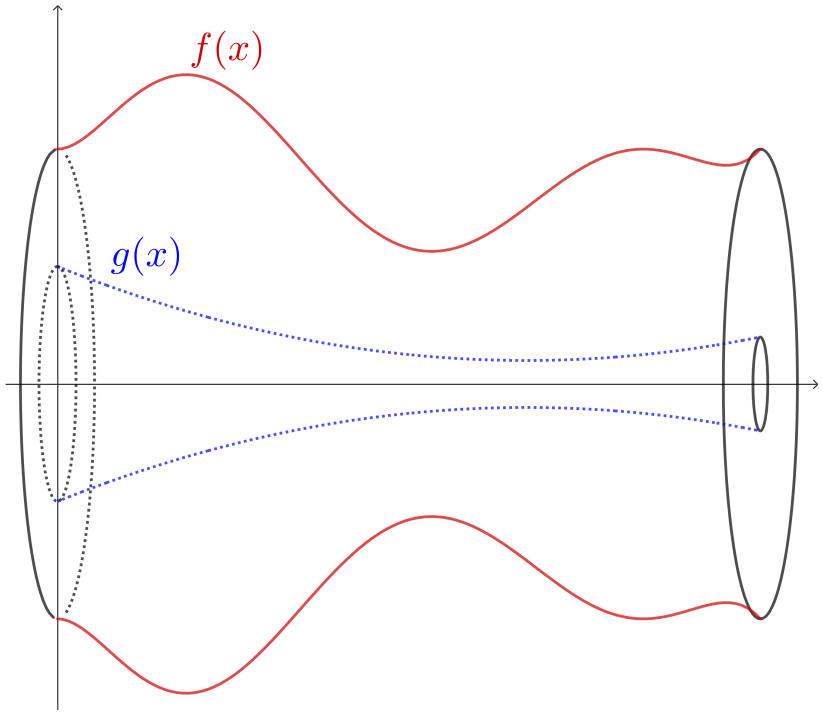


Figure 6.3.3

(a) How is a single generic slice on this solid different than the one in Figure 6.3.2?

(b) Find a formula for the area of the face of the cross-sectional slice.

(c) Use the slice-and-sum process to create an integral expression representing the volume of this solid.

Activity 6.3.4 Volumes by Disks/Washers.

Consider the region bounded between the curves $y = 4 + 2x - x^2$ and $y = \dfrac{4}{x+1}$. We will create a 3-dimensional solid by revolving this region around the x-axis.

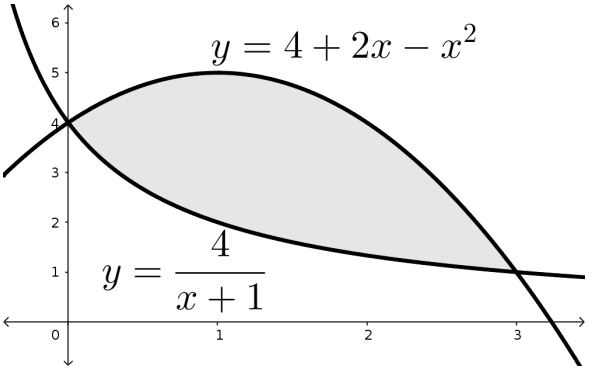


Figure 6.3.5

(a) Visualize the solid you'll create when you revolve this region around the x-axis.

(b) Draw a single rectangle in your region, standing perpendicular to the x-axis.

(c) Let's use this rectangle to visualize the kth slice of this 3-dimensional solid. What does the "face" of it look like?

(d) Find the area of the face of the kth slice.

(e) Set up the integral representing the volume of the solid.

(f) Can you describe how you would antidifferentiate and evaluate this integral?

Activity 6.3.6 Another Volume.

Now let's consider another region: this time, the one bounded between the curves $y = x$ and $y = 3\sqrt{x}$. We will, again, create a 3-dimensional solid by revolving this region around the y-axis.

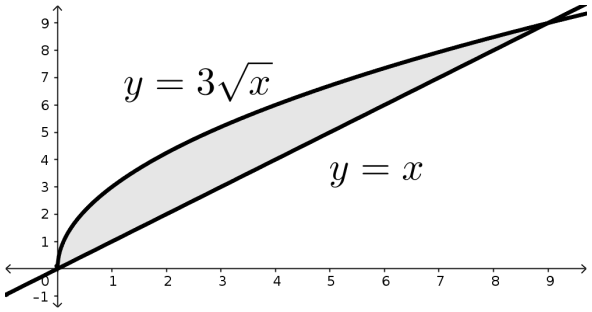


Figure 6.3.7

(a) Visualize the solid you'll create when you revolve this region around the y-axis.

(b) Draw a single rectangle in your region standing perpendicular to the y-axis.

(c) Let's use this rectangle to visualize the kth slice of this 3-dimensional solid. What does the "face" of it look like?

(d) Find the area of the face of the kth slice.

(e) Set up the integral representing the volume of this solid.

Activity 6.3.8 Volume by Shells.

Let's consider the region bounded by the curves $y = x^3$ and $y - x + 6$ as well as the line $y = 0$. You might remember this region from Activity 6.2.9. This time, we'll create a 3-dimensional solid by revolving the region around the x-axis

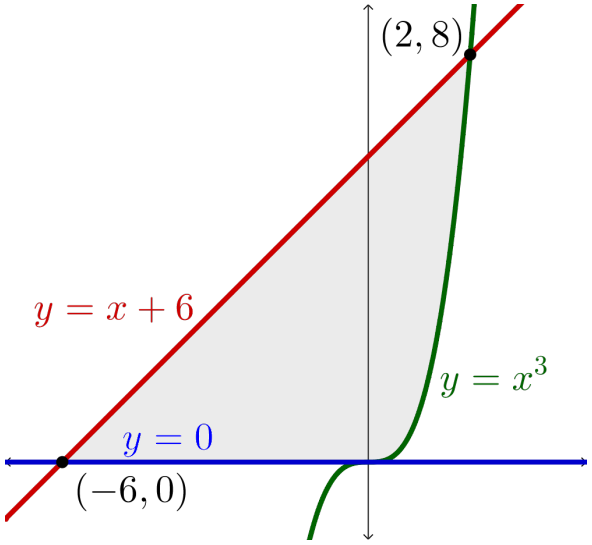


Figure 6.3.9

(a) Sketch one or two rectangles that are *perpendicular* to the x-axis. Then set up an integral expression to find the volume of the solid using them.

(b) Now, draw a single rectangle in the region that is *parallel* to the axis of revolution. Use this rectangle to visualize the kth slice of this 3-dimensional solid. What does that single rectangle create when it is revolved around the x-axis?

(c) Set up the integral expression representing the volume of the solid.

(d) Confirm that your volumes are the same, no matter your approach to setting it up.

Practice Problems

1. We say that the volume of a solid can be thought of as $\int_{x=a}^{x=b} A(x)\,dx$ where $A(x)$ is a function describing the cross-sectional area of our solid at an x-value between $x = a$ and $x = b$. Explain how this integral formula gets built, referencing the slice-and-sum (Riemann sum) method.

2. Explain the differences and similarities between the disk and washer methods for finding volumes of solids of revolution.

3. When do we integrate with regard to x (using a dx in our integral and writing our functions with x-value inputs) and when do we integrate with regard to y (using a dy in our integral and writing our functions with y-value inputs) when we're finding volumes using disks and washers? How do we know?

4. For each of the solids described below, set up an integral using the *disk/washer method* that describes the volume of the solid. It will be helpful to visualize the region, a rectangle on that region, as well as the rectangle revolved around the axis of revolution.

(a) The region bounded by the curve $y = 2x$ and the lines $y = 0$ and $x = 3$, revolved around the x-axis.

(b) The region bounded by the curve $y = e^{-2x}$ and the x-axis between $x = 0$ and $x = \ln(2)$, revolved around the x-axis.

(c) The region bounded by the curves $y = \ln(x)$ and $y = \sqrt{x}$ between $y = 0$ and $y = 1$, revolved around the y-axis.

(d) The region bounded by the curves $y = 2x + 1$ and $y = x$ between $x = 0$ and $x = 3$, revolved around the x-axis.

(e) The region bounded by the curve $y = x^3$, the x-axis, and the line $x = 2$, revolved around the y-axis.

(f) The region bounded by the curve $y = x^3$ and the y-axis between $y = 0$ and $y = 2$, revolved around the y-axis.

5. Explain where the pieces of the shell formula come from. How is this different than using disks/washers?

6. Say we're revolving a region around the x-axis to create a solid. Using the disk/washer method, we will integrate with respect to x. Using the shell method, we integrate with respect to y. Explain the difference, and why this difference occurs.

7. For each of the solids described below, set up an integral using the *shell method* that describes the volume of the solid. It will be helpful to visualize the region, a rectangle on that region, as well as the rectangle revolved around the axis of revolution.

(a) The region bounded by the curve $y = 3x$ and the lines $x = 0$ and $y = 5$, revolved around the y-axis.

(b) The region bounded by the curve $y = \sqrt{x}$ and the x-axis between $x = 0$ and $x = 9$, revolved around the x-axis.

(c) The region bounded by the curves $y = 2 - x^2$ and $y = x$ and the line $x = 0$ revolved around the y-axis.

(d) The region bounded by the curves $y = \sin(x^2) + 2$ and $y = x$ from $x = 0$ to $x = 1$, revolved around the y-axis.

(e) The region bounded by the curves $y = x^2 - 6x + 10$ and $y = 2 + 4x - x^2$ revolved around the y-axis.

(f) The region bounded by the curves $y = \sqrt{2x}$ and $y = 4 - x$ and the x-axis between $x = 0$ and $x = 4$, revolved around the x-axis.

8. Pick at least 2 integrals from Exercise 4 to rewrite using shells instead. What about those regions did you look for to choose which ones to rewrite and which ones to not?

9. Pick at least 2 integrals from Exercise 7 to rewrite using disks/washers instead. What about those regions did you look for to choose which ones to rewrite and which ones to not?

10. For each of the following solids, set up an integral expression using either the disk/washer method or the shell method. You don't need to evaluate them, but you should do some careful thinking about how you set these up, especially as you choose between methods and what variable you are integrating with.

(a) The region bounded by the curves $y = x^2 + 1$ and $y = x^3 + 1$ in the first quadrant, revolved around the x-axis.

(b) The region bounded by the curves $y = x^2 + 1$ and $y = x^3 + 1$ in the first quadrant, revolved around the y-axis.

(c) The region bounded by the curves $y = \dfrac{1}{x}$ and $y = 1 - (x-1)^2$ in the first quadrant, revolved around the x-axis.

6.4 More Volumes: Shifting the Axis of Revolution

Activity 6.4.1 What Changes (in the Washer Method) with a New Axis?

Let's revisit Activity 6.3.14 Volumes by Disks/Washers, and ask some more follow-up questions. First, we'll tinker with the solid we created: instead of revolving around the x-axis, let's revolve the same solid around the horizontal line $y = -3$.

(a) What changes, if any, do you have to make to the rectangle you drew in Activity 6.3.14?

(b) What changes, if any, do you have to make to the area of the "face" kth washer?

(c) What changes, if any, do you have to make to the eventual volume integral for this solid?

Activity 6.4.2 What Changes (in the Shell Method) with a New Axis?

Let's revisit Activity 6.3.23 Volume by Shells, and ask some more follow-up questions about the shell method. Again, we'll tinker with the solid we created: instead of revolving around the x-axis, let's revolve the same solid around the horizontal line $y = 9$.

(a) What changes, if any, do you have to make to the rectangle you drew in Activity 6.3.23?

(b) What changes, if any, do you have to make to the area of the rectangle formed by "unrolling" up kth cylinder?

(c) What changes, if any, do you have to make to the eventual volume integral for this solid?

Activity 6.4.3 More Shifted Axes.

We're going to spend some time constructing *several* different volume integrals in this activity. We'll consider the same region each time, but make changes to the axis of revolution. For each, we'll want to think about what kind of method we're using (disks/washers or shells) and how the different axis of revolution gets implemented into our volume integral formulas.

Let's consider the region bounded by the curves $y = \cos(x) + 3$ and $y = \dfrac{x}{2}$ between $x = 0$ and $x = 2\pi$.

(a) Let's start with revolving this around the x-axis and thinking about the solid formed. While you set up your volume integral, think carefully about which method you'll be using (disks/washers or shells) as well as which variable you are integrating with regard to (x or y).

(b) Now, let's create a different solid by revolving this region around the y-axis. Set up a volume integral, and continue to think carefully about which method you'll be using (disks/washers or shells) as well as which variable you are integrating with regard to (x or y).

(c) We'll start shifting our axis of revolution now. We'll revolve the same region around the horizontal line $y = -1$ to create a solid. Set up an integral expression to calculate the volume.

(d) Now, revolve the region around the line $y = 5$ to create a solid of revolution, and write down the integral representing the volume.

(e) Let's change things up. Revolve the region around the vertical line $x = -1$ to create a new solid. Set up an integral representing the volume of that solid.

(f) We'll do one more solid. Let's revolve this region around the line $x = 7$. Set up an integral representing the volume.

Practice Problems

1. Consider the integral formula for computing volumes of a solid of revolution using the disk/washer method. What part of this integral formula represents the radius/radii of any circle(s)? Why is the radius represented using the function output from the curve(s) defining the region?

2. Consider the integral formula for computing volumes of a solid of revolution using the shell method. What part of this integral formula represents the radius/radii of any circle(s)? Why is the radius not represented using the function output from the curve(s) defining the region?

3. For each of the solids described below, set up an integral expression using disks/washers representing the volume of the solid.

(a) The region bounded by the curve $y = 1 - \sqrt{x}$ in the first quadrant, revolved around $x = 2$.

(b) The region bounded by the curve $y = 1 - \sqrt{x}$ in the first quadrant, revolved around $x = -1$.

(c) The region bounded by the curve $y = 1 - \sqrt{x}$ in the first quadrant, revolved around $y = -2$.

(d) The region bounded by the curve $y = 1 - \sqrt{x}$ in the first quadrant, revolved around $y = 3$.

4. For each of the solids described below, set up an integral expression using shells representing the volume of the solid.

(a) The region bounded by the curves $y = \sqrt{x}$ and $y = x$ in the first quadrant, revolved around the line $x = 2$.

(b) The region bounded by the curves $y = \sqrt{x}$ and $y = x$ in the first quadrant, revolved around the line $x = -1$.

(c) The region bounded by the curves $y = \sqrt{x}$ and $y = x$ in the first quadrant, revolved around the line $y = -2$.

(d) The region bounded by the curves $y = \sqrt{x}$ and $y = x$ in the first quadrant, revolved around the line $y = 3$.

6.5 Arc Length and Surface Area

Activity 6.5.1 Measuring Distance.

(a) Consider the following right-triangle with the normal names of side lengths.

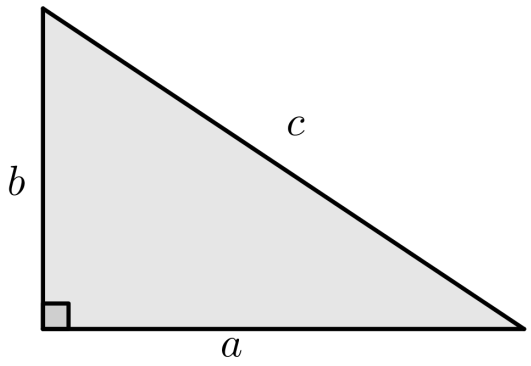


Figure 6.5.2

How do we use the Pythagorean Theorem to find the length of c?

(b) Consider the two points (x_1, y_1) and (x_2, y_2) below.

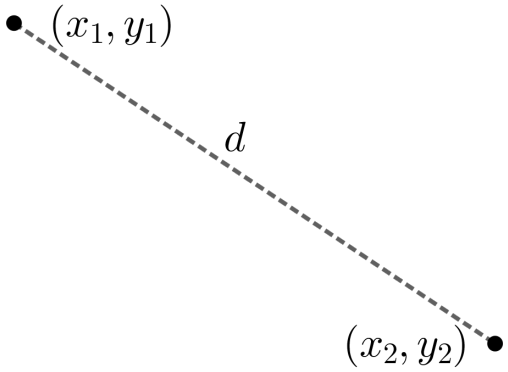


Figure 6.5.3

How do we use the distance formula to find the length of the line connecting the two points, d?

(c) How are these two things the same? How are they different?

Practice Problems

1. The formula for arc length of the function $f(x)$ on the interval $[a, b]$ is $L = \int_{x=a}^{x=b} \sqrt{1 + f'(x)^2}\, dx$. Explain how this definition is built, using the slice-and-sum method. Make sure to explain how we the Pythagorean Theorem is involved.

2. Why do we use $f'(x)$ in the formula for arc length?

3. For each of the following curves on intervals, evaluate the arc length.

(a) $y = -6x + 2$ on $[1, 5]$

(b) $y = \dfrac{x^{3/2}}{3}$ on $[0, 60]$

(c) $y = \dfrac{x^4}{4} + \dfrac{1}{8x^2}$ on $[1, 3]$

4. For each of the following curves on intervals, set up an integral representing the arc length. Do not evaluate.

(a) $y = \tan^2(x)$ on $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

(b) $y = \ln(x^2)$ on $[1, e]$

(c) $y = \sqrt{x + 1}$ on $[0, 8]$

5. Why is the formula for arc length seemingly involved in the integral formula for surface area of a solid of revolution?

6. In the integral formula $A = \int_{x=a}^{x=b} 2\pi f(x)\sqrt{1 + f'(x)^2}\, dx$, what does $f(x)$ represent? What about 2π?

7. For each of the following curves and intervals, find the surface area of the solid formed when the curve is revolved around the x-axis.

(a) $y = 2x + 3$ on $[0, 3]$

(b) $y = 4\sqrt{x}$ on $[4, 9]$

(c) $y = \dfrac{x^4}{4} + \dfrac{1}{8x^2}$ on $[1, 3]$

8. For each of the following curves and intervals, set up the surface area of the solid formed when the curve is revolved around the x-axis. Do not evaluate the integral.

(a) $y = \sin\left(e^{x^2}\right) + 1$ on $[0, 2]$

(b) $y = \ln(x^2)$ on $[1, e]$

6.6 Other Applications of Integrals

Practice Problems

1. Consider a thin wire whose density is variable and is written as $\rho(x)$ on some interval. For each of the following density functions (and their intervals), build and evaluate an integral representing the mass of the wire.

(a) $\rho(x) = \dfrac{2x+1}{10}$ on $[0, 10]$

(b) $\rho(x) = x^2 - 2$ on $[100, 200]$

2. Consider a spring that requires 100 N of force to stretch the spring 0.3 m from its equilibrium position.

(a) Using Hooke's Law, find the spring constant k such that $F(x) = kx$.

(b) How much work is required to stretch the spring 0.4 m from its equilibrium position?

(c) How much work is required to stretch the spring another 0.1 m after this?

3. Consider a spring that is stretched 0.1 m from its equilibrium position. It requires 10 N to stretch it an additional 0.2 m.

(a) Using Hooke's Law, find the spring constant k such that $F(x) = kx$.

(b) How much work is required to stretch the spring 1 m from its equilibrium position?

(c) How much work is required to compress the spring 0.2 m from is equilibrium position?

4. For each tank, assume the density of the liquid is ρ, and set up and evaluate the integral representing the work required to pump the liquid out of the top of the tank.

(a) A square-based tank that is 2 meters tall with a 0.5 m $\times$ 0.5 m base, full of liquid.

(b) A cylindrical tank where the base has a radius of 1 m and the height is 3 m, half-full of liquid.

(c) A frustum-shaped tank where the bottom radius is 4 m, the top radius is 1 m, the height is 2 m, and the tank is filled to a height of $\frac{2}{3}$ m.

Chapter 7

Techniques for Antidifferentiation

7.1 Improper Integrals

Activity 7.1.1 Remembering a Theme so Far.

(a) Let's say that we want to find what the y-values of some function $f(x)$ are when the x-values are "infinitely close to" some value, $x = a$. Since there is no single x-value that is "infinitely close to" a that we can evaluate $f(x)$ at, we need to do something else. How do we do this?

(b) Let's say that we want to find the rate of change of some function instantaneously at a point with $x = a$. We can't find a rate of change unless we have two points, since we need to find some differences in the outputs and inputs. How do we do this?

(c) Suppose you want to find the total area, covered by an infinite number of infinitely thin rectangles. You have a formula for finding the dimensions and areas for some finite number of rectangles, but how do we get an infinite number of them?

(d) Can you find the common calculus theme in each of these scenarios?

Activity 7.1.2 Remembering the Fundamental Theorem of Calculus.

We want to think about generalizing our notion of integrals a bit. So in this activity, section, we're going to think about some of the requirements for the Fundamental Theorem of Calculus and try to loosen them up a bit to see what happens. We'll try to construct meaningful approaches to these situations that fit our overall goals of calculating area under a curve.
This practice, in general, is a really good and common mathematical process: taking some result and playing with the requirements or assumptions to see what else can happen. So it might feel like we're just fiddling with the "What if?" questions, but what we're actually doing is good mathematics!

(a) What does the Fundamental Theorem of Calculus say about evaluating the definite integral $\int_{x=a}^{x=b} f(x)\,dx$?

(b) What do we need to be true about our setup, our function, etc. for us to be able to apply this technique to evaluate $\int_{x=a}^{x=b} f(x)\,dx$?

We are going to introduce the idea of "Improper Integrals" as kind-of-but-not-quite definite integrals that we can evaluate. They are going to violate the requirements for the Fundamental Theorem of Calculus, but we'll work to salvage them in meaningful ways.

Activity 7.1.3 Approximating Improper Integrals.

In this activity, we're going to look at two improper integrals:

1. $\displaystyle\int_{x=2}^{\infty} \frac{1}{(x+1)^2}\,dx$
2. $\displaystyle\int_{x=-1}^{x=2} \frac{1}{(x+1)^2}\,dx$

(a) First, let's just clarify to ourselves what it means for an integral to be improper. Why are each of these integrals improper? Be specific!

(b) Let's focus on $\displaystyle\int_{x=2}^{\infty} \frac{1}{(x+1)^2}\,dx$ first. We're going to look at the slightly different integral:

$$\int_{x=2}^{x=t} \frac{1}{(x+1)^2}\,dx.$$

As long as t is some real number with $t > 2$, then our function is continuous and bounded on $[2, t]$, and so we can evaluate this integral:

$$\int_{x=2}^{x=t} \frac{1}{(x+1)^2}\,dx = F(t) - F(2)$$

where $F(x)$ is an antiderivative of $f(x) = \dfrac{1}{(x+1)^2}$.

Find and antiderivative, $F(x)$.

(c) Now we're going to evaluate some areas for different values of t. Use your antiderivative $F(x)$ from above!

- Let's start with making $t = 99$. So we're going to evaluate:

$$\int_{x=2}^{x=99} \frac{1}{(x+1)^2}\,dx = F(99) - F(2)$$

- Now let $t = 999$. Evaluate:

$$\int_{x=2}^{x=999} \frac{1}{(x+1)^2}\,dx = F(999) - F(2)$$

- Now let $t = 9999$. Evaluate:

$$\int_{x=2}^{x=9999} \frac{1}{(x+1)^2}\,dx = F(9999) - F(2)$$

(d) Based on what you found, what do you *think* is happening when $t \to \infty$ to the definite integral

$$\int_{x=2}^{x=t} \frac{1}{(x+1)^2}\,dx = F(t) - F(2)?$$

(e) Ok, we're going to switch our focus to the other improper integral, $\displaystyle\int_{x=-1}^{x=2} \frac{1}{(x+1)^2}\,dx$. again, we'll look at a slightly different integral:

$$\int_{x=t}^{x=2} \frac{1}{(x+1)^2}\,dx.$$

As long as t is some real number with $-1 < t < 2$, then our function is continuous and bounded on $[t, 2]$, and so we can evaluate this integral:

$$\int_{x=t}^{x=2} \frac{1}{(x+1)^2}\, dx = F(2) - F(t)$$

where $F(x)$ is an antiderivative of $f(x) = \dfrac{1}{(x+1)^2}$. We can use the same antiderivative as before!

We're going to evaluate this intergal for different values of t again, but this time we'll use values that are close to -1, but slightly bigger, since we want to be in the interval $[-1, 2]$.

- Let's start with making $t = -\frac{9}{10}$. So we're going to evaluate:

$$\int_{x=-9/10}^{x=2} \frac{1}{(x+1)^2}\, dx = F(2) - F\left(-\frac{9}{10}\right)$$

- Now let $t = -\frac{99}{100}$. Evaluate:

$$\int_{x=-99/100}^{x=2} \frac{1}{(x+1)^2}\, dx = F(2) - F\left(-\frac{99}{100}\right)$$

- Now let $t = -\frac{999}{1000}$. Evaluate:

$$\int_{x=-999/1000}^{x=2} \frac{1}{(x+1)^2}\, dx = F(2) - F\left(-\frac{999}{1000}\right)$$

(f) Based on what you found, what do you *think* is happening when $t \to -1^+$ to the definite integral

$$\int_{x=t}^{x=2} \frac{1}{(x+1)^2}\, dx = F(2) - F(t)?$$

We can think about putting this a bit more generally into limit notation, but we'll get to this later.

Practice Problems

1. Explain what it means for an integral to be improper. What kinds of issues are we looking at?

2. Give an example of an integral that is improper due to an unbounded or infinite interval of integration (infinite width).

3. Give an example of an integral that is improper due to an unbounded integrand (infinite height).

4. What does it mean for an improper integral to converge? How does this connect with limits?

5. What does it mean for an improper integral to diverge? How does this connect with limits?

6. Why do we need to use limits to evaluate improper integrals?

7. For each of the following improper integrals:

- Explain why the integral is improper. Be specific, and point out the issues in detail.
- Set up the integral using the correct limit notation.
- Antidifferentiate and evaluate the limit.
- Explain whether the integral converges or diverges.

(a) $\displaystyle\int_{x=0}^{\infty} \frac{1}{\sqrt{x+1}}\,dx$

(b) $\displaystyle\int_{x=0}^{\infty} e^{-2x}\,dx$

(c) $\displaystyle\int_{x=-1}^{x=3} \frac{1}{x+1}\,dx$

(d) $\displaystyle\int_{-\infty}^{x=0} \sqrt{e^x}\,dx$

(e) $\displaystyle\int_{x=2}^{x=8} \frac{5}{(x-2)^3}\,dx$

(f) $\displaystyle\int_{x=1}^{x=12} \frac{1}{\sqrt[5]{12-x}}\,dx$

8. One of the big ideas in probability is that for a curve that defines a probability density function, the area under the curve needs to be 1. What value of k makes the function $\dfrac{kx}{(x^2+3)^{5/4}}$ a valid probability distribution on the interval $[0,\infty)$?

9. Let's consider the integral $\displaystyle\int_{x=1}^{\infty} \frac{\sqrt{x^2+1}}{x^2}\,dx$. This is a difficult integral to evaluate!

(a) First, compare $\sqrt{x^2+1}$ to $\sqrt{x^2}$ using an inequality: which one is bigger?

(b) Second, use this inequality to compare the function $\dfrac{\sqrt{x^2+1}}{x^2}$ to $\dfrac{1}{x}$ for $x>0$: which one is bigger? Again, use your inequality from above to help!

(c) Now compare $\displaystyle\int_{x=1}^{\infty} \frac{\sqrt{x^2+1}}{x^2}\,dx$ to $\displaystyle\int_{x=1}^{\infty} \frac{1}{x}\,dx$. Which one is bigger?

(d) Explain how we can use this result to make a conclusion about whether our integral, $\displaystyle\int_{x=1}^{\infty} \frac{\sqrt{x^2+1}}{x^2}\,dx$ converges or diverges.

7.2 More on u-Substitution

Activity 7.2.1 Recapping u-Substitution.

We're going to consider a few integrals, and work through each of the questions for all integrals.

1. $\displaystyle\int \left(\frac{1}{\sqrt{4-3x}}\right)\,dx$
2. $\displaystyle\int x^2(5+x^3)^7\,dx$
3. $\displaystyle\int \left(\frac{\sin(x)}{\cos(x)}\right)\,dx$
4. $\displaystyle\int \left(xe^{-x^2}\right)\,dx$

(a) For each integral, explain why u-substitution is a good choice. How can you tell, just by looking at the integral, that this strategy will be a reasonable thing to try?

(b) For each integral, explain your choice of u, and what that means for how we define du.

(c) For each integral, is your definition of du present in the integrand function? How do you go about making this substitution when the integrand function isn't set up perfectly?

(d) Finish the substitution and integration, and substitute back to the original variable.

Activity 7.2.2 Turn Around Problems.

The two integrals that we're going to look at are "just" some u-substitution problems, but I like to call integrals like these **turn-around** problems. We'll see why!

(a) Consider the integral:

$$\int x\sqrt[3]{x+5}\,dx.$$

First, explain why u-substitution is reasonable here.

(b) Identify du for your chosen u-substitution. When you substitute, you should notice that there are some extra bits in this integrand function that have not been assigned to be translated over to be written in terms of u. Which parts?

(c) We need to think about how to write x in terms of u. Luckily, we already have everything we need! We have defined a link between the x variable and the u variable. We defined it as u being written as some function of x, but can we "turn around" that link to write x in terms of u?

(d) Substitute the integral to be fully written in terms of u.

(e) Before antidifferentiating, compare this integral with the original one. Specifically thinking about how we might multiply, describe the differences between the integrals with regard to composition and rewriting our integrand.

Then, go ahead and use this nicely rewritten version to antidifferentiate and substitute back to x.

(f) Apply this same strategy to the following integral:

$$\int \frac{x}{x+5}\,dx.$$

This integral might be a bit trickier to find the composition in order to identify the u-substitution! Give some things a try!

(g) Compare your integral in terms of x with the substituted version, in terms of u. Why was the second one so much easier to think about or rewrite?

Practice Problems

1. Why do we say that u-substitution undoes the Chain Rule? Explain how the process of our variable substitution accomplishes that.

2. For each of the following indefinite integrals, identify a useful substitution and antidifferentiate.

(a) $\displaystyle\int \frac{x^2+2}{x^3+6x-5}\,dx$

(b) $\displaystyle\int \sec(4x-1)\tan(4x-1)\,dx$

(c) $\displaystyle\int \left(\frac{x+5}{x-1}\right)\,dx$

(d) $\displaystyle\int (x+5)\sqrt{x-1}\,dx$

(e) $\displaystyle\int (x+5)^{51}(x-1)^2\,dx$

(f) $\displaystyle\int \frac{x+1}{\sqrt{x-3}}\,dx$

(g) $\displaystyle\int \frac{1}{(x+2)\sqrt{x+1}}\,dx$

7.3 Manipulating Integrands

Activity 7.3.1 A Negative Exponent.

Let's think about this integral:

$$\int \frac{1}{1+e^{-x}}\,dx.$$

(a) Is there any composition in this integral? Pick it out, and either explain or show that using this to guide your substitution will not be helpful.

(b) What does e^{-x} mean? What does $\frac{1}{e^{-x}}$ mean?

(c) Rewrite the integral, specifically focusing on the negative exponent. You should find that the function looks worse! How can you clean that up?

(d) Why is this new integral set up so much better for the purpose of u-substitution? How could we tell this just by looking at the initial integral?

Activity 7.3.2 Integrating a Rational Function Three Ways.

We're going to think about the integral:

$$\int \left(\frac{x^2 + 3x - 1}{x - 1} \right) \, dx.$$

Let's find 3 different ways of integrating this. This is kind of misleading, since we're actually going to look at 2, since we've already used u-substitution to integrate this in Example 7.2.3.

(a) Let's just notice some things about this rational function.

- Are there any vertical asymptotes? How do you know where to find them?
- Are there any horizontal asymptotes? How do you know that there *aren't*?
- When you zoom really far out on the graph of this function, it looks like a different kind of function. What kind of function? Why is that?

(b) Now we're going to rewrite the function itself: $\dfrac{x^2 + 3x - 1}{x - 1}$ means we're dividing $(x^2 + 3x - 1)$ by $(x - 1)$. So let's do the division!

$$x - 1 \overline{\big|\; x^2 \quad +3x \quad -1 \;\big.}$$

(c) Rewrite your integral using this new version of the function. Notice that we haven't done any calculus or antidifferentiating yet. Explain why this new version of this integrand function is easier to antidifferentiate. What do you get?

(d) Let's approach this integral differently. We said earlier that this function is really an "almost" linear function in disguise: when we divide the quadratic numerator by a linear denominator, we expect a linear function to be left over. In the long division, we saw this happen! We ended with a linear function and some remainder.

Let's try to uncover this linear function. If we're looking to find what linear functions multiply together to get $(x^2 + 3x - 1)$, then we can try factoring!

$$\frac{x^2 + 3x - 1}{x - 1} = \frac{(\qquad\qquad)(\qquad\qquad)}{x - 1}$$

In order for this factoring to be useful, we want to be able to "cancel" out the $x - 1$ factor in the denominator. We're really only interested in what linear factor will multiply by $(x - 1)$ to get $(x^2 + 3x - 1)$.

$$\frac{x^2 + 3x - 1}{x - 1} = \frac{(x - 1)(\qquad\qquad)}{x - 1}$$

First, explain why there is no linear function factor that accomplishes this.

(e) What if we're able to "almost" factor this?

If there *was* a linear factor that multiplied by $(x - 1)$ to get $(x^2 + 3x - 1)$, then the linear portions would multiply together to get x^2. What does this mean about the first linear term of our factor?

(f) What does the constant term of our missing factor need to be? We are hoping that whatever it is can multiply by x (from $x - 1$) and combine with the $-x$ (from the constant -1 multiplied by x in our missing factor) to match the $3x$ in $x^2 + 3x - 1$.

What is it?

(g) Note that we have *not* factored $(x^2 + 3x - 1)$! We *almost* did: we found two factors:

$$(x - 1)(\quad) = x^2 + 3x + \quad .$$

How far off is the actual polynomial that we are working with, $x^2 + 3x - 1$?

Write $x^2 + 3x - 1$ as your two factors, plus or minus some remainder.

(h) You should get the same thing that we got from using long division! Great! The rest of the integral will work the same.

Before we end, though, compare this antiderivative to the one we got in Example 7.2.3. It's different. Why? Is this a problem?

Activity 7.3.3 Comparing Two Very Similar Integrals.

We're going to compare these two integrals:

$$\int \frac{x+2}{x^2+4x+5}\,dx \qquad\qquad \int \frac{2}{x^2+4x+5}\,dx$$

(a) Describe why $u = x^2 + 4x + 5$ is such a useful choice for the first integral, but not for the second. How do the differences in these two integrals influence this substitution, even though the denominators are the same?

(b) Why would it be useful to have a *linear* substitution rule (instead of the *quadratic* one that we picked) for the second integral? Why would that match the structure of the numerator better?

Go ahead and integrate the first integral.

(c) We're going to write the denominator, $x^2 + 4x + 5$ in a different way, in order to get a linear function composed into something familiar.

Complete the square for this polynomial: that is, find some linear factor $(x + h)$ and a real number b such that $(x+h)^2+b = x^2+4x+5$. This should feel familiar, since we already haves tried to force polynomials to factor cleverly in Activity 7.3.4.

(d) There is an intuitive substitution to pick, since we now have more obvious composition. Pick it. What kind of integral do we end up with and how do we antidifferentiate? Complete this problem!

Practice Problems

1. Use polynomial division or some clever factoring to rewrite and find the following indefinite integrals or evaluate the following definite integrals.

(a) $\displaystyle\int \left(\frac{x+4}{x-3}\right)\,dx$

(b) $\displaystyle\int \left(\frac{x^2+4}{x-4}\right)\,dx$

(c) $\displaystyle\int \left(\frac{t^2+t+6}{t^2+1}\right)\,dt$

(d) $\displaystyle\int_{x=2}^{x=4} \left(\frac{x^3+1}{x-1}\right)\,dx$

(e) $\displaystyle\int_{x=0}^{x=1} \left(\frac{x^4+1}{x^2+1}\right)\,dx$

2. Complete the square in order to find the following indefinite integrals.

(a) $\displaystyle\int \left(\frac{1}{x^2-2x+10}\right)\,dx$

(b) $\displaystyle\int \left(\frac{x}{x^2+4x+8}\right)\,dx$

(c) $\displaystyle\int \left(\frac{2x}{x^4+6x^2+10}\right)\,dx$

3. Find the following indefinite integrals.

(a) $\displaystyle\int \left(\frac{1}{x^{-1}+1}\right)\,dx$

(b) $\displaystyle\int \left(\frac{\sin(\theta)+\tan(\theta)}{\cos^2(\theta)}\right)\,d\theta$

(c) $\displaystyle\int \left(\frac{1-x}{1-\sqrt{x}}\right)\,dx$

(d) $\displaystyle\int \left(\frac{1}{1-\sin^2(\theta)}\right)\,d\theta$

(e) $\displaystyle\int \left(\frac{x^{2/3}-x^3}{x^{1/4}}\right)\,dx$

(f) $\displaystyle\int \left(\frac{4+x}{\sqrt{1-x^2}}\right)\,dx$

7.4 Integration By Parts

Activity 7.4.1 Discovering the Integration by Parts Formula.

The product rule for derivatives says that:

$$\frac{d}{dx}\left(u(x)\cdot v(x)\right) = \qquad + \qquad .$$

We know that we intend to "undo" the product rule, so let's try to reframe the product rule from a rule about derivatives to a rule about antiderivatives.

(a) Antidifferentiate the product rule by antidifferentiating each side of the equation.

$$\int \left(\frac{d}{dx}\left(u\cdot v\right)\right)\, dx = \int \qquad + \qquad dx$$

$$\qquad = \int \qquad dx + \int \qquad dx$$

(b) On the right side, we have two integrals. Since each of them has a product of functions (one function and a derivative of another), we can isolate one of them in this equation and create a formula for how to antidifferentiate a product of functions! Solve for $\int uv'\, dx$.

(c) Look back at this formula for $\int uv'\, dx$. Explain how this is really the product rule for derivatives (without just undoing all of the steps we have just done).

Activity 7.4.2 Picking the Parts for Integration by Parts.

Let's consider the integral:

$$\int x \sin(x)\, dx.$$

We'll investigate how to set up the integration by parts formula with the different choices for the parts.

(a) We'll start with selecting $u = x$ and $dv = \sin(x)\, dx$. Fill in the following with the rest of the pieces:

$u = x$ $\qquad$ $v =$

$du =$ $\qquad$ $dv = \sin(x)\, dx$

(b) Now, set up the integration by parts formula using your labeled pieces. Notice that the integration by parts formula gives us another integral. Don't worry about antidifferentiating this yet, let's just set up the pieces.

(c) Let's swap the pieces and try the setup with $u = \sin(x)$ and $dv = x\, dx$. Fill in the following with the rest of the pieces:

$u = \sin(x)$ $\qquad$ $v =$

$du =$ $\qquad$ $dv = x\, dx$

(d) Now, set up the integration by parts formula using this setup.

(e) Compare the two results we have. Which setup do you think will be easier to move forward with? Why?

(f) Finalize your work with the setup you have chosen to find $\int x \sin(x)\, dx$.

What made things so much better when we chose $u = x$ compared to $dv = x\, dx$? We know that the new integral from our integration by parts formula will be built from the new pieces, the derivative we find from u and the antiderivative we pick from dv. So when we differentiate $u = x$, we get a constant, compared to antidifferentiating $dv = x\, dx$ and getting another power function, but with a larger exponent. We know this will be combined with a $\cos(x)$ function no matter what (since the derivative and antiderivatives of $\sin(x)$ only will differ in their sign). So picking the version that gets that second integral to be built from a trig function and a constant is going to be much nicer than a trig function and a power function. It was nice to pick x to be the piece that we found the derivative of!

Activity 7.4.3 Picking the Parts for Integration by Parts.

This time we'll look at a very similar integral:

$$\int x \ln(x)\, dx.$$

Again, we'll set this up two different ways and compare them.

(a) We'll start with selecting $u = x$ and $dv = \ln(x)\, dx$. Fill in the following with the rest of the pieces:

$u = x$ $\qquad$ $v =$ $\qquad$
$du =$ $\qquad$ $dv = \ln(x)\, dx$

(b) Ok, so here we *have to* swap the pieces and try the setup with $u = \ln(x)$ and $dv = x\, dx$, since we only know how to differentiate $\ln(x)$. Fill in the following with the rest of the pieces:

$u = \ln(x)$ $\qquad$ $v =$ $\qquad$
$du =$ $\qquad$ $dv = x\, dx$

(c) Now set up the integration by parts formula using this setup.

(d) Why was it fine for us to antidifferentiate x in this example, but not in Activity 7.4.2?

(e) Finish this work to find $\int x \ln(x)\, dx$.

So here, we didn't actually get much choice. We couldn't pick $u = x$ in order to differentiate it (and get a constant to multiply into our second integral) since we don't know how to antidifferentiate $\ln(x)$ (yet: once we know how, it might be fun to come back to this problem and try it again with the parts flipped). But we can also notice that it ended up being fine to antidifferentiate x: the increased power from our power rule didn't really matter much when we combined it with the derivative of the logarithm, since the derivative of the log is *also a power function*! So we were able to combine those easily and actually integrate that second integral.

Activity 7.4.4 Squared Trig Functions.

Let's look at two integrals. We'll talk about both at the same time, since they're similar.

$$\int \sin^2(x)\,dx \qquad\qquad \int \cos^2(x)\,dx$$

(a) What does it mean to "square" a trig function? Write these integrals in a different way, where the meaning of the "squared" exponent is more clear. What do you notice about the structure of these integrals, the operation in the integrand function? What does this mean about our choice of integration technique?

(b) If you were to use integration by parts on these integrals, does your choice of u and dv even matter here? Why not?

(c) Apply the integration by parts formula to each. What do you notice?

(d) Instead of applying another round of integration by parts to the resulting integral, use the Pythagorean identities to rewrite these integrals:

$$\sin^2(x) = 1 - \cos^2(x)$$
$$\cos^2(x) = 1 - \sin^2(x)$$

(e) You should notice that, in your equation for the integration of $\sin^2(x)\,dx$, you have another copy of $\int \sin^2(x)\,dx$. Similarly, in your equation for the integration of $\cos^2(x)\,dx$, you have another copy of $\int \cos^2(x)\,dx$.

Replace these integrals with a variable, like I (for "integral"). Can you "solve" for this variable (integral)?

Practice Problems

1. Explain how we build the Integration by Parts formula, as well as what the purpose of this integration strategy is.

2. How do you choose options for u and dv? What are some good strategies to think about?

3. Let's say that you make a choice for u and dv and begin working through the Integration by Parts strategy. How can you tell if you've made a poor choice for your parts? Can you *always* tell?

4. In this problem, we'll consider the integral $\int \sin^2(x)\,dx$. We'll integrate this in two different ways!

(a) We know that:

$$\int \sin^2(x)\,dx = \int \sin(x)\sin(x)\,dx.$$

Use the Integration by Parts strategy, and especially note that you can solve for the integral (Solving for the Integral).

(b) We can use a trigonometric identity to rewrite the integral:

$$\sin^2(x) = \frac{1-\cos(2x)}{2}.$$

So we have:

$$\int \sin^2(x)\,dx = \int \frac{1}{2} - \frac{1}{2}\cos(2x)\,dx.$$

Use u-substitution.

(c) Were your answers the same or different? Should they be the same? Why or why not? Are they connected somehow?

5. Integrate the following.

(a) $\int 3x\sin(x)\,dx$

(b) $\int 5xe^x\,dx$

(c) $\int x^2e^{-x}\,dx$

(d) $\int x^2\ln(x)\,dx$

(e) $\int x^2\cos(x)\,dx$

(f) $\int x^3e^{-x}\,dx$

(g) $\int x\sin(x)\cos(x)\,dx$

(h) $\int e^x\sin(x)\,dx$

(i) $\int \sin^{-1}(x)\,dx$

(j) $\int \tan^{-1}(x)\,dx$

6. Evaluate the following definite integrals.

(a) $\displaystyle\int_{x=1}^{x=e} x\ln(x)\ dx$

(b) $\displaystyle\int_{x=0}^{x=\pi/4} x\cos(2x)\ dx$

(c) $\displaystyle\int_{x=0}^{x=\ln(5)} xe^x\ dx$

7. For these next problems, we'll use $x = u^2$ and $dx = 2u\ du$ to substitute into the integral as written. Then use Integration by Parts.

(a) $\displaystyle\int \sin(\sqrt{x})\ dx$

(b) $\displaystyle\int e^{\sqrt{x}}\ dx$

8. In Example 7.4.6, we found that

$$\int \ln(x)\ dx = x\ln(x) - x + C.$$

Let's find this result again, but in a different way.

(a) We'll start with a u-substitution. Let $u = \ln(x)$. Solve for x, and find a substitution for dx.

(b) Apply your substitution above. You should have an integral for which you can use integration by parts! Integrate.

(c) Use this same technique to integrate the following:

$$\int (\ln(x))^3\ dx$$

7.5 Integrating Powers of Trigonometric Functions

Activity 7.5.1 Compare and Contrast.

Let's do a quick comparison of two integrals, keeping the above examples in mind. Consider these two integrals:

$$\int \sin^4(x)\cos(x)\,dx \qquad\qquad \int \sin^4(x)\cos^3(x)\,dx$$

(a) Consider the first integral, $\int \sin^4(x)\cos(x)\,dx$. Think about and set up a good technique for antidifferentiating. Without actually solving the integral, explain why this technique will work.

(b) Now consider the second integral, $\int \sin^4(x)\cos^3(x)\,dx$. Does the same integration strategy work here? What happens when you apply the same thing?

(c) We know that $\sin(x)$ and $\cos(x)$ are related to each other through derivatives (each is the derivative of the other, up to a negative). Is there some other connection that we have between these functions? We might especially notice that we have a $\cos^2(x)$ left over in our integral. Can we write this in terms of $\sin(x)$, so that we can write it in terms of u?

(d) Why would this strategy not have worked if we were looking at the integrals $\int \sin^4(x)\cos^2(x)\,dx$ or $\int \sin^4(x)\cos^4(x)\,dx$? What, specifically, did we need in order to use this combination of substitution and trigonometric identity to solve the integral?

Activity 7.5.2 Compare and Contrast (Again).

We're going to do another Compare and Contrast, but this time we're only going to consider one integral:

$$\int \sec^4(x)\tan^3(x)\,dx.$$

We're going to employ another strategy, similar to the one for Integrating Powers of Sine and Cosine.

(a) Before you start thinking about this integral, let's build the relevant version of the Pythagorean Identity that we'll use. Our standard version of this is:

$$\sin^2(x)+\cos^2(x)=1.$$

Since we want a version that connects $\tan(x)$, which also is written as $\frac{\sin(x)}{\cos(x)}$, with $\sec(x)$, or $\frac{1}{\cos(x)}$, let's divide everything in the Pythagorean Identity by $\cos^2(x)$:

$$\frac{\sin^2(x)}{\quad}+\frac{\cos^2(x)}{\quad}=\frac{1}{\quad}$$

$$\quad+\quad=\quad$$

(b) Now start with the integral. We're going to use two different processes here, two different u-substitutions. First, set $u=\tan(x)$. Complete the substitution and solve the integral.

(c) Now try the integral again, this time using $u=\sec(x)$ as your substitution.

(d) For each of these integrals, why were the exponents set up *just right* for u-substitution each time? How does the structure of the derivatives of each function play into this?

(e) Which substitution would be best for the integral $\int \sec^4(x)\tan^4(x)\,dx$. Why?

(f) Which substitution would be best for the integral $\int \sec^3(x)\tan^3(x)\,dx$. Why?

Practice Problems

1. For an integral $\int \sin^a(x)\cos^b(x)\,dx$, how do you know whether to use $u = \sin(x)$ or $u = \cos(x)$ as the substitution?

2. For an integral $\int \tan^a(x)\sec^b(x)\,dx$, how do you know whether to use $u = \tan(x)$ or $u = \sec(x)$ as the substitution?

3. Integrate the following.

(a) $\int \sin^3(x)\cos^2(x)\,dx$

(b) $\int \sin^2(x)\cos^3(x)\,dx$

(c) $\int \sin^3(x)\cos^3(x)\,dx$

(d) $\int \tan^4(x)\sec^4(x)\,dx$

(e) $\int \tan^3(x)\sec^3(x)\,dx$

(f) $\int \tan^3(x)\sec^4(x)\,dx$

4. Integrate the following.

(a) $\int \sin^{3/4}(x)\cos^5(x)\,dx$

(b) $\int \tan^5(x)\sec^{-1/2}(x)\,dx$

5. Consider the integral $\int \sin^2(x)\,dx$.

(a) Use the trigonometric identity:

$$\sin^2(x) = \frac{1-\cos(2x)}{2}$$

to integrate.

(b) Use integration by parts to integrate.

(c) Which of these techniques do you think was easier to implement and use? Why is that?

6. Consider the integral $\int \cos^4(x)\,dx$.

(a) Use the trigonometric identity:

$$\cos^2(x) = \frac{1+\cos(2x)}{2}$$

to integrate.

(b) Use integration by parts to integrate.

(c) Which of these techniques did you prefer? Why?

7. Integrate the following integrals.

(a) $\int \tan^2(x)\, dx$

(b) $\int \sec^3(x)\, dx$

(c) $\int \tan^5(x)\, dx$

(d) $\int \sin^5(x)\, dx$

7.6 Trigonometric Substitution

Activity 7.6.1 Difference of Squares.

Consider the integral:

$$\int \sqrt{1-x^2}\, dx.$$

(a) First, convince yourself that a normal u-substitution will not be an effective strategy for integration in this case. Why not?

(b) Second, convince yourself that $\sqrt{1-x^2} \neq \sqrt{1}-\sqrt{x^2}$. Why can we not distribute roots across sums and differences like this? When *can* we "distribute" roots across multiple things?

(c) Our goal, then, is to utilize a substitution (using trigonometric functions) to somehow transform this difference of squared terms under the square root into a single product of squared things under the square root.

Which trigonometric identities from our list of them above utilize differences of thing squared, and equate them to a single term?

Can you use the order of the subtraction to help guide which substitution we should use?

(d) When we do a variable substitution in an integral, we are not only finding a way of transforming x to be in terms of some other variable (in this case, θ). We also need to transform the differential, dx. Based on your substitution of $x = T(\theta)$, what is dx?

(e) Perform your substitution! Use your substitution $x = T(\theta)$ and $dx = T'(\theta)\, d\theta$. Note that we have picked this substitution with a very specific goal: we are hoping to notice a Pythagorean Identity.

After you have performed your substitution, apply the relevant Pythagorean Identity to the **radicand**: the bit of our function underneath the radical or root. What integral are we left with (in terms of θ)?

This new integral is something we can antidifferentiate now! We already have done this one in Activity 7.4.8 Squared Trig Functions. So we can end up with:

$$\begin{aligned}\int \sqrt{1-x^2}\, dx &= \int \cos^2(\theta)\, d\theta \\ &= \frac{\theta + \sin(\theta)\cos(\theta)}{2} + C\end{aligned}$$

It is up to us, now, to translate this antiderivative family to be written in terms of x. We can utilize our substitution to do this, but let's first think about how this variable substitution works a bit more.

Activity 7.6.2 Trig Substitution Geoemtry.

We're going to consider three triangles, and we're going to fill in side lengths. In each of these, we'll assume that the lengths x and a are real numbers and are positive.

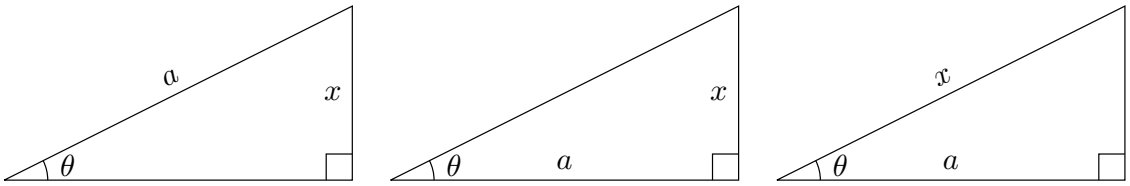


Figure 7.6.3 Three triangles to guide our trigonometric substitutions.

(a) Use the Pythagorean theorem to label the missing side length in each of the three triangles.

(b) For each triangle, explain how you can tell which side length represents the hypotenuse when you see the lengths x, a, and then the missing lengths you found above: $\sqrt{x^2-a^2}$, $\sqrt{a^2-x^2}$, or $\sqrt{x^2+a^2}$.

(c) For each triangle, find a trigonometric function of θ that connects lengths x and a to each other.

Solve each for x to reveal the relevant substitution.

(d) For each substitution, find the corresponding substitution for the differential, dx.

Activity 7.6.4 Practicing Trigonometric Substitution.

Let's look at three integrals, and practice the kind of thinking we'll need to use to apply trigonometric substitution to them.

1. $\displaystyle\int \frac{\sqrt{x^2-9}}{x}\,dx$
2. $\displaystyle\int \frac{2}{(4-x^2)^{3/2}}\,dx$
3. $\displaystyle\int \frac{1}{x^2\sqrt{x^2+1}}\,dx$

For each integral, do the following:

(a) Identify the term (or terms) that signify that trigonometric substitution might be a reasonable strategy.

(b) Use that portion of the integral to compare three side lengths of a triangle. Which one is the largest (and so must represent the length of the hypotenuse)?

(c) Construct the triangle, label an angle θ, and use a trigonometric function to connect the two single-term side lengths. (Feel free to change the angle you label in order to use the sine, secant, or tangent functions instead of their co-functions).

(d) Define your substitution (for both x and the differential dx), and identify the Pythagorean Identity that will be relevant for the integral.

(e) Substitute and antidifferentiate!

(f) Use your triangle to substitute your antiderivative back in terms of x.

Practice Problems

1. Explain how trigonometric substitution helps to convert sums or differences of squares to products of squares. Why is this helpful? When is it helpful?

2. Draw a right triangle with $\sqrt{x^2-4}$ as one of the non-hypotenuse side lengths. What is the length of the hypotenuse? What about the other side length? What would be an appropriate substitution for an integral containing $\sqrt{x^2-4}$?

3. Draw a right triangle with $\sqrt{4-x^2}$ as one of the non-hypotenuse side lengths. What is the length of the hypotenuse? What about the other side length? What would be an appropriate substitution for an integral containing $\sqrt{4-x^2}$?

4. Draw a right triangle with $\sqrt{x^2+4}$ as the hypotenuse. What are the lengths of the other two sides? What would be an appropriate substitution for an integral containing $\sqrt{x^2+4}$?

5. Integrate the following using an appropriate trigonometric substitution.

(a) $\displaystyle\int \frac{x^2}{\sqrt{16-x^2}}\,dx$

(b) $\displaystyle\int \frac{\sqrt{1-x^2}}{x^2}\,dx$

(c) $\displaystyle\int \frac{1}{(9x^2+1)^{3/2}}\,dx$

(d) $\displaystyle\int \frac{\sqrt{x^2-1}}{x}\,dx$

(e) $\displaystyle\int \sqrt{49-x^2}\,dx$

(f) $\displaystyle\int \frac{1}{x(x^2-1)^{3/2}}\,dx$ (for $x>1$)

(g) $\displaystyle\int \frac{x^3}{\sqrt{4+x^2}}\,dx$

(h) $\displaystyle\int \frac{x^2}{(x^2+81)^2}\,dx$

6. Complete the square and then integrate.

(a) $\displaystyle\int \frac{1}{x^2-8x+62}\,dx$

(b) $\displaystyle\int \frac{x^2-8x+16}{(-x^2+8x+9)^{3/2}}\,dx$

7.7 Partial Fractions

Activity 7.7.1 Comparing Rational Integrands.

We're going to compare three integrals:

$$\int \frac{2}{x^2+4x+5}\,dx$$

$$\int \frac{2}{x^2+4x+3}\,dx$$

$$\int \left(\frac{1}{x+1}-\frac{1}{x+3}\right)\,dx$$

(a) Start with the first integral:

$$\int \frac{2}{x^2+4x+5}\,dx.$$

How would you approach integrating this?

(b) Try the same tactic on the second integral:

$$\int \frac{2}{x^2+4x+3}\,dx.$$

You don't need to complete this integral, but think about how you might proceed.

(c) Think about the third integral:

$$\int \left(\frac{1}{x+1}-\frac{1}{x+3}\right)\,dx.$$

How would you integrate this?

(d) The third integral is unique from the other two in that it is has two terms. Let's combine them together to see how we could write this integral to compare it more closely to the other two.

Subtract $\dfrac{1}{x+1}-\dfrac{1}{x+3}$ using common denominators and compare your rewritten integral to the other two.

(e) Which of these integrals and/or representations of an integral is easiest to work with? Which one is most annoying to work with? Why?

Activity 7.7.2 First Examples of Partial Fractions.

(a) Consider the integral:

$$\int \frac{6x-16}{x^2-4x+3}\,dx.$$

First, confirm that this would be *very* annoying to try to use u-substitution on, even though we have a linear numerator and quadratic denominator.

(b) Notice that the denominator can be factored:

$$\int \frac{6x-16}{(x-3)(x-1)}\,dx.$$

If this integrand function were a sum of two "smaller" rational functions, what would their denominators be? What do we know about their numerators?

(c) Use some variables (it's typical to use capital letters like A, B, C, etc.) to represent the numerators, and then add the partial fractions together. What do you get? How does this rational function compare to $\dfrac{6x-16}{(x-3)(x-1)}$?

(d) Set up an equation connecting the numerators, and solve for your unknown variables. What are the two fractions that added together to get $\dfrac{6x-16}{(x-3)(x-1)}$?

(e) Antidifferentiate to solve the integral $\displaystyle\int \frac{6x-16}{(x-3)(x-1)}\,dx$.

(f) Let's do the same thing with a new integral:

$$\int \frac{3x^2-2x+3}{x(x^2+1)}\,dx.$$

What are the partial fraction forms? What kinds of numerators should we expect to see? Use variables to represent these.

(g) Add the partial fractions together and set up an equation for the numerators to solve. What are the two partial fractions after you solve for the unknown coefficients?

(h) Antidifferentiate and solve the integral $\displaystyle\int \frac{3x^2-2x+3}{x(x^2+1)}\,dx$.

Practice Problems

1. Why do we use partial fraction decomposition on some integrals of rational functions? Give an example and explain why it is helpful in your example.

2. For each rational function described, write out the corresponding partial fraction forms.

(a) $\dfrac{p(x)}{(x-4)(x+2)(x-1)}$ where $p(x)$ is some polynomial with degree less than 3.

(b) $\dfrac{p(x)}{(x+1)^2(3x-5)^3}$ where $p(x)$ is some polynomial with degree less than 5.

(c) $\dfrac{p(x)}{(x^2+1)(x^2+2x+5)}$ where $p(x)$ is some polynomial with degree less than 4.

(d) $\dfrac{p(x)}{x^4-1}$ where $p(x)$ is some polynomial with degree less than 4.

3. Consider the following integral, with the partial fraction forms written out:

$$\int \frac{x^3+6x^2-x}{(x-2)(x+1)(x^2+1)}\,dx = \int \left(\frac{A}{x-2}+\frac{B}{x+1}+\frac{Cx+D}{x^2+1}\right)\,dx.$$

(a) Write an equation connecting the numerators.

(b) Find (and use) a specific x-value to input into the equation to solve for A.

(c) Find (and use) a specific x-value to input into the equation to solve for B.

(d) Why can you not use this strategy to solve for coefficients C or D?

(e) Find the cubic terms (you will need to do some multiplication) on both sides of your equation. Use these to solve for C.

(f) Find the constant terms (you will need to do some multiplication) on both sides of your equation. Use these to solve for D.

(g) Integrate!

4. Explain why partial fractions is not an appropriate technique for the following integral:

$$\int \frac{x^2+x}{x^2-x+1}\,dx.$$

How should we approach this integral, instead? Use this to practice some other strategies for integrating rational functions!

5. Integrate the following.

(a) $\displaystyle\int \frac{2}{(x-1)(x+3)}\,dx$

(b) $\displaystyle\int \frac{4x+1}{(x-4)(x+5)}\,dx$

(c) $\displaystyle\int \frac{2x^2-15x+32}{x(x^2-8x+64)}\,dx$

(d) $\displaystyle\int \frac{1}{(x+2)(x-2)}\,dx$

(e) $\displaystyle\int \frac{20x}{(x-1)(x^2+4x+5)}\,dx$

(f) $\displaystyle\int \frac{x^2}{(x-2)^3}\,dx$

6. In the problems we are looking at in this section, we're limiting ourselves to, at most, irreducible quadratic factors in the denominator. In problems with simple linear factors, repeated linear factors, or irreducible quadratic factors, what types of antiderivative functions do you expect to see? Explain.

7. For each of the following integrals, we will do some preliminary work before using partial fractions to integrate. Really, we'll perform a specific u-substitution that will give us some resulting integral to use partial fractions on.

(a) $\displaystyle\int \frac{4e^{2x}}{(e^{2x}+3)(e^{2x}-5)}\,dx$ where we use $u = e^{2x}$.

(b) $\displaystyle\int \sqrt{e^x+1}\,dx$ where we use $u = \sqrt{e^x+1}$.

(c) $\displaystyle\int \frac{\sqrt{x}+3}{\sqrt{x}(x-1)}\,dx$ where we use $u = \sqrt{x}$.

Chapter 8

Infinite Series

8.1 Introduction to Infinite Sequences

Activity 8.1.1 Building our First Sequences.

We might already have some familiarity with sequences. Here, we'll focus less on some of the detailed mechanics and just think about these sequences as functions.

(a) Describe a sequence of numbers where you use a consistent rule/function to build each term (each number) based only on the *previous term* in the sequence. You will need to decide on some first term to start your sequence.

(b) Describe a different sequence of numbers using the same rule to generate new terms/numbers from the previous one. What do you need to do to make these two sequences different from each other?

(c) Describe a new sequence of numbers where you use a consistent rule/function to build each term based on its position in the sequence (i.e. the first term will be some rule/function based on the input 1, the second will be based on 2, you'll use 3 to get the third term, etc.). We will call the position of each term in the sequence the *index*.

(d) Describe another, new, sequence of numbers where you use a consistent rule/function to build each term based on its index. This time, make the terms get smaller in size as the index increases.

Activity 8.1.2 Returning to our First Sequences.

Let's return back to the four sequences we created in Activity 8.1.1.

(a) For each of the sequences, how are we going to define them? Explicit formulas? Recursion relations? How do you know?

(b) Now, for each sequence, define the sequence formally using either an explicit formula or recursion relation, whichever matches with how you described the sequence in Activity 8.1.1.

Activity 8.1.3 Describing These Sequences.

Let's look at the sequences from Example 8.1.5. Go through the following tasks for each sequence.

(a) What do you think each sequence is "counting towards" (if anything)?

(b) Can you show that the sequence is counting towards what you think it is with a limit (or show that it's not counting towards anything)?

Activity 8.1.4 Write the Sequence Rules.

We'll look at some sequences by writing out the first handful of terms. From there, our goal is to write out more terms and eventually define each sequence fully.
For each sequence, write an explicit formula and a recursion relation to define the sequence. You can choose whether to start your index at $k = 0$ or $k = 1$.

(a) $\{a_k\} = \left\{4, \frac{2}{3}, \frac{1}{9}, \frac{1}{54}, ...\right\}$

(b) $\{b_k\} = \left\{\frac{3}{5}, \frac{2}{5}, \frac{5}{17}, \frac{3}{13}, \frac{7}{37}, ..., \right\}$

(c) $\{c_k\} = \left\{\frac{1}{5}, \frac{3}{5}, 1, \frac{7}{5}, ...\right\}$

(d) What kinds of connections do you notice between the explicit formulas and the recursion relations for these sequences?

Practice Problems

1. What is the index of an infinite sequence? How does the index relate to terms of the infinite sequence?

2. Explain the differences between how an explicit formula and a recursion relation can define an infinite sequence.

3. Why does a recursion relation defining a sequence rely on an initial value?

4. Use the explicit formulas write out the first 8 terms of each sequence.

(a) $\{a_k\} = \left\{\dfrac{1}{k}\right\}_{k=1}^{\infty}$

(b) $\{b_k\} = \left\{\dfrac{k+2}{k^2+1}\right\}_{k=1}^{\infty}$

(c) $\{c_k\} = \left\{\sin\left(\dfrac{k\pi}{4}\right)\right\}_{k=0}^{\infty}$

(d) $\{d_k\} = \left\{\dfrac{1}{2^k}\right\}_{k=0}^{\infty}$

(e) $\{j_k\} = \left\{\dfrac{(-1)^k}{k^2}\right\}_{k=1}^{\infty}$

(f) $\{\ell_k\} = \left\{\dfrac{k+1}{k!}\right\}_{k=0}^{\infty}$

5. For each of the following recursion relations, write out the first 8 terms of the sequence.

(a) $a_k = \dfrac{a_{k-1}+1}{2}$; $a_1 = 3$

(b) $b_k = \dfrac{2}{b_{k-1}}$; $b_1 = 4$

(c) $c_k = 2c_{k-1} - 2$; $c_1 = 3$

(d) $d_k = \dfrac{d_{k-1}}{3}$; $d_1 = 2$

(e) $j_k = 3(j_{k-1})^2 - 1$; $j_1 = 1$

(For this one, only find the first 6 terms. You'll see why.)

(f) $\ell_k = \dfrac{1}{(\ell_{k-1})^2}$; $\ell_1 = 2$

(Again, only find the first 6 terms for this one.)

6. For each of the following sequences, write down a possibility for an explicit formula defining the sequence as well as a recursion relation (with initial value) defining the sequence.

(a) $\{a_k\} = \{4, 5, 6, 7, ...\}$

(b) $\{b_k\} = \{0, 2, 4, 6, ...\}$

(c) $\{c_k\} = \left\{\dfrac{3}{4}, \dfrac{1}{2}, \dfrac{1}{3}, \dfrac{2}{9}, ...\right\}$

(d) $\{d_k\} = \left\{\dfrac{1}{8}, \dfrac{3}{16}, \dfrac{1}{4}, \dfrac{5}{16}, ...\right\}$

7. For each of the following sequences, describe whether they converge or diverge. If they converge, show what they converge to using limit work.

(a) $\{a_k\} = \left\{3 + \dfrac{(-1)^{k+1}}{k}\right\}_{k=1}^{\infty}$

(b) $\{b_k\} = \left\{\sqrt[k]{k}\right\}_{k=1}^{\infty}$

(c) $\{c_k\} = \left\{\dfrac{k!}{e^k}\right\}_{k=0}^{\infty}$

(d) $\{d_k\} = \left\{\left(1 + \dfrac{2}{k}\right)^k\right\}_{k=1}^{\infty}$

(e) $\{j_k\} = \left\{\dfrac{k^3 + 1}{k^2 + \ln(k)}\right\}_{k=1}^{\infty}$

(f) $\{\ell_k\} = \{k \ln(1/k)\}_{k=1}^{\infty}$

8. Compare a sequence to a function. What are some similarities? What are some differences? Carefully discuss this in terms of different representations of sequences/functions.

9. Choose through the sequences in any of the problems above to find one to fit each of the descriptions below. Explain your choices for each.

(a) Monotonic non-decreasing.

(b) Monotonic non-increasing.

(c) Bounded. What are the upper and lower bounds?

8.2 Introduction to Infinite Series

Activity 8.2.1 How Do We Think About Infinite Series?

Let's consider the following sequence:

$$\left\{\frac{9}{10^k}\right\}_{k=1}^{\infty}$$

(a) Write out the first 5 terms of the sequence.

(b) What does this sequence converge to? Show this with a limit!

(c) Now we'll construct a new sequence, this time by adding things up. We're going to be working with the sequence $\{S_n\}_{n=1}^{\infty}$ where

$$S_n = \sum_{k=1}^{k=n} \left(\frac{9}{10^k}\right).$$

Write out the first five terms of this sequence: S_1, S_2, S_3, S_4, S_5.

(d) Can you come up with an explicit formula for S_n?

(e) Does $\{S_n\}$ converge or diverge? Use a limit to find what it converges to!

(f) What do you think this means for the infinite series $\sum_{k=1}^{\infty}\left(\frac{9}{10^k}\right)$? Does the infinite series converge or diverge?

This is hopefully a nice little introduction to how we'll think about infinite series: we'll consider, instead, the sequence of sums where we add up more and more terms. This is also a nice first example, because we really just showed that

$$0.999... = 1$$

since

$$\begin{aligned}\sum_{k=1}^{\infty}\left(\frac{9}{10^k}\right) &= \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + ... \\ &= 0.9 + 0.09 + 0.009 + ... \\ &1.\end{aligned}$$

But more importantly, we now have a good strategy for thinking about infinite series as sequences of *partial sums*.

Practice Problems

1. What does it mean for an infinite series $\sum_{k=1}^{\infty} a_k$ to converge? What does it mean for this series to diverge?

2. What is a partial sum, S_n, of the infinite series $\sum_{k=1}^{\infty} a_k$? Why do we care about this?

3. For some infinite series $\sum_{k=1}^{\infty} b_k$, we consider the partial sum

$$S_n = \sum_{k=1}^{n} b_k.$$

For each of the following descriptions of S_n, explain what conclusions we can make about the infinite series. Be specific, and justify your answers.

(a) $S_n = \dfrac{3n^2}{2n^2+1}$

(b) $\{S_n\}$ is increasing and bounded.

(c) $0 \leq S_n \leq 1$ for all $n \geq 1$.

(d) $4 - \dfrac{1}{n} \leq S_n \leq 4 + \dfrac{1}{n}$ for all $n \geq 1$.

4. Consider the infinite series $\sum_{k=1}^{\infty} c_k = 0.6 + 0.06 + 0.006 + 0.0006 + ...$

(a) Write out an explicit formula for the sequence of terms $\{c_k\} = \{0.6, 0.06, 0.006, 0.0006, ...\}$.

(b) Write out the partial sums S_1, S_2, S_3, and S_4.

(c) Write out an explicit formula for the nth partial sum, S_n.

(d) Does the sequence $\{S_n\}$ converge or diverge? Explain. If it converges, what does it converge to?

(e) Does the series $\sum_{k=1}^{\infty} c_k$ converge or diverge? Explain. If it converges, what does it converge to.

5. Consider the infinite series $\sum_{k=1}^{\infty} \dfrac{1}{k(k+1)}$.

(a) Use Partial Fraction Decomposition to rewrite this series.

(b) Write out the first five terms, and the partial sum S_5.

(c) Write out the explicit formula for the nth partial sum, S_n.

(d) Does the sequence $\{S_n\}$ converge or diverge? Explain. If it converges, what does it converge to?

(e) Does the series $\sum_{k=1}^{\infty} \dfrac{1}{k(k+1)}$ converge or diverge? Explain. If it converges, what does it converge to?

8.3 The Divergence Test and the Harmonic Series

Activity 8.3.1 Investigating the Harmonic Series.

(a) Write out the first several terms of the harmonic series, terms from $\left\{\frac{1}{k}\right\}_{k=1}^{\infty}$. Write however many you need to get a feel for how the terms work.

(b) Can you find out how many terms you would have to go "into" the series before the term was less than 0.00000001?

(c) Can you do this same kind of thing, no matter how small? For instance, how many terms would you have to go into the series before the term was less than some real number ε where $\varepsilon > 0$?

(d) Remind/explain/convince yourself that what we've really done is show that $\lim_{k\to\infty} \frac{1}{k} = 0$. This isn't a new or terribly interesting fact, but make sure that you understand why the argument above shows this.

(e) Let's do something very similar, but with $\left\{\sum_{k=1}^{n} \frac{1}{k}\right\}_{n=1}^{\infty}$, the sequence of partial sums, instead. Write out the first few partial sums. There's no specific number that you *need* to write, but make sure to write enough partial sums to get a feel for how the partial sums work.

(f) Can you find out how many terms you need to add up until the partial sum is larger than 1?

(g) Can you find out how many terms you need to add up until the partial sum is larger than 5?

(h) Can you find out how many terms you need to add up until the partial sum is larger than 10?

(i) Do you think that for any positive number S, we can always find some partial sum $\sum_{k=1}^{n} \frac{1}{k} > S$? What do you think this would mean about

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k}?$$

To actually show that for any $S > 0$ we could always find an $n > 1$ where

$$\sum_{k=1}^{n} \frac{1}{k} > S$$

is an extremely difficult task! We will show that the Harmonic Series diverges in a different way, but for now I want us to notice these contradictory results: we have a series whose terms get small, but whose partial sums do not seem to converge.

We have $\frac{1}{k} \to 0$ but it seems like $\lim_{n\to\infty} S_n$ does not exist. Is this behavior special to the Harmonic Series? Is this something we should make note of? Is there some other connection between the terms of a series and the behavior of the partial sums of the series that we need to note?

Activity 8.3.2 Thinking about the Terms of a Series.

(a) What does it mean for the series $\sum_{k=1}^{\infty} a_k$ to converge?

(b) Calculate the difference between $S_{n+1} = \sum_{k=1}^{n+1} a_k$ and $S_n = \sum_{k=1}^{n} a_k$, $S_{n+1} - S_n$.

(c) If the sequence of partial sums for a series, $\{S_n\}_{n=1}^{\infty}$ converges, then what should we expect to happen with the values of S_{n+1} and S_n? What does this mean for the difference $S_{n+1} - S_n$?

(d) Now consider a series that we know diverges, the Harmonic Series. Find the limit,

$$\lim_{k\to\infty} \frac{1}{k}.$$

Practice Problems

1. Give an example of a series that diverges due to the Divergence Test. Explain.

2. Give an example of a series that fails the Divergence Test. Do you know if this series converges or diverges? Explain.

3. Apply the Divergence Test to each of the following series. Explain the conclusions of the test for each series.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{k^2+1}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{e^k}\right)$

(c) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{1}{k\ln(k)}\right)$

(d) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{k}{\ln(k)}\right)$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\sqrt[k]{k}\right)$

(f) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{k^2+k}{e^k}\right)$

(g) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^2+1}\right)$

(h) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k!}{k^2}\right)$

(i) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^k(k+1)}{k+\frac{1}{k}}\right)$

8.4 The Integral Test

Activity 8.4.1 Integrals and Infinite Series.

We're going to work with a graph of a continuous function, and we're going to start with a couple of conditions:

1. Our function will be continuous wherever it's defined.
2. Our function will be decreasing on its domain.
3. All of the function outputs will be positive.

Let's not worry about picking a specific function for this, but we will visualize a graph of one that meets these three requirements.

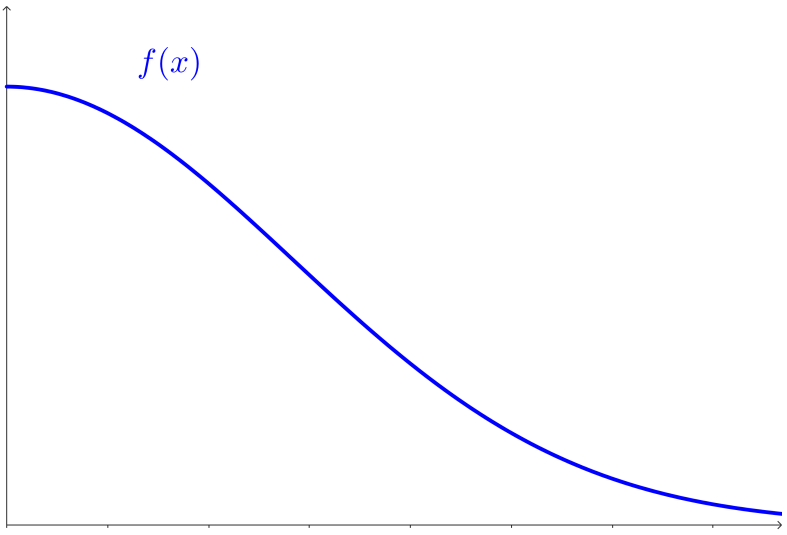


Figure 8.4.2

We can then visualize the sequence of terms, $a_k = f(k)$ for $k = 0, 1, 2, ...,$.

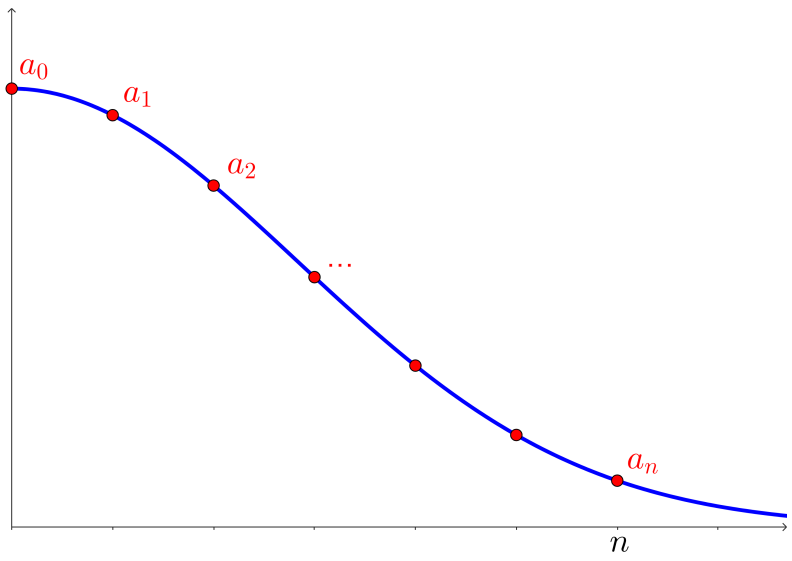


Figure 8.4.3

(a) How does the partial sum, $\sum_{k=0}^{n} a_k$ compare to the Riemann sum for $f(x)$ from $x = 0$ to $x = n$ with $n + 1$ rectangles?

(b) We're going to visualize the accumulation of $f(x)$ from $x = 0$ to $x = n$ by thinking about the integral:

$$\int_{x=0}^{x=n} f(x)\, dx.$$

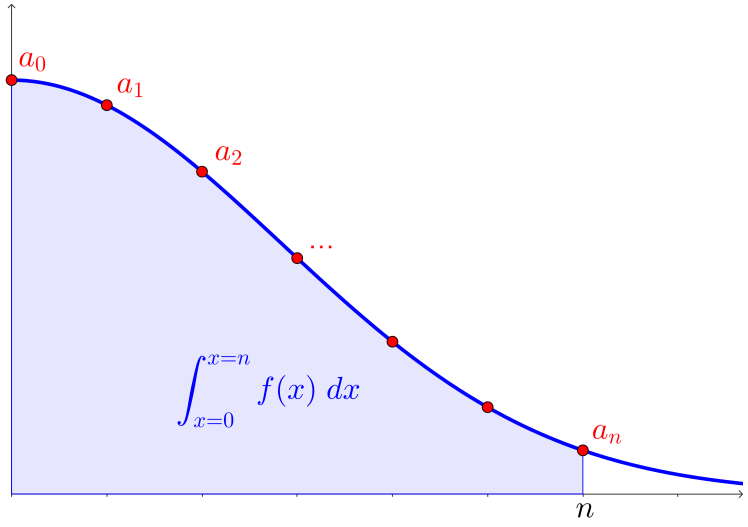


Figure 8.4.4

How does this area compare to the Riemann sum you thought of above? Compare them with an inequality and make sure you can explain why this has to be true.

(c) Remove the first term of the series, a_0, and instead think of the sum $\sum_{k=1}^{n} a_k$. Can you still think of this as a Riemann sum to approximate the area from the integral $\int_{x=0}^{x=n} f(x)\,dx$?

How does this new Riemann sum compare to the area formed by the integral? Compare them with an inequality and make sure you can explain why this has to be true.

(d) We have thought about two sums, and we can connect them:

$$\sum_{k=0}^{n} a_k = a_0 + \sum_{k=1}^{n} a_k.$$

Use the sums to bound the integral:

$$\quad < \int_{x=0}^{x=n} f(x)\,dx < \quad$$

(e) Similarly, use the integral to bound the sum:

$$\quad < \sum_{k=0}^{n} a_k < \quad$$

These bounds are going to be super useful! Discovering them is the main task for finding the connections between improper integrals and infinite series. These inequalities might seem kind of strange at first, but we're going to apply a limit to everything as $n \to \infty$, and then think about our definitions of convergence (Definition 7.1.7 and Definition 8.2.3).

Practice Problems

1. What are the conditions of the Integral Test? What are the conclusions of the Integral Test?

2. For each of the following series, apply the Integral Test and interpret the conclusion of the test.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{k^2+1}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{e^k}\right)$

(c) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{1}{k\ln(k)}\right)$

(d) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^2+k}{e^k}\right)$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^2+1}\right)$

3. For what values of p (with $p > 0$) will the following series converge?

$$\sum_{k=1}^{\infty}\frac{1}{k^p}$$

4. For what values of r (with $r > 0$) will the following series converge?

$$\sum_{k=0}^{\infty} r^k$$

8.5 Alternating Series and Conditional Convergence

Activity 8.5.1 Which is More Likely to Converge?

We're going to try to think about what might be different when we analyze an alternating series compared to a series with only positive (or non-negative) terms.
Let's say that $\{a_k\}_{k=0}^{\infty}$ is some sequence of positive real numbers. Now let's consider the two series:

$$\sum_{k=0}^{\infty} a_k \qquad \text{vs} \qquad \sum_{k=0}^{\infty} (-1)^k a_k$$

(a) Let's first consider the sequences of terms: $\{a_k\}$ compared with $\{(-1)^k a_k\}$. Is either of these more or less likely to converge? Does this tell us anything about whether or not the corresponding series converges?

(b) Now let's think of the partial sums:

$$\left\{\sum_{k=0}^{n} a_k\right\}_{n=0}^{\infty} \qquad \text{vs} \qquad \left\{\sum_{k=0}^{n} (-1)^k a_k\right\}_{n=0}^{\infty}$$

Is either of these sequences more or less likely to converge? Does this tell us anything about whether or not the corresponding series converges?

(c) Now make a conjecture about which infinite series is more likely to converge:

$$\sum_{k=0}^{\infty} a_k \qquad \text{vs} \qquad \sum_{k=0}^{\infty} (-1)^k a_k$$

Remember that $a_k > 0$ for $k = 0, 1, 2, ...$, so the only differences are the changes in sign.

Activity 8.5.2 Approximating an Alternating Series.

Let's look, again, at the picture of the partial sums of an alternating series in Figure 8.5.3. We're going to assume that the series converges, which means that:

- $\lim_{n\to\infty} S_n$ exists.
- $\lim_{n\to\infty} a_n = 0$.

Let's add to our figure.

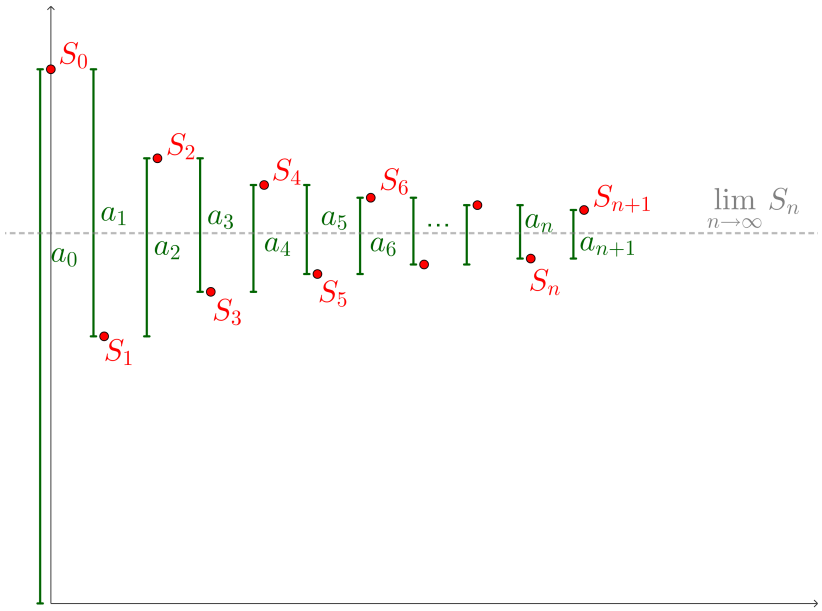


Figure 8.5.3

(a) Why are the even-indexed partial sums sitting above the odd-indexed partial sums?

(b) Why are the even-indexed partial sums sitting above the horizontal line, $\lim_{n\to\infty} S_n$?

(c) Why are the odd-indexed partial sums sitting below the horizontal line, $\lim_{n\to\infty} S_n$?

(d) If we were trying to approximate the value of $\lim_{n\to\infty} S_n$, how can we use the partial sums to build an interval that approximates the value?

Activity 8.5.4 The Alternating Harmonic Series Converges.

Let's consider the alternating harmonic series, as written below:

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + -...$$

(a) First, confirm that this series does converge, using the Alternating Series Test.

(b) Find $S_6 = \sum_{k=1}^{k=6} \frac{(-1)^{k+1}}{k}$.

(c) Use Theorem 8.5.7 Approximations of Alternating Series to create a bound on this estimation. Report an interval that you know that the actual value that the alternating harmonic series converges to is in.

(d) Use technology to find S_{1000}. Compare this value to $\ln(2)$. Would it surprise you for someone to claim that the alternating harmonic series converges to $\ln(2)$?

Activity 8.5.5 The Alternating Harmonic Series Converges (Again).

Now let's consider a new, rearrranged, version of the alternating harmonic series:

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + ...$$

We can write this in summation notation as:

$$\sum_{k=1}^{\infty} \left(\frac{1}{2k-1} - \frac{1}{4k-2} - \frac{1}{4k} \right).$$

(a) First, confirm that all of the terms from the alternating harmonic series will eventually show up in this series. Convince yourself that we are truly adding (and subtracting) all of the same values with just a different order.

(b) Does this new series converge? Check, using the Alternating Series Test.

(c) Add up the first few terms of this series to find the value of a partial sum. You can choose how many terms you add. Does it look like it will also converge to ln(2)?

(d) Use technology to add up many terms of this series. Can you convince yourself what this series converges to?

Practice Problems

1. What is an "alternating series?" What are some reasons that we need to consider these as a separate class of infinite series?

2. Explain the differences between the Divergence Test and the Alternating Series Test. What do each tell us about the infinite series? How do we do each of the tests?

3. Give an example of an alternating series that converges conditionally. Explain your reasoning, and show that it converges conditionally.

4. Give an example of an alternating series that converges absolutely. Explain your reasoning, and show that it converges absolutely.

5. Explain how an alternating series could converge conditionally. What about alternating series makes this even possible? Can a positive-term series converge conditionally? Why or why not?

6. For each of the following alternating series, determine whether the series converges or diverges.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}\tan^{-1}(k)}{k^4}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}}{\sqrt{k^2+3}}\right)$

(c) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k}}{\sqrt[k]{k}}\right)$

(d) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{(-1)^{k+1}}{\sqrt[3]{k^2-k}}\right)$

(e) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{(-1)^{k}}{\sqrt{k^3+1}}\right)$

(f) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-2)^{k+1}}{k^k}\right)$

(g) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}k}{k^5-k^2+1}\right)$

(h) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{\cos(\pi(k-1))}{k^2\ln(k+1)}\right)$

(i) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{\cos(\pi k)}{e^k+\pi^k}\right)$

7. In Section 8.4, Practice Problem 2, you used the Integral Test to see if some series converged or diverged. Now, consider the alternating series below: test to see if they diverge, converge conditionally, or converge absolutely.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}(k)}{k^2+1}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}(k)}{e^k}\right)$

(c) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{(-1)^k}{k\ln(k)}\right)$

(d) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}(k^2+k)}{e^k}\right)$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{(-1)^{k+1}}{k^2+1}\right)$

8. In Section 8.4, Practice Problem 3, you found the values of p (with $p>0$) such that the series $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^p}$ converges.

What are the values of p (with $p>0$) such that

$$\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k^p}$$

converges conditionally?

9. In Section 8.4, Practice Problem 4, you found the values of r (with $r>0$) such that the series $\displaystyle\sum_{k=0}^{\infty}\frac{1}{r^k}$ converges.

Are there any values of r (with $r>0$) such that

$$\sum_{k=0}^{\infty}\frac{(-1)^k}{r^k}$$

converges conditionally?

8.6 Common Series Types

Activity 8.6.1 Building a Convergence Formula for Geometric Series.

We're going to think of constructing two different ways of thinking about how much area of of a circle has been shaded. We can pretend we have a circle with area that is 1, where the radius is $r = \sqrt{\frac{1}{\pi}}$, giving

$$A = \pi \left(\sqrt{\frac{1}{\pi}} \right)^2 = 1.$$

Then we can describe the areas we're looking at as almost a percentages of the total area.

(a) We are going to split our circle into two parts, with r amount of the area left unshaded and so $1 - r$ area shaded. We'll shade in some angular sector.

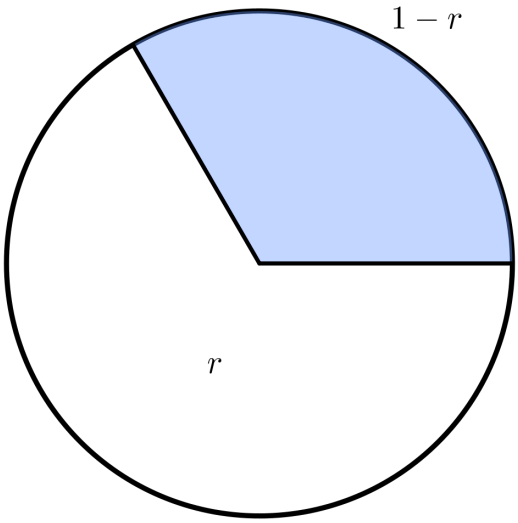


Figure 8.6.2

This part is easy: how much of the area is shaded?

(b) This next step will set the stage for how we think about this problem now: we're going to divide the remaining white area up into the same proportional pieces: we'll shade in a ratio of $1 - r$ of the remaining white space and leave a ratio of r of the white space unshaded.

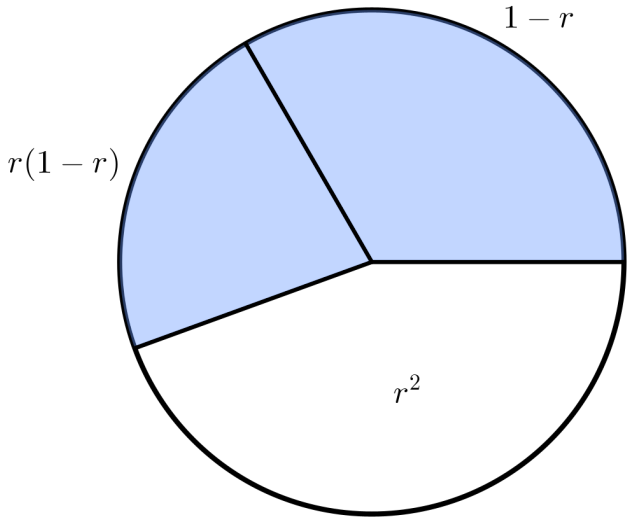


Figure 8.6.3

Can you describe two ways of calculating the total amount of shaded area?

(c) We'll repeat the process: shade in more, where the ratio of shaded area to unshaded area is $1-r$ to r.

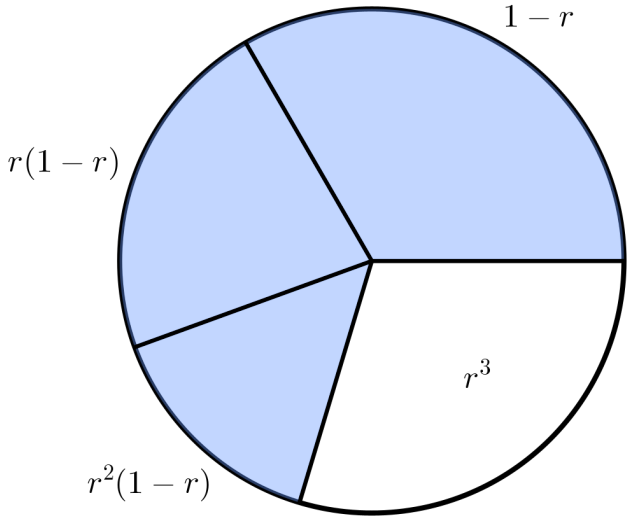


Figure 8.6.4

Can you describe two ways of calculating the total amount of shaded area?

(d) Now we're going to repeat this process until we've done it a total of $n+1$ times.

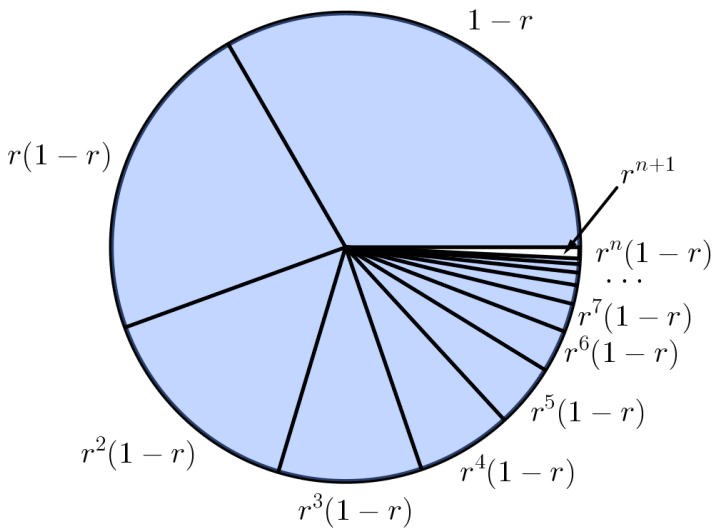


Figure 8.6.5

Can you describe two ways of calculating the total amount of shaded area?

(e) In the limit as $n \to \infty$, how much of the area is shaded in?

Notice that $(1-r)$ is likely a common factor in one of your ways of calculating this area. Convince yourself, then, that:

$$\begin{aligned}\lim_{n\to\infty} \sum_{k=0}^{n} r^k &= \lim_{n\to\infty} (1 + r + r^2 + ... + r^n) \\ &= \frac{1}{1-r}\end{aligned}$$

Activity 8.6.6 (Im)Possible Combinations.

When we have thought about infinite series, we have thought about three different mathematical objects: the sequence of terms of the series, the sequence of partial sums of the series, and the infinite series itself. As a reminder, if we had an infinite series

$$\sum_{k=1}^{\infty} a_k$$

we can say that:

- $\{a_k\}_{k=1}^{\infty}$ is the sequence of terms of the series
- $S_n = \sum_{k=1}^{n} a_k$ is a partial sum and $\{S_n\}_{n=1}^{\infty}$ is the sequence of partial sums of the series

For each of these three objects—the terms, the partial sums, and the series—we have some notion of what it means for that object to converge or diverge.
Consider the following table of all of the different combinations of convergence and divergence of the three objects. For each combination, decide whether this combination is possible or impossible. If it is possible, give an example of an infinite series whose terms, partial sums, and the series itself converge/diverge appropriately. If it is impossible, give an explanation of why.

Table 8.6.7 (Im)Possible Combinations

$\{a_k\}_{k=1}^{\infty}$	$\{S_n\}_{n=1}^{\infty}$	$\sum_{k=1}^{\infty} a_k$	**(Im)Possible?**	**Example or Explanation**
Converges	Converges	Converges		
Converges	Converges	Diverges		
Converges	Diverges	Converges		
Converges	Diverges	Diverges		
Diverges	Converges	Converges		
Diverges	Converges	Diverges		
Diverges	Diverges	Converges		
Diverges	Diverges	Diverges		

Practice Problems

1. For each of the following series, classify each of the series by type (geometric, p-series, or neither) and whether it converges or diverges (as well as why).

For any geometric series that converges, find what it converges to.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{3^k}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^3}\right)$

(c) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{4^k}\right)$

(d) $\displaystyle\sum_{k=0}^{\infty}\left(e^{-2k}\right)$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt{k-2}}\right)$

(f) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt[5]{k^7}}\right)$

(g) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt[7]{k^5}}\right)$

(h) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{(-1)^k 3^{k+8}}{5^{k+2}}\right)$

2. This is a fun problem! We'll consider a special cake with a weird twist. The cake is made of stacked, cylindrical tiers, each with a height of 1 unit. The radius of the tiers is $\frac{1}{k}$ for $k = 1, 2, ...$, meaning the bottom tier has a radius of 1, the second tier has a radius of 1/2, the third has a radius of 1/3, etc. The cake, somehow, has an infinite number of tiers.

(a) We want to ice the sides of the cake, and want to know how much icing to use to cover the sides of each tier. Create an infinite series to represent the area of the outside edges. Does the series converge or diverge? Explain.

(b) We want to figure out how much batter needed to bake the cake. Create an infinite series representing the volume of the tiers. Does the series converge or diverge? Explain.

8.7 Comparison Tests

Activity 8.7.1 Comparisons to Bound Partial Sums.

This activity is mostly going to be thinking about proof mechanisms, and so it might be helpful to review Activity 8.4.2 Integrals and Infinite Series. If you want to see more, then the proof of Integral Test will provide some further details on why the inequalities we built were useful.

(a) In the Integral Test, how did we guarantee that our sequence of partial sums was monotonic?

(b) How did we know that, as long as the corresponding integral converged, then our sequence of partial sums was bounded?

(c) How did we know that, as long as our integral diverged, then our sequence of partial sums had to diverge as well?

(d) What happens if we swap out the integral we're connecting our series to with a different series?

For these inequalities to be useful, what do we need to be true?

Activity 8.7.2 (Un)Helpful Comparisons.

We're going to consider a handful of infinite series, here:

1. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(3^k)}$
2. $\displaystyle\sum_{k=3}^{\infty} \frac{\sqrt{k+1}}{k-2}$

(a) Pick a series that is reasonable to use as a comparison for each of the series listed. Remember, we want:

- A series that is recognizable (probably a Geometric Series or a p-Series), so that we know the behavior of it: we need to know whether the series we're comparing to converges or diverges!
- A series that is similar enough to the series we're working with that we can construct an inequality comparing the term structure. It can be hard to compare functions that are seemingly unrelated to each other!
- A series that has terms that are either larger or smaller than our series, depending on whether we are trying to show that our series converges or diverges.

(b) Build the comparison from the series we start with to the one you picked. What kinds of conclusions can you make?

(c) We're going to change the series we're considering to two slightly different series:

(a) $\displaystyle\sum_{k=1}^{\infty} \frac{2k}{3^k}$

(b) $\displaystyle\sum_{k=3}^{\infty} \frac{\sqrt{k-1}}{k+2}$

How do these small changes impact the inequalities you built?

(d) How do these changes in the inequalities change the conclusions we can draw from the Direct Comparison Test?

(e) What do you *think* is happening with these series: do you think that these small changes are enough to change the behavior of the series (i.e. whether it converges or diverges)?

Activity 8.7.3 Ratios for Comparison.

Let's start with some functions: we'll consider $f(x)$ and $g(x)$ as two functions that are continuous when $x \geq 0$ with $f(x) \to 0$ and $g(x) \to 0$ as $x \to \infty$.

All of this is so that we can think about $\dfrac{f(x)}{g(x)}$ and know that we have an indeterminate form. We could put the requirement of differentiability on these functions (so that we could think about L'Hôpital's Rule), but we don't need to do that.

We're going to now consider the limit:

$$\lim_{x\to\infty} \frac{f(x)}{g(x)}.$$

(a) What would the limit $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ look like if $g(x) \to 0$ at a faster rate than $f(x)$ does? In this case, we might say that:

$$f(x) >> g(x).$$

(b) What would the limit $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ look like if $f(x) \to 0$ at a faster rate than $g(x)$ does? In this case, we might say that:

$$g(x) >> f(x).$$

(c) If the functions $f(x)$ and $g(x)$ eventually act equivalently, then what does the limit $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ look like?

(d) If the function $f(x)$ eventually acts like some scaled version of $g(x)$, then what does the limit $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ look like?

Practice Problems

1. For a series $\sum a_k$ where all of the terms a_k are positive, explain whether the series converges, diverges, or if we cannot tell based on the following comparisons.

(a) $a_k < \frac{1}{k^2}$ for all k .

(b) $a_k > \frac{1}{k^2}$ for all k .

(c) $a_k < \frac{1}{k}$ for all k .

(d) $a_k > \frac{1}{k}$ for all k .

(e) $a_k < 1$ for all k .

(f) $a_k > 1$ for all k .

2. For each pair of series, compare the terms of each with either $<$ or $>$. What does the Direct Comparison Test say about the first series in each pair (if anything)?

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt{k^3+1}}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^{3/2}}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^2 4^k}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{4^k}\right)$

(c) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{k+1}{(k-1)^2}\right)$ compared with $\displaystyle\sum_{k=2}^{\infty}\left(\frac{1}{k}\right)$

(d) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k!}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^2}\right)$

3. What is an advantage of using the Limit Comparison Test instead of the Direct Comparison Test? Demonstrate this with a specific example.

4. For each pair of series, attempt to compare the terms with either $<$ or $>$. Explain why the Direct Comparison Test fails (or explain why the comparison is difficult), and then apply the Limit Comparison Test, and explain the conclusions.

(a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt{k+1}}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k^{1/2}}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k-1}{(k+1)^2}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k}\right)$

(c) $\displaystyle\sum_{k=1}^{\infty}\left(\sqrt{\frac{k}{k^2+1}}\right)$ compared with $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\sqrt{k}}\right)$

5. For each series below, determine whether the series converges or diverges using either the Direct or Limit Comparison Test.

(a) $\displaystyle\sum_{k=2}^{\infty}\left(\frac{4k}{k^3-1}\right)$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{2^k}\right)$

(c) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{3}{\sqrt{k}+\sqrt[3]{k}}\right)$

(d) $\displaystyle\sum_{k=2}^{\infty}\left(\sqrt[4]{\frac{k}{k^4-1}}\right)$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\ln(k)}\right)$

(f) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{2}{k^k}\right)$

(g) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^3-k+1}{k^{9/2}+4k^2+1}\right)$

(h) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^{3/2}+4}{k^2-k+\sqrt{k}}\right)$

8.8 The Ratio and Root Tests

Activity 8.8.1 Reminder about Geometric Series.

We are going to build some convergence tests that try to link some infinite series to the family of geometric series and show that even though a series is *not* geometric, it might act enough like one to be considered "eventually geometric-ish."
But first, what does it mean for a series to be a geometric series?

(a) Describe a defining characteristic of a geometric series. What makes it geometric?

(b) Can you describe this characteristic in another way? For instance, if you described a geometric series using a characteristic about the Explicit Formula, can you describe the same characteristic in the context of the Recursion Relation instead? Or vice versa?

(c) Write out a generalized and simplified form of the term a_k of a geometric series explicitly and recursively. In each case, solve for r, the ratio between terms.

Activity 8.8.2 When Are These Tests Useful?

We're going to look at a couple of small examples where we can rewrite some expressions into friendlier forms, and try to connect these rewriting strategies to the Ratio and Root Tests.

(a) Rewrite the following expression into a friendlier form. Explain why this new form is friendlier.

$$\sqrt[k]{\frac{2^{k+1}}{7^{3k}}}$$

(b) Rewrite the following expression into a friendlier form. Explain why this new form is friendlier.

$$\sqrt[k]{\frac{(k+1)^k}{4^{2k+1}}}$$

(c) Rewrite the following expression into a friendlier form. Explain why this new form is friendlier.

$$\frac{(5^{k+2})(6^{k-3})}{(5^{k-1})(6^{k+1})}$$

(d) Rewrite the following expression into a friendlier form. Explain why this new form is friendlier.

$$\frac{103!}{99!}$$

(e) Rewrite the following expression into a friendlier form. Explain why this new form is friendlier.

$$\frac{(2k+4)!}{(2k+2)!}$$

(f) Why do you think the Ratio Test especially will be useful for series whose terms include factorials and exponentials?

Why do you think the Root Test will be useful for series whose terms include exponentials and functions raised to functions (of k)?

Practice Problems

1. For the Ratio Test, we note that if $\lim_{k\to\infty}\left|\frac{a_{k+1}}{a_k}\right| < 1$, then the series $\sum a_k$ converges. Why is this? Explain some general intuition for why the Ratio Test guarantees convergence in this case.

2. For the Root Test, we note that if $\lim_{k\to\infty}\sqrt[k]{|a_k|} < 1$, then the series $\sum a_k$ converges. Why is this? Explain some general intuition for why the Root Test guarantees convergence in this case.

3. Note that in both the Ratio and Root Tests above, that we use the absolute value of the terms. Is it possible, then, for the Ratio or Root Test to tell us that a series converges conditionally?

4. Simplify each of the following expressions.

(a) $\dfrac{104!}{103!}$

(b) $\dfrac{22!}{20!}$

(c) $\sqrt[k]{\dfrac{2^k}{e^{k^2}}}$

(d) $\dfrac{(k+4)!}{(k+3)!}$

(e) $\dfrac{3^{k+2}}{3^k}$

(f) $\dfrac{2^{k+1}(k)!}{2^k(k+1)!}$

5. For the following series, use the apply the Ratio or Root Test (whichever is more applicable) and interpret the results.

(a) $\displaystyle\sum_{k=1}^{\infty}\frac{(k+1)^2}{k!}$

(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k!}\right)^k$

(c) $\displaystyle\sum_{k=2}^{\infty}\frac{k^2}{e^k}$

(d) $\displaystyle\sum_{k=1}^{\infty}\frac{(-1)^k}{(3k)^k}$

(e) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{\ln(k+1)}\right)^k$

(f) $\displaystyle\sum_{k=1}^{\infty}\frac{k}{\pi^k}$

(g) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k}{\pi}\right)^k$

(h) $\displaystyle\sum_{k=1}^{\infty}\frac{(-1)^{k+1}k^k}{k!}$

(i) $\displaystyle\sum_{k=1}^{\infty} \frac{k!}{(2k+1)!}$

6. For what values of p (with $p > 0$) will the following series converge?

$$\sum_{k=1}^{\infty} \frac{k^p}{k!}$$

7. For each of the following infinite series, use any methods we've learned about so far to determine whether the series converges or diverges. If it converges and you are able to tell what it converges to, say so and explain. If you cannot tell what it converges to, say so and explain. Clearly state the test or method you use or the type of series.

(a) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{k^2-1}{(k^3-k+1)^4} \right)$

(b) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{1}{k \ln(k^4)} \right)$

(c) $\displaystyle\sum_{k=0}^{\infty} \left(\frac{(-1)^k + 2^k}{e^{2k}} \right)$

(d) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{1}{k^2-1} \right)$

(e) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{1}{k\sqrt{k^2-1}} \right)$

(f) $\displaystyle\sum_{k=0}^{\infty} \left(\frac{4^k(k^2)}{k!} \right)$

(g) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{\ln(k)}{k+2} \right)$

8.9 Exploration: Infinite Series

Exploration 8.9.1 Interesting Infinite Series.

(a) Consider the series

$$\sum_{k=1}^{\infty} \frac{1}{D(k)k^p}$$

where $D(k)$ is the function that returns the number of digits in the number k (so $D(9) = 1$ and $D(435) = 3$), and p is a real number.

For what values of p does this series converge?

(b) Consider the series

$$\sum_{k=1}^{\infty} \frac{1}{F(k)k^p}$$

where $F(k)$ is the function that returns the kth Fibbonacci number, and p is a real number. As a reminder, the Fibbonacci sequence is:

$$1, 1, 2, 3, 5, 8, 13, 21...$$

where $F(k) = F(k-1) + F(k-2)$.

For what values of p does this series converge?

Chapter 9

Power Series

9.1 Polynomial Approximations of Functions

Activity 9.1.1 Build a Polynomial.

We're going to use the formula in Definition 9.1.1 to construct two different polynomials that approximate two different approximations. Then, we'll use them to approximate things!

(a) We're going to start with approximating the function $f(x) = \sin(x)$ centered at $x = 0$. Let's choose to look at a 5th degree polynomial.

This means we'll need to find the first five derivatives of $f(x) = \sin(x)$. Then, we'll evaluate our function and the five derivatives at the center. After that, we can divide by the relevant factorial in order to create the coefficients of our polynomial.

Fill out the following chart to produce these coefficients.

Table 9.1.2 Coefficients for Polynomial Approximation

k	$f^{(k)}(x)$	$f^{(k)}(a)$	$\frac{f^{(k)}(a)}{k!}$
$k = 0$	$f(x) = \sin(x)$	$f(0) =$	
$k = 1$	$f'(x) =$	$f'(0) =$	
$k = 2$	$f''(x) =$	$f''(0) =$	
$k = 3$	$f'''(x) =$	$f'''(0) =$	
$k = 4$	$f^{(4)}(x) =$	$f^{(4)}(0) =$	
$k = 5$	$f^{(5)}(x) =$	$f^{(5)}(0) =$	

(b) Now we can use these coefficients to construct the polynomial! These coefficients should all be on power functions in the form $(x - a)^k$ for $k = 0, 1, ..., 5$. These (added together) will form your polynomial, $p_5(x)$.

(c) Approximate $f(1) = \sin(1)$ using your polynomial.

(d) Let's repeat this for another function. Let's build a 5th degree polynomial approximation of $g(x) = e^x$ centered at $x = 0$. We can construct the coefficients in the same way.

Table 9.1.3 Coefficients for Polynomial Approximation

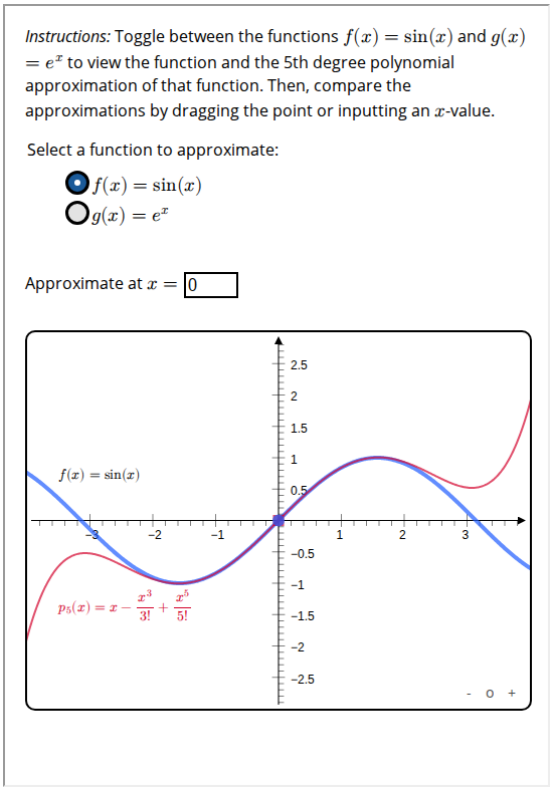
k	$g^{(k)}(x)$	$g^{(k)}(a)$	$\frac{g^{(k)}(a)}{k!}$
$k=0$	$g(x)=e^x$	$g(0)=$	
$k=1$	$g'(x)=$	$g'(0)=$	
$k=2$	$g''(x)=$	$g''(0)=$	
$k=3$	$g'''(x)=$	$g'''(0)=$	
$k=4$	$g^{(4)}(x)=$	$g^{(4)}(0)=$	
$k=5$	$g^{(5)}(x)=$	$g^{(5)}(0)=$	

(e) And now, again, we can use these coefficients to construct the polynomial! These coefficients should all be on power functions in the form $(x-a)^k$ for $k=0,1,...,5$. These (added together) will form your polynomial, $p_5(x)$.

(f) Approximate $g(-3)=\dfrac{1}{e^3}$ using your polynomial.

Activity 9.1.4 How Good Are Our Approximations?

We're going to think more carefully about our approximations of $\sin(1)$ and e^{-3} from Activity 9.1.2. In order for us to do this, let's visualize the function and the 5th degree polynomial for it.



Standalone
In Context
Embed

(a) How good of a job did the polynomial approximation do when approximating $\sin(1)$? How can you tell, visually?

(b) How good of a job did the polynomial approximation do when approximating e^{-3}? How can you tell, visually?

(c) How does the relationship between the "center" and the x-value that we're approximating at impact the accuracy of our approximation?

(d) How do you think you could make these approximations better (without changing the center)?

Activity 9.1.5 Partial Sums of What?

Let's revisit our 5th degree polynomial approximations from Activity 9.1.2.

$$\sin(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$$
$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!}$$

These approximations work well for x-values that are close to 0, but we will not be more formal than that.

(a) Make a conjecture about what the 7th degree polynomial approximations are for each of these functions.

What about the 15th degree?

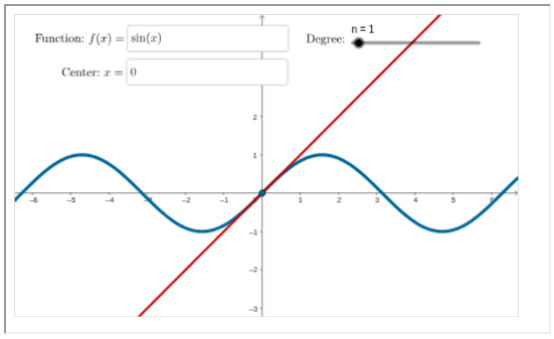
(b) Make a conjecture about what the general formula would be for these terms. If you were to write these out using summation notation, what would they look like?

(c) Why does the polynomial approximation for the sine function only have odd-exponent terms?

(d) Make a conjecture about what a polynomial approximation for $f(x) = \cos(x)$ centered at $x = 0$ would be.

Activity 9.1.6 How Do These Polynomials Converge?

We're going to end here by thinking about these polynomials as some partial sum from an infinite series. If there is an infinite series, we should be prepared to think about convergence!
We're going to think about convergence in the same way that we have already: as an end behavior limit of the partial sums. So let's spend our time investigating this end behavior by visualizing polynomial approximations as the degree increases.



Standalone
In Context
Embed

(a) What happens to the polynomial approximation of $\sin(x)$ centered at $x = 0$ as the degree $n \to \infty$?

(b) Does this behavior change if we centered our approximation elsewhere?

(c) What happens to the polynomial approximation of e^x centered at $x = 0$ as the degree $n \to \infty$?

(d) Does this behavior change if we centered our approximation elsewhere?

(e) What happens to the polynomial approximation of $\cos(x)$ centered at $x = 0$ as the degree $n \to \infty$?

(f) Does this behavior change if we centered our approximation elsewhere?

(g) What happens to the polynomial approximation of $\ln(x)$ centered at $x = 2$ as the degree $n \to \infty$?

(h) Describe the difference in what you're seeing with the log function compared to the other functions we've thought about. Describe how the polynomial approximations converge: do they converge to the log function? How? More importantly, *where*?

(i) Does this behavior change if we centered our approximation elsewhere?

Practice Problems

1. Consider a polynomial approximation of some function $f(x)$ centered at 0. Explain how the degree of the polynomial approximation impacts the accuracy of the approximation.

2. Consider a function $f(x)$ with a a second degree polynomial, $p(x) = \frac{f''(a)}{2}(x-a)^2 + f'(a)(x-a) + f(a)$. We want to approximate $f(b)$ using the polynomial by evaluating $p(b)$. Explain how the location of b in relation to the center a impacts the accuracy of the approximation.

3. Say we want to approximate $\sqrt[3]{7}$ using a third degree polynomial.

(a) Which function, $f(x)$, will you use to build the polynomial?

(b) What is an appropriate value for the center, a, for your polynomial? Why is this value appropriate?

(c) Build the polynomial approximation, $p(x)$, to approximate $f(x)$.

(d) Evaluate your polynomial at $x = 7$. What does this tell you about the value of $\sqrt[3]{7}$?

4. Consider the function $f(x) = \sin(x)$. Our goal is to approximate $\sin(0.023)$.

(a) Build a third degree polynomial approximation of $f(x)$ centered at 0 and use it to approximate $\sin(0.023)$.

(b) Build a fifth degree polynomial approximation of $f(x)$ centered at 0 and use it to approximate $\sin(0.023)$.

(c) What would you do to get a better approximation of the value $\sin(0.023)$?

(d) What could you do to get the exact value of $\sin(0.023)$ using polynomials?

5. For each of the following functions, find a fourth degree polynomial to approximate the given value.

(a) Approximate $\sqrt{1.05}$ using $f(x) = \sqrt{1+x}$.

(b) Approximate $\ln(0.95)$ using $f(x) = \ln(x)$.

(c) Approximate $e^{-0.12}$ using $f(x) = e^{-x}$.

(d) Approximate $\cos(-0.15)$ using $f(x) = \cos(x)$.

(e) Approximate $\frac{1}{1.02}$ using $f(x) = \frac{1}{x+1}$.

9.2 Power Series Convergence

Activity 9.2.1 Polynomial Division.

We're going to do some fiddling with polynomials, and hopefully use this as a bridge to connect how we think of polynomials and power series with how we think about our traditional infinite series and the notions of convergence that we've already built.

(a) We're going to factor some polynomials, but we might end up using some division. First, we'll confirm some factors that we already know.

$$x^2 - 1 = (x-1)(x+1)$$

We'll confirm this by using division.

$$x - 1 \overline{\big|\ x^2 \qquad -1}$$

(b) Now let's factor $x^3 - 1$. If the factors for this polynomial isn't as familiar, it might be helpful to know that $(x-1)$ is also a factor of $x^3 - 1$.

$$x - 1 \overline{\big|\ x^3 \qquad -1}$$

(c) Let's try another one. Complete the following division.

$$x - 1 \overline{\big|\ x^4 \qquad -1}$$

(d) Can you generalize this? Find the formula for $\dfrac{x^n - 1}{x - 1}$ for some positive integer n.

$$x - 1 \overline{\big|\ x^n \qquad -1}$$

(e) Now that we have good evidence that

$$\sum_{k=0}^{\infty} x^k = \frac{x^n - 1}{x - 1},$$

We can apply a limit as $n \to \infty$.

$$\begin{aligned} \sum_{k=0}^{\infty} x^k &= \lim_{n\to\infty} \sum_{k=0}^{n} x^k \\ &= \lim_{n\to\infty} \frac{x^n - 1}{x - 1} \end{aligned}$$

For what values of x will this limit exist?

(f) Revisit Theorem 8.6.7 Geometric Series Convergence Criteria. Is there any difference for what we've just done compared to this result that we already know?

Activity 9.2.2 Some Power Series and their Convergence.

Let's consider a couple of power series and apply some convergence tests to them in order for us to find out how it might converge.

(a) Consider the power series:

$$\sum_{k=1}^{\infty} \frac{(x-1)^k}{k^2}.$$

In order for us to apply the Ratio Test, we'll actually need to consider the positive-term version:

$$\sum_{k=1}^{\infty} \frac{|x-1|^k}{k^2}.$$

Apply the Ratio Test. What do you get in the limit of the ratio between terms?

(b) What kind of result from the Ratio Test guarantees convergence for the series? What are the x-values that guarantee convergence?

(c) The Ratio Test is inconclusive when the limit is equal to 1. What x-values does this happen at? Consider the power series evaluated at each of these x-values. Do these series converge or diverge?

(d) Consider the power series:

$$\sum_{k=1}^{\infty} \frac{(x-1)^k}{\sqrt{k}}.$$

Find the interval of x-values for which this series converges and test the end points of the interval in the same way as earlier. If it differs, explain why.

Practice Problems

1. Describe the following terms in your own words, and explain how they relate to a power series.

(a) Center.

(b) Interval of Convergence.

(c) Radius of Convergence.

2. Does a power series ever diverge everywhere? Explain why not, using an example.

3. For a power series that converges for $\left|\frac{x-3}{4}\right| < 1$ and diverges elsewhere, find the center of the power series, the radius of convergence, and interval of convergence.

4. For a power series that converges for $|4(x+1)| < 1$ and diverges elsewhere, find the center of the power series, the radius of convergence, and interval of convergence.

5. For each of the following power series, determine the interval of convergence. Be specific on which convergence test you are applying and how you are finding the interval of convergence.

(a) $\displaystyle\sum_{k=0}^{\infty}(3x)^k$

(b) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{x}{3}\right)^k$

(c) $\displaystyle\sum_{k=0}^{\infty}\frac{(x-3)^k}{k!}$

(d) $\displaystyle\sum_{k=0}^{\infty}\frac{(x-3)^k}{k}$

(e) $\displaystyle\sum_{k=1}^{\infty}\frac{2^k(x-1)^k}{k}$

(f) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{x}{k}\right)^k$

(g) $\displaystyle\sum_{k=0}^{\infty}(-1)^k(3x-4)^k$

(h) $\displaystyle\sum_{k=0}^{\infty}\left(\frac{3x}{5}\right)^{k/2}$

(i) $\displaystyle\sum_{k=1}^{\infty}\frac{(x-1)^k 3^k}{(k+1)^k}$

9.3 How to Build Taylor Series

Activity 9.3.1 Constructing a Taylor Series Directly.

Let's work on this! To be honest, the goal here is to realize how difficult this can be.
Let's consider the function $f(x) = \tan(x)$. We're going to build a Taylor polynomial approximating $f(x)$ centered at $x = 0$.

(a) We're going to build a 4th degree polynomial approximating $f(x) = \tan(x)$. In order to do this, we'll need to find the coefficients for the 5 terms.

Table 9.3.2 Coefficients for Polynomial Approximation

k	$f^{(k)}(x)$	$f^{(k)}(a)$	$\frac{f^{(k)}(a)}{k!}$
$k = 0$	$f(x) = \tan(x)$	$f(0) =$	
$k = 1$	$f'(x) =$	$f'(0) =$	
$k = 2$	$f''(x) =$	$f''(0) =$	
$k = 3$	$f'''(x) =$	$f'''(0) =$	
$k = 4$	$f^{(4)}(x) =$	$f^{(4)}(0) =$	
$k = 5$	$f^{(5)}(x) =$	$f^{(5)}(0) =$	

(b) Now we can use these coefficients to construct the polynomial! These coefficients should all be on power functions in the form $(x - a)^k$ for $k = 0, 1, ..., 5$. These (added together) will form your polynomial, $p_5(x)$.

(c) Can you find a general explicit formula for the terms? If you're having a hard time with this, what could you do to try to make this easier? Why is this difficult?

We're going to stop there. This series is famously difficult to work with. We won't generate the whole Taylor series this way, and this strategy really doesn't work well for functions like this (where the derivatives are pretty annoying to find).

Activity 9.3.3 Connecting to Another Series.

Let's start with some "known" series.

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \qquad \text{on } (-1,1)$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \qquad \text{on } (-\infty,\infty)$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \qquad \text{on } (-\infty,\infty)$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \qquad \text{on } (-\infty,\infty)$$

We're going to build a Taylor series for a logarithmic function:

$$y = \ln(1+x).$$

(a) In order to start, we're going to find a Taylor series for $g(x) = \dfrac{1}{1+x}$. Which of the "known" series above is this most similar to? How is it different?

(b) Find a Taylor series for $g(x) = \dfrac{1}{1+x}$ by using Theorem 9.2.6. Can you connect this series to the known series by multiplying or composing something?

(c) Notice that the function we're trying to get a Taylor series for, $y = \ln(1+x)$, is an antiderivative of $g(x) = \dfrac{1}{1+x}$. Antidifferentiate the series you found above, using Theorem 9.2.7 Differentiating and Integrating Power Series. We know that at $x = 0$, $y = 0$, so use this information to find specific constant that we need when we antidifferentiate.

(d) What is the interval of convergence for this new series? How does that relate to the interval of convergence of the "known" series you started with?

Activity 9.3.4 Connecting to Yet Another Series.

Here, again, are some *updated* "known" series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \qquad \text{on } (-1,1)$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \qquad \text{on } (-\infty,\infty)$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \qquad \text{on } (-\infty,\infty)$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \qquad \text{on } (-\infty,\infty)$$

$$\ln(1+x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{k+1}}{k+1} \qquad \text{on } (-1,1]$$

We're going to build a Taylor series for the inverse tangent function:

$$y = \tan^{-1}(x).$$

(a) In order to start, we're going to find a Taylor series for $g(x) = \dfrac{1}{1+x^2}$. Which of the "known" series above is this most similar to? How is it different?

(b) Find a Taylor series for $g(x) = \dfrac{1}{1+x^2}$ by using Theorem 9.2.6. Can you connect this series to the known series by multiplying or composing something?

(c) Notice that the function we're trying to get a Taylor series for, $y = \tan^{-1}(x)$, is an antiderivative of $g(x) = \dfrac{1}{1+x^2}$. Antidifferentiate the series you found above, using Theorem 9.2.7 Differentiating and Integrating Power Series. We know that at $x = 0$, $y = 0$, so use this information to find specific constant that we need when we antidifferentiate.

(d) What is the interval of convergence for this new series? How does that relate to the interval of convergence of the "known" series you started with?

Practice Problems

1. What is the difference between a Taylor Polynomial approximation for $f(x)$ centered at $x = a$ and the Taylor Series for $f(x)$ centered at $x = a$?

2. Given the geometric power series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$ for $|x| < 1$, find the Taylor series representations for the following functions. Make sure to explain, for each, how it connects to the geometric Taylor series. Also, report the interval of convergence for each.

(a) $\frac{x^3}{1-x}$

(b) $\frac{x}{1+x^4}$

(c) $\frac{1}{x^2-1} = -\frac{1}{1-x^2}$

(d) $\frac{1}{(1-x)^2}$

3. or each of the following functions, explain how they are related to $f(x) = \sin(x)$. Then, use the connections you describe to find the Taylor Series for the function given by relating it to the Taylor Series for $\sin(x)$.

(a) $\sin(x^2)$

(b) $2x\sin(x)$

(c) $2x\cos(x^2)$

4. Find a Taylor Series for each of the following functions. Report the center and interval of convergence for each.

(a) $f(x) = xe^x$

(b) $f(x) = e^{\sqrt{x}}$

(c) $f(x) = e^{x+2}$

(d) $f(x) = \cos(2x)$

(e) $f(x) = x^3\tan^{-1}(x^2)$

(f) $f(x) = \sqrt[5]{x^7}\ln(x+1)$

9.4 How to Use Taylor Series

Activity 9.4.1 Approximating π and Other Values.

Let's pick a couple of values that are just based on functions with Known Taylor Series and approximate them.

(a) If we note that $\tan^{-1}(1) = \dfrac{\pi}{4}$, then we can say that $\pi = 4\tan^{-1}(1)$.

Find an infinite series that converges to $\tan^{-1}(1)$, and use that to construct a series that converges to π.

(b) If we note that $\dfrac{1}{e} = e^{-1}$, then find an infinite series that converges to $\dfrac{1}{e}$.

(c) Find an infinite series that converges to $\sin(1)$.

(d) Note that each of these three series are alternating series! We can approximate them using a partial sum, and then we can get an error bound for that partial sum by looking at the size of the next term in the infinite series (Theorem 8.5.7 Approximations of Alternating Series).

Approximate the value of each infinite series using a partial sum with the same number of terms. You can pick the number of terms you use. Then, compare the margin of error. Which approximations are most/least accurate? Why do you think that is?

Activity 9.4.2 Integrating using Taylor Series.

Let's consider the definite integral:

$$\int_{x=0}^{x=1} \sin(x^2)\, dx.$$

We know this function is continuous everywhere, and so it is continuous on the interval $[0, 1]$. So the Fundamental Theorem of Calculus applies. All we need to do is find an antiderivative, evaluate it at the end-points of the interval, and subtract.
Great! Easy!
Except that we can't write out the family of antiderivatives for this function.
So let's convert over to the Taylor series context and solve our problem there.

(a) Create a Taylor series for the function $f(x) = \sin(x^2)$.

(b) Find the interval of convergence for this Taylor series. Specifically, we want to know if this series converges to $\sin(x^2)$ on the interval $[0, 1]$, since that's the interval we're integrating over.

(c) If $f(x) = \sin(x^2)$, find a Taylor series representation of $F(x)$, an antiderivative of $f(x) = \sin(x^2)$.

(d) Evaluate this antidervative at the endpoints and subtract:

$$F(1) - F(0).$$

(e) Use Theorem 8.5.7 Approximations of Alternating Series to approximate the value of $\int_{x=0}^{x=1} \sin(x^2)\, dx$ with a maximum error of 10^{-5}.

Practice Problems

1. We can define hyperbolic trigonometric functions in the following way:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$
$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

(a) Find a Taylor Series representation for $\sinh(x)$ centered at $x = 0$.

(b) Find a Taylor Series representation for $\cosh(x)$ centered at $x = 0$.

(c) Show that $\sinh(x) + \cosh(x) = e^x$ using the definition of the hyperbolic trigonometric functions above.

(d) Show that $\sinh(x) + \cosh(x) = e^x$ using the Taylor Series.

2. Use a Taylor series to estimate the value of the integral

$$\int_{x=0}^{x=1} e^{-x^2}\, dx$$

within a maximum error margin of 0.001.

9.5 Exploration: Power Series

Exploration 9.5.1 The Basel Problem.

We're going to follow along this proof, and for most of the time it might seem like we're fiddling with unrelated results and functions. By the end, we'll tie what we're doing together to show that:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

(a) First, consider the function $f(x) = \begin{cases} \dfrac{\sin(x)}{x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases}$.

Find a Taylor series representation for this function.

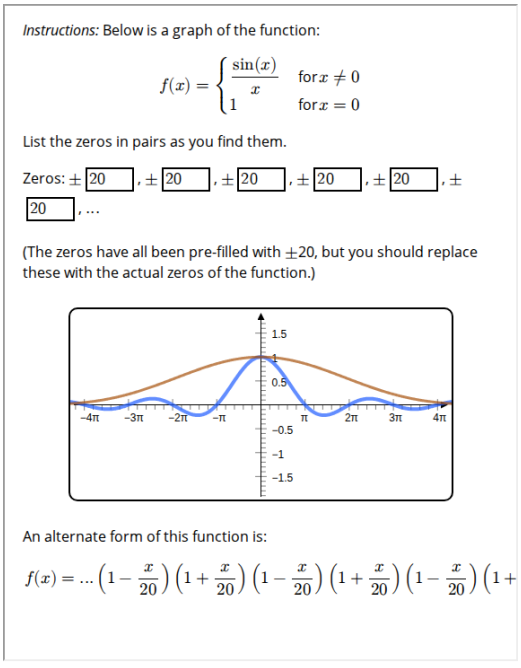
(b) We're going to consider this function again, but in a different form. We're going to think about this infinite polynomial, but we'll try to write it by thinking about the factors that we expect to see, specifically when we think of the zeros of the function.

So first, note that the zeros of this function, $\dfrac{\sin(x)}{x}$ must occur when $\sin(x) = 0$. What are the x-values that make $\sin(x) = 0$?

(c) We'll use these zeros to write out the function's factors, but we're also going to make use of the fact that $f(0) = 1$. This means that if $x = c$ is a zero of the function, then the corresponding factor should be written as $\left(1 - \dfrac{x}{c}\right)$. So, if $c_1, c_2, c_3, c_4, \ldots$ are all zeros, then we should write the function as:

$$f(x) = \left(1 - \frac{x}{c_1}\right)\left(1 - \frac{x}{c_2}\right)\left(1 - \frac{x}{c_3}\right)\left(1 - \frac{x}{c_4}\right)\ldots$$

Use the zeros of the $\dfrac{\sin(x)}{x}$ function that you found earlier to write out the factors of this.


Instructions: Below is a graph of the function:

$$f(x) = \begin{cases} \dfrac{\sin(x)}{x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases}$$

List the zeros in pairs as you find them.

Zeros: $\pm$ 20 , $\pm$ 20 , $\pm$ 20 , $\pm$ 20 , $\pm$ 20 , $\pm$ 20 , ...

(The zeros have all been pre-filled with ± 20, but you should replace these with the actual zeros of the function.)

An alternate form of this function is:

$f(x) = \ldots\left(1 - \frac{x}{20}\right)\left(1 + \frac{x}{20}\right)\left(1 - \frac{x}{20}\right)\left(1 + \frac{x}{20}\right)\left(1 - \frac{x}{20}\right)\left(1 +$

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(d) Note that we have pairs of factors that are in the form $\left(1-\frac{x}{c}\right)\left(1+\frac{x}{c}\right)$. This is hopefully a very recognizable factoring pattern! It's the difference of squares! What do you get when you multiply these?

This should allow you to write your factored function from above in a slightly different way, by combining the pairs of factors using this multiplication. What does this look like?

(e) We are almost complete. We now have represented the function

$$f(x)\begin{cases}\frac{\sin(x)}{x} & \text{for } x \neq 0\\ 1 & \text{for } x = 0\end{cases}$$

as an infinite sum

$$1-\frac{x^2}{3!}+\frac{x^4}{5!}-\frac{x^6}{7!}+-...$$

and also as an infinite product

$$\left(1-\frac{x^2}{\pi^2}\right)\left(1-\frac{x^2}{4\pi^2}\right)\left(1-\frac{x^2}{9\pi^2}\right)...$$

We can note that the constant term is 1 in each, since it is the product of all of the constant terms in the infinite product.

We know that the quadratic terms should be $-\frac{x^2}{6}$ (from the infinite sum). Can you find what the quadratic term from the infinite product would be?

(f) You should have shown that:

$$-\frac{x^2}{6}=-\frac{x^2}{\pi^2}-\frac{x^2}{4\pi^2}-\frac{x^2}{9\pi^2}-...$$

What is the sum of $1+\frac{1}{4}+\frac{1}{9}+...\frac{1}{k^2}+...$?

(g) Can you use this same strategy to find our what the sum $\sum_{k=1}^{\infty}\frac{1}{k^4}$ converges to?

Note that we're really asking what the degree 4 terms are in each representation of the function $f(x)$.

Exploration 9.5.2 Normal Distribution.

(a) Write an integral that represents the probability of a random variable with a normal distribution taking on a value up to 2 standard deviations above the mean.

(b) Based on the Empirical Rule, what do you expect the value of this integral to be?

(c) Construct a Taylor series representing the function $e^{-\frac{x^2}{2}}$.

(d) In order to estimate your integral, we will need to antidifferentiate $e^{-\frac{x^2}{2}}$. This function does not have an **elementary antiderivative**, meaning we cannot express an antiderivative in terms of the traditional functions we have named and the traditional operations we have defined. Antidifferentiate your Taylor series representation of $e^{-\frac{x^2}{2}}$, and call this $F(x)$.

(e) Evaluate your integral using this new representation an antiderivative, $F(x)$.

(f) This is an alternating series, and so we can approximate this value to whatever degree of accuracy we'd like using Theorem 8.5.7 Approximations of Alternating Series.

How many terms do you need, in order to approximate this probability within 0.01?

(g) Construct, evaluate, and approximate the integral representing the probability that a normally distributed random variable takes on a value between 1 and 1.5 standard deviations above the mean (using the same number of terms as above).

Chapter 10

Parametric and Polar Curves

10.1 Parametric Curves

Activity 10.1.1 A Refresher on Functions.

This is mostly a discussion activity. There's not a lot to *do*, other than think and share your thoughts with others. If you're not in a classroom setting, this can be hard! Find someone else with a relatively similar math background, and talk about this with them!

(a) When you think of a function, what do you think of? What do you imagine in your mind when you hear the word "function" (in a mathematical context)? Is it based on a definition, a visualization, an example, or some other more loosely connected set of phrases?

(b) What do you remember about variables? For instance, what is their purpose, their relationship, or their use in defining or visualizing a function?

(c) How do you visualize a function? What does the picture represent, or how would you explain it to others? What are the important aspects of the visualization that we might use to control what these pictures look like, or what do we try to keep consistent when we visualize these functions? Why do we do this?

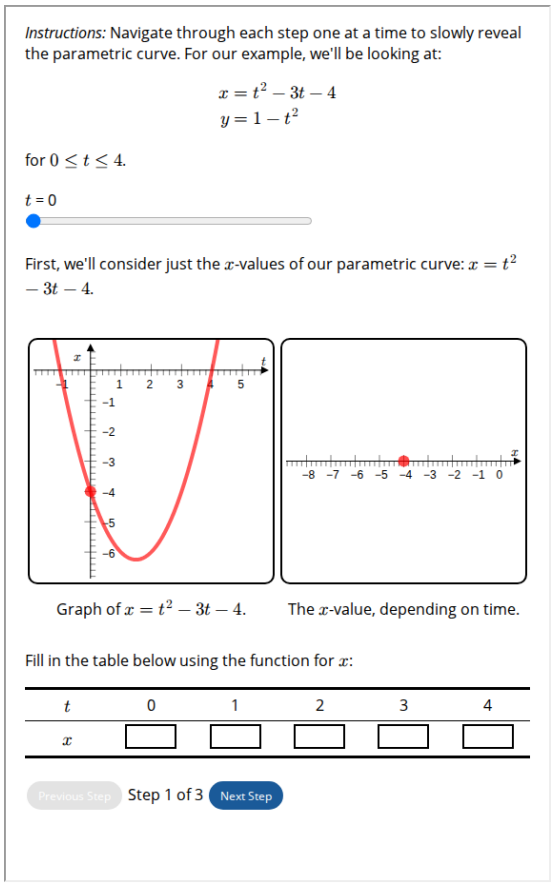
(d) How might you define a "point on the function?"

Activity 10.1.2 Our First Parametric Curve.

We'll begin by exploring the parametrically defined curve:

$$x = t^2 - 3t - 4$$
$$y = 1 - t^2$$

for $0 \leq t \leq 4$.
For each task, we'll use the following visualization:



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(a) In Step 1, how does the graph of $x = t^2 - 3t - 4$ give us information about the horizontal movement of our parametric curve?

(b) Fill in the table with the x-values at different values of t. Describe the horizontal motion of the parametric curve using this information.

(c) In Step 2, how does the graph of $y = 1 - t^2$ give us information about the vertical movement of our parametric curve?

(d) Fill in the table with the y-values at different values of t. Describe the vertical motion of the parametric curve using this information.

(e) In Step 3, confirm that the point follows the path you constructed.

Activity 10.1.3 Explore Some Curves.

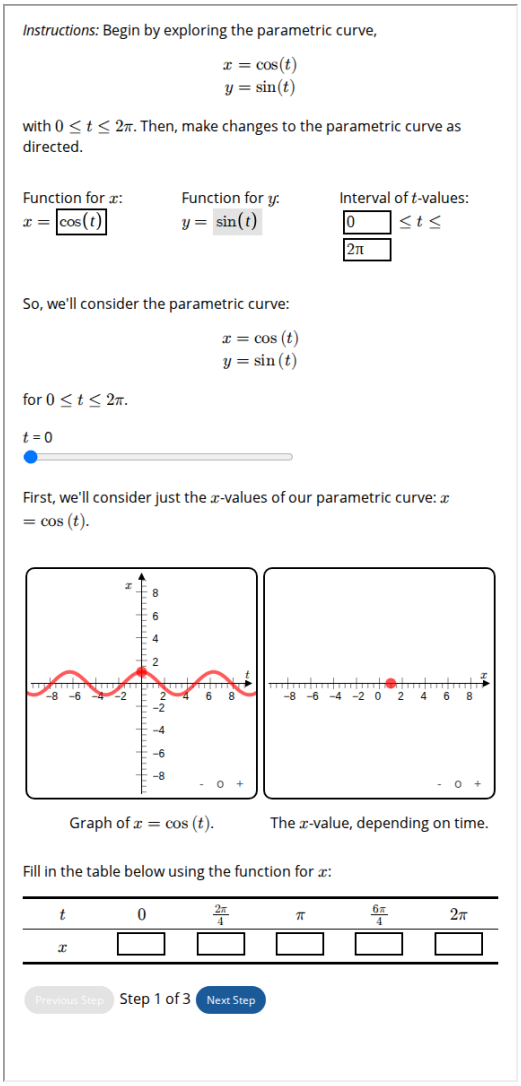
We'll start off with a family of parametric curves, where we can change the functions and hopefully predict some of the curve's behavior.

(a) Let's start by thinking about the parametric curve:

$$x = \cos(t)$$
$$y = \sin(t)$$

for $0 \leq t \leq 2\pi$.

Before you visualize any of this, think about the behavior of the x-values and y-values. What do you think they'll do? What will this curve look like?Visualize the movement of the x-values, the y-values, and the complete parametric curve using the graphing tool below.



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(b) What do you think would happen when we multiply the function for x by a factor of 3?

In the graphing tool above, change the function for x to:

$$x = 3\cos(t).$$

Explain the changes you're seeing.

(c) What do you think would happen when we multiply the input of the cosine function by 3? In the graphing tool above, change the function for x to:

$$x = \cos(3t).$$

Explain the changes you're seeing.

(d) Explore the parametric curve:

$$x = \cos(3t)$$
$$y = \sin(2t)$$

for $0 \leq t \leq 2\pi$.

Explain to yourself why the curve looks the way that it does.

(e) Create a parametric curve for a circle, but reverse the orientation.

(f) Create a parametric curve for a circle centered at the point $(2, -1)$ with radius 3.

(g) What do you think the following parametric curve will look like:

$$x = \cos^3(t)$$
$$y = \sin(t)$$

for $0 \leq t \leq 2\pi$?

Activity 10.1.4 Eliminate the Parameter.

Consider the parametric curve

$$x = \sqrt{t+1}$$
$$y = \frac{t}{3} + 1$$

for $0 \leq t \leq 15$.

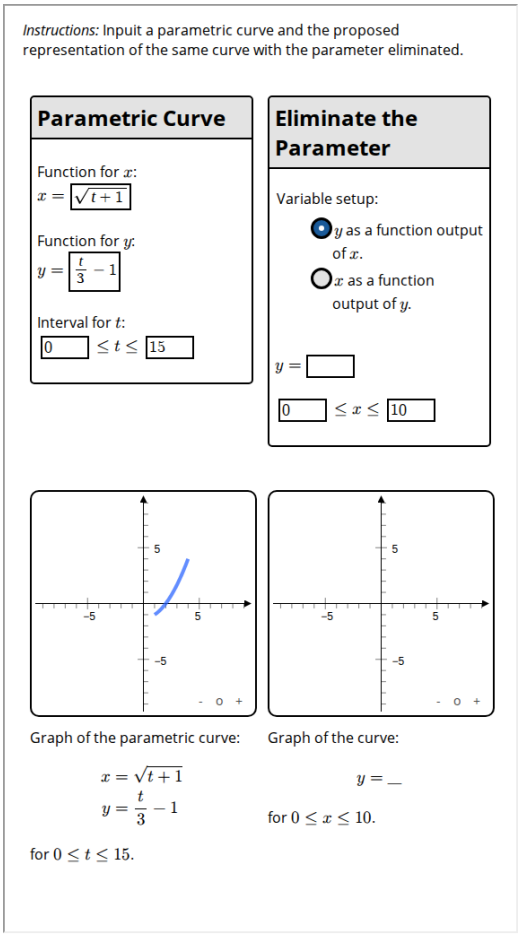
(a) Solve $x = \sqrt{t+1}$ for t.

(b) Substitute this expression for t into $y = \frac{t}{3} - 1$.

(c) Start over, and instead solve $y = \frac{t}{3} - 1$ for t.

(d) Now substitute this expression for t into $x = \sqrt{t+1}$.

(e) Use the graphing tool below to check to see if these curves are all the same. You'll need to find appropriate intervals of x-values or y-values along the way.



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Activity 10.1.5 Eliminate More Parameters.

For each parametrically-defined curve, eliminate the parameter to create a function $y = f(x)$ or $x = f(y)$.
Feel free to use the interactive graphing tool in Activity 10.1.7.

(a) $\begin{aligned} x &= t^2 - 1 \\ y &= (3t+1)^3 \end{aligned}$ for $0 \leq t \leq 2$.

(b) $\begin{aligned} x &= \ln(t+1) \\ y &= \sin(t) \end{aligned}$ for $0 \leq t$.

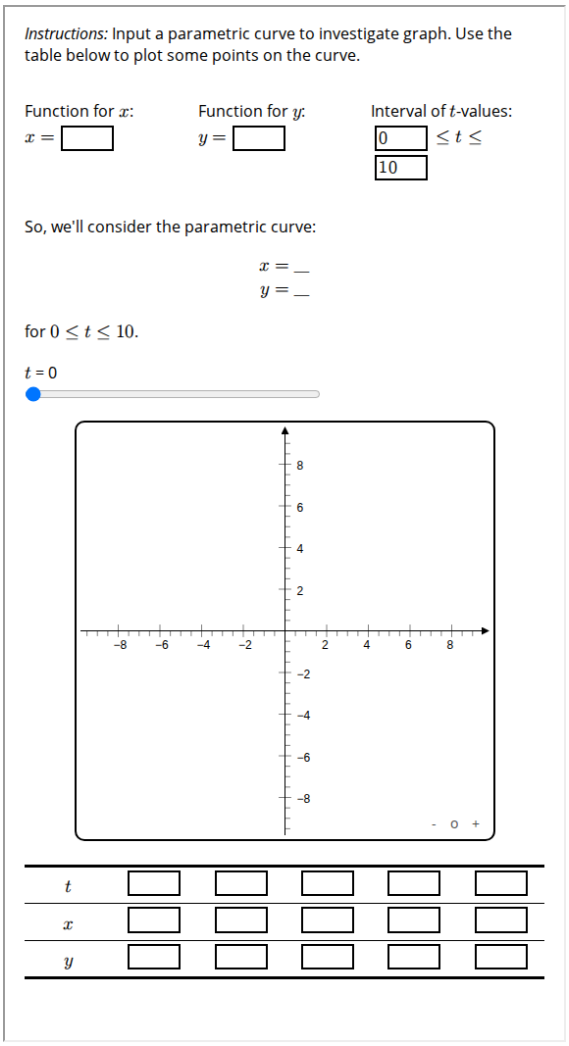
(c) $\begin{aligned} x &= \cos^2(t) \\ y &= (t+1)^{3/2} \end{aligned}$ for $0 \leq t$.

(d) $\begin{aligned} x &= \tan^{-1}(x) \\ y &= t^2 \end{aligned}$ for $-2 \leq t \leq 2$.

Practice Problems

1. Explain how a curve written in parametric form is different than one written in standard function notation.

2. For each of the following parametric curves, plot the curves in the graphing utility below. Then, find at least 5 different points on the curve, plot them, and describe the curve.



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(a) $\begin{aligned} x &= t - 4 \\ y &= 3 + t - t^2 \end{aligned}$ for $0 \le t \le 10$.

(b) $\begin{aligned} x &= t + 3 \\ y &= 5 - 2t \end{aligned}$ for $0 \le t \le 5$.

(c) $\begin{aligned} x &= 5\cos^2(t) \\ y &= 5\sin^2(t) \end{aligned}$ for $0 \le t \le 2\pi$.

(d) $\begin{aligned} x &= \sqrt{t+4} \\ y &= 5t + 1 \end{aligned}$ for $0 \le t \le 5$.

3. Eliminate the parameter, t, in each of the parametric curves, expressing the curve either as a function $y = f(x)$ or $x = f(y)$.

(a) $\begin{aligned} x &= t - 4 \\ y &= 3 + t - t^2 \end{aligned}$ for $0 \leq t \leq 10$.

(b) $\begin{aligned} x &= t + 3 \\ y &= 5 - 2t \end{aligned}$ for $0 \leq t \leq 5$.

(c) $\begin{aligned} x &= 5\cos^2(t) \\ y &= 5\sin^2(t) \end{aligned}$ for $0 \leq t \leq 2\pi$.

(d) $\begin{aligned} x &= \sqrt{t+4} \\ y &= 5t + 1 \end{aligned}$ for $0 \leq t \leq 5$.

4. In Activity 10.1.6 Explore Some Curves, we saw the we can define a circle parametrically by

$$\begin{aligned} x &= \cos(t) \\ y &= \sin(t) \end{aligned}$$

for $0 \leq t \leq 2\pi$.

(a) Remind yourself why this parametric definition makes sense, based on how we think about trig functions on a unit circle.

(b) Eliminate the parameter to create an equation with x and y related to each other.

(c) How does the function $f(t) = \cos(5t)$ compare to the function $g(t) = \cos(t)$? How about $\ell(t) = \sin(5t)$ compared to the function $m(t) = \sin(t)$?

(d) The parametric curve

$$\begin{aligned} x &= \cos(5t) \\ y &= \sin(5t) \end{aligned}$$

for $0 \leq t \leq 2\pi$ looks exactly the same as the unit circle we defined earlier.

How is it different? What does the coefficient on t do?

(e) Create a parametric curve that looks like the unit circle but has an opposite orientation (direction).

10.2 Polar Curves

Activity 10.2.1 A Refresher on Graphing Functions.

This is, again, mostly a discussion activity. There's not much to *do*, but you should think about these questions and share your thoughts with others.

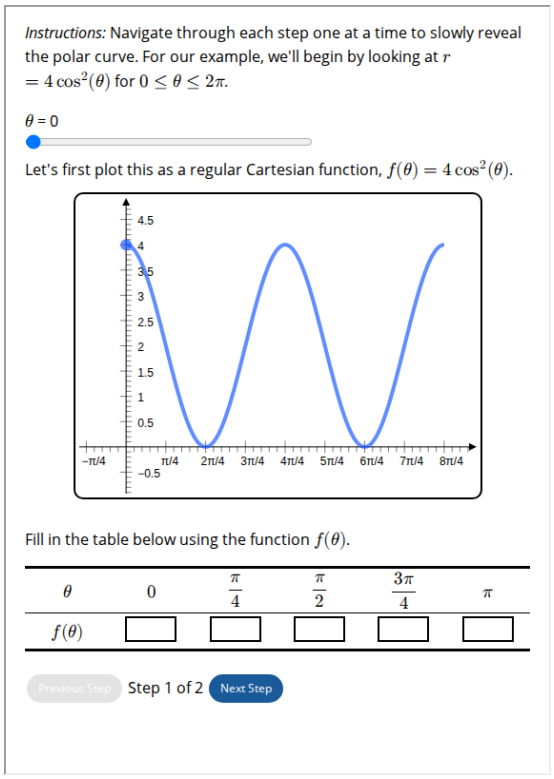
(a) How do we plot a typical function (in the form $y = f(x)$) on a coordinate grid? How do you think about the different variables, and how do we use them to plot a point?

(b) How do we represent the coordinates of a point? What does each component of a point tell us about its location?

(c) Does the "grid" that we use match with the way you might think about locations of things in your life? Where else do humans use a grid like this to describe locations?

Activity 10.2.2 First Polar Curve.

We'll begin by exploring the polar curve, $r = 4\cos^2(\theta)$ for $0 \leq \theta \leq 2\pi$.
For each task, we'll use the following visualization:



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(a) In Step 1, explain the shape and behavior of the function based on your knowledge of trigonometric functions. How do you know, based on the function being specifically $4\cos^2(\theta)$ that it will look like this?

(b) Fill in the table with the function outputs $f(\theta)$ for different values of θ.

(c) In Step 2, why do the points that we filled in end up where they are on the polar curve? Explain how we use the coordinates you input in the previous step to plot these polar points.

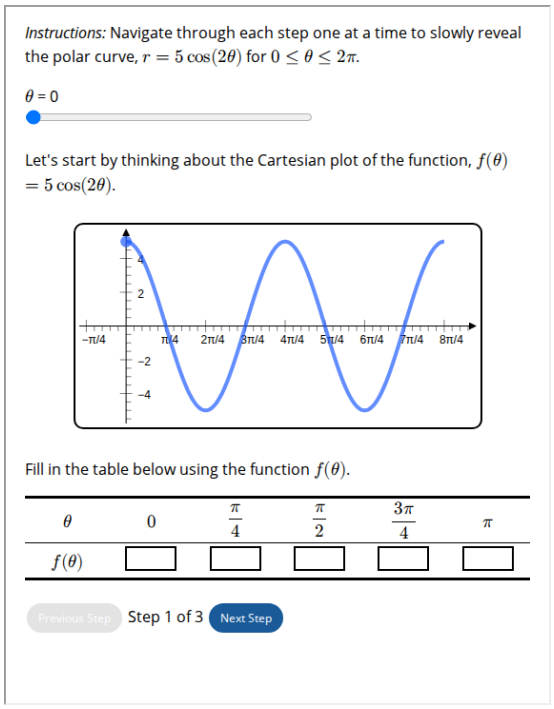
(d) Explain why the polar point has this shape/behavior. Refer back to explanation of the Cartesian curve in Step 1, and explain how these graphical features manifest themselves in the polar curve.

(e) Fill in the table with the r outputs at different values of θ. Describe why these points are plotted where they are, including the quadrants and distances from the origin/pole.

(f) Why do the two points $(4, 0)$ and $(4, 2\pi)$ seem to overlap in the graph? Why does it look like this is only a single point?

Activity 10.2.3 A Polar Flower.

We'll now explore a polar flower, $r = 5\cos(2\theta)$.
For each task, we'll use the following visualization:



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(a) In Step 1, explain how the changes to the cosine function (multiplying the input, θ by 2 and multiplying the cosine function by 5) changes the graph from a standard cosine function's graph. Where are the zeros, compared to where they typically are?

(b) Fill in the table with the function outputs $f(\theta)$ for different values of θ. Note that the points have been colored based on the sign on the output.

(c) In Step 2, explain why there only looks to be one grey point (when $r = 0$, when there were more plotted in Step 1.)

(d) Think about how θ changes. As θ increases, where should the points be traveling? Describe this in terms of quadrants on the graph if you'd like. Does this match the movement of the points on the polar function?

(e) Fill in the table with the r outputs at different values of θ. Describe why these points are plotted where they are, including the quadrants and distances from the origin/pole.

(f) In Step 3, confirm that the polar curve $r = 5\cos(2\theta)$ doesn't follow the counter-clockwise path that we expect when θ increases.

Compare the path that $r = 5\cos(2\theta)$ travels with the path that $r = |5\cos(2\theta)|$ travels as θ increases. What is happening to the curve when $r = 5\cos(2\theta)$ is negative?

(g) What, then, do you think happens to a polar point (r, θ) when $r < 0$? How do we plot it?

Activity 10.2.4 Converting the Coordinates of a Point.

We're going to think about the two coordinate systems (and converting between them) by considering a point and collecting information. For simplicity, we'll visualize a point in the first quadrant.

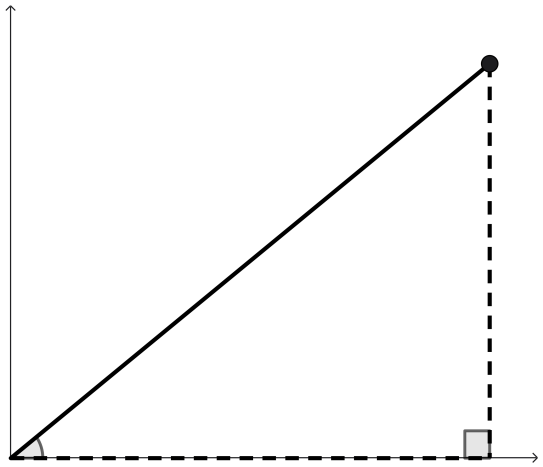


Figure 10.2.5

(a) If the point is written in Cartesian coordinates, (x, y), label x and y in the image.

(b) If the point is written in polar coordinates, (r, θ), label θ and r in the image.

(c) Assume that we know the values of x and y. Come up with an equation linking r, x, and y in order for us to find the value of r.

(d) Now come up with some equation linking θ, x, and y in order for us to find the value of θ.

(e) Now imagine that we know the values of r and θ instead. Come up with an equation connecting x, r, and θ in order for us to find the value of x.

(f) Come up with an equation connecting y, r, and θ in order for us to find the value of y.

Activity 10.2.6 Converting Coordinates.

Fill in the following table, converting points written in the Cartesian coordinate system to the polar coordinate system and points in the polar coordinate system to the Cartesian coordinate system. It might be helpful to sketch the point first!
Then, find a second polar representation for each point.

Table 10.2.7

Cartesian	**Polar**	**Second Polar**
	$\left(3, \frac{7\pi}{4}\right)$	
	$\left(\frac{1}{2}, \frac{\pi}{6}\right)$	
$(-4, 4\sqrt{3})$		
$(-3, -5)$		
	$\left(-1, -\frac{2\pi}{3}\right)$	

Practice Problems

1. Explain how a curve written in polar coordinates is different than one written in Cartesian coordinates.

2. Convert the following Cartesian coordinates to polar coordinates.

(a) $(4, 0)$

(b) $(-5, 5)$

(c) $(3, -3)$

(d) $(-\sqrt{3}, -1)$

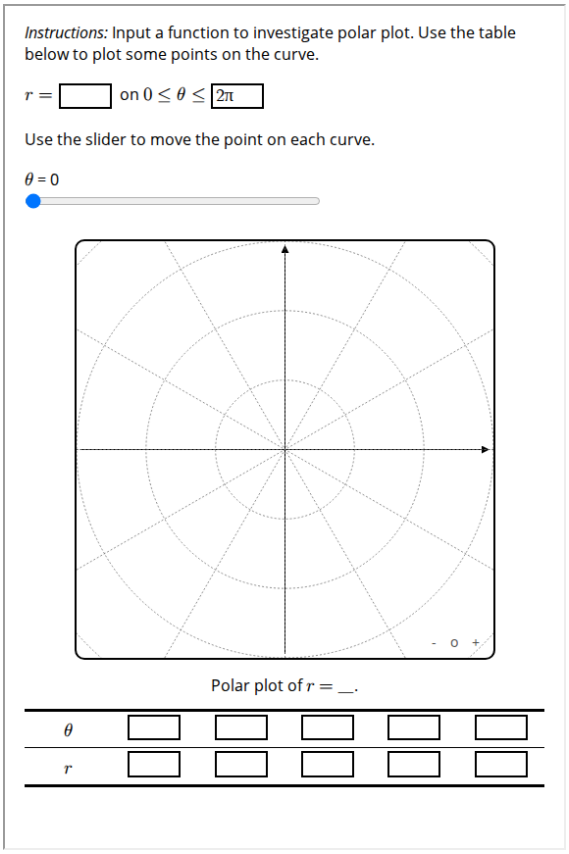
(e) $(0, -3)$

(f) $(-4, 4\sqrt{3})$

3. For each point above, find a second, different, representation of the same point. Explain how you did this.

4. Explain why there are an infinite number of possible polar coordinates corresponding with each pair of Cartesian coordinates above.

5. For each of the following polar curves, plot the curve in the graphing utility below. Then, find at least 5 different points on the polar curve, plot them, and describe the behavior of the curve.



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(a) $r = \sin(\theta) + \cos(\theta)$

(b) $r = \sin^2(\theta)$

(c) $r = \cos(4\theta) + \theta$

(d) $r = \sin^2(-\theta)$

(e) $r = \theta^2 \sin(\theta)$

10.3 Calculus with Parametric Curves

Activity 10.3.1 Slopes on a Cycloid.

Consider the cycloid formed by a circle with radius $r = 1$:

$$x = t - \sin(t)$$
$$y = 1 - \cos(t)$$

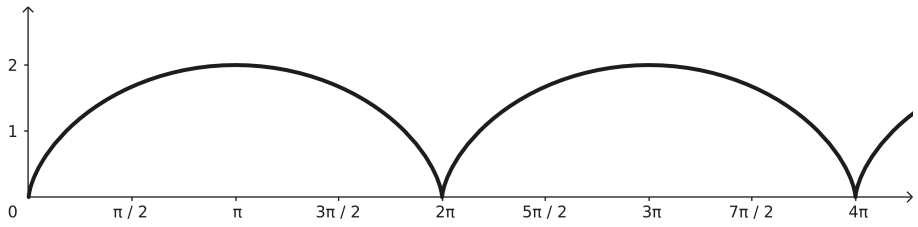


Figure 10.3.2 The cycloid.

(a) First, make a conjecture: where do you think the cycloid will have horizontal tangent lines? What about vertical tangent lines, or other points where the derivative doesn't exist?

(b) Find $\frac{dy}{dt}$ and $\frac{dx}{dt}$, and construct $\frac{dy}{dx}$.

(c) Find the values of t where $\frac{dy}{dx} = 0$.

(d) Find the values of t where $\frac{dy}{dx}$ does not exist.

(e) Use these values of t to find the points on the cycloid where there are horizontal tangent lines or points where the derivative doesn't exist. Do they match what you conjectured?

Activity 10.3.3 Building a Parametric Arc Length Formula.

Before we begin, it might be helpful to remind yourself how we build the arclength formula in Section 6.5 Arc Length and Surface Area.

Let's drop back in to that derivation with some small changes. Because we're dealing with a parametric curve, we'll use t as our input variable, and so we'll eventually need a Δt to turn into a differential dt in the integral. This means that we'll have non-uniform Δx distances, since they'll be based on the changes in the input, t. So we'll actually use Δx_k to represent the change in the x-variable on the kth subinterval.

So, our length on the kth subinterval is:

$$\ell_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

(a) In order to build the arc length formula in Section 6.5, we factored out $\sqrt{\Delta x^2}$ in order to end up with a Δx that turned into the differential dx in the integral. This time, factor out a Δt^2 under the square root.

(b) Now, create a Riemann sum and let $n \to \infty$ (and, correspondingly, $\Delta t \to 0$). Note that you should end up with some differentials instead of deltas!

Practice Problems

1. Explain how (and why) we implement the chain rule to find $\frac{dy}{dx}$ for the parametric curve:

$$x = f(t)$$
$$y = g(t)$$

2. For each of the parametric curves below, find the slope of the line tangent to the curve at the indicated value.

(a) The parametric curve

$$x = 4 - 3t^2$$
$$y = 6 + \frac{1}{t}$$

at $t = 2$.

(b) The parametric curve

$$x = t - \sin(t)$$
$$y = 1 - \cos(t)$$

ar $t = \frac{3\pi}{4}$.

(c) The parametric curve

$$x = \sin^3(t)$$
$$y = \cos(t)$$

at $t = \frac{7\pi}{6}$.

3. Why, when we calculate the area under a parametric curve, do we need to consider x'?

4. Set up and evaluate a definite integral to calculate the following areas.

(a) The area bounded by the parametric curve

$$x = \sin^3(t)$$
$$y = \cos(t)$$

in the first quadrant and the x-axis.

(b) The area bounded underneath the curve

$$x = \frac{t^2}{10}$$
$$y = 1 - \cos(t)$$

from $t = 0$ to $t = 2\pi$.

(c) The area bounded between the parametric curve

$$x = 5 - t + \frac{1}{t+1}$$
$$y = t^2$$

and the x-axis from $t = 0$ to $t = 1$.

5. Explain how the formula for finding the length of a parametric curve is similar (and different) from the arc length formula in Definition 6.5.7.

6. Set up an integral representing the length of the following parametric curves.

(a) The parametric curve

$$x = \sqrt{t} + 1$$
$$y = t^2 - t + 3$$

on the interval $0 \leq t \leq 4$.

(b) The parametric curve

$$x = \cos(t)$$
$$y = \sin(t)$$

on the interval $0 \leq t \leq 2\pi$.

(c) The parametric curve

$$x = \cos^3(t)$$
$$y = \sin^3(t)$$

on the interval $\leq t \leq \pi$.

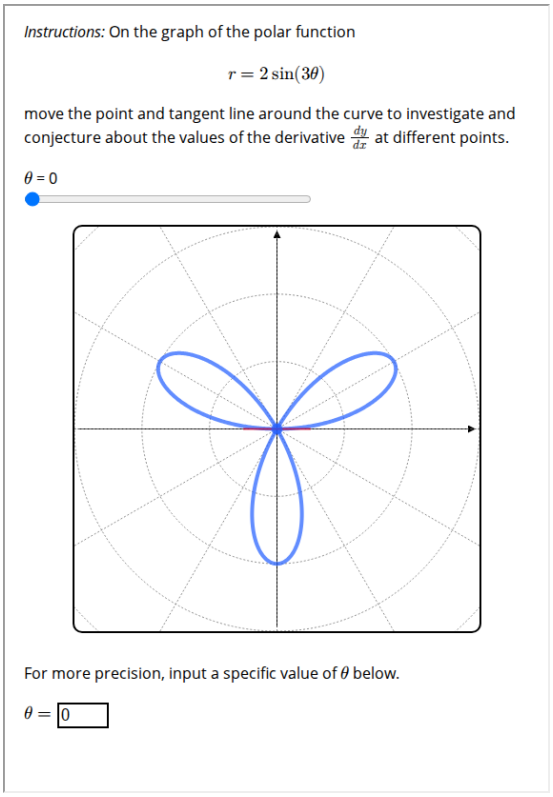
10.4 Calculus with Polar Curves

Activity 10.4.1 Slopes on a Polar Flower.

Consider the polar flower defined by

$$r = 2\sin(3\theta)$$

for $0 \leq \theta \leq \pi$. For this activity, refer to the graphing utility below.



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(a) How many points on this polar flower do you think have horizontal tangent lines?

(b) How many points on this polar flower do you think have vertical tangent lines?

(c) Make a conjecture about the slope of the tangent line when $\theta = \dfrac{5\pi}{6}$ by looking at the graph only.

(d) Find an expression for $\frac{dy}{dx}$.

(e) Pick one of the values of θ where you thought this curve might have a horizontal or vertical tangent line. Evaluate $\frac{dy}{dx}$ at that value, and check to see if you were correct.

(f) Evaluate $\frac{dy}{dx}$ when $\theta = \dfrac{5\pi}{6}$. How close was your guess at the slope?

Activity 10.4.2 Slice-and-Sum for Polar Areas.

Remember slice-and-sum? Here's the general procedure:

- We'll divide up the interval of inputs into n (equally-sized) subintervals. We'll pick one value from each subinterval.
- One each slice, we'll use the representative input variable to calculate the measurement we're interested in. In this case, it's an area.
- Add up the areas of the n slices, and the sum should be an approximation of the actual area we're interested in.
- Use a limit as $n \to \infty$ to produce an integral our of a Riemann sum!

(a) The area of a sector of a circle with angle α is:

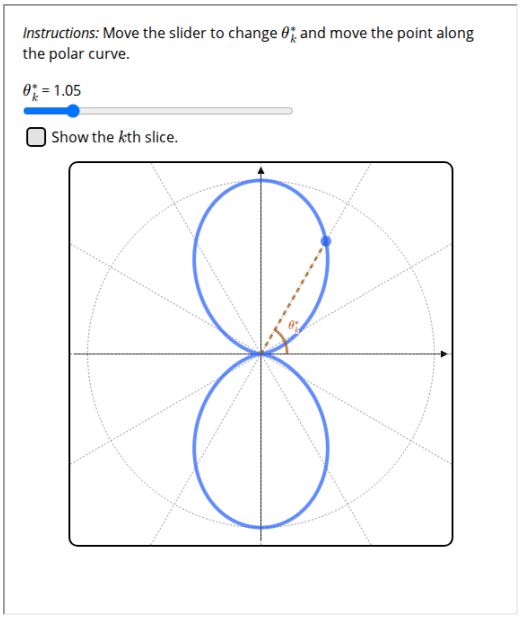
$$A = \frac{1}{2}\alpha r^2.$$

Convince yourselves that this is true, using appropriate test angles for:

- A circle.
- A semicircle.
- A quarter of a circle.

(b) The reason that we're thinking about sectors of circles is that when we slice an interval of angles in a polar coordinate system, a bunch of rays extending away from the pole.

When do this for a polar curve, we'll pick a point in the subinterval and assume that the whole subinterval has a constant function output, just like we did with the rectangle. Take a look at the graphing utility below, and get a feel for how we'll find the kth slice.



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(c) Calculate the area of the kth slice, A_k.

(d) Add these up to approximate the total area.

(e) Introduce a limit to find an integral formula for the area trapped inside a polar curve from $\theta = \alpha$ to $\theta = \beta$.